Ответы и официальные разборы

Том 1. Официальные оригиналы

Оригинальная вёрстка сохранена вместе с формулами, рисунками и авторскими пояснениями. В сборниках могут встречаться и условия, и ответы, и критерии оценивания.

Соответствие отдельной задаче проверяйте по олимпиаде, году, классу, номеру и варианту. Если этих данных недостаточно, документ не объявляется точным решением конкретной задачи.

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APMO 2026 – Problems and Solutions

Problem 1. Find all quadruples of positive integers (a, p, m, o) with $o \geq 2$ such that

$$a! + p! = m^o + 26.$$

Answer: $(a, p, m, o) \in \{(4, 3, 2, 2), (3, 4, 2, 2), (5, 3, 10, 2), (3, 5, 10, 2)\}$.

Solution.

Without loss of generality, assume that $a \geq p$. First, observe that if $p \geq 4$, then $m^o = a! + p! - 26 \equiv 2 \pmod 4$, which forces $o = 1$, a contradiction. Hence, it suffices to consider the remaining cases $p = 1, 2, 3$.

Case $p = 1$**.** Then $a! = m^o + 25$. Clearly $a \geq 5$, which implies that $5 \mid m$ and m is odd. If $o = 2$, or more generally if o is even, then $m^o + 25 \equiv 2 \pmod 4$, which is impossible. Thus assume $o \geq 3$. Then $a! = m^o + 25 \equiv 5^2 \pmod{5^3}$, which yields $10 \leq a \leq 14$. Note that $5^3 \mid m^o = a! - 25$. However, a direct check shows that $5^3 \nmid a! - 25$ for all $10 \leq a \leq 14$, which is again a contradiction.

Case $p = 2$**.** Then $a! = m^o + 24$. Clearly $a \geq 5$. If $a = 5$, then $m^o = 120 - 24 = 96$, which is impossible. If $a \geq 6$, then $m^o = a! - 24 \equiv 3 \pmod 9$, forcing $o = 1$, again a contradiction.

Case $p = 3$**.** Then $a! = m^o + 20$. Clearly $a \geq 4$. If $a = 4$, then $m^o = 4! - 20 = 4$, so $m = o = 2$. If $a = 5$, then $m^o = 5! - 20 = 100$, giving $m = 10$ and $o = 2$. Now suppose that $a \geq 6$. Then $2^4 \mid a! = m^o + 20$, which implies that m is even. However, if $4 \mid m$ or $o \geq 3$, then $m^o + 20 \equiv 4 \pmod 8$, and if $o = 2$ and $m \equiv 2 \pmod 4$, then $m^o + 20 = m^2 + 20 \equiv 4 + 20 \equiv 8 \pmod{16}$, contradiction.

Therefore, the solutions are

$$(a, p, m, o) \in \{(4, 3, 2, 2), (3, 4, 2, 2), (5, 3, 10, 2), (3, 5, 10, 2)\}.$$

Problem 2. Let $n \geq 2$ be an integer. Let $A_1, A_2, \cdots, A_{2^n}$ be the 2^n subsets of an n-element set in some order. Prove that

$$|A_1 \cap A_2| + |A_2 \cap A_3| + \cdots + |A_{2^n-1} \cap A_{2^n}| + |A_{2^n} \cap A_1| \geq 2^{n-2}.$$

Solution 1.

Indices are reduced modulo 2^n throughout. Set, $a_i := |A_i \cap A_{i+1}|$ and $c_i = |A_{i-1}| + 2|A_i| + |A_{i+1}| - 2n$. Note that $a_i = |A_i| + |A_{i+1}| - |A_i \cup A_{i+1}| \geq |A_i| + |A_{i+1}| - n$. It follows that $a_i + a_{i-1} \geq c_i$. In particular, if $c_i \geq 1$ then

$$a_{i-1} + a_i \geq \frac{1}{2}(c_i + 1). \tag{1}$$

Notice that, because $a_i, a_{i-1} \geq 0$, (1) is trivially satisfied whenever $c_i < 0$. We now verify that (1) also holds in the case $c_i = 0$, and hence in all cases. Indeed, if either $a_i > |A_i| + |A_{i+1}| - n$ or $a_{i-1} > |A_i| + |A_{i+1}| - n$ then $a_i + a_{i-1} > c_i$, thus $a_i + a_{i-1} \geq 1$ and (1) holds. Otherwise, we must have $|A_{i-1} \cup A_i| = |A_i \cup A_{i+1}| = n$, which implies that both A_{i-1} and A_{i+1} contain the complement of A_i. Since they are distinct one of them strictly contain the complement of A_i, therefore has a non-empty intersection with A_i. Consequently, $a_i + a_{i-1} \geq 1$, which shows that (1) holds in this case as well.

Consequently,

$$\begin{aligned}\sum_{i=1}^{2^n} a_i &= \frac{1}{2}\sum_{i=1}^{2^n}(a_{i-1} + a_i) \geq \frac{1}{4}\sum_{i=1}^{2^n}(c_i + 1) = \frac{1}{4}\sum_{i=1}^{2^n} c_i + 2^{n-2} \\ &= \sum_{i=1}^{2^n} |A_i| - 2^n n + 2^{n-2} = \sum_{k=0}^{n} k\binom{n}{k} - 2^{n-1}n + 2^{n-2} = 2^{n-2}.\end{aligned}$$

The last equality follows from the well-know identity $\sum_{k=0}^{n} k\binom{n}{k} = 2^{n-1}n$.

Solution 2.

Let indices of the A_i be taken modulo 2^n. Consider any element x of the n-element set. The number of i for which x is in both A_i and A_{i+1} equals the number of i for which x is in neither of those sets (since there are equal numbers of sets containing and not containing x, and equal numbers of transitions in each direction, from containing to not containing x and vice versa, as we go through the cycle of sets). Thus the required sum equals

$$\frac{1}{2}\sum_{i=1}^{2^n}\Big(|A_i \cap A_{i+1}| + |\overline{A_i} \cap \overline{A_{i+1}}|\Big). \tag{2}$$

A term of this sum is zero only for two sets that are complements, so no two consecutive terms of it can be zero, so the sum is at least 2^{n-2}.

Comments. It is possible to construct examples for which equality holds..

Problem 3 . Let $\mathbb{R}_+$ denote the set of positive real numbers. Determine all the functions $f : \mathbb{R}_+ \to \mathbb{R}$ such that for all $x, y, z \in \mathbb{R}_+$,

$$|x - y| < |y - z| \text{ if and only if } |f(x) - f(y)| < |f(y) - f(z)|. \tag{1}$$

Answer: All functions of the form $f(x) = ax + b$ for fixed real numbers a, b with $a \neq 0$.

Solution.

It is clear that (1) holds for all functions f of the form $f(x) = ax + b$ with $a \neq 0$. We shall show that these are the only ones that satisfy (1). Suppose (1) holds for f.

Note that, from (1) it's obvious that $|x - y| > |y - z|$ if and only if $|f(x) - f(y)| > |f(y) - f(z)|$. It follows that

$$|x - y| = |z - y| \text{ implies } |f(x) - f(y)| = |f(z) - f(y)|. \tag{2}$$

Claim 1. *f is one-to-one.*

Proof. Suppose that $f(y) = f(z)$ for some $y \neq z$. By choosing $x = y$, (1) implies $|f(x) - f(y)| < |f(y) - f(z)| = 0$, a contradiction.

□

Claim 2. *For all $x, z \in \mathbb{R}_+$, we have*

$$f\left(\frac{x+z}{2}\right) = \frac{f(x) + f(z)}{2}. \tag{3}$$

Proof. Since (3) holds trivially for $x = z$, we assume that $x \neq z$. By Claim 1, $f(x) \neq f(z)$. Set $y = \frac{x+z}{2}$, then $|y-z| = |x-z|$. It follows from (2) that $|f(y)-f(x)| = |f(y)-f(z)|$, which implies $f(y) = \frac{f(x)+f(z)}{2}$ (since $f(x) \neq f(z)$). □

Claim 3. *f is monotone.*

Proof. Assume f is not monotone. Then there exist $y < x < z$ such that $f(x) - f(y)$ and $f(z) - f(x)$ have different signs. Hence $|f(y)-f(z)| < \max(|f(x)-f(y)|, |f(z)-f(x)|)$, whereas (1) gives $|f(y)-f(z)| > \max(|f(x) - f(y)|, |f(z) - f(x)|)$, a contradiction. □

Note that for any constants $a \neq 0, b \in \mathbb{R}$, the function f satisfies (1) if and only if $af + b$ does. Thus, without loss of generality, we may assume that $f(1) = 1$ and $f(2) = 2$. We will prove that this forces $f(x) = x$ for all $x \in \mathbb{R}_+$. Given $f(1) = 1, f(2) = 2$, (3) yields $f(3) = 3$, $f\left(\frac{3}{2}\right) = \frac{3}{2}$, $f\left(\frac{1}{2}\right) = \frac{1}{2}$. By a

straightforward induction argument, it follows that $f(n) = n$ for all positive integers n, and more generally $f\left(\frac{n}{2^k}\right) = \frac{n}{2^k}$ for all positive integers n and non-negative integers k. Since the numbers of the form $\frac{n}{2^k} = \frac{n}{2^k}$ is a dense subset of $\mathbb{R}_+$, the monotonicity of f implies that $f(x) = x$ for all $x \in \mathbb{R}_+$.

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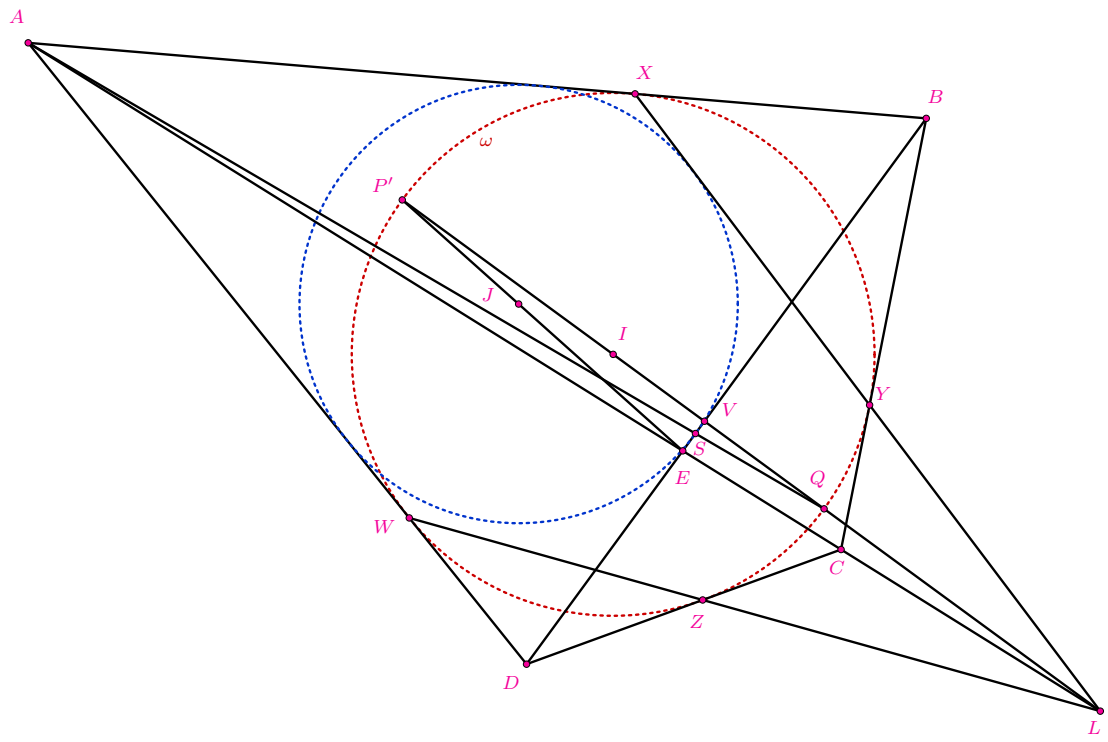
Problem 4. Let $ABCD$ be a quadrilateral with an incircle ω of centre I. The diagonals AC and BD intersect at E. Let J be the incentre of triangle ABD. The extension of the ray EJ intersects ω at P. Prove that $PI \perp BD$.

Solution.



Draw the diameter $P'Q$ of ω such that $P'Q \perp BD$ and $P'A < QA$. We claim that P' actually coincide with P, in other words, E, J, P' are collinear.

Let S be the contact point of the incircle (J) of the triangle ABD with BD. Since A is the center of the external homothety of the circles (I), (J), the points A, Q, S are collinear.

Let X, Y, Z, W be the contact points of AB, BC, CD, DA with (I) respectively. Let L be the intersection point of XY and ZW. By Pascal's theorem (for $\begin{pmatrix} W & Y & Z \\ X & Z & Y \end{pmatrix}$ and $\begin{pmatrix} W & X & Y \\ X & W & Z \end{pmatrix}$), L lies on AC. It follows that BD is the polar of L with respect to the circle (I). Therefore, if we denote by V the intersection point of IL and BD then P', V, Q, L form a harmonic range. We deduce that

$$\overline{IV} \cdot \overline{IL} = IP'^2 = IQ^2 \quad \text{and} \quad \overline{VI} \cdot \overline{VL} = \overline{VP'} \cdot \overline{VQ} = \mathcal{P}_{V/(I)}.$$

It follows that $\frac{\overline{VI}}{\overline{VP'}} = \frac{\overline{VQ}}{\overline{VL}}$, and hence $\frac{\overline{P'I}}{\overline{P'V}} = \frac{\overline{LQ}}{\overline{LV}}$. Consequently,

$$(IV, P'\infty) = (QV, L\infty).$$

Since inversion conserves cross-ratios, we obtain $(QV, L\infty) = (QL, VI)$. Hence $(IV, P'\infty) = (QL, VI)$, and therefore $J(IV, P'\infty) = A(QL, VI)$, which gives

$$J(IV, P'S) = A(SE, VJ).$$

On the other hand, we have

$$A(SE, VJ) = J(SE, VA) = J(AV, ES) = J(IV, ES).$$

Combining these equalities yields $J(IV, P'S) = J(IV, ES)$, which implies that the points E, J, P' are collinear, as desired.

Problem 5. Let n be a positive integer. There exist $n(n+1)$ rooms arranged in a $(n+1) \times n$ grid. For each two adjacent rooms, there is a door between them. Find the number of ways to choose a subset of doors and lock them such that there are two rooms S and G satisfying the following properties:

(i) S is in the first row and G is in the $(n+1)$-th row.

(ii) One can reach G from S only through unlocked doors.

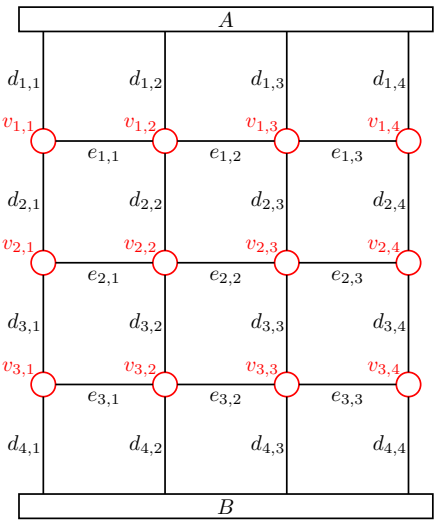
Answer: 2^{2n^2-2}.

Solution.

Without loss of generality, we may assume that every door between two rooms in the first row and the $(n+1)$-th row does not be locked. Consider the following graph $\mathcal{G}$:

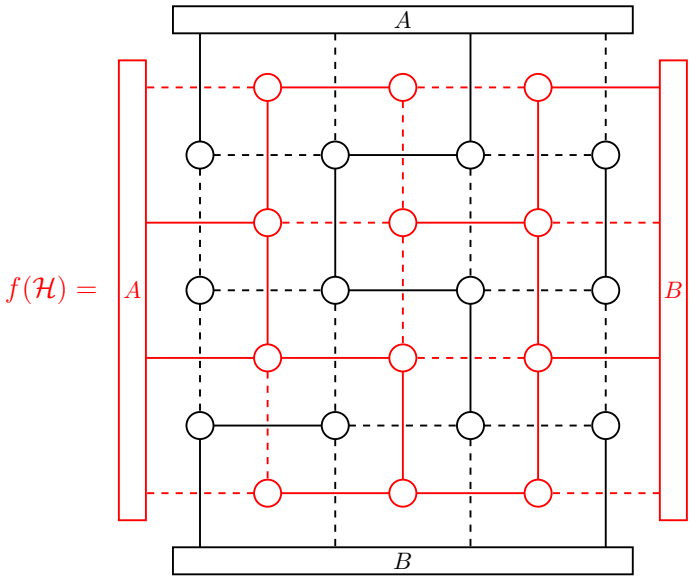
- The vertex set of $\mathcal{G}$ consists of $n(n-1)+2$ points A, B, $v_{i,j}$ $(1 \leqq i \leqq n-1, 1 \leqq j \leqq n)$.
- The edge set of $\mathcal{G}$ consists of $n^2+(n-1)^2$ edges $d_{i,j}$ $(1 \leqq i,j \leqq n)$, $e_{i,j}$ $(1 \leqq i,j \leqq n-1)$.
- $d_{1,j}$ joins A and $v_{1,j}$ for all $1 \leqq j \leqq n$, and $d_{n,j}$ joins $v_{n-1,j}$ and B for all $1 \leqq j \leqq n$.
- $d_{i,j}$ joins $v_{i-1,j}$ and $v_{i,j}$ for all integers i, j with $2 \leqq i \leqq n-1$ and $1 \leqq j \leqq n$.
- $e_{i,j}$ joins $v_{i,j}$ and $v_{i,j+1}$ for all integers i, j with $1 \leqq i,j \leqq n-1$.

For example, the following figure illustrates $\mathcal{G}$ for $n=4$.

$\mathcal{G} =$ 

Rooms and doors between them correspond to vertices and edges of the graph as follows. Rooms on the first row correspond to A, and that on the $(n+1)$-th row correspond to B. For all integers i, j with $2 \leqq i \leqq n$ and $1 \leqq j \leqq n$, the room in row i and column j corresponds to $v_{i-1,j}$. Each door between two rooms corresponds to the edge between the corresponding vertex. The problem can be reframed as determining the number of spanning subgraphs $\mathcal{H}$ which has a path from A to B.

Let $S(\mathcal{G})$ be the set of spanning subgraphs of $\mathcal{G}$. Define $f : S(\mathcal{G}) \to S(\mathcal{G})$ such that $d_{i,j} \in f(\mathcal{H})$ if and only if $d_{j,i} \notin \mathcal{H}$ and $e_{i,j} \in f(\mathcal{H})$ if and only if $e_{j,i} \notin \mathcal{H}$. For example, the following figure illustrates $\mathcal{H}$ and $f(\mathcal{H})$ for $n = 4$. Black lines correspond to $\mathcal{H}$ and red lines correspond to $f(\mathcal{H})$.



We say that $\mathcal{H} \in S(\mathcal{G})$ is *achievable* if there is a path from A to B. Now, we prove two lemmas.

Lemma 1. *For each $\mathcal{H} \in S(\mathcal{G})$, we have $f(f(\mathcal{H})) = \mathcal{H}$. In particular, f is a bijection from $S(\mathcal{G})$ to $S(\mathcal{G})$.*

Proof. For all integers i, j with $1 \leqq i, j \leqq n$, we have $d_{i,j} \in \mathcal{H}$ if and only if $d_{j,i} \notin f(\mathcal{H})$, which is equivalent to $d_{i,j} \in f(f(\mathcal{H}))$. Similarly, $e_{i,j} \in \mathcal{H}$ if and only if $e_{i,j} \in f(f(\mathcal{H}))$ for all integers i, j with $1 \leqq i, j \leqq n-1$. Since $\mathcal{H}$ and $f(f(\mathcal{H}))$ are spanning subgraphs of $\mathcal{G}$, we obtain $\mathcal{H} = f(f(\mathcal{H}))$. □

Lemma 2. *$\mathcal{H}$ is achievable if and only if $f(\mathcal{H})$ is not achievable.*

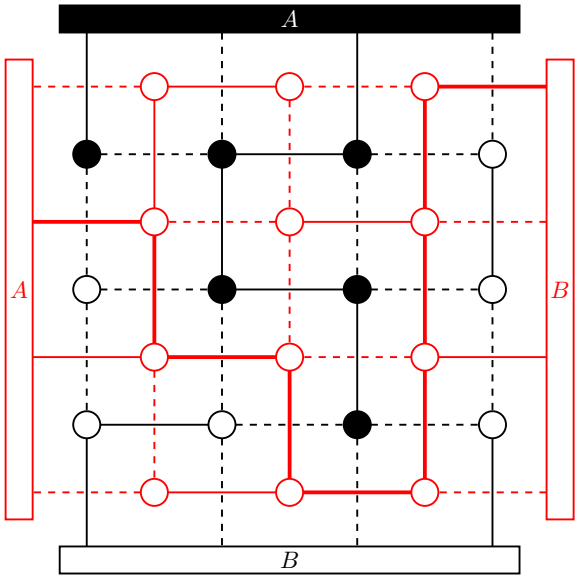
Proof. Let us denote $\overline{d_{i,j}} = d_{j,i}$ and $\overline{e_{i,j}} = e_{j,i}$. Suppose $\mathcal{H}$ is achievable. Then there are edges $p_1, p_2, \ldots, p_k$ such that $(p_1, p_2, \ldots, p_k)$ is a path from A to B. If there is a path from A to B in $f(\mathcal{H})$, then the path must pass through one of $\overline{p_1}, \overline{p_2}, \ldots, \overline{p_k}$. This contradicts the fact that $\overline{p_1}, \overline{p_2}, \ldots, \overline{p_k}$ are not contained in $f(\mathcal{H})$.

Conversely, assume that $\mathcal{H}$ is not achievable. Let V be the vertex set of $\mathcal{G}$ and let V_A be the connected component of $\mathcal{H}$ containing A. Denote by E the subset of the edge set of $\mathcal{G}$ such that $e \in E$ if one of the

two endpoints of e is in V_A and the other is in $V \setminus V_A$. Then every edge $e \in E$ is not an edge of $\mathcal{H}$. Define a set $\overline{E}$ of edges of $\mathcal{G}$ by

$$\overline{E} = \{\overline{d_{i,j}}; d_{i,j} \in E\} \cup \{\overline{e_{i,j}}; e_{i,j} \in E\}.$$

Then $\overline{E}$ is a subset of the edge set of $f(\mathcal{H})$, and we can find a path from A to B in $\overline{E}$ by connecting these edges. Hence, $f(\mathcal{H})$ is achievable. In the following figure, black dots represent vertices in V_A and white dots represent vertices in $V \setminus V_A$. The thick red line is a path from A to B in $f(\mathcal{H})$.



□

Using these lemmas, we see that

$$\begin{aligned}\#\{\mathcal{H} \in S(\mathcal{G}) \mid \mathcal{H} \text{ is achievable}\} &= \#\{\mathcal{H} \in S(\mathcal{G}) \mid f(\mathcal{H}) \text{ is achievable}\} \\ &= \#S(\mathcal{G}) - \#\{\mathcal{H} \in S(\mathcal{G}) \mid f(\mathcal{H}) \text{ is not achievable}\} \\ &= 2^{2n^2-2n+1} - \#\{\mathcal{H} \in S(\mathcal{G}) \mid \mathcal{H} \text{ is achievable}\}\end{aligned}$$

Therefore, $\#\{\mathcal{H} \in S(\mathcal{G}) \mid \mathcal{H} \text{ is achievable}\} = 2^{2n^2-2n}$ and considering the state of doors in the first row and the $(n+1)$-th row, we obtain that the answer is $2^{2n^2-2n} \cdot 2^{2n-2} = 2^{2n^2-2}$.

Remark. The converse direction in the proof of Lemma 2 can be explained in a more geometric way as follows. Place a unit square with center at each vertex in V_A (the set A it self is considered as n vertices). Then we get a connected polynomio P. Since $\mathcal{H}$ is not achievable P does not contain B. The boundary of P acts as a connected path from the leftmost square to the rightmost square. This path (which corresponds to the thick red line) connects A to B in the graph $f(\mathcal{H})$.

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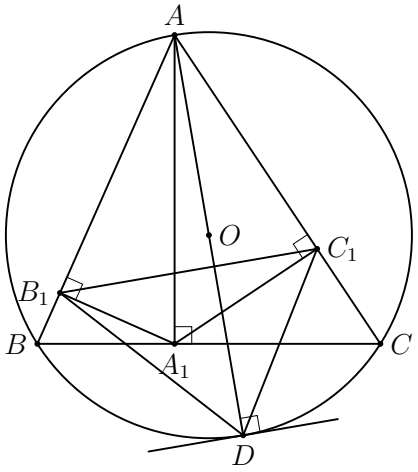
Problem 1

Let ABC be an acute triangle inscribed in a circle Γ. Let A_1 be the orthogonal projection of A onto BC so that AA_1 is an altitude. Let B_1 and C_1 be the orthogonal projections of A_1 onto AB and AC, respectively. Point P is such that quadrilateral AB_1PC_1 is convex and has the same area as triangle ABC. Is it possible that P strictly lies in the interior of circle Γ? Justify your answer.

Answer: No.

Solution

First notice that, since angles $\angle AA_1B_1$ and $\angle AA_1C_1$ are both right, the points B_1 and C_1 lie on the circle with AA_1 as a diameter. Therefore, $AC_1 = AA_1 \sin\angle AA_1C_1 = AA_1 \sin(90^\circ - \angle A_1AC) = AA_1 \sin\angle C$, similarly $AB_1 = AA_1 \sin\angle B$, and $B_1C_1 = AA_1 \sin\angle A$. Hence; triangles AC_1B_1 and ABC are similar.



Let O be the circumcenter of ABC and AD be one of its diameters. Since $\angle OAC = \frac{1}{2}(180^\circ - \angle AOC) = 90^\circ - \angle B = 90^\circ - \angle AC_1B_1$, it follows that AD is perpendicular to B_1C_1. Let $AD = 2R$; recall that, from the law of sines, $\frac{BC}{\sin\angle A} = 2R \iff BC = 2R\sin\angle A$. The area of quadrilateral AB_1DC_1 is

$$\frac{B_1C_1 \cdot AD}{2} = \frac{AA_1 \sin\angle A \cdot 2R}{2} = \frac{AA_1 \cdot BC}{2},$$

which is indeed the area of ABC.

Since B_1 and C_1 are fixed points, the loci of the points P such that AB_1PC_1 is a convex quadrilateral with the same area as ABC is a line parallel to B_1C_1. That is, perpendicular to AD. Since the area of AB_1DC_1 is the same as the area of ABC, this locus is the line perpendicular to AD through D, which is tangent to the circumcircle of ABC. Therefore, it is not possible that the point P lies inside the circumcircle of ABC.

Problem 2

Let α and β be positive real numbers. *Emerald* makes a trip in the coordinate plane, starting off from the origin $(0, 0)$. Each minute she moves one unit up or one unit to the right, restricting herself to the region $|x-y| < 2025$, in the coordinate plane. By the time she visits a point (x, y) she writes down the integer $\lfloor x\alpha+y\beta \rfloor$ on it. It turns out that Emerald wrote each non-negative integer exactly once. Find all the possible pairs (α, β) for which such a trip would be possible.

Answer: (α, β) such that $\alpha + \beta = 2$.

Solution

Let (x_n, y_n) be the point that Emerald visits after n minutes. Then $(x_{n+1}, y_{n+1}) \in \{(x_n + 1, y_n), (x_n, y_n + 1)\}$. Either way, $x_{n+1} + y_{n+1} = x_n + y_n + 1$, and since $x_0 + y_0 = 0 + 0 = 0$, $x_n + y_n = n$.

The n-th number would be then

$$z_n = \lfloor x_n\alpha + (n - x_n)\beta \rfloor \implies n\beta + x_n(\alpha - \beta) - 1 < z_n < n\beta + x_n(\alpha - \beta),$$

in which

$$-2025 < x_n - y_n < 2025 \iff \frac{n - 2025}{2} < x_n < \frac{n + 2025}{2}.$$

Suppose without loss of generality that $\alpha \geq \beta$. Then

$$n\beta + \frac{n - 2025}{2}(\alpha - \beta) - 1 < z_n < n\beta + \frac{n + 2025}{2}(\alpha - \beta),$$

which reduces to

$$\left| z_n - \frac{\alpha + \beta}{2} n \right| < \frac{2025}{2}(\alpha - \beta) + 1.$$

On the other hand, $z_{n+1} = \lfloor x_{n+1}\alpha + y_{n+1}\beta \rfloor \in \{\lfloor x_n\alpha + y_n\beta + \alpha \rfloor, \lfloor x_n\alpha + y_n\beta + \beta \rfloor\}$, which implies $z_{n+1} \geq z_n$. Since every non-negative integer appears exactly once, in increasing order, it follows that $z_n = n$.

Therefore, for all positive integers n,

$$\left| n - \frac{\alpha + \beta}{2} n \right| < \frac{2025}{2}(\alpha - \beta) + 1,$$

which can only be possible if $\alpha + \beta = 2$; otherwise, the left hand side would be unbounded.

If $\alpha + \beta = 2$, consider $x_n = \left\lceil \frac{n}{2} \right\rceil$ and $y_n = \left\lfloor \frac{n}{2} \right\rfloor$. If n is even,

$$z_n = \left\lfloor \frac{n}{2}\alpha + \frac{n}{2}\beta \right\rfloor = n;$$

if n is odd,

$$z_n = \left\lfloor \frac{n + 1}{2}\alpha + \frac{n - 1}{2}\beta \right\rfloor = n + \left\lfloor \frac{\alpha - \beta}{2} \right\rfloor,$$

which equals n because $0 < \beta \leq \alpha < \alpha + \beta = 2 \implies 0 \leq \alpha - \beta < 2$.

Problem 3

Let $P(x)$ be a non-constant polynomial with integer coefficients such that $P(0) \neq 0$. Let $a_1, a_2, a_3, \ldots$ be an infinite sequence of integers such that $P(i-j)$ divides $a_i - a_j$ for all distinct positive integers i, j. Prove that the sequence $a_1, a_2, a_3, \ldots$ must be constant, that is, a_n equals a constant c for all n positive integer.

Solution

Let $a_0 = P(0) \neq 0$ be the independent coefficient, i.e., the constant term of $P(x)$. Then there are infinitely many primes p such that p divides $P(k)$ but p does not divide k. In fact, since $P(k) - a_0$ is a multiple of k, $\gcd(P(k), k) = \gcd(k, a_0) \leq a_0$ is bounded, so pick, say, k with prime factors each larger than a_0.

Since $P(k)$ divides $a_{i+k} - a_i$, p divides $a_{i+k} - a_i$. Moreover, since $P(k+p) \equiv P(k) \equiv 0 \pmod{p}$, p also divides $a_{i+k+p} - a_i$. Therefore, $a_i \bmod p$ is periodic with periods $k+p$ and k. By *Bezout's theorem*, $\gcd(k+p, k) = 1$ is also a period, that is, p divides $a_{i+1} - a_i$ for all i and p such that $p \mid P(k)$ and $p \nmid k$ for some k. Since there are infinitely many such primes p, $a_{i+1} - a_i$ is divisible by infinitely many primes, which implies $a_{i+1} = a_i$, that is, the sequence is constant.

Problem 4
Let $n \geq 3$ be an integer. There are n cells on a circle, and each cell is assigned either 0 or 1 . There is a rooster on one of these cells, and it repeats the following operations:

- If the rooster is on a cell assigned 0, it changes the assigned number to 1 and moves to the next cell counterclockwise.
- If the rooster is on a cell assigned 1, it changes the assigned number to 0 and moves to the cell after the next cell counterclockwise.

Prove that the following statement holds true after sufficiently many operations:

> If the rooster is on a cell C, then the rooster would go around the circle exactly three times, stopping again at C. Moreover, every cell would be assigned the same number as it was assigned right before the rooster went around the circle 3 times.

Solution 1
Reformulate the problem as a n-string of numbers in $\{0, 1\}$ and a position at which the action described in the problem is performed, and add 1 or 2 modulo n to the position according to the action. Say that a *lap* is complete for each time the position resets to 0 or 1. We will prove that the statement claim holds after at most two laps, after which the n-tuple cycles every three laps.
Say the rooster *stops* at a position in a certain lap if it performs an action at that position on that lap; otherwise, the rooster *bypasses* that position. We start with some immediate claims:

- The rooster has to stop at at least one of each two consecutive positions.
- The rooster stops at every position preceded by a 0. Indeed, if the numbers preceding that position are 00 then the rooster will definitely stop at the second zero, and it the numbers preceding that position are 10 then the rooster will either stop at 1 and go directly to the position or bypass 1 and stop at the second zero, and then stop at the position.
- Therefore, if the rooster bypasses a position, then it is preceded by a 1, and that 1 must be changed to a 0. This means that the rooster never bypasses a position in two consecutive laps.
- The rooster bypasses every position preceded by 01. Indeed, the rooster stops at either 1 or at 0, after which it will move to 1; at any rate, it stops at 1 and bypasses the position.

Our goal is to prove that, eventually, for every three consecutive laps, each position is bypassed exactly once. Then each position changes states exactly twice, so it gets back to its initial state after three laps. The following two lemmata achieve this goal:

Lemma 1. *If the rooster stops at a certain position in two laps in a row, it bypasses it on the next lap, except for the n-string* $1010\ldots10$*, for which the problem statement holds.*

Proof. If the rooster stopped at a position in lap t, then it is preceded by either (A) a 0 that was changed to 1, (B) a 11 that was changed to 01, or (C) a 10 in which the rooster stopped at 1. In case (A), the position must be preceded by 11 in the lap $t + 1$, which becomes 01, so the rooster will bypass the position in the lap $t + 2$. In case (B), the position will be bypassed in lap $t + 1$.
Now we deal with case (C): suppose that the position was preceded by m occurrences of 10, that is, $(10)^m$, on lap t and take $m \leq \frac{n}{2}$ maximal. The rooster stopped at the 1 from each occurrence of 10, except possibly the first one.

First, suppose that $n \geq 2m+2$. After lap t, $(10)^m$ becomes either $(00)^m$ or $11(00)^{m-1}$ in lap $t+1$. In the latter case, initially we had $1(10)^m$, which became $011(00)^{m-1}$. It will then become $a01(11)^{m-1}$ in lap $t+2$, in which the rooster will bypass the position. In the former case, $(10)^m$ becomes $(00)^m$, so it was either $00(10)^m$ or $11(10)^m$ in lap t, which becomes respectively $a1(00)^m$ and $01(00)^m$ in lap $t+1$, respectively. In the second sub-case, it becomes $b001(11)^{m-1}$ in lap $t+2$, and the position will be bypassed. In the first sub-case, it must be $11(00)^m$ after which it becomes either $01(11)^m$ or $1001(11)^{m-1}$. In any case, the position will be bypassed in lap $t+2$.
If $n = 2m+1$, the possible configurations are

$$(10)^m0 \to (00)^m1 \to (11)^m0 \to (10)^m0,$$

the rooster stops at the 1 from the first 10 because it was preceded by a 0.

$$(10)^m1 \to (00)^m0 \to 0(11)^m \to 1(01)^m,$$

or

$$(10)^m1 \to 11(00)^{m-1}0 \to 01(11)^{m-1}1 \to (10)^m1,$$

In any case, the position is bypassed in lap $t+2$.
If $n = 2m$, the entire configuration is $(10)^m$, $m \geq 2$. If the rooster did not stop at the first 1, it becomes $11(00)^{m-1}$ in the lap $t+1$, then $01(11)^{m-1}$ in the lap $t+2$, so the position is bypassed in this last lap. If the rooster stopped at the first 1, it becomes $(00)^m$, then $(11)^m$, then $(01)^m$, then $10(00)^{m-1}$, then $(11)^m$, and then it cycles between $(11)^m$, $(01)^m$ and $10(00)^{m-1}$.
So, apart from this specific string, the rooster will stop at most two laps in a row at each position. □

Lemma 2. *If the rooster bypasses one position on a lap, then it stops at that position on the next two laps, with the same exception as lemma 1.*

Proof. The position must be preceded by 1 in lap t. If it is preceded by 11, it changes to 10 in lap $t+1$. Then it becomes 00 because the 1 was already skipped in the previous lap, and the rooster will stop at the position in the lap $t+2$.
Now suppose that the position was preceded by $(01)^m$ on lap t and take $m \leq \frac{n}{2}$ maximal. It becomes $10(00)^{m-1}$ or $(00)^m$ in lap $t+1$. In the former case, in lap $t+2$ it becomes either $00(11)^{m-1}$, after which the rooster stops at the position again, or $(11)^m$, which we'll study later. In the former case, $(00)^m$ becomes $(11)^m$ or $01(11)^{m-1}$. In the latter case, the 0 was bypassed, so it must be sopped in the next lap, becoming $(10)^m$. In the $(11)^m$ case, in order to bypass the position in lap $t+2$, it must become $(10)^m$. All in all, the preceding terms are $(01)^m$. Then, either $10(00)^{m-1}$ or $(00)^m$, then either $(11)^m$ or $01(11)^{m-1}$, then $(10)^m$. Then the second term in $(01)^m$ is 1, then 0, then 1, and then 0, that is, it changed three times. So we fall under the exception to lemma 1. □

The result then immediately follows from lemmata 1 and 2.

Solution 2

Define positions, laps, stoppings, and bypassing as in Solution 1. This other pair of lemmata also solves the problem.

Lemma 3. *There is a position and a lap in which the rooster stops twice and bypasses once (in some order) in the next three laps.*

Proof. There is a position j the rooster stops for infinitely many times. Each time it stops at j, it changes between stopping and bypassing $j+1$. So the rooster stops and bypasses $j+1$ infinitely many times. Then there is a lap in which the rooster stops at $j+1$, and bypasses it in the next. As in solution 1, it cannot bypass $j+1$ two times in a row, so it stops at $j+1$ in the next lap. □

Lemma 4. *If the rooster stops twice and bypasses once (in some order) at some position in three consecutive laps, it also stops twice and bypasses once at the next position (in some order) in the same laps (or in the next laps, in case the lap changes from one position to the other).*

Proof. When the rooster bypasses the position, it must stop at the next one. In the two times it stops at the position, the cell has different numbers on it, so the rooster will stop once and bypass once at the next position. □

By lemmata 3 and 4, there would be a moment that the rooster stops twice and bypasses once (in some order) at any position after this moment. After this moment, if the rooster is in a cell C in lap t, we know that it stopped; stopped-bypassed; and bypassed-stopped at C in laps $t, t+1, t+2$. Since it stopped twice and bypassed once (in some order) in laps $t+1, t+2, t+3$, it must stop at C in lap $t+3$. Moreover, the rooster stopped twice and bypassed once (in some order) each cell between stopping at C in laps t and $t+3$, so every cell has the same assigned number before and after going around the circle three times.

Solution 3

Let us reformulate the problem in terms of Graphs: we have a directed graph G with $V = \{v_1, v_2, \ldots, v_n\}$ representing positions and $E = \{v_i \to v_{i+1}, v_i \to v_{i+2} \mid 1 \le i \le n\}$ representing moves. Indices are taken mod n. Note that each vertex has in-degree and out-degree both equal to 2. We say that the edge $v_i \to v_{i+1}$ is *active* and $v_i \to v_{i+2}$ is *inactive* if the number on v_i is 0, or the edge $v_i \to v_{i+2}$ is *active* and $v_i \to v_{i+1}$ is *inactive* if the number on v_i is 1. The rooster then traces an infinite trail on the graph in the following manner;

- it starts at v_1;
- if the rooster is at v_i, it uses the active edge going out of v_i to continue the path, and changes the number on v_i.

Take the first vertex that appears 3 times on the rooster's trail, and suppose without any loss of generality that it is v_1 (otherwise, just ignore the path before the first time that this vertex appears). Consider the sub-trail from the first to the third time v_1 appears: $C = v_1 \to \cdots \to v_1 \to \cdots \to v_1$. Trail C induces a circuit C^* on G that passes through v_1 twice.
In between two occurrences of v_1, the rooster must have completed at least one lap. Vertex v_1 appears three times at C, so C contains at least two laps. Since (as per Solution 1) no vertex is bypassed twice in a row, C^* must contain every vertex. Moreover, v_1 is the first (and only) vertex that appeared three times on the rooster's path, so each vertex appear at most twice on C^*.

Lemma 5. *Take V' to be the subset of V containing the vertices that only appears once on C^* and E' to be the edges of G that don't appear on C^*. So $G' = (V', E')$ is a simple cycle or $V' = E' = \emptyset$.*

Proof. We will first prove that each edge in C^* appears exactly one. Suppose, for the sake of contradiction, that there is an edge $u \to v$ that appears twice on C^*. So u must appear times in C, because the active edge going out of u changes at each visit to it. So $u = v_1$, but the rooster only goes out of v_1 twice in C, which is a contradiction.
Now, since C^* does not have repeated edges, all four edges through v are in C^* if v is visited twice in this cycle. So the edges on E' cannot pass through vertices in $V \setminus V'$, then G' is well-defined.
Since $v \in V'$ is visited once on C^*, it has $\deg_{in}(v) = \deg_{out}(v) = 1$ in G', so G' is a union of disjoint simple cycles. Each of these cycles completes at least one lap, and v_1 is skipped on these laps, so all cycles must use the edge $v_n \to v_2$. But the cycles are disjoint, so there is at most one cycle. □

Lemma 6. *The rooster eventually traverses a contiguous* Eulerian *circuit of G.*

Proof. If $V' = E' = \emptyset$, C^* is already an Eulerian circuit traversed in C. If not, take u_1 to be the first vertex of C that is in V'. Let $U = u_1 \to \cdots \to u_k \to u_1$ be the cycle determined by G' by Lemma 5, $C_1 = v_1 \to \cdots \to u_1$ be the sub-path of C from the first v_1 to u_1, and $C_2 = u_1 \to \cdots \to v_1$ be the rest of C.
Each vertex on $V \setminus V'$ has been changed twice, so they would be in their initial states after C; every vertex on V' has been changed only once, and therefore are not on their initial states. Let us trace the rooster's trail after it traversed C. By the minimality of u_1, all the edges of C_1 are active after C, so the rooster traverses C_1 after C. Moreover, each u_i was visited exactly once in C and has not yet used the edge $u_i \to u_{i+1}$. Since all their states have changed, all edges of U are active after C. Therefore, the rooster traverse U after C_1, flipping all the vertices in U to their initial states. Then, since every state in U was changed, the rooster traverses C_2 instead of U. The trail is then $C \to C_1 \to U \to C_2$, and hen $C_1 \to U \to C_2$ is an Eulerian circuit. □

Having this in mind, after the rooster completes an Eulerian circuit, it has passed through each vertex twice and returned to its initial vertex, so the state of the rooster and of the edges are the same before and after the Eulerian cycle. The rooster will then traverse the Eulerian circuit repeatedly.
The edges $v_i \to v_{i+1}$ move forward by one position and the edges $v_i \to v_{i+2}$ move forward by two positions. Since all edges are used, each time the rooster traverses the Eulerian circuit, it moves forward a total of $n(1 + 2) = 3n$ positions, which corresponds to three laps. The proof is now complete.

Comment: Solution 1 presents a somewhat brute-force case analysis; the main ideas are

- Lemma 1: The rooster stops at one position in at most two consecutive laps, with one notable exception.
- Lemma 2: If the rooster bypasses one position then it stops at it in the next two laps, with the same notable exception.

Solution 2 has a recursive nature; the main ideas are

- Lemma 3: Proving that one position changes states exactly two out of three laps.
- Lemma 4: Given that this happens to one position, it also happens to the next position in the same laps, so it happens to every position.

Solution 3 cleverly identifies the moves the rooster performs within a graph theoretical framework. The main goal is to prove that the rooster traces an Eulerian circuit in a graph.

- Lemma 5: Analyzing what happens after the rooster first visits a position three times.
- Lemma 6: Proving that the rooster traces an Eulerian circuit.

Problem 5
Consider an infinite sequence $a_1, a_2, \ldots$ of positive integers such that

$$100!(a_m + a_{m+1} + \cdots + a_n) \quad \text{is a multiple of} \quad a_{n-m+1}a_{n+m}$$

for all positive integers m, n such that $m \leq n$.
Prove that the sequence is either bounded or linear.
Observation: A sequence of positive integers is *bounded* if there exists a constant N such that $a_n < N$ for all $n \in \mathbb{Z}_{>0}$. A sequence is *linear* if $a_n = n \cdot a_1$ for all $n \in \mathbb{Z}_{>0}$.

Solution
Let $c = 100!$. Suppose that $n \geq m+2$. Then $a_{m+n} = a_{(m+1)+(n-1)}$ divides both $c(a_m + a_{m+1} + \cdots + a_{n-1} + a_n)$ and $c(a_{m+1} + \cdots + a_{n-1})$, so it also divides the difference $c(a_m + a_n)$. Notice that if $n = m+1$ then a_{m+n} divides $c(a_m + a_{m+1}) = c(a_m + a_n)$, and if $n = m$ then a_{m+n} divides both ca_m and $2ca_m = c(a_m + a_n)$. In either cases; a_{m+n} divides $c(a_m + a_n)$.
Analogously, one can prove that if $m > n$, $a_{m-n} = a_{(m-1)-(n-1)}$ divides $c(a_m - a_n)$, as it divides both $c(a_{n+1} + \cdots + a_m)$ and $c(a_n + \cdots + a_{m-1})$.
From now on, drop the original divisibility statement and keep the statements "a_{m+n} divides $c(a_m + a_n)$" and "a_{m-n} divides $c(a_m - a_n)$." Now, all conditions are linear, and we can suppose without loss of generality that there is no integer $D > 1$ that divides every term of the sequence; if there is such an integer D, divide all terms by D.
Having this in mind, notice that $a_m = a_{m+n-n}$ divides $c(a_{m+n} - a_n)$ and also $c(a_{m+n} - a_m - a_n)$; analogously, a_n also divides $c(a_{m+n} - a_m - a_n)$, and since a_{m+n} divides $c(a_m + a_n)$, it also divides $c(a_{m+n} - a_m - a_n)$. Therefore, $c(a_{m+n} - a_m - a_n)$ is divisible by a_m, a_n, and a_{m+n}, and therefore also by $\operatorname{lcm}(a_m, a_n, a_{m+n})$. In particular, $ca_{m+n} \equiv c(a_m + a_n) \pmod{\operatorname{lcm}(a_m, a_n)}$.
Since a_{m+n} divides $c(a_m + a_n)$, $a_{m+n} \leq c(a_m + a_n)$.
From now on, we divide the problem in two cases.

Case 1: there exist m, n such that $\operatorname{lcm}(a_m, a_n) > c^2(a_m + a_n)$.
If $\operatorname{lcm}(a_m, a_n) > c^2(a_m + a_n) > c(a_m + a_n)$ then both $c(a_m + a_n)$ and ca_{m+n} are less than $ca_{m+n} \leq c^2(a_m + a_n) < \operatorname{lcm}(a_m, a_n)$. This implies $ca_{m+n} = c(a_m + a_n) \iff a_{m+n} = a_m + a_n$.
Now we can extend this further: since $\gcd(a_m, a_{m+n}) = \gcd(a_m, a_m + a_n) = \gcd(a_m, a_n)$, it follows that

$$\begin{aligned}\operatorname{lcm}(a_m, a_{m+n}) &= \frac{a_m a_{m+n}}{\gcd(a_m, a_n)} = \frac{a_{m+n}}{a_n}\operatorname{lcm}(a_m, a_n) > \frac{c^2(a_m + a_n)^2}{a_n} \\ &> \frac{c^2(2a_m a_n + a_n^2)}{a_n} = c^2(2a_m + a_n) = c^2(a_m + a_{m+n}).\end{aligned}$$

We can iterate this reasoning to obtain that $a_{km+n} = ka_m + a_n$, for all $k \in \mathbb{Z}_{>0}$. In fact, if the condition $\operatorname{lcm}(a_m, a_n) > c^2(a_m + a_n)$ holds for the pair (n, m), then it also holds for the pairs $(m+n, m), (2m+n, m), \ldots, ((k-1)m+n, m)$, which implies $a_{km+n} = a_{(k-1)m+n} + a_m = a_{(k-2)m+n} + 2a_m = \cdots = a_n + ka_m$.
Similarly, $a_{m+kn} = a_m + ka_n$.
Now, $a_{m+n+mn} = a_{n+(n+1)m} = a_{m+(m+1)n} \implies a_n + (n+1)a_m = a_m + (m+1)a_n \iff ma_n = na_m$. If $d = \gcd(m, n)$ then $\frac{n}{d}a_m = \frac{m}{d}a_n$.
Therefore, since $\gcd(\frac{m}{d}, \frac{n}{d}) = 1$, $\frac{m}{d}$ divides a_m and $\frac{n}{d}$ divides a_n, which means that

$$a_n = \frac{m}{d} \cdot t = \frac{t}{d}m \quad \text{and} \quad a_m = \frac{n}{d} \cdot t = \frac{t}{d}n, \quad \text{for some } t \in \mathbb{Z}_{>0},$$

which also implies

$$a_{km+n} = \frac{t}{d}(km + n) \quad \text{and} \quad a_{m+kn} = \frac{t}{d}(m + kn), \quad \text{for all } k \in \mathbb{Z}_{>0}.$$

Now let's prove that $a_{kd} = tk = \frac{t}{d}(kd)$ for all $k \in \mathbb{Z}_{>0}$. In fact, there exist arbitrarily large positive integers R, S such that $kd = Rm - Sn = (n+(R+1)m) - (m+(S+1)n)$ (for instance, let $u, v \in \mathbb{Z}$ such that $kd = mu - nv$ and take $R = u + Qn$ and $S = v + Qm$ for Q sufficiently large.)
Let $x = n + (R+1)m$ and $y = m + (S+1)n$. Then $kd = x - y \iff x = y + kd$, $a_x = \frac{t}{d}x$, and $a_y = \frac{t}{d}y = \frac{t}{d}(x - kd) = a_x - tk$. Thus a_x divides $c(a_y + a_{kd}) = c(a_{kd} + a_x - tk)$, and therefore also $c(a_{kd} - tk)$. Since $a_x = \frac{t}{d}x$ can be arbitrarily large, $a_{kd} = tk = \frac{t}{d}(kd)$. In particular, $a_d = t$, so $a_{kd} = ka_d$.
Since $ka_d = a_{kd} = a_{1+(kd-1)}$ divides $c(a_1 + a_{kd-1})$, $b_k = a_{kd-1}$ is unbounded. Pick $p > a_{kd-1}$ a large prime and consider $a_{pd} = pa_d$. Then

$$\operatorname{lcm}(a_{pd}, a_{kd-1}) \geq \operatorname{lcm}(p, a_{kd-1}) = pa_{kd-1}.$$

We can pick a_{kd-1} and p large enough so that their product is larger than a particular linear combination of them, that is,

$$\operatorname{lcm}(p, a_{kd-1}) = pa_{kd-1} > c^2(pa_d + a_{kd-1}) = c^2(a_{pd} + a_{kd-1}).$$

Then all the previous facts can be applied, and since $\gcd(pd, kd-1) = 1$, $a_k = a_{k\gcd(pd,kd-1)}$ $= ka_{\gcd(pd,kd-1)} = ka_1$, that is, the sequence is linear. Also, since a_1 divides all terms, $a_1 = 1$.

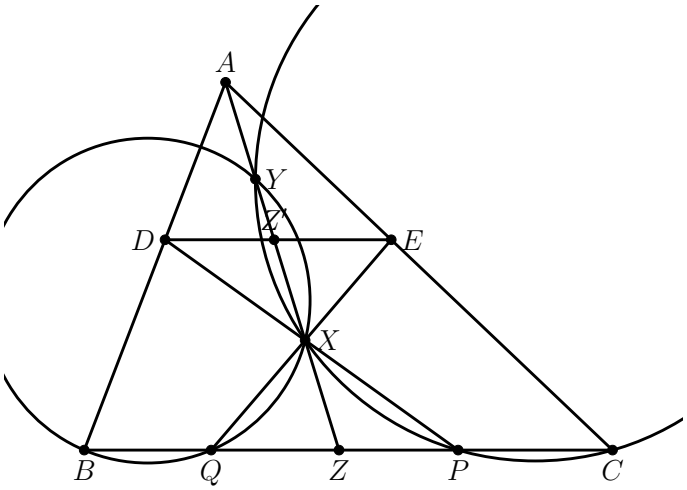
Case 2: $\operatorname{lcm}(a_m, a_n) \leq c^2(a_m + a_n)$ *for all* m, n.
Suppose that $a_m \leq a_n$; then $\operatorname{lcm}(a_m, a_n) = Ma_n \leq c^2(a_m + a_n) \leq 2c^2a_n \implies M \leq 2c^2$, that is, the factor in the smaller term that is not in the larger term is at most $2c^2$.
We prove that in this case the sequence must be bounded. Suppose on the contrary; then there is a term a_m that is divisible by a large prime power p^d. Then every larger term a_n is divisible by a factor larger than $\frac{p^d}{2c^2}$. So we pick $p^d > (2c^2)^2$, so that every large term a_n is divisible by the prime power $p^e > 2c^2$. Finally, fix a_k for any k. It follows from $a_{k+n} \mid c(a_k + a_n)$ that (a_n) is unbounded, so we can pick a_n and a_{k+n} large enough such that both are divisible by p^e. Hence any a_k is divisible by p, which is a contradiction to the fact that there is no $D > 1$ that divides every term in the sequence.

APMO 2024 – Problems and Solutions

Problem 1

Let ABC be an acute triangle. Let D be a point on side AB and E be a point on side AC such that lines BC and DE are parallel. Let X be an interior point of $BCED$. Suppose rays DX and EX meet side BC at points P and Q, respectively such that both P and Q lie between B and C. Suppose that the circumcircles of triangles BQX and CPX intersect at a point $Y \neq X$. Prove that points A, X, and Y are collinear.

Solution 1

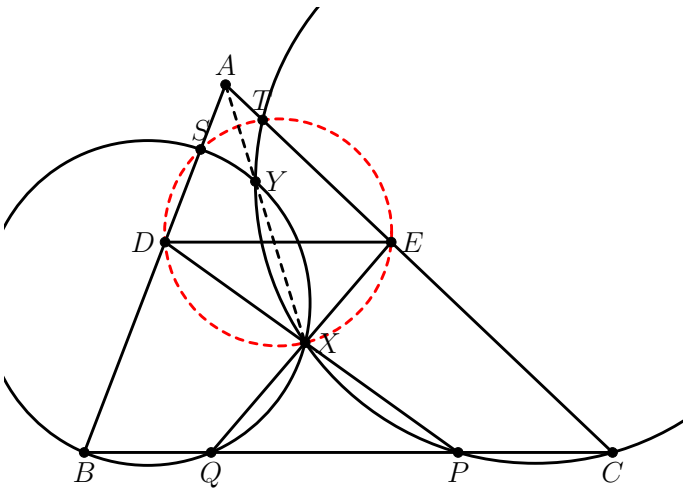


Let ℓ be the radical axis of circles BQX and CPX. Since X and Y are on ℓ, it is sufficient to show that A is on ℓ. Let line AX intersect segments BC and DE at Z and Z', respectively. Then it is sufficient to show that Z is on ℓ. By $BC \parallel DE$, we obtain

$$\frac{BZ}{ZC} = \frac{DZ'}{Z'E} = \frac{PZ}{ZQ},$$

thus $BZ \cdot QZ = CZ \cdot PZ$, which implies that Z is on ℓ.

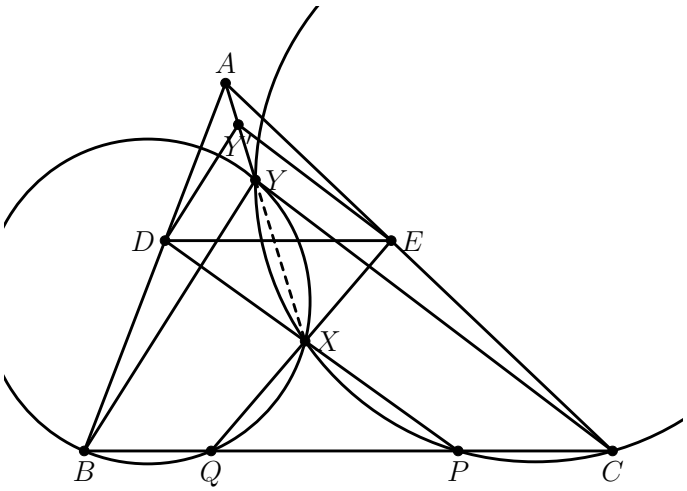
Solution 2



Let circle BQX intersect line AB at a point S which is different from B. Then $\angle DEX = \angle XQC = \angle BSX$, thus S is on circle DEX. Similarly, let circle CPX intersect line AC at a point T which is different from C. Then T is on circle DEX. The power of A with respect to the circle DEX is $AS \cdot AD = AT \cdot AE$. Since $\frac{AD}{AB} = \frac{AE}{AC}$, $AS \cdot AB = AT \cdot AC$. Then A is in the radical axis of circles BQX and CPX, which implies that three points A, X and Y are collinear.

Solution 3

Consider the (direct) homothety that takes triangle ADE to triangle ABC, and let Y' be the image of Y under this homothety; in other words, let Y' be the intersection of the line parallel to BY through D and the line parallel to CY through E.



The homothety implies that A, Y, and Y' are collinear, and that $\angle DY'E = \angle BYC$. Since $BQXY$ and $CPXY$ are cyclic,

$$\angle DY'E = \angle BYC = \angle BYX + \angle XYC = \angle XQP + \angle XPQ = 180^\circ - \angle PXQ = 180^\circ - \angle DXE,$$

which implies that $DY'EX$ is cyclic. Therefore

$$\angle DY'X = \angle DEX = \angle PQX = \angle BYX,$$

which, combined with $DY' \parallel BY$, implies $Y'X \parallel YX$. This proves that X, Y, and Y' are collinear, which in turn shows that A, X, and Y are collinear.

Problem 2

Consider a 100×100 table, and identify the cell in row a and column b, $1 \leq a, b \leq 100$, with the ordered pair (a, b). Let k be an integer such that $51 \leq k \leq 99$. A *k-knight* is a piece that moves one cell vertically or horizontally and k cells to the other direction; that is, it moves from (a, b) to (c, d) such that $(|a - c|, |b - d|)$ is either $(1, k)$ or $(k, 1)$. The k-knight starts at cell $(1, 1)$, and performs several moves. A *sequence of moves* is a sequence of cells $(x_0, y_0) = (1, 1)$, (x_1, y_1), (x_2, y_2), $\ldots$, (x_n, y_n) such that, for all $i = 1, 2, \ldots, n$, $1 \leq x_i, y_i \leq 100$ and the k-knight can move from (x_{i-1}, y_{i-1}) to (x_i, y_i). In this case, each cell (x_i, y_i) is said to be *reachable*. For each k, find $L(k)$, the number of reachable cells.

Answer: $L(k) = \begin{cases} 100^2 - (2k - 100)^2 & \text{if } k \text{ is even} \\ \frac{100^2 - (2k-100)^2}{2} & \text{if } k \text{ is odd} \end{cases}$.

Solution

Cell (x, y) is directly reachable from another cell if and only if $x - k \geq 1$ or $x + k \leq 100$ or $y - k \geq 1$ or $y + k \leq 100$, that is, $x \geq k+1$ or $x \leq 100-k$ or $y \geq k+1$ or $y \leq 100-k$ $(*)$. Therefore the cells (x, y) for which $101 - k \leq x \leq k$ and $101 - k \leq y \leq k$ are unreachable. Let S be this set of unreachable cells in this square, namely the square of cells (x, y), $101 - k \leq x, y \leq k$.

If condition $(*)$ is valid for both (x, y) and $(x \pm 2, y \pm 2)$ then one can move from (x, y) to $(x \pm 2, y \pm 2)$, if they are both in the table, with two moves: either $x \leq 50$ or $x \geq 51$; the same is true for y. In the first case, move $(x, y) \to (x + k, y \pm 1) \to (x, y \pm 2)$ or $(x, y) \to (x \pm 1, y + k) \to (x \pm 2, y)$. In the second case, move $(x, y) \to (x - k, y \pm 1) \to (x, y \pm 2)$ or $(x, y) \to (x \pm 1, y - k) \to (x \pm 2, y)$.

Hence if the table is colored in two colors like a chessboard, if $k \leq 50$, cells with the same color as $(1, 1)$ are reachable. Moreover, if k is even, every other move changes the color of the occupied cell, and all cells are potentially reachable; otherwise, only cells with the same color as $(1, 1)$ can be visited. Therefore, if k is even then the reachable cells consists of all cells except the center square defined by $101 - k \leq x \leq k$ and $101 - k \leq y \leq k$, that is, $L(k) = 100^2 - (2k - 100)^2$; if k is odd, then only half of the cells are reachable: the ones with the same color as $(1, 1)$, and $L(k) = \frac{1}{2}(100^2 - (2k - 100)^2)$.

Problem 3
Let n be a positive integer and $a_1, a_2, \ldots, a_n$ be positive real numbers. Prove that

$$\sum_{i=1}^{n} \frac{1}{2^i} \left(\frac{2}{1+a_i}\right)^{2^i} \geq \frac{2}{1+a_1a_2\ldots a_n} - \frac{1}{2^n}.$$

Solution
We first prove the following lemma:

Lemma 1. *For k positive integer and $x, y > 0$,*

$$\left(\frac{2}{1+x}\right)^{2^k} + \left(\frac{2}{1+y}\right)^{2^k} \geq 2\left(\frac{2}{1+xy}\right)^{2^{k-1}}.$$

The proof goes by induction. For $k = 1$, we have

$$\left(\frac{2}{1+x}\right)^{2} + \left(\frac{2}{1+y}\right)^{2} \geq 2\left(\frac{2}{1+xy}\right),$$

which reduces to

$$xy(x-y)^2 + (xy-1)^2 \geq 0.$$

For $k > 1$, by the inequality $2(A^2+B^2) \geq (A+B)^2$ applied at $A = \left(\frac{2}{1+x}\right)^{2^{k-1}}$ and $B = \left(\frac{2}{1+y}\right)^{2^{k-1}}$ followed by the induction hypothesis

$$\begin{aligned} 2\left(\left(\frac{2}{1+x}\right)^{2^k} + \left(\frac{2}{1+y}\right)^{2^k}\right) &\geq \left(\left(\frac{2}{1+x}\right)^{2^{k-1}} + \left(\frac{2}{1+y}\right)^{2^{k-1}}\right)^2 \\ &\geq \left(2\left(\frac{2}{1+xy}\right)^{2^{k-2}}\right)^2 = 4\left(\frac{2}{1+xy}\right)^{2^{k-1}}, \end{aligned}$$

from which the lemma follows.
The problem now can be deduced from summing the following applications of the lemma, multiplied by the appropriate factor:

$$\begin{aligned} \frac{1}{2^n}\left(\frac{2}{1+a_n}\right)^{2^n} + \frac{1}{2^n}\left(\frac{2}{1+1}\right)^{2^n} &\geq \frac{1}{2^{n-1}}\left(\frac{2}{1+a_n\cdot 1}\right)^{2^{n-1}} \\ \frac{1}{2^{n-1}}\left(\frac{2}{1+a_{n-1}}\right)^{2^{n-1}} + \frac{1}{2^{n-1}}\left(\frac{2}{1+a_n}\right)^{2^{n-1}} &\geq \frac{1}{2^{n-2}}\left(\frac{2}{1+a_{n-1}a_n}\right)^{2^{n-2}} \\ \frac{1}{2^{n-2}}\left(\frac{2}{1+a_{n-2}}\right)^{2^{n-2}} + \frac{1}{2^{n-2}}\left(\frac{2}{1+a_{n-1}a_n}\right)^{2^{n-2}} &\geq \frac{1}{2^{n-3}}\left(\frac{2}{1+a_{n-2}a_{n-1}a_n}\right)^{2^{n-3}} \\ &\cdots \\ \frac{1}{2^k}\left(\frac{2}{1+a_k}\right)^{2^k} + \frac{1}{2^k}\left(\frac{2}{1+a_{k+1}\ldots a_{n-1}a_n}\right)^{2^k} &\geq \frac{1}{2^{k-1}}\left(\frac{2}{1+a_k\ldots a_{n-2}a_{n-1}a_n}\right)^{2^{k-1}} \\ &\cdots \\ \frac{1}{2}\left(\frac{2}{1+a_1}\right)^{2} + \frac{1}{2}\left(\frac{2}{1+a_2\ldots a_{n-1}a_n}\right)^{2} &\geq \frac{2}{1+a_1\ldots a_{n-2}a_{n-1}a_n}. \end{aligned}$$

Comment: Equality occurs if and only if $a_1 = a_2 = \cdots = a_n = 1$.

Comment: The main motivation for the lemma is trying to "telescope" the sum

$$\frac{1}{2^n}+\sum_{i=1}^{n}\frac{1}{2^i}\left(\frac{2}{1+a_i}\right)^{2^i},$$

that is,

$$\frac{1}{2}\left(\frac{2}{1+a_1}\right)^{2}+\cdots+\frac{1}{2^{n-1}}\left(\frac{2}{1+a_{n-1}}\right)^{2^{n-1}}+\frac{1}{2^n}\left(\frac{2}{1+a_n}\right)^{2^n}+\frac{1}{2^n}\left(\frac{2}{1+1}\right)^{2^n}$$

to obtain an expression larger than or equal to

$$\frac{2}{1+a_1a_2\ldots a_n}.$$

It seems reasonable to obtain a inequality that can be applied from right to left, decreases the exponent of the factor $1/2^k$ by 1, and multiplies the variables in the denominator. Given that, the lemma is quite natural:

$$\frac{1}{2^k}\left(\frac{2}{1+x}\right)^{2^k}+\frac{1}{2^k}\left(\frac{2}{1+y}\right)^{2^k}\geq\frac{1}{2^{k-1}}\left(\frac{2}{1+xy}\right)^{2^{i-1}},$$

or

$$\left(\frac{2}{1+x}\right)^{2^k}+\left(\frac{2}{1+y}\right)^{2^k}\geq 2\left(\frac{2}{1+xy}\right)^{2^{k-1}}.$$

Problem 4

Prove that for every positive integer t there is a unique permutation $a_0, a_1, \ldots, a_{t-1}$ of $0, 1, \ldots, t-1$ such that, for every $0 \le i \le t-1$, the binomial coefficient $\binom{t+i}{2a_i}$ is odd and $2a_i \neq t+i$.

Solution

We constantly make use of *Kummer's theorem* which, in particular, implies that $\binom{n}{k}$ is odd if and only if k and $n-k$ have ones in different positions in binary. In other words, if $S(x)$ is the set of positions of the digits 1 of x in binary (in which the digit multiplied by 2^i is in position i), $\binom{n}{k}$ is odd if and only if $S(k) \subseteq S(n)$. Moreover, if we set $k < n$, $S(k)$ is a proper subset of $S(n)$, that is, $|S(k)| < |S(n)|$.

We start with a lemma that guides us how the permutation should be set.

Lemma 1.

$$\sum_{i=0}^{t-1} |S(t+i)| = t + \sum_{i=0}^{t-1} |S(2i)|.$$

The proof is just realizing that $S(2i) = \{1+x, x \in S(i)\}$ and $S(2i+1) = \{0\} \cup \{1+x, x \in S(i)\}$, because $2i$ in binary is i followed by a zero and $2i+1$ in binary is i followed by a one. Therefore

$$\begin{aligned}\sum_{i=0}^{t-1} |S(t+i)| &= \sum_{i=0}^{2t-1} |S(i)| - \sum_{i=0}^{t-1} |S(i)| = \sum_{i=0}^{t-1} |S(2i)| + \sum_{i=0}^{t-1} |S(2i+1)| - \sum_{i=0}^{t-1} |S(i)| \\ &= \sum_{i=0}^{t-1} |S(i)| + \sum_{i=0}^{t-1} (1 + |S(i)|) - \sum_{i=0}^{t-1} |S(i)| = t + \sum_{i=0}^{t-1} |S(i)| = t + \sum_{i=0}^{t-1} |S(2i)|.\end{aligned}$$

The lemma has an immediate corollary: since $t+i > 2a_i$ and $\binom{t+i}{2a_i}$ is odd for all i, $0 \le i \le t-1$, $S(2a_i) \subset S(t+i)$ with $|S(2a_i)| \le |S(t+i)| - 1$. Since the sum of $|S(2a_i)|$ is t less than the sum of $|S(t+i)|$, and there are t values of i, equality must occur, that is, $|S(2a_i)| = |S(t+i)| - 1$, which in conjunction with $S(2a_i) \subset S(t+i)$ means that $t+i-2a_i = 2^{k_i}$ for every i, $0 \le i \le t-1$, $k_i \in S(t+i)$ (more precisely, $\{k_i\} = S(t+i) \setminus S(2a_i)$.)

In particular, for $t+i$ odd, this means that $t+i-2a_i = 1$, because the only odd power of 2 is 1. Then $a_i = \frac{t+i-1}{2}$ for $t+i$ odd, which takes up all the numbers greater than or equal to $\frac{t-1}{2}$. Now we need to distribute the numbers that are smaller than $\frac{t-1}{2}$ (call these numbers *small*).

If $t+i$ is even then by *Lucas' Theorem* $\binom{t+i}{2a_i} \equiv \binom{\frac{t+i}{2}}{a_i} \pmod 2$, so we pair numbers from $\lceil t/2 \rceil$ to $t-1$ (call these numbers *big*) with the small numbers.

Say that a set A is *paired* with another set B whenever $|A| = |B|$ and there exists a bijection $\pi\colon A \to B$ such that $S(a) \subset S(\pi(a))$ and $|S(a)| = |S(\pi(a))| - 1$; we also say that a and $\pi(a)$ are paired. We prove by induction in t that $A_t = \{0, 1, 2, \ldots, \lfloor t/2 \rfloor - 1\}$ (the set of small numbers) and $B_t = \{\lceil t/2 \rceil, \ldots, t-2, t-1\}$ (the set of big numbers) can be uniquely paired.

The claim is immediate for $t = 1$ and $t = 2$. For $t > 2$, there is exactly one power of two in B_t, since $t/2 \le 2^a < t \iff a = \lceil \log_2(t/2) \rceil$. Let 2^a be this power of two. Then, since $2^a \ge t/2$, no number in A_t has a one in position a in binary. Since for every number x, $2^a \le x < t$, $a \in S(x)$ and $a \notin S(y)$ for all $y \in A_t$, x can only be paired with $x - 2^a$, since $S(x)$ needs to be stripped of exactly one position. This takes cares of $x \in B_t$, $2^a \le x < t$, and $y \in A_t$, $0 \le y < t - 2^a$.

Now we need to pair the numbers from $A' = \{t - 2^a, t - 2^a + 1, \ldots, \lfloor t/2 \rfloor - 1\} \subset A$ with the numbers from $B' = \{\lceil t/2 \rceil, \lceil t/2 \rceil + 1, \ldots, 2^a - 1\} \subset B$. In order to pair these $t - 2(t - 2^a) = 2^{a+1} - t < t$ numbers, we use the induction hypothesis and a bijection between $A' \cup B'$ and $B_{2^{a+1}-t} \cup A_{2^{a+1}-t}$. Let $S = S(2^a - 1) = \{0, 1, 2, \ldots, a-1\}$. Then take a pair x, y, $x \in A_{2^{a+1}-t}$ and $y \in B_{2^{a+1}-t}$ and biject it with $2^a - 1 - x \in B'$ and $2^a - 1 - y \in A'$. In fact,

$$0 \le x \le \left\lfloor \frac{2^{a+1} - t}{2} \right\rfloor - 1 = 2^a - \left\lceil \frac{t}{2} \right\rceil - 1 \iff \left\lceil \frac{t}{2} \right\rceil \le 2^a - 1 - x \le 2^a - 1$$

and

$$\left\lceil \frac{2^{a+1}-t}{2} \right\rceil = 2^a - \left\lfloor \frac{t}{2} \right\rfloor \le y \le 2^{a+1}-t-1 \iff t-2^a \le 2^a-1-y \le \left\lfloor \frac{t}{2} \right\rfloor - 1.$$

Moreover, $S(2^a-1-x) = S \setminus S(x)$ and $S(2^a-1-y) = S \setminus S(y)$ are complements with respect to S, and $S(x) \subset S(y)$ and $|S(x)| = |S(y)| - 1$ implies $S(2^a-1-y) \subset S(2^a-1-x)$ and $|S(2^a-1-y)| = |S(2^a-1-x)| - 1$. Therefore a pairing between A' and B' corresponds to a pairing between $A_{2^{a+1}-t}$ and $B_{2^{a+1}-t}$. Since the latter pairing is unique, the former pairing is also unique, and the result follows.
We illustrate the bijection by showing the case $t = 23$:

$$A_{23} = \{0, 1, 2, \ldots, 10\}, \quad B_{23} = \{12, 13, 14, \ldots, 22\}.$$

The pairing is

$$\begin{pmatrix} 12 & 13 & 14 & 15 & 16 & 17 & 18 & 19 & 20 & 21 & 22 \\ 8 & 9 & 10 & 7 & 0 & 1 & 2 & 3 & 4 & 5 & 6 \end{pmatrix},$$

in which the bijection is between

$$\begin{pmatrix} 12 & 13 & 14 & 15 \\ 8 & 9 & 10 & 7 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 3 & 2 & 1 & 0 \\ 7 & 6 & 5 & 8 \end{pmatrix} \to \begin{pmatrix} 5 & 6 & 7 & 8 \\ 1 & 2 & 3 & 0 \end{pmatrix}.$$

Problem 5

Line ℓ intersects sides BC and AD of cyclic quadrilateral $ABCD$ in its interior points R and S respectively, and intersects ray DC beyond point C at Q, and ray BA beyond point A at P. Circumcircles of the triangles QCR and QDS intersect at $N \neq Q$, while circumcircles of the triangles PAS and PBR intersect at $M \neq P$. Let lines MP and NQ meet at point X, lines AB and CD meet at point K and lines BC and AD meet at point L. Prove that point X lies on line KL.

Solution 1

We start with the following lemma.

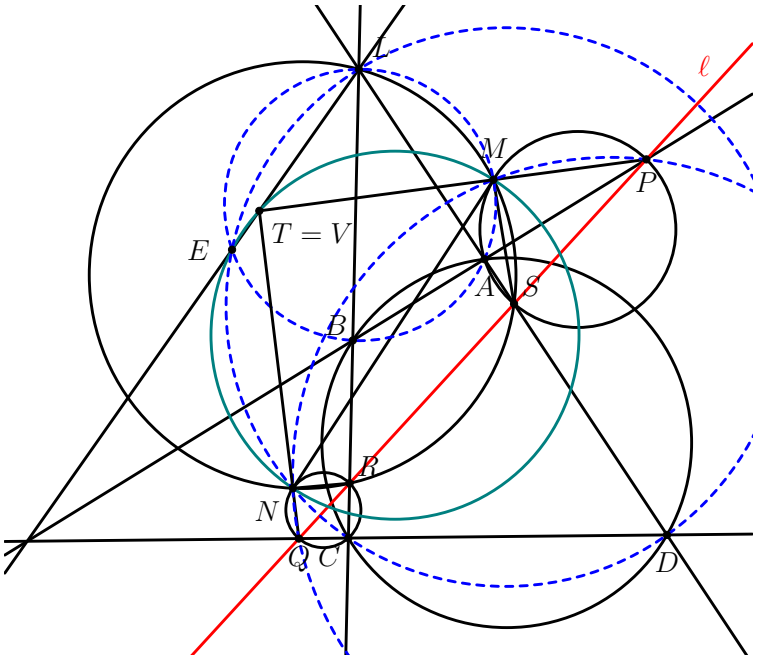
Lemma 1. *Points M, N, P, Q are concyclic.*

Point M is the Miquel point of lines $AP = AB$, $PS = \ell$, $AS = AD$, and $BR = BC$, and point N is the Miquel point of lines $CQ = CD$, $RC = BC$, $QR = \ell$, and $DS = AD$. Both points M and N are on the circumcircle of the triangle determined by the common lines AD, ℓ, and BC, which is LRS.

Then, since quadrilaterals $QNRC$, $PMAS$, and $ABCD$ are all cyclic, using directed angles (modulo 180°)

$$\begin{aligned}\measuredangle NMP &= \measuredangle NMS + \measuredangle SMP = \measuredangle NRS + \measuredangle SAP = \measuredangle NRQ + \measuredangle DAB = \measuredangle NRQ + \measuredangle DCB \\ &= \measuredangle NRQ + \measuredangle QCR = \measuredangle NRQ + \measuredangle QNR = \measuredangle NQR = \measuredangle NQP,\end{aligned}$$

which implies that $MNQP$ is a cyclic quadrilateral.



Let E be the Miquel point of $ABCD$ (that is, of lines AB, BC, CD, DA). It is well known that E lies in the line t connecting the intersections of the opposite lines of $ABCD$. Let lines NQ and t meet at T. If $T \neq E$, using directed angles, looking at the circumcircles of LAB (which contains, by definition, E and M), APS (which also contains M), and $MNQP$,

$$\measuredangle TEM = \measuredangle LEM = \measuredangle LAM = \measuredangle SAM = \measuredangle SPM = \measuredangle QPM = \measuredangle QNM = \measuredangle TNM,$$

that is, T lies in the circumcircle ω of EMN. If $T = E$, the same computation shows that $\measuredangle LEM = \measuredangle ENM$, which means that t is tangent to ω.

Now let lines MP and t meet at V. An analogous computation shows, by looking at the circumcircles of LCD (which contains E and N), CQR, and $MNQP$, that V lies in ω as well, and that if $V = E$ then t is tangent to ω.
Therefore, since ω meet t at T, V, and E, either $T = V$ if both $T \neq E$ and $V \neq E$ or $T = V = E$. At any rate, the intersection of lines MP and NQ lies in t.

Solution 2
Barycentric coordinates are a viable way to solve the problem, but even the solution we have found had some clever computations. Here is an outline of this solution.

Lemma 2. *Denote by* $\operatorname{pow}_\omega X$ *the power of point* X *with respect to circle* ω. *Let* Γ_1 *and* Γ_2 *be circles with different centers. Considering* ABC *as the reference triangle in barycentric coordinates, the radical axis of* Γ_1 *and* Γ_2 *is given by*

$$(\operatorname{pow}_{\Gamma_1} A - \operatorname{pow}_{\Gamma_2} A)x + (\operatorname{pow}_{\Gamma_1} B - \operatorname{pow}_{\Gamma_2} B)y + (\operatorname{pow}_{\Gamma_1} C - \operatorname{pow}_{\Gamma_2} C)z = 0.$$

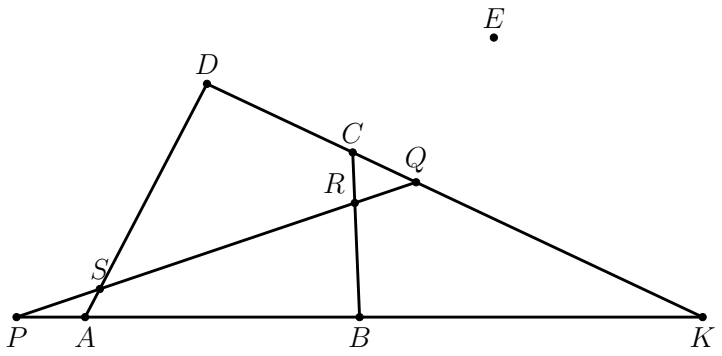
Proof: Let Γ_i have the equation $\Gamma_i(x, y, z) = -a^2yz - b^2zx - c^2xy + (x + y + z)(r_ix + s_iy + t_iz)$. Then $\operatorname{pow}_{\Gamma_i} P = \Gamma_i(P)$. In particular, $\operatorname{pow}_{\Gamma_i} A = \Gamma_i(1, 0, 0) = r_i$ and, similarly, $\operatorname{pow}_{\Gamma_i} B = s_i$ and $\operatorname{pow}_{\Gamma_i} C = t_i$.
Finally, the radical axis is

$$\begin{aligned}
&\operatorname{pow}_{\Gamma_1} P = \operatorname{pow}_{\Gamma_2} P \\
\iff &\Gamma_1(x, y, z) = \Gamma_2(x, y, z) \\
\iff &r_1x + s_1y + t_1z = r_2x + s_2y + t_2z \\
\iff &(\operatorname{pow}_{\Gamma_1} A - \operatorname{pow}_{\Gamma_2} A)x + (\operatorname{pow}_{\Gamma_1} B - \operatorname{pow}_{\Gamma_2} B)y + (\operatorname{pow}_{\Gamma_1} C - \operatorname{pow}_{\Gamma_2} C)z = 0.
\end{aligned}$$

We still use the Miquel point E of $ABCD$. Notice that the problem is equivalent to proving that lines MP, NQ, and EK are concurrent. The main idea is writing these three lines as radical axes. In fact, by definition of points M, N, and E:

- MP is the radical axis of the circumcircles of PAS and PBR;
- NQ is the radical axis of the circumcircles of QCR and QDS;
- EK is the radical axis of the circumcircles of KBC and KAD.

Looking at these facts and the diagram, it makes sense to take triangle KQP the reference triangle. Because of that, we do not really need to draw circles nor even points M and N, as all powers can be computed directly from points in lines KP, KQ, and PQ.



Associate P with the x-coordinate, Q with the y-coordinate, and K with the z-coordinate. Applying the lemma, the equations of lines PM, QN, and EK are

- MP: $(KA \cdot KP - KB \cdot KP)x + (QS \cdot QP - QR \cdot QP)y = 0$
- NQ: $(KC \cdot KQ - KD \cdot KQ)x + (PR \cdot PQ - PS \cdot PQ)z = 0$
- MP: $(-QC \cdot QK + QD \cdot QK)y + (PB \cdot PK - PA \cdot PK)z = 0$

These equations simplify to

- MP: $(AB \cdot KP)x + (PQ \cdot RS)y = 0$
- NQ: $(-CD \cdot KQ)x + (PQ \cdot RS)z = 0$
- MP: $(CD \cdot KQ)y + (AB \cdot KP)z = 0$

Now, if $u = AB \cdot KP$, $v = PQ \cdot RS$, and $w = CD \cdot KQ$, it suffices to show that

$$\begin{vmatrix} u & v & 0 \\ -w & 0 & v \\ 0 & w & u \end{vmatrix} = 0,$$

which is a straightforward computation.

APMO 2023 – Problems and Solutions

Problem 1

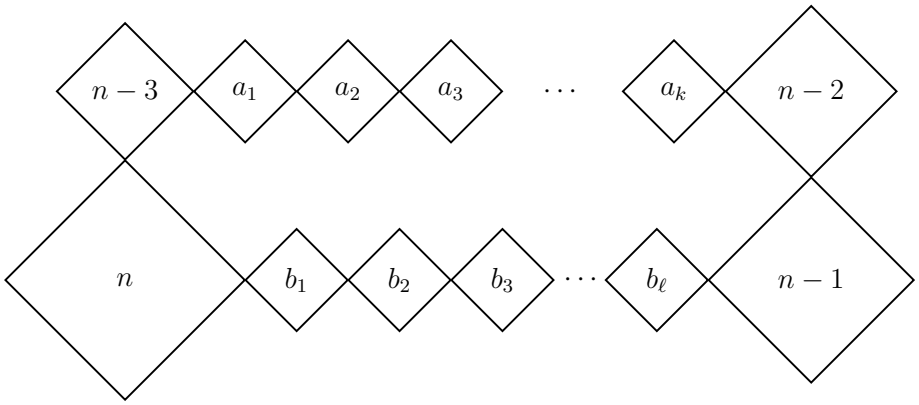
Let $n \geq 5$ be an integer. Consider n squares with side lengths $1, 2, \ldots, n$, respectively. The squares are arranged in the plane with their sides parallel to the x and y axes. Suppose that no two squares touch, except possibly at their vertices.

Show that it is possible to arrange these squares in a way such that every square touches exactly two other squares.

Solution 1

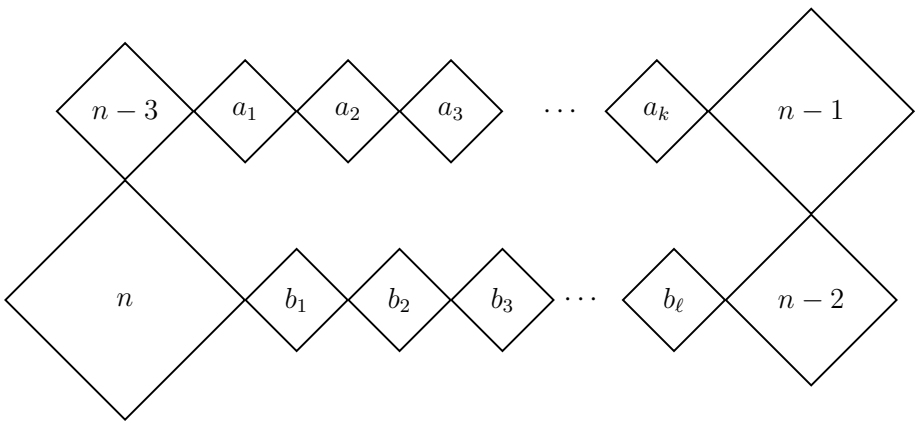
Set aside the squares with sidelengths $n-3$, $n-2$, $n-1$, and n and suppose we can split the remaining squares into two sets A and B such that the sum of the sidelengths of the squares in A is 1 or 2 units larger than the sum of the sidelengths of the squares in B.

String the squares of each set A, B along two parallel diagonals, one for each diagonal. Now use the four largest squares along two perpendicular diagonals to finish the construction: one will have sidelengths n and $n-3$, and the other, sidelengths $n-1$ and $n-2$. If the sum of the sidelengths of the squares in A is 1 unit larger than the sum of the sidelengths of the squares in B, attach the squares with sidelengths $n-3$ and $n-1$ to the A-diagonal, and the other two squares to the B-diagonal. The resulting configuration, in which the A and B-diagonals are represented by unit squares, and the sidelengths a_i of squares from A and b_j of squares from B are indicated within each square, follows:



Since $(a_1 + a_2 + \cdots + a_k)\sqrt{2} + \frac{((n-3)+(n-2))\sqrt{2}}{2} = (b_1 + b_2 + \cdots + b_\ell + 2)\sqrt{2} + \frac{(n+(n-1))\sqrt{2}}{2}$, this case is done.

If the sum of the sidelengths of the squares in A is 1 unit larger than the sum of the sidelengths of the squares in B, attach the squares with sidelengths $n-3$ and $n-2$ to the A-diagonal, and the other two squares to the B-diagonal. The resulting configuration follows:

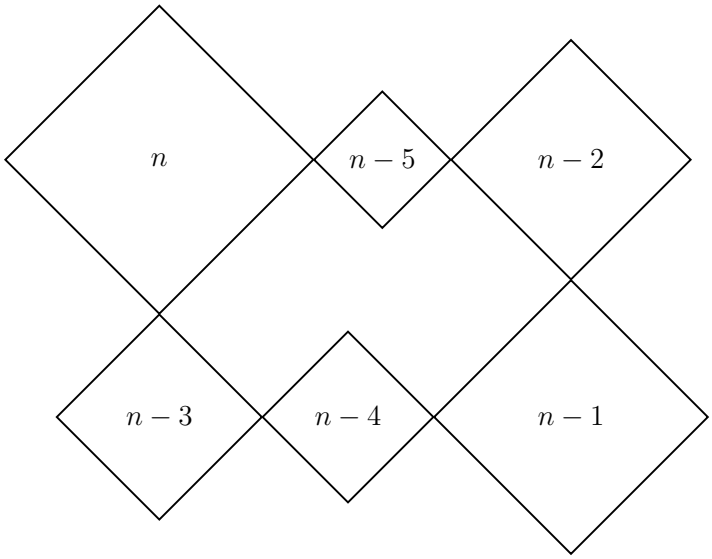


Since $(a_1 + a_2 + \cdots + a_k)\sqrt{2} + \frac{((n-3)+(n-1))\sqrt{2}}{2} = (b_1 + b_2 + \cdots + b_\ell + 1)\sqrt{2} + \frac{(n+(n-2))\sqrt{2}}{2}$, this case is also done.
In both cases, the distance between the A-diagonal and the B-diagonal is $\frac{((n-3)+n)\sqrt{2}}{2} = \frac{(2n-3)\sqrt{2}}{2}$.
Since $a_i, b_j \leq n-4$, $\frac{(a_i+b_j)\sqrt{2}}{2} < \frac{(2n-4)\sqrt{2}}{2} < \frac{(2n-3)\sqrt{2}}{2}$, and therefore the A- and B-diagonals do not overlap.
Finally, we prove that it is possible to split the squares of sidelengths 1 to $n-4$ into two sets A and B such that the sum of the sidelengths of the squares in A is 1 or 2 units larger than the sum of the sidelengths of the squares in B. One can do that in several ways; we present two possibilities:

- *Direct construction:* Split the numbers from 1 to $n-4$ into several sets of four consecutive numbers $\{t, t+1, t+2, t+3\}$, beginning with the largest numbers; put squares of sidelengths t and $t+3$ in A and squares of sidelengths $t+1$ and $t+2$ in B. Notice that $t + (t+3) = (t+1) + (t+2)$. In the end, at most four numbers remain.
  - If only 1 remains, put the corresponding square in A, so the sum of the sidelengths of the squares in A is one unit larger that those in B;
  - If 1 and 2 remains, put the square of sidelength 2 in A and the square of sidelength 1 in B (the difference is 1);
  - If 1, 2, and 3 remains, put the squares of sidelengths 1 and 3 in A, and the square of sidelength 2 in B (the difference is 2);
  - If 1, 2, 3, and 4 remains, put the squares of sidelengths 2 and 4 in A, and the squares of sidelengths 1 and 3 in B (the difference is 2).
- *Indirect construction:* Starting with A and B as empty sets, add the squares of sidelengths $n-4, n-3, \ldots, 2$ to either A or B in that order such that at each stage the difference between the sum of the sidelengths in A and the sum of the sidelengths of B is minimized. By induction it is clear that after adding an integer j to one of the sets, this difference is at most j. In particular, the difference is 0, 1 or 2 at the end. Finally adding the final 1 to one of the sets can ensure that the final difference is 1 or 2. If necessary, flip A and B.

Solution 2

Solve the problem by induction in n. Construct examples for $n = 5, 6, 7, 8, 9, 10$ (one can use the constructions from the previous solution, for instance). For $n > 10$, set aside the six larger squares and arrange them in the following fashion:



By the induction hypothesis, one can arrange the remaining $n-6$ squares away from the six larger squares, so we are done.

Problem 2

Find all integers n satisfying $n \geq 2$ and $\frac{\sigma(n)}{p(n)-1} = n$, in which $\sigma(n)$ denotes the sum of all positive divisors of n, and $p(n)$ denotes the largest prime divisor of n.

Answer: $n = 6$.

Solution

Let $n = p_1^{\alpha_1} \cdot \ldots \cdot p_k^{\alpha_k}$ be the prime factorization of n with $p_1 < \ldots < p_k$, so that $p(n) = p_k$ and $\sigma(n) = (1 + p_1 + \cdots + p_1^{\alpha_1}) \ldots (1 + p_k + \cdots + p_k^{\alpha_k})$. Hence

$$p_k - 1 = \frac{\sigma(n)}{n} = \prod_{i=1}^{k} \left(1 + \frac{1}{p_i} + \cdots + \frac{1}{p_i^{\alpha_i}}\right) < \prod_{i=1}^{k} \frac{1}{1 - \frac{1}{p_i}} = \prod_{i=1}^{k} \left(1 + \frac{1}{p_i - 1}\right) \leq \prod_{i=1}^{k} \left(1 + \frac{1}{i}\right) = k+1,$$

that is, $p_k - 1 < k+1$, which is impossible for $k \geq 3$, because in this case $p_k - 1 \geq 2k - 2 \geq k+1$. Then $k \leq 2$ and $p_k < k + 2 \leq 4$, which implies $p_k \leq 3$.

If $k = 1$ then $n = p^\alpha$ and $\sigma(n) = 1 + p + \cdots + p^\alpha$, and in this case $n \nmid \sigma(n)$, which is not possible. Thus $k = 2$, and $n = 2^\alpha 3^\beta$ with $\alpha, \beta > 0$. If $\alpha > 1$ or $\beta > 1$,

$$\frac{\sigma(n)}{n} > \left(1 + \frac{1}{2}\right)\left(1 + \frac{1}{3}\right) = 2.$$

Therefore $\alpha = \beta = 1$ and the only answer is $n = 6$.

Comment: There are other ways to deal with the case $n = 2^\alpha 3^\beta$. For instance, we have $2^{\alpha+2} 3^\beta = (2^{\alpha+1} - 1)(3^{\beta+1} - 1)$. Since $2^{\alpha+1} - 1$ is not divisible by 2, and $3^{\beta+1} - 1$ is not divisible by 3, we have

$$\begin{cases} 2^{\alpha+1} - 1 = 3^\beta \\ 3^{\beta+1} - 1 = 2^{\alpha+2} \end{cases} \iff \begin{cases} 2^{\alpha+1} - 1 = 3^\beta \\ 3 \cdot (2^{\alpha+1} - 1) - 1 = 2 \cdot 2^{\alpha+1} \end{cases} \iff \begin{cases} 2^{\alpha+1} = 4 \\ 3^\beta = 3 \end{cases},$$

and $n = 2^\alpha 3^\beta = 6$.

Problem 3

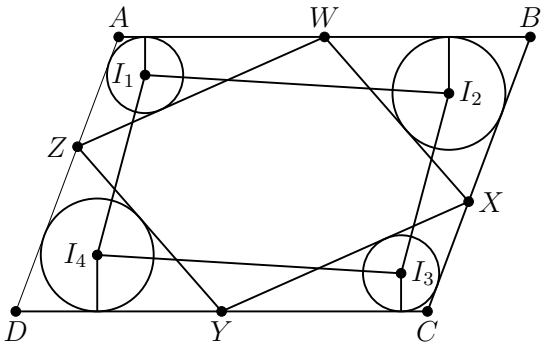
Let $ABCD$ be a parallelogram. Let W, X, Y, and Z be points on sides AB, BC, CD, and DA, respectively, such that the incenters of triangles AWZ, BXW, CYX and DZY form a parallelogram. Prove that $WXYZ$ is a parallelogram.

Solution

Let the four incenters be I_1, I_2, I_3, and I_4 with inradii r_1, r_2, r_3, and r_4 respectively (in the order given in the question). Without loss of generality, let I_1 be closer to AB than I_2. Let the acute angle between I_1I_2 and AB (and hence also the angle between I_3I_4 and CD) be θ. Then

$$r_2 - r_1 = I_1I_2 \sin\theta = I_3I_4 \sin\theta = r_4 - r_3,$$

which implies $r_1 + r_4 = r_2 + r_3$. Similar arguments show that $r_1 + r_2 = r_3 + r_4$. Thus we obtain $r_1 = r_3$ and $r_2 = r_4$.



Now let's consider the possible positions of W, X, Y, Z. Suppose $AZ \neq CX$. Without loss of generality assume $AZ > CX$. Since the incircles of AWZ and CYX are symmetric about the centre of the parallelogram $ABCD$, this implies $CY > AW$. Using similar arguments, we have

$$CY > AW \implies BW > DY \implies DZ > BX \implies CX > AZ,$$

which is a contradiction. Therefore $AZ = CX \implies AW = CY$ and $WXYZ$ is a parallelogram.

Comment: There are several ways to prove that $r_1 = r_3$ and $r_2 = r_4$. The proposer shows the following three alternative approaches:

Using parallel lines: Let O be the centre of parallelogram $ABCD$ and P be the centre of parallelogram $I_1I_2I_3I_4$. Since AI_1 and CI_3 are angle bisectors, we must have $AI_1 \parallel CI_3$. Let ℓ_1 be the line through O parallel to AI_1. Since $AO = OC$, ℓ_1 is halfway between AI_1 and CI_3. Hence P must lie on ℓ_1.

Similarly, P must also lie on ℓ_2, the line through O parallel to BI_2. Thus P is the intersection of ℓ_1 and ℓ_2, which must be O. So the four incentres and hence the four incircles must be symmetric about O, which implies $r_1 = r_3$ and $r_2 = r_4$.

Using a rotation: Let the bisectors of $\angle DAB$ and $\angle ABC$ meet at X and the bisectors of $\angle BCD$ and $\angle CDA$ meet at Y. Then I_1 is on AX, I_2 is on BX, I_3 is on CY, and I_4 is on DY. Let O be the centre of $ABCD$. Then a 180 degree rotation about O takes $\triangle AXB$ to $\triangle CYD$. Under the same transformation I_1I_2 is mapped to a parallel segment $I_1'I_2'$ with I_1' on CY and I_2' on DY. Since $I_1I_2I_3I_4$ is a parallelogram, $I_3I_4 = I_1I_2$ and $I_3I_4 \parallel I_1I_2$. Hence $I_1'I_2'$ and I_3I_4 are parallel, equal length segments on sides CY, DY and we conclude that $I_1' = I_3$, $I_2' = I_4$. Hence the centre of $I_1I_2I_3I_4$ is also O and we establish that by rotational symmetry that $r_1 = r_3$ and $r_2 = r_4$.

Using congruent triangles: Let AI_1 and BI_2 intersect at E and let CI_3 and DI_4 intersect at F. Note that $\triangle ABE$ and $\triangle CDF$ are congruent, since $AB = CD$ and corresponding pairs of angles are equal (equal opposite angles parallelogram $ABCD$ are each bisected).

Since $AI_1 \parallel CI_3$ and $I_1I_2 \parallel I_4I_3$, $\angle I_2I_1E = \angle I_4I_3F$. Similarly $\angle I_1I_2E = \angle I_3I_4F$. Furthermore $I_1I_2 = I_3I_4$. Hence triangles I_2I_1E and $I4I_3F$ are also congruent.
Hence $ABEI_1I_2$ and $DCFI_3I_4$ are congruent. Therefore, the perpendicular distance from I_1 to AB equals the perpendicular distance from I_3 to CD, that is, $r_1 = r_3$. Similarly $r_2 = r_4$.

Problem 4

Let $c > 0$ be a given positive real and $\mathbb{R}_{>0}$ be the set of all positive reals. Find all functions $f\colon \mathbb{R}_{>0} \to \mathbb{R}_{>0}$ such that

$$f((c+1)x + f(y)) = f(x+2y) + 2cx \quad \text{for all } x, y \in \mathbb{R}_{>0}.$$

Answer: $f(x) = 2x$ for all $x > 0$.

Solution 1

We first prove that $f(x) \geq 2x$ for all $x > 0$. Suppose, for the sake of contradiction, that $f(y) < 2y$ for some positive y. Choose x such that $f((c+1)x + f(y))$ and $f(x+2y)$ cancel out, that is,

$$(c+1)x + f(y) = x + 2y \iff x = \frac{2y - f(y)}{c}.$$

Notice that $x > 0$ because $2y - f(y) > 0$. Then $2cx = 0$, which is not possible. This contradiction yields $f(y) \geq 2y$ for all $y > 0$.

Now suppose, again for the sake of contradiction, that $f(y) > 2y$ for some $y > 0$. Define the following sequence: a_0 is an arbitrary real greater than $2y$, and $f(a_n) = f(a_{n-1}) + 2cx$, so that

$$\begin{cases} (c+1)x + f(y) = a_n \\ x + 2y = a_{n-1} \end{cases} \iff x = a_{n-1} - 2y \quad \text{and} \quad a_n = (c+1)(a_{n-1} - 2y) + f(y).$$

If $x = a_{n-1} - 2y > 0$ then $a_n > f(y) > 2y$, so inductively all the substitutions make sense.

For the sake of simplicity, let $b_n = a_n - 2y$, so $b_n = (c+1)b_{n-1} + f(y) - 2y \quad (*)$. Notice that $x = b_{n-1}$ in the former equation, so $f(a_n) = f(a_{n-1}) + 2cb_{n-1}$. Telescoping yields

$$f(a_n) = f(a_0) + 2c\sum_{i=0}^{n-1} b_i.$$

One can find b_n from the recurrence equation $(*)$: $b_n = \left(b_0 + \frac{f(y)-2y}{c}\right)(c+1)^n - \frac{f(y)-2y}{c}$, and then

$$\begin{aligned} f(a_n) &= f(a_0) + 2c\sum_{i=0}^{n-1}\left(\left(b_0 + \frac{f(y)-2y}{c}\right)(c+1)^i - \frac{f(y)-2y}{c}\right) \\ &= f(a_0) + 2\left(b_0 + \frac{f(y)-2y}{c}\right)((c+1)^n - 1) - 2n(f(y)-2y). \end{aligned}$$

Since $f(a_n) \geq 2a_n = 2b_n + 4y$,

$$\begin{aligned} f(a_0) + 2\left(b_0 + \frac{f(y)-2y}{c}\right)((c+1)^n - 1) - 2n(f(y)-2y) &\geq 2b_n + 4y \\ = 2\left(b_0 + \frac{f(y)-2y}{c}\right)(c+1)^n - 2\frac{f(y)-2y}{c}, \end{aligned}$$

which implies

$$f(a_0) + 2\frac{f(y)-2y}{c} \geq 2\left(b_0 + \frac{f(y)-2y}{c}\right) + 2n(f(y)-2y),$$

which is not true for sufficiently large n.

A contradiction is reached, and thus $f(y) = 2y$ for all $y > 0$. It is immediate that this function satisfies the functional equation.

Solution 2

After proving that $f(y) \geq 2y$ for all $y > 0$, one can define $g(x) = f(x) - 2x$, $g\colon \mathbb{R}_{>0} \to \mathbb{R}_{\geq 0}$, and our goal is proving that $g(x) = 0$ for all $x > 0$. The problem is now rewritten as

$$\begin{aligned} & g((c+1)x + g(y) + 2y) + 2((c+1)x + g(y) + 2y) = g(x+2y) + 2(x+2y) + 2cx \\ \iff & g((c+1)x + g(y) + 2y) + 2g(y) = g(x+2y). \end{aligned} \tag{1}$$

This readily implies that $g(x+2y) \geq 2g(y)$, which can be interpreted as $z > 2y \implies g(z) \geq 2g(y)$, by plugging $z = x + 2y$.

Now we prove by induction that $z > 2y \implies g(z) \geq 2m \cdot g(y)$ for any positive integer $2m$. In fact, since $(c+1)x + g(y) + 2y > 2y$, $g((c+1)x + g(y) + 2y) \geq 2m \cdot g(y)$, and by (**??**),

$$g(x+2y) \geq 2m \cdot g(y) + 2g(y) = 2(m+1)g(y),$$

and we are done by plugging $z = x + 2y$ again.

The problem now is done: if $g(y) > 0$ for some $y > 0$, choose a fixed $z > 2y$ arbitrarily and and integer m such that $m > \frac{g(z)}{2g(y)}$. Then $g(z) < 2m \cdot g(y)$, contradiction.

Problem 5

There are n line segments on the plane, no three intersecting at a point, and each pair intersecting once in their respective interiors. Tony and his $2n-1$ friends each stand at a distinct endpoint of a line segment. Tony wishes to send Christmas presents to each of his friends as follows:

First, he chooses an endpoint of each segment as a "sink". Then he places the present at the endpoint of the segment he is at. The present moves as follows:

- If it is on a line segment, it moves towards the sink.
- When it reaches an intersection of two segments, it changes the line segment it travels on and starts moving towards the new sink.

If the present reaches an endpoint, the friend on that endpoint can receive their present. Prove Tony can send presents to exactly n of his $2n-1$ friends.

Solution 1

Draw a circle that encloses all the intersection points between line segments and extend all line segments until they meet the circle, and then move Tony and all his friends to the circle. Number the intersection points with the circle from 1 to $2n$ anticlockwise, starting from Tony (Tony has number 1). We will prove that the friends eligible to receive presents are the ones on even-numbered intersection points.

First part: at most n friends can receive a present.

The solution relies on a well-known result: the n lines determine regions inside the circle; then it is possible to paint the regions with two colors such that no regions with a common (line) boundary have the same color. The proof is an induction on n: the fact immediately holds for $n=0$, and the induction step consists on taking away one line ℓ, painting the regions obtained with $n-1$ lines, drawing ℓ again and flipping all colors on exactly one half plane determined by ℓ.

Now consider the line starting on point 1. Color the regions in red and blue such that neighboring regions have different colors, and such that the two regions that have point 1 as a vertex are red on the right and blue on the left, from Tony's point of view. Finally, assign to each red region the clockwise direction and to each blue region the anticlockwise direction. Because of the coloring, every boundary will have two directions assigned, but the directions are the same since every boundary divides regions of different colors. Then the present will follow the directions assigned to the regions: it certainly does for both regions in the beginning, and when the present reaches an intersection it will keep bordering one of the two regions it was dividing.

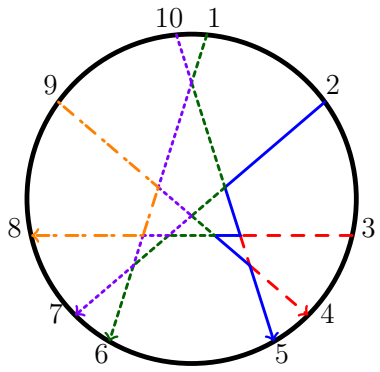
To finish this part of the problem, consider the regions that share a boundary with the circle. The directions alternate between outcoming and incoming, starting from 1 (outcoming), so all even-numbered vertices are directed as incoming and are the only ones able to receive presents.

Second part: all even-numbered vertices can receive a present.

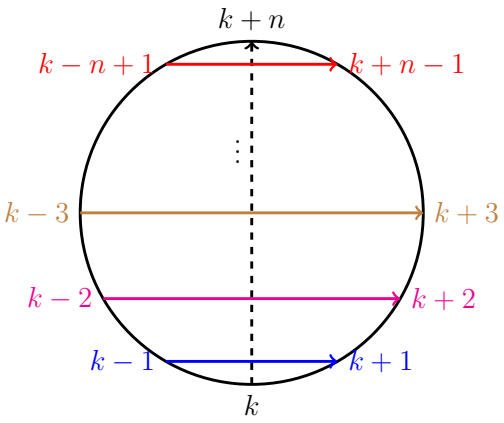
First notice that, since every two chords intersect, every chord separates the endpoints of each of the other $n-1$ chords. Therefore, there are $n-1$ vertices on each side of every chord, and each chord connects vertices k and $k+n$, $1 \le k \le n$.

We prove a stronger result by induction in n: let k be an integer, $1 \le k \le n$. Direct each chord from i to $i+n$ if $1 \le i \le k$ and from $i+n$ to i otherwise; in other words, the sinks are $k+1, k+2, \ldots, k+n$. Now suppose that each chord sends a present, starting from the vertex opposite to each sink, and all presents move with the same rules. Then $k-i$ sends a present to $k+i+1$, $i=0,1,\ldots,n-1$ (indices taken modulo $2n$). In particular, for $i=k-1$, Tony, in vertex 1, send a present to vertex $2k$. Also, the n paths the presents make do not cross (but they may touch.) More formally, for all i, $1 \le i \le n$, if one path takes a present from $k-i$ to $k+i+1$, separating the circle into two regions, all paths taking a present from $k-j$ to $k+j+1$, $j<i$, are completely contained in one region, and all paths taking a present from

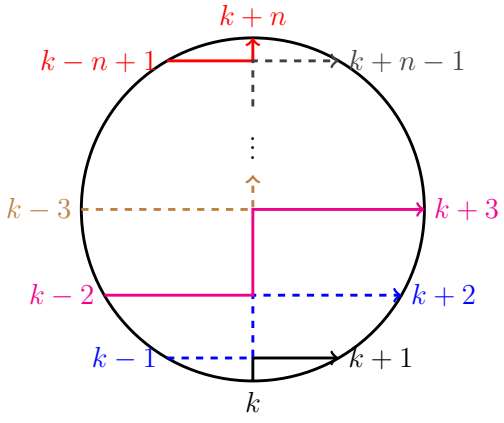
$k-j$ to $k+j+1$, $j>i$, are completely contained in the other region. For instance, possible[1] paths for $k=3$ and $n=5$ follow:



The result is true for $n=1$. Let $n>1$ and assume the result is true for less chords. Consider the chord that takes k to $k+n$ and remove it. Apply the induction hypothesis to the remaining $n-1$ lines: after relabeling, presents would go from $k-i$ to $k+i+2$, $1\le i\le n-1$ if the chord were not there.
Reintroduce the chord that takes k to $k+n$. From the induction hypothesis, the chord intersects the paths of the presents in the following order: the i-th path the chord intersects is the the one that takes $k-i$ to $k+i$, $i=1,2,\ldots,n-1$.



Paths without chord $k\to k+n$



Corrected paths with chord $k\to k+n$

Then the presents cover the following new paths: the present from k will leave its chord and take the path towards $k+1$; then, for $i=1,2,\ldots,n-1$, the present from $k-i$ will meet the chord from k to $k+n$, move towards the intersection with the path towards $k+i+1$ and go to $k+i+1$, as desired. Notice that the paths still do not cross. The induction (and the solution) is now complete.

Solution 2

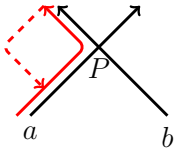
First part: at most n friends can receive a present.

Similarly to the first solution, consider a circle that encompasses all line segments, extend the lines, and use the endpoints of the chords instead of the line segments, and prove that each chord connects vertices k and $k+n$. We also consider, even in the first part, n presents leaving from n outcoming vertices.

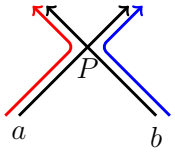
First we prove that a present always goes to a sink. If it does not, then it loops; let it first enter the loop at point P after turning from chord a to chord b. Therefore after it loops once,

[1]The paths do not depend uniquely on k and n; different chord configurations and vertex labelings may change the paths.

it must turn to chord b at P. But P is the intersection of a and b, so the present should turn from chord a to chord b, which can only be done in one way – the same way it came in first. This means that some part of chord a before the present enters the loop at P is part of the loop, which contradicts the fact that P is the first point in the loop. So no present enters a loop, and every present goes to a sink.



There are no loops



No two paths cross

The present paths also do not cross: in fact, every time two paths share a point P, intersection of chords a and b, one path comes from a to b and the other path comes from b to a, and they touch at P. This implies the following sequence of facts:

- Every path divides the circle into two regions with paths connecting vertices within each region.
- All n presents will be delivered to n different persons; that is, all sinks receive a present. This implies that every vertex is an endpoint of a path.
- The number of chord endpoints inside each region is even, because they are connected within their own region.

Now consider the path starting at vertex 1, with Tony. It divides the circle into two regions with an even number of vertices in their interior. Then there is an even number of vertices between Tony and the recipient of his present, that is, their vertex is an even numbered one.
Second part: all even-numbered vertices can receive a present.
The construction is the same as the in the previous solution: direct each chord from i to $i+n$ if $1 \leq i \leq k$ and from $i+n$ to i otherwise; in other words, the sinks are $k+1$, $k+2$, …, $k+n$. Then, since the paths do not cross, k will send a present to $k+1$, $k-1$ will send a present to $k+2$, and so on, until 1 sends a present to $(k+1)+(k-1)=2k$.

APMO 2022 Solution

1. Find all pairs (a, b) of positive integers such that a^3 is a multiple of b^2 and $b-1$ is a multiple of $a-1$.

   Note: An integer n is said to be a multiple of an integer m if there is an integer k such that $n = km$.

   Solution

   Solution 1.1

   By inspection, we see that the pairs (a, b) with $a = b$ are solutions, and so too are the pairs $(a, 1)$. We will see that these are the only solutions.

   - *Case 1.* Consider the case $b < a$. Since $b-1$ is a multiple of $a-1$, it follows that $b = 1$. This yields the second set of solutions described above.
   - *Case 2.* This leaves the case $b \geq a$. Since the positive integer a^3 is a multiple of b^2, there is a positive integer c such that $a^3 = b^2c$.

     Note that $a \equiv b \equiv 1$ modulo $a-1$. So we have

     $$1 \equiv a^3 = b^2c \equiv c \pmod{a-1}.$$

     If $c < a$, then we must have $c = 1$, hence, $a^3 = b^2$. So there is a positive integer d such that $a = d^2$ and $b = d^3$. Now $a-1 \mid b-1$ yields $d^2-1 \mid d^3-1$. This implies that $d+1 \mid d(d+1)+1$, which is impossible.

     If $c \geq a$, then $b^2c \geq b^2a \geq a^3 = b^2c$. So there's equality throughout, implying $a = c = b$. This yields the first set of solutions described above.

   Therefore, the solutions described above are the only solutions.

   □

   Solution 1.2

   We will start by showing that there are positive integers x, c, d such that $a = x^2cd$ and $b = x^3c$. Let $g = \gcd(a, b)$ so that $a = gd$ and $b = gx$ for some coprime d and x. Then, $b^2 \mid a^3$ is equivalent to $g^2x^2 \mid g^3d^3$, which is equivalent to $x^2 \mid gd^3$. Since x and d are coprime, this implies $x^2 \mid g$. Hence, $g = x^2c$ for some c, giving $a = x^2cd$ and $b = x^3c$ as required.

   Now, it remains to find all positive integers x, c, d satisfying

   $$x^2cd - 1 \mid x^3c - 1.$$

   That is, $x^3c \equiv 1 \pmod{x^2cd-1}$. Assuming that this congruence holds, it follows that $d \equiv x^3cd \equiv x \pmod{x^2cd-1}$. Then, either $x = d$ or $x - d \geq x^2cd - 1$ or $d - x \geq x^2cd - 1$.

   - If $x = d$ then $b = a$.
   - If $x - d \geq x^2cd - 1$, then $x - d \geq x^2cd - 1 \geq x - 1 \geq x - d$. Hence, each of these inequalities must in fact be an equality. This implies that $x = c = d = 1$, which implies that $a = b = 1$.
   - If $d - x \geq x^2cd - 1$, then $d - x \geq x^2cd - 1 \geq d - 1 \geq d - x$. Hence, each of these inequalities must in fact be an equality. This implies that $x = c = 1$, which implies that $b = 1$.

   Hence the only solutions are the pairs (a, b) such that $a = b$ or $b = 1$. These pairs can be checked to satisfy the given conditions.

□

Solution 1.3

All answers are (n, n) and $(n, 1)$ where n is any positive integer. They all clearly work.

To show that these are all solutions, note that we can easily eliminate the case $a = 1$ or $b = 1$. Thus, assume that $a, b \neq 1$ and $a \neq b$. By the second divisibility, we see that $a - 1 \mid b - a$. However, $\gcd(a, b) \mid b - a$ and $a - 1$ is relatively prime to $\gcd(a, b)$. This implies that $(a - 1)\gcd(a, b) \mid b - a$, which implies $\gcd(a, b) \mid \dfrac{b-1}{a-1} - 1$.

The last relation implies that $\gcd(a, b) < \dfrac{b-1}{a-1}$, since the right-hand side are positive. However, due to the first divisibility,

$$\gcd(a, b)^3 = \gcd(a^3, b^3) \geq \gcd(b^2, b^3) = b^2.$$

Combining these two inequalities, we get that

$$b^{\frac{2}{3}} < \frac{b-1}{a-1} < 2\frac{b}{a}.$$

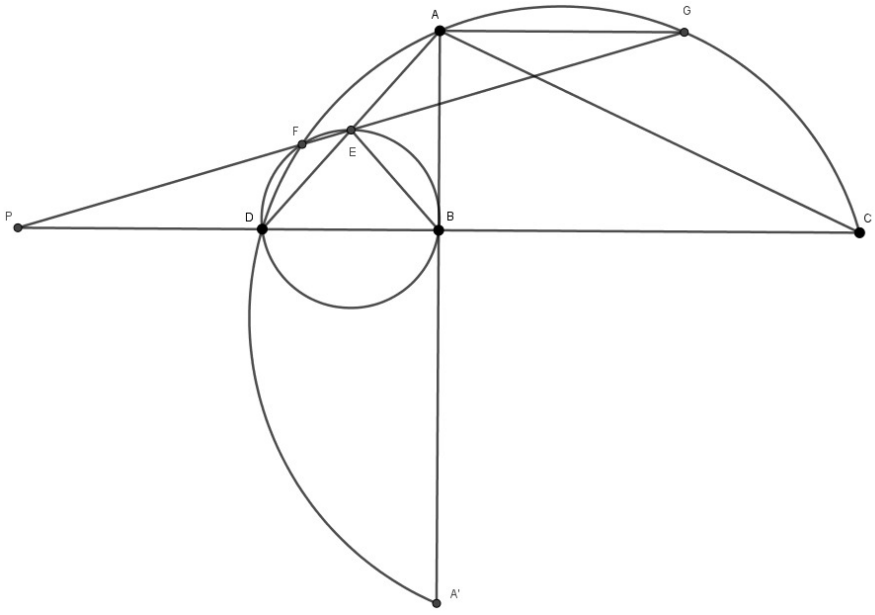
This implies $a < 2b^{\frac{1}{3}}$. However, $b^2 \mid a^3$ gives $b \leq a^{\frac{3}{2}}$. This forces

$$a < 2(a^{\frac{3}{2}})^{\frac{1}{3}} = 2\sqrt{a} \implies a < 4.$$

Extracting $a = 2, 3$ by hand yields no additional solution.

□

2. Let ABC be a right triangle with $\angle B = 90°$. Point D lies on the line CB such that B is between D and C. Let E be the midpoint of AD and let F be the second intersection point of the circumcircle of $\triangle ACD$ and the circumcircle of $\triangle BDE$. Prove that as D varies, the line EF passes through a fixed point.



Solution

Solution 2.1

Let the line EF intersect the line BC at P and the circumcircle of $\triangle ACD$ at G distinct from F. We will prove that P is the fixed point.

First, notice that $\triangle BED$ is isosceles with $EB = ED$. This implies $\angle EBC = \angle EDP$.

Then, $\angle DAG = \angle DFG = \angle EBC = \angle EDP$ which implies $AG \parallel DC$. Hence, $AGCD$ is an isosceles trapezoid.

Also, $AG \parallel DC$ and $AE = ED$. This implies $\triangle AEG \cong \triangle DEP$ and $AG = DP$.

Since B is the foot of the perpendicular from A onto the side CD of the isosceles trapezoid $AGCD$, we have $PB = PD + DB = AG + DB = BC$, which does not depend on the choice of D. Hence, the initial statement is proven.

Solution 2.2

Set up a coordinate system where BC is along the positive x-axis, BA is along the positive y-axis, and B is the origin. Take $A = (0, a)$, $B = (0, 0)$, $C = (c, 0)$, $D = (-d, 0)$ where $a, b, c, d > 0$. Then $E = (-\frac{d}{2}, \frac{a}{2})$. The general equation of a circle is

$$x^2 + y^2 + 2fx + 2gy + h = 0 \tag{1}$$

Substituting the coordinates of A, D, C into (1) and solving for f, g, h, we find that the equation of the circumcircle of $\triangle ADC$ is

$$x^2 + y^2 + (d - c)x + (\frac{cd}{a} - a)y - cd = 0. \tag{2}$$

Similarly, the equation of the circumcircle of $\triangle BDE$ is

$$x^2 + y^2 + dx + (\frac{d^2}{2a} - \frac{a}{2})y = 0. \tag{3}$$

Then (3)$-$(2) gives the equation of the line DF which is

$$cx + \frac{a^2 + d^2 - 2cd}{2a}y + cd = 0. \tag{4}$$

Solving (3) and (4) simultaneously, we get

$$F = \left(\frac{c(d^2 - a^2 - 2cd)}{a^2 + (d - 2c)^2}, \frac{2ac(c - d)}{a^2 + (d - 2c)^2}\right),$$

and the other solution $D = (-d, 0)$.

From this we obtain the equation of the line EF which is $ax + (d - 2c)y + ac = 0$. It passes through $P(-c, 0)$ which is independent of d.

3. Find all positive integers $k < 202$ for which there exists a positive integer n such that

$$\left\{\frac{n}{202}\right\} + \left\{\frac{2n}{202}\right\} + \cdots + \left\{\frac{kn}{202}\right\} = \frac{k}{2},$$

where $\{x\}$ denote the fractional part of x.

Note: $\{x\}$ denotes the real number k with $0 \le k < 1$ such that $x - k$ is an integer.

Solution

Denote the equation in the problem statement as (*), and note that it is equivalent to the condition that the average of the remainders when dividing $n, 2n, \ldots, kn$ by 202 is 101. Since $\left\{\frac{in}{202}\right\}$ is invariant in each residue class modulo 202 for each $1 \le i \le k$, it suffices to consider $0 \le n < 202$.

If $n = 0$, so is $\left\{\frac{in}{202}\right\}$, meaning that (*) does not hold for any k. If $n = 101$, then it can be checked that (*) is satisfied if and only if $k = 1$. From now on, we will assume that $101 \nmid n$.

For each $1 \leq i \leq k$, let $a_i = \left\lfloor \frac{in}{202} \right\rfloor = \frac{in}{202} - \left\{\frac{in}{202}\right\}$. Rewriting (*) and multiplying the equation by 202, we find that

$$n(1 + 2 + \ldots + k) - 202(a_1 + a_2 + \ldots + a_k) = 101k.$$

Equivalently, letting $z = a_1 + a_2 + \ldots + a_k$,

$$nk(k + 1) - 404z = 202k.$$

Since n is not divisible by 101, which is prime, it follows that $101 \mid k(k + 1)$. In particular, $101 \mid k$ or $101 \mid k + 1$. This means that $k \in \{100, 101, 201\}$. We claim that all these values of k work.

- If $k = 201$, we may choose $n = 1$. The remainders when dividing n, $2n$, $\ldots$, kn by 202 are 1, 2, $\ldots$, 201, which have an average of 101.
- If $k = 100$, we may choose $n = 2$. The remainders when dividing n, $2n$, $\ldots$, kn by 202 are 2, 4, $\ldots$, 200, which have an average of 101.
- If $k = 101$, we may choose $n = 51$. To see this, note that the first four remainders are 51, 102, 153, 2, which have an average of 77. The next four remainders (53, 104, 155, 4) are shifted upwards from the first four remainders by 2 each, and so on, until the 25th set of the remainders (99, 150, 201, 50) which have an average of 125. Hence, the first 100 remainders have an average of $\frac{77 + 125}{2} = 101$. The 101th remainder is also 101, meaning that the average of all 101 remainders is 101.

In conclusion, all values $k \in \{1, 100, 101, 201\}$ satisfy the initial condition.

□

4. Let n and k be positive integers. Cathy is playing the following game. There are n marbles and k boxes, with the marbles labelled 1 to n. Initially, all marbles are placed inside one box. Each turn, Cathy chooses a box and then moves the marbles with the smallest label, say i, to either any empty box or the box containing marble $i + 1$. Cathy wins if at any point there is a box containing only marble n.
Determine all pairs of integers (n, k) such that Cathy can win this game.

Solution

We claim Cathy can win if and only if $n \leq 2^{k-1}$.

First, note that each non-empty box always contains a consecutive sequence of labeled marbles. This is true since Cathy is always either removing from or placing in the lowest marble in a box. As a consequence, every move made is reversible.

Next, we prove by induction that Cathy can win if $n = 2^{k-1}$. The base case of $n = k = 1$ is trivial. Assume a victory can be obtained for m boxes and 2^{m-1} marbles. Consider the case of $m + 1$ boxes and 2^m marbles. Cathy can first perform a sequence of moves so that only marbles $2^{m-1}, \ldots, 2^m$ are left in the starting box, while keeping one box, say B, empty. Now move the marble 2^{m-1} to box B, then reverse all of the initial moves while treating B as the starting box. At the end of that, we will have marbles $2^{m-1} + 1$, $\ldots$, 2^m in the starting box, marbles 1, 2, $\ldots$, 2^{m-1} in box B, and $m - 1$ empty boxes. By repeating the original sequence of moves on marbles $2^{m-1} + 1, \ldots, 2^m$, using the m boxes that are not box B, we can reach a state where only marble 2^m remains in the starting box. Therefore

a victory is possible if $n = 2^{k-1}$ or smaller.

We now prove by induction that Cathy loses if $n = 2^{k-1} + 1$. The base case of $n = 2$ and $k = 1$ is trivial. Assume a victory is impossible for m boxes and $2^{m-1}+1$ marbles. For the sake of contradiction, suppose that victory is possible for $m + 1$ boxes and $2^m + 1$ marbles. In a winning sequence of moves, consider the last time a marble $2^{m-1} + 1$ leaves the starting box, call this move X. After X, there cannot be a time when marbles $1, \ldots, 2^{m-1} + 1$ are all in the same box. Otherwise, by reversing these moves after X and deleting marbles greater than $2^{m-1} + 1$, it gives us a winning sequence of moves for $2^{m-1} + 1$ marbles and m boxes (as the original starting box is not used here), contradicting the inductive hypothesis. Hence starting from X, marbles 1 will never be in the same box as any marbles greater than or equal to $2^{m-1} + 1$.

Now delete marbles $2, \ldots, 2^{m-1}$ and consider the winning moves starting from X. Marble 1 would only move from one empty box to another, while blocking other marbles from entering its box. Thus we effectively have a sequence of moves for $2^{m-1} + 1$ marbles, while only able to use m boxes. This again contradicts the inductive hypothesis. Therefore, a victory is not possible if $n = 2^{k-1} + 1$ or greater.

□

5. Let a, b, c, d be real numbers such that $a^2 + b^2 + c^2 + d^2 = 1$. Determine the minimum value of $(a - b)(b - c)(c - d)(d - a)$ and determine all values of (a, b, c, d) such that the minimum value is achieved.

Solution

The minimum value is $\boxed{-\frac{1}{8}}$. There are eight equality cases in total. The first one is

$$\left(\frac{1}{4} + \frac{\sqrt{3}}{4}, -\frac{1}{4} - \frac{\sqrt{3}}{4}, \frac{1}{4} - \frac{\sqrt{3}}{4}, -\frac{1}{4} + \frac{\sqrt{3}}{4}\right).$$

Cyclic shifting all the entries give three more quadruples. Moreover, flipping the sign $((a, b, c, d) \to (-a, -b, -c, -d))$ all four entries in each of the four quadruples give four more equality cases.

Solution 5.1

Since the expression is cyclic, we could WLOG $a = \max\{a, b, c, d\}$. Let

$$S(a, b, c, d) = (a - b)(b - c)(c - d)(d - a)$$

Note that we have given (a, b, c, d) such that $S(a, b, c, d) = -\frac{1}{8}$. Therefore, to prove that $S(a, b, c, d) \geq -\frac{1}{8}$, we just need to consider the case where $S(a, b, c, d) < 0$.

- **Exactly** 1 **of** $a - b, b - c, c - d, d - a$ **is negative.**
  Since $a = \max\{a, b, c, d\}$, then we must have $d - a < 0$. This forces $a > b > c > d$. Now, let us write
  $$S(a, b, c, d) = -(a - b)(b - c)(c - d)(a - d)$$
  Write $a - b = y, b - c = x, c - d = w$ for some positive reals $w, x, y > 0$. Plugging to the original condition, we have
  $$(d + w + x + y)^2 + (d + w + x)^2 + (d + w)^2 + d^2 - 1 = 0 \ (*)$$
  and we want to prove that $wxy(w + x + y) \leq \frac{1}{8}$. Consider the expression $(*)$ as a quadratic in d:
  $$4d^2 + d(6w + 4x + 2y) + ((w + x + y)^2 + (w + x)^2 + w^2 - 1) = 0$$

Since d is a real number, then the discriminant of the given equation has to be non-negative, i.e. we must have

$$\begin{aligned}4 &\geq 4((w+x+y)^2+(w+x)^2+w^2)-(3w+2x+y)^2\\ &= (3w^2+2wy+3y^2)+4x(w+x+y)\\ &\geq 8wy+4x(w+x+y)\\ &= 4(x(w+x+y)+2wy)\end{aligned}$$

However, AM-GM gives us

$$wxy(w+x+y) \leq \frac{1}{2}\left(\frac{x(w+x+y)+2wy}{2}\right)^2 \leq \frac{1}{8}$$

This proves $S(a,b,c,d) \geq -\frac{1}{8}$ for any $a,b,c,d \in \mathbb{R}$ such that $a>b>c>d$. Equality holds if and only if $w=y$, $x(w+x+y)=2wy$ and $wxy(w+x+y)=\frac{1}{8}$. Solving these equations gives us $w^4=\frac{1}{16}$ which forces $w=\frac{1}{2}$ since $w>0$. Solving for x gives us $x(x+1)=\frac{1}{2}$, and we will get $x=-\frac{1}{2}+\frac{\sqrt{3}}{2}$ as $x>0$. Plugging back gives us $d=-\frac{1}{4}-\frac{\sqrt{3}}{4}$, and this gives us

$$(a,b,c,d)=\left(\frac{1}{4}+\frac{\sqrt{3}}{4},-\frac{1}{4}+\frac{\sqrt{3}}{4},\frac{1}{4}-\frac{\sqrt{3}}{4},-\frac{1}{4}-\frac{\sqrt{3}}{4}\right)$$

Thus, any cyclic permutation of the above solution will achieve the minimum equality.

- **Exactly 3 of $a-b, b-c, c-d, d-a$ are negative** Since $a=\max\{a,b,c,d\}$, then $a-b$ has to be positive. So we must have $b<c<d<a$. Now, note that

$$\begin{aligned}S(a,b,c,d) &= (a-b)(b-c)(c-d)(d-a)\\ &= (a-d)(d-c)(c-b)(b-a)\\ &= S(a,d,c,b)\end{aligned}$$

Now, note that $a>d>c>b$. By the previous case, $S(a,d,c,b)\geq -\frac{1}{8}$, which implies that

$$S(a,b,c,d)=S(a,d,c,b)\geq -\frac{1}{8}$$

as well. Equality holds if and only if

$$(a,b,c,d)=\left(\frac{1}{4}+\frac{\sqrt{3}}{4},-\frac{1}{4}-\frac{\sqrt{3}}{4},\frac{1}{4}-\frac{\sqrt{3}}{4},-\frac{1}{4}+\frac{\sqrt{3}}{4}\right)$$

and its cyclic permutation.

Solution 5.2

The minimum value is $\boxed{-\frac{1}{8}}$. There are eight equality cases in total. The first one is

$$\left(\frac{1}{4}+\frac{\sqrt{3}}{4},-\frac{1}{4}-\frac{\sqrt{3}}{4},\frac{1}{4}-\frac{\sqrt{3}}{4},-\frac{1}{4}+\frac{\sqrt{3}}{4}\right).$$

Cyclic shifting all the entries give three more quadruples. Moreover, flipping the sign ($(a,b,c,d)\to(-a,-b,-c,-d)$) all four entries in each of the four quadruples give four more equality cases. We then begin the proof by the following optimization:

Claim 1. In order to get the minimum value, we must have $a+b+c+d=0$.

Proof. Assume not, let $\delta = \frac{a+b+c+d}{4}$ and note that

$$(a-\delta)^2 + (b-\delta)^2 + (c-\delta)^2 + (d-\delta)^2 < a^2 + b^2 + c^2 + d^2,$$

so by shifting by δ and scaling, we get an even smaller value of $(a-b)(b-c)(c-d)(d-a)$.

□

The key idea is to substitute the variables

$$\begin{aligned} x &= ac + bd \\ y &= ab + cd \\ z &= ad + bc, \end{aligned}$$

so that the original expression is just $(x-y)(x-z)$. We also have the conditions $x, y, z \geq -0.5$ because of:

$$2x + (a^2 + b^2 + c^2 + d^2) = (a+c)^2 + (b+d)^2 \geq 0.$$

Moreover, notice that

$$0 = (a+b+c+d)^2 = a^2 + b^2 + c^2 + d^2 + 2(x+y+z) \implies x + y + z = \frac{-1}{2}.$$

Now, we reduce to the following optimization problem.

Claim 2. Let $x, y, z \geq -0.5$ such that $x + y + z = -0.5$. Then, the minimum value of

$$(x-y)(x-z)$$

is $-1/8$. Moreover, the equality case occurs when $x = -1/4$ and $\{y, z\} = \{1/4, -1/2\}$.

Proof. We notice that

$$\begin{aligned} (x-y)(x-z) + \frac{1}{8} &= \left(2y + z + \frac{1}{2}\right)\left(2z + y + \frac{1}{2}\right) + \frac{1}{8} \\ &= \frac{1}{8}(4y + 4z + 1)^2 + \left(y + \frac{1}{2}\right)\left(z + \frac{1}{2}\right) \geq 0. \end{aligned}$$

The last inequality is true since both $y + \frac{1}{2}$ and $z + \frac{1}{2}$ are not less than zero.

The equality in the last inequality is attained when either $y + \frac{1}{2} = 0$ or $z + \frac{1}{2} = 0$, and $4y + 4z + 1 = 0$. This system of equations give $(y, z) = (1/4, -1/2)$ or $(y, z) = (-1/2, 1/4)$ as the desired equality cases.

□

Note: We can also prove (the weakened) Claim 2 by using Lagrange Multiplier, as follows. We first prove that, in fact, $x, y, z \in [-0.5, 0.5]$. This can be proved by considering that

$$-2x + (a^2 + b^2 + c^2 + d^2) = (a-c)^2 + (b-d)^2 \geq 0.$$

We will prove the Claim 2, only that in this case, $x, y, z \in [-0.5, 0.5]$. This is already sufficient to prove the original question. We already have the bounded domain $[-0.5, 0.5]^3$, so the global minimum must occur somewhere. Thus, it suffices to consider two cases:

- If the global minimum lies on the boundary of $[-0.5, 0.5]^3$. Then, one of x, y, z must be -0.5 or 0.5. By symmetry between y and z, we split to a few more cases.

- If $x = 0.5$, then $y = z = -0.5$, so $(x-y)(x-z) = 1$, not the minimum.
- If $x = -0.5$, then both y and z must be greater or equal to x, so $(x-y)(x-z) \geq 0$, not the minimum.
- If $y = 0.5$, then $x = z = -0.5$, so $(x-y)(x-z) = 0$, not the minimum.
- If $y = -0.5$, then $z = -x$, so

$$(x-y)(x-z) = 2x(x+0.5),$$

which obtain the minimum at $x = -1/4$. This gives the desired equality case.

- If the global minimum lies in the interior $(-0.5, 0.5)^3$, then we apply Lagrange multiplier:

$$\frac{\partial}{\partial x}(x-y)(x-z) = \lambda \frac{\partial}{\partial x}(x+y+z) \implies 2x - y - z = \lambda.$$
$$\frac{\partial}{\partial y}(x-y)(x-z) = \lambda \frac{\partial}{\partial y}(x+y+z) \implies z - x = \lambda.$$
$$\frac{\partial}{\partial z}(x-y)(x-z) = \lambda \frac{\partial}{\partial z}(x+y+z) \implies y - x = \lambda.$$

Adding the last two equations gives $\lambda = 0$, or $x = y = z$. This gives $(x-y)(x-z) = 0$, not the minimum.

Having exhausted all cases, we are done.

APMO 2021 Solution

1. Prove that for each real number $r > 2$, there are exactly two or three positive real numbers x satisfying the equation $x^2 = r\lfloor x \rfloor$.

   Note: $\lfloor x \rfloor$ denotes the largest integer less than or equal to x

   Solution Let $r > 2$ be a real number. Let x be a positive real number such that $x^2 = r\lfloor x \rfloor$ with $\lfloor x \rfloor = k$. Since $x > 0$ and $x^2 = rk$, we also have $k > 0$. From $k \le x < k+1$, we get $k^2 \le x^2 = rk < k^2 + 2k + 1 \le k^2 + 3k$, hence $k \le r < k+3$, or $r - 3 < k \le r$. There are at most three positive integers in the interval $(r-3, r]$. Thus there are at most three possible values for k. Consequently, there are at most three positive solutions to the given equation.

   Now suppose that k is a positive integer in the interval $[r-2, r]$. There are at least two such positive integer. Observe that $k \le \sqrt{rk} \le \sqrt{(k+2)k} < k+1$ and so $rk = r\lfloor \sqrt{rk} \rfloor$. We conclude that the equation $x^2 = r\lfloor x \rfloor$ has at least two positive solutions, namely $x = \sqrt{rk}$ with $k \in [r-2, r]$.

2. For a polynomial P and a positive integer n, define P_n as the number of positive integer pairs (a, b) such that $a < b \le n$ and $|P(a)| - |P(b)|$ is divisible by n.

   Determine all polynomial P with integer coefficients such that for all positive integers n, $P_n \le 2021$.

   Solution There are two possible families of solutions:

   - $P(x) = x + d$, for some integer $d \ge -2022$.
   - $P(x) = -x + d$, for some integer $d \le 2022$.

   Suppose P satisfies the problem conditions. Clearly P cannot be a constant polynomial. Notice that a polynomial P satifies the conditions if and only if $-P$ also satisfies them. Hence, we may assume the leading coefficient of P is positive. Then, there exists positive integer M such that $P(x) > 0$ for $x \ge M$.

   Lemma 1. For any positive integer n, the integers $P(1), P(2), \ldots, P(n)$ leave pairwise distinct remainders upon division by n.

   Proof. Assume for contradiction that this is not the case. Then, for some $1 \le y < z \le n$, there exists $0 \le r \le n-1$ such that $P(y) \equiv P(z) \equiv r \pmod{n}$. Since $P(an+b) \equiv P(b) \pmod{n}$ for all a, b integers, we have $P(an+y) \equiv P(an+z) \equiv r \pmod{n}$ for any integer a. Let A be a positive integer such that $An \ge M$, and let k be a positive integer such that $k > 2A + 2021$. Each of the $2(k-A)$ integers $P(An+y), P(An+z), P((A+1)n+y), P((A+1)n+z), \ldots, P((k-1)n+y), P((k-1)n+z)$ leaves one of the k remainders
   $$r, n+r, 2n+r, \ldots, (k-1)n+r$$
   upon division by kn. This implies that at least $2(k-A) - k = k - 2A$ (possibly overlapping) pairs leave the same remainder upon division by kn. Since $k - 2A > 2021$ and all of the $2(k-A)$ integers are positive, we find more than 2021 pairs a, b with $a < b \le kn$ for which $|P(b)| - |P(a)|$ is divisible by kn - hence, $P_{kn} > 2021$, a contradiction.

   □

Next, we show that P is linear. Assume that this is not the case, i.e., $\deg P \geq 2$. Then we can find a positive integer k such that $P(k) - P(1) \geq k$. This means that among the integers $P(1), P(2), \ldots, P(P(k) - P(1))$, two of them, namely $P(k)$ and $P(1)$, leave the same remainder upon division by $P(k) - P(1)$ - contradicting the lemma (by taking $n = P(k) - P(1)$). Hence, P must be linear.

We can now write $P(x) = cx + d$ with $c > 0$. We prove that $c = 1$ by two ways.

Solution 1 If $c \geq 2$, then $P(1)$ and $P(2)$ leave the same remainder upon division by c, contradicting the Lemma. Hence $c = 1$.

□

Solution 2 Suppose $c \geq 2$. Let n be a positive integer such that $n > 2cM$, $n\left(1 - \frac{3}{2c}\right) > 2022$ and $2c|n$. Notice that for any positive integers i such that $\frac{3n}{2c} + i < n$, $P\left(\frac{3n}{2c} + i\right) - P\left(\frac{n}{2c} + i\right) = n$. Hence, $\left(\frac{n}{2c} + i, \frac{3n}{2c} + i\right)$ satifies the condition in the question for all positive integers i such that $\frac{3n}{2c} + i < n$. Hence, $P_n > 2021$, a contradiction. Then, $c = 1$.

□

If $d \leq -2023$, then there are at least 2022 pairs $a < b$ such that $P(a) = P(b)$, namely $(a, b) = (1, -2d - 1), (2, -2d - 2), ..., (-d - 1, -d + 1)$. This implies that $d \geq -2022$.

Finally, we verify that $P(x) = x + d$ satisfies the condition for any $d \geq -2022$. Fix a positive integer n. Note that $||P(b)| - |P(a)|| < n$ for all positive integers $a < b \leq n$, so the only pairs a, b for which $|P(b)| - |P(a)|$ could be divisible by n are those for which $|P(a)| = |P(b)|$. When $d \geq -2022$, there are indeed at most 2021 such pairs.

3. Let $ABCD$ be a cyclic convex quadrilateral and Γ be its circumcircle. Let E be the intersection of the diagonals AC and BD, let L be the center of the circle tangent to sides AB, BC, and CD, and let M be the midpoint of the arc BC of Γ not containing A and D. Prove that the excenter of triangle BCE opposite E lies on the line LM.

Solution 1

Let L be the intersection of the bisectors of $\angle ABC$ and $\angle BCD$. Let N be the E-excenter of $\triangle BCE$. Let $\angle BAC = \angle BDC = \alpha, \angle DBC = \beta$ and $\angle ACB = \gamma$.

We have the following:

$$\begin{aligned}
\angle CBL = \frac{1}{2}\angle ABC = 90^\circ - \frac{1}{2}\alpha - \frac{1}{2}\gamma \text{ and } \angle BCL = 90^\circ - \frac{1}{2}\alpha - \frac{1}{2}\beta,\\
\angle CBN = 90^\circ - \frac{1}{2}\beta \text{ and } \angle BCN = 90^\circ - \frac{1}{2}\gamma,\\
\angle MBL = \angle MBC + \angle CBL = 90^\circ - \frac{1}{2}\gamma \text{ and } \angle MCL = 90^\circ - \frac{1}{2}\beta,\\
\angle LCN = \angle LBN = 180^\circ - \frac{1}{2}(\alpha + \beta + \gamma).
\end{aligned}$$

Applying the sine rule to $\triangle MBL$ and $\triangle MCL$ we obtain

$$\frac{MB}{ML} = \frac{MC}{ML} = \frac{\sin\angle BLM}{\sin\angle MBL} = \frac{\sin\angle CLM}{\sin\angle MCL}.$$

It follows that

$$\frac{\sin\angle BLM}{\sin\angle CLM} = \frac{\sin\angle MBL}{\sin\angle MCL} = \frac{\cos(\gamma/2)}{\cos(\beta/2)}. \tag{1}$$

Now

$$\frac{\sin\angle BLM}{\sin\angle MLC}\cdot\frac{\sin\angle LCN}{\sin\angle NCB}\cdot\frac{\sin\angle NBC}{\sin\angle NBL}=\frac{\cos(\gamma/2)}{\cos(\beta/2)}\cdot\frac{\sin(90^\circ-\frac{1}{2}\beta)}{\sin(90^\circ-\frac{1}{2}\gamma)}=1.$$

Hence LM, BN, CN are concurrent and therefore L, M, N are collinear.

Alternative proof

We proceed similarly as above until the equation (1).

We use the following lemma.

Lemma: If $\pi>\alpha,\beta,\gamma,\delta>0$, $\alpha+\beta=\gamma+\delta<\pi$, and $\frac{\sin\alpha}{\sin\beta}=\frac{\sin\gamma}{\sin\delta}$, then $\alpha=\gamma$ and $\beta=\delta$.

Proof of Lemma: Let $\theta=\alpha+\beta=\gamma+\delta$. Then $\frac{\sin(\theta-\beta)}{\sin\beta}=\frac{\sin(\theta-\delta)}{\sin\delta}$.

$$\begin{aligned}
&\iff \sin(\theta-\beta)\sin\delta=\sin(\theta-\delta)\sin\beta\\
&\iff (\sin\theta\cos\beta-\sin\beta\cos\theta)\sin\delta=(\sin\theta\cos\delta-\sin\delta\cos\theta)\sin\beta\\
&\iff \sin\theta\cos\beta\sin\delta=\sin\theta\cos\delta\sin\beta\\
&\iff \sin\theta\sin(\beta-\delta)=0
\end{aligned}$$

Since $0<\theta<\pi$, then $\sin\theta\neq 0$. Therefore, $\sin(\beta-\delta)=0$, and we must have $\beta=\delta$.

Applying the sine rule to $\triangle NBL$ and $\triangle NCL$ we obtain

$$\frac{NB}{NL}=\frac{\sin\angle BLN}{\sin\angle LBN},$$
$$\frac{NC}{NL}=\frac{\sin\angle CLN}{\sin\angle LCN}.$$

Since $\angle LBN=\angle LCN$, it follows that

$$\frac{\sin\angle BLN}{\sin\angle CLN}=\frac{NB}{NC}=\frac{\sin\angle BCN}{\sin\angle CBN}=\frac{\cos(\gamma/2)}{\cos(\beta/2)}=\frac{\sin\angle BLM}{\sin\angle CLM}.$$

By the lemma, it is concluded that $\angle BLM=\angle BLN$ and $\angle CLM=\angle CLN$. Therefore, L, M, N are collinear.

Solution 2

Denote by N the excenter of triangle BCE opposite E. Since BL bisects $\angle ABC$, we have $\angle CBL=\frac{\angle ABC}{2}$. Since M is the midpoint of arc BC, we have $\angle MBC=\frac{1}{2}(\angle MBC+\angle MCB)$ It follows by angle chasing that

$$\begin{aligned}
\angle MBL&=\angle MBC+\angle CBL=\frac{1}{2}(\angle MBC+\angle MCB+\angle ABC)\\
&=\frac{1}{2}(\angle MBA+\angle MCB)=90^\circ-\frac{\angle BCE}{2}=\angle BCN.
\end{aligned}$$

Denote by X and Y the second intersections of lines BM and CM with the circumcircle of BCL, respectively. Since $\angle MBC=\angle MCB$, we have $BC\parallel XY$. It suffices to show that $BN\parallel XL$ and $CN\parallel YL$. Indeed, from this it follows that $\triangle BCN\sim\triangle XYL$, and therefore a homothety with center M that maps B to X and C to Y also maps N to L, implying that N lies on the line LM.

By symmetry, it suffices to show that $CN\parallel YL$, which is equivalent to showing that $\angle BCN=\angle XYL$. But we have $\angle BCN=\angle MBL=\angle XBL=\angle XYL$, completing the proof.

4. Given a 32×32 table, we put a mouse (facing up) at the bottom left cell and a piece of cheese at several other cells. The mouse then starts moving. It moves forward except that when it reaches a piece of cheese, it eats a part of it, turns right, and continues moving forward. We say that a subset of cells containing cheese is good if, during this process, the mouse tastes each piece of cheese exactly once and then falls off the table. Show that:

(a) No good subset consists of 888 cells.

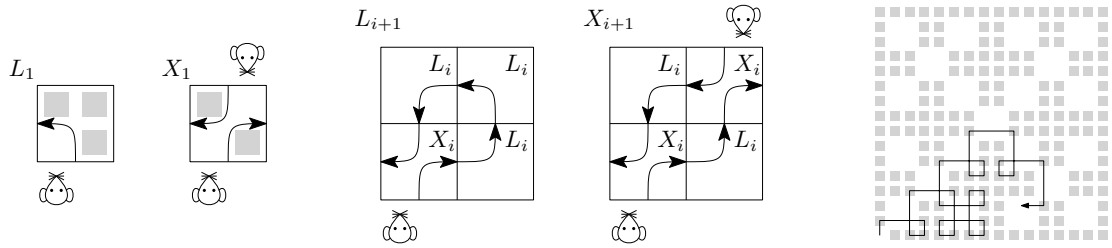
(b) There exists a good subset consisting of at least 666 cells.

Solution.

(a) For the sake of contradiction, assume a good subset consisting of 888 cells exists. We call those *cheese-cells* and the other ones *gap-cells*. Observe that since each cheese-cell is visited once, each gap-cell is visited at most twice (once vertically and once horizontally). Define a finite sequence s whose i-th element is C if the i-th step of the mouse was onto a cheese-cell, and G if it was onto a gap-cell. By assumption, s contains 888 C's. Note that s does not contain a contiguous block of 4 (or more) C's. Hence s contains at least $888/3 = 296$ such C-blocks and thus at least 295 G's. But since each gap-cell is traversed at most twice, this implies there are at least $\lceil 295/2 \rceil = 148$ gap-cells, for a total of $888 + 148 = 1036 > 32^2$ cells, a contradiction.

(b) Let L_i, X_i be two $2^i \times 2^i$ tiles that allow the mouse to "turn left" and "cross", respectively. In detail, the "turn left" tiles allow the mouse to enter at its bottom left cell facing up and to leave at its bottom left cell facing left. The "cross" tiles allow the mouse to enter at its top right facing down and leave at its bottom left facing left, while also to enter at its bottom left facing up and leave at its top right facing right.

(a) Basic tiles (b) Inductive construction (c) 16×16



Note that given two $2^i \times 2^i$ tiles L_i, X_i we can construct larger $2^{i+1} \times 2^{i+1}$ tiles L_{i+1}, X_{i+1} inductively as shown on in (b). The construction works because the path intersects itself (or the other path) only inside the smaller X-tiles where it works by induction.

For a tile T, let $|T|$ be the number of pieces of cheese in it. By straightforward induction, $|L_i| = |X_i| + 1$ and $|L_{i+1}| = 4 \cdot |L_i| - 1$. From the initial condition $|L_1| = 3$. We now easily compute $|L_2| = 11$, $|L_3| = 43$, $|L_4| = 171$, and $|L_5| = 683$. Hence we get the desired subset.

Another proof of (a).

Let X_N be the largest possible density of cheese-cells in a good subset on an $N \times N$ table. We will show that $X_N \leq 4/5 + o(1)$. Specifically, this gives $X_{32} \leq 817/1024$. We look at the (discrete analogue) of the winding number of the trajectory of the mouse. Since the mouse enters and leaves the table, for every 4 right turns in its trajectory there has to be a self-crossing. But each self-crossing requires a different empty square, hence $X_N \leq 4/5$.

5. Determine all functions $f : \mathbb{Z} \to \mathbb{Z}$ such that $f(f(a) - b) + bf(2a)$ is a perfect square for all integers a and b.

Solution 1.

There are two families of functions which satisfy the condition:

(1) $f(n) = \begin{cases} 0 & \text{if } n \text{ is even, and} \\ \text{any perfect square} & \text{if } n \text{ is odd} \end{cases}$

(2) $f(n) = n^2$, for every integer n.

It is straightforward to verify that the two families of functions are indeed solutions. Now, suppose that f is any function which satisfies the condition that $f(f(a)-b)+bf(2a)$ is a perfect square for every pair (a,b) of integers. We denote this condition by (*). We will show that f must belong to either Family (1) or Family (2).

Claim 1. $f(0)=0$ and $f(n)$ is a perfect square for every integer n.

Proof. Plugging $(a,b)\to(0,f(0))$ in (*) shows that $f(0)(f(0)+1)=z^2$ for some integer z. Thus, $(2f(0)+1-2z)(2f(0)+1+2z)=1$. Therefore, $f(0)$ is either -1 or 0.

Suppose, for sake of contradiction, that $f(0)=-1$. For any integer a, plugging $(a,b)\to(a,f(a))$ implies that $f(a)f(2a)-1$ is a square. Thus, for each $a\in\mathbb{Z}$, there exists $x\in\mathbb{Z}$ such that $f(a)f(2a)=x^2+1$ This implies that any prime divisor of $f(a)$ is either 2 or is congruent to 1 (mod 4), and that $4\nmid f(a)$, for every $a\in\mathbb{Z}$.

Plugging $(a,b)\to(0,3)$ in (*) shows that $f(-4)-3$ is a square. Thus, there is $y\in\mathbb{Z}$ such that $f(-4)=y^2+3$. Since $4\nmid f(-4)$, we note that $f(-4)$ is a positive integer congruent to 3 (mod 4), but any prime dividing $f(-4)$ is either 2 or is congruent to 1 (mod 4). This gives a contradiction. Therefore, $f(0)$ must be 0.

For every integer n, plugging $(a,b)\to(0,-n)$ in (*) shows that $f(n)$ is a square.

Replacing b with $f(a)-b$, we find that for all integers a and b,

$$f(b)+(f(a)-b)f(2a)\text{ is a square.}\tag{**}$$

Now, let S be the set of all integers n such that $f(n)=0$. We have two cases:

- Case 1: S is unbounded from above.
  We claim that $f(2n)=0$ for any integer n. Fix some integer n, and let $k\in S$ with $k>f(n)$. Then, plugging $(a,b)\mapsto(n,k)$ in (**) gives us that $f(k)+(f(n)-k)f(2n)=(f(n)-k)f(2n)$ is a square. But $f(n)-k<0$ and $f(2n)$ is a square by Claim 1. This is possible only if $f(2n)=0$.
  In summary, $f(n)=0$ whenever n is even and Claim 1 shows that $f(n)$ is a square whenever n is odd.
- Case 2: S is bounded from above.
  Let T be the set of all integers n such that $f(n)=n^2$. We show that T is unbounded from above. In fact, we show that $\frac{p+1}{2}\in T$ for all primes p big enough.
  Fix a prime number p big enough, and let $n=\frac{p+1}{2}$. Plugging $(a,b)\mapsto(n,2n)$ in (**) shows us that $f(2n)(f(n)-2n+1)$ is a square for any integer n. For p big enough, we have $2n\notin S$, so $f(2n)$ is a non-zero square. As a result, when p is big enough, $f(n)$ and $f(n)-2n+1=f(n)-p$ are both squares. Writing $f(n)=k^2$ and $f(n)-p=m^2$ for some $k,m\geq 0$, we have

  $$(k+m)(k-m)=k^2-m^2=p\implies k+m=p,k-m=1\implies k=n,m=n-1.$$

  Thus, $f(n)=k^2=n^2$, giving us $n=\frac{p+1}{2}\in T$.
  Next, for all $k\in T$ and $n\in\mathbb{Z}$, plugging $(a,b)\mapsto(n,k)$ in (**) shows us that $k^2+(f(n)-k)f(2n)$ is a square. But that means $(2k-f(2n))^2-(f(2n)^2-4f(n)f(2n))=4(k^2+(f(n)-k)f(2n))$ is also a square. When k is large enough, we have $|f(2n)^2-4f(n)f(2n)|+1<|2k-f(2n)|$. As a result, we must have $f(2n)^2=4f(n)f(2n)$ and thus $f(2n)\in\{0,4f(n)\}$ for all integers n.
  Finally, we prove that $f(n)=n^2$ for all integers n. Fix n, and take $k\in T$ big enough such that $2k\notin S$. Then, we have $f(k)=k^2$ and $f(2k)=4f(k)=4k^2$. Plugging $(a,b)\mapsto(k,n)$ to (**) shows us that $f(n)+(k^2-n)4k^2=(2k^2-n)^2+(f(n)-n^2)$ is a square. Since T is unbounded from above, we can take $k\in T$ such that $2k\notin S$ and also $|2k^2-n|>|f(n)-n^2|$. This forces $f(n)=n^2$, giving us the second family of solution.

Another approach of Case 1.

Claim 2. One of the following is true.

(i) For every integer n, $f(2n) = 0$.

(ii) There exists an integer $K > 0$ such that for every integer $n \geq K$, $f(n) > 0$.

Proof. Suppose that there exists an integer $\alpha \neq 0$ such that $f(2\alpha) > 0$. We claim that for every integer $n \geq f(\alpha) + 1$, we have $f(n) > 0$.

For every $n \geq f(\alpha) + 1$, plugging $(a, b) \to (\alpha, f(\alpha) - n)$ in (*) shows that $f(n) + (f(\alpha) - n)f(2\alpha)$ is a square, and in particular, is non-negative. Hence, $f(n) \geq (n - f(\alpha))f(2\alpha) > 0$, as desired.

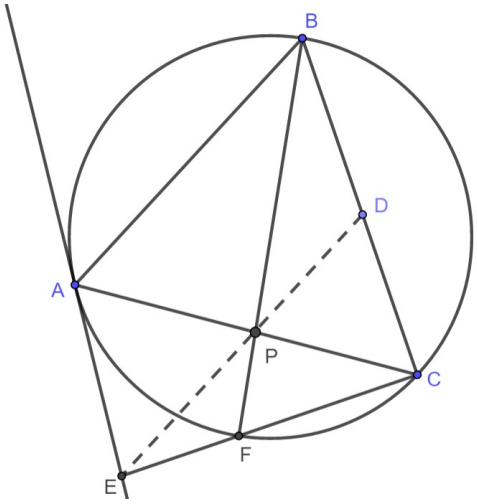
If f belongs to Case (i), Claim 1 shows that f belongs to Family (1).

If f belongs to Case (ii), then S is bounded from above. From Case 2 we get $f(n) = n^2$.

APMO 2020 Solution

1. Let Γ be the circumcircle of ΔABC. Let D be a point on the side BC. The tangent to Γ at A intersects the parallel line to BA through D at point E. The segment CE intersects Γ again at F. Suppose B, D, F, E are concyclic. Prove that AC, BF, DE are concurrent.

Solution 1 From the conditions, we have



$$\begin{aligned}\angle CBA &= 180^\circ - \angle EDB = 180^\circ - \angle EFB \\ &= 180^\circ - \angle EFA - \angle AFB \\ &= 180^\circ - \angle CBA - \angle ACB = \angle BAC.\end{aligned}$$

Let P be the intersection of AC and BF. Then we have

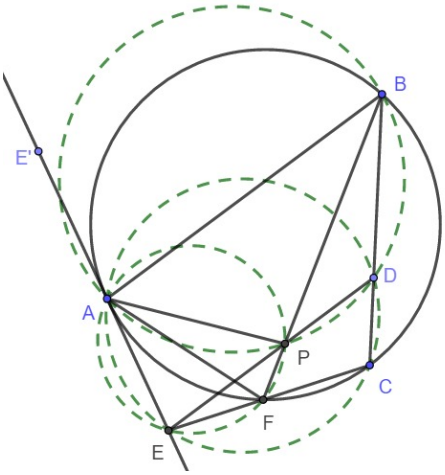
$$\angle PAE = \angle CBA = \angle BAC = \angle BFC.$$

This implies A, P, F, E are concyclic. It follows that

$$\angle FPE = \angle FAE = \angle FBA,$$

and hence AB and EP are parallel. So E, P, D are collinear, and the result follows.

Solution 2
Let E' be any point on the extension of EA. From $\angle AED = \angle E'AB = \angle ACD$, points A, D, C, E are concyclic.



Let P be the intersection of BF and DE. From $\angle AFP = \angle ACB = \angle AEP$, the points A, P, F, E are concyclic. In addition, from $\angle EPA = \angle EFA = \angle DBA$, points A, B, D, P are concyclic.
By considering the radical centre of $(BDFE)$, $(APFE)$ and $(BDPA)$, we find that the lines BD, AP, EF are concurrent at C. The result follows.

2. Show that $r = 2$ is the largest real number r which satisfies the following condition:
If a sequence $a_1, a_2, \ldots$ of positive integers fulfills the inequalities

$$a_n \leq a_{n+2} \leq \sqrt{a_n^2 + ra_{n+1}}$$

for every positive integer n, then there exists a positive integer M such that $a_{n+2} = a_n$ for every $n \geq M$.

Solution 1. First, let us assume that $r > 2$, and take a positive integer $a \geq 1/(r-2)$.
Then, if we let $a_n = a + \lfloor n/2 \rfloor$ for $n = 1, 2, \ldots$, the sequence a_n satisfies the inequalities

$$\sqrt{a_n^2 + ra_{n+1}} \geq \sqrt{a_n^2 + ra_n} \geq \sqrt{a_n^2 + \left(2 + \frac{1}{a}\right) a_n} \geq a_n + 1 = a_{n+2},$$

but since $a_{n+2} > a_n$ for any n, we see that r does not satisfy the condition given in the problem.

Now we show that $r = 2$ does satisfy the condition of the problem. Suppose $a_1, a_2, \ldots$ is a sequence of positive integers satisfying the inequalities given in the problem, and there exists a positive integer m for which $a_{m+2} > a_m$ is satisfied.

By induction we prove the following assertion:

(†) $\quad a_{m+2k} \leq a_{m+2k-1} = a_{m+1}$ holds for every positive integer k.

The truth of (†) for $k = 1$ follows from the inequalities below

$$2a_{m+2} - 1 = a_{m+2}^2 - (a_{m+2} - 1)^2 \leq a_m^2 + 2a_{m+1} - (a_{m+2} - 1)^2 \leq 2a_{m+1}.$$

Let us assume that (†) holds for some positive integer k. From

$$a_{m+1}^2 \leq a_{m+2k+1}^2 \leq a_{m+2k-1}^2 + 2a_{m+2k} \leq a_{m+1}^2 + 2a_{m+1} < (a_{m+1} + 1)^2,$$

it follows that $a_{m+2k+1} = a_{m+1}$ must hold. Furthermore, since $a_{m+2k} \leq a_{m+1}$, we have

$$a_{m+2k+2}^2 \leq a_{m+2k}^2 + 2a_{m+2k+1} \leq a_{m+1}^2 + 2a_{m+1} < (a_{m+1} + 1)^2,$$

from which it follows that $a_{m+2k+2} \leq a_{m+1}$, which proves the assertion (†).

We can conclude that for the value of m with which we started our argument above, $a_{m+2k+1} = a_{m+1}$ holds for every positive integer k. Therefore, in order to finish the proof, it is enough to show that a_{m+2k} becomes constant after some value of k. Since every a_{m+2k} is a positive integer less than or equal to a_{m+1}, there exists $k = K$ for which a_{m+2K} takes the maximum value. By the monotonicity of a_{m+2k}, it then follows that $a_{m+2k} = a_{m+2K}$ for all $k \geq K$.

Solution 2

We only give an alternative proof of the assertion (†) in solution 1. Let $\{a_n\}$ be a sequence satisfying the inequalities given in the problem. We will use the following key observations:

(a) If $a_{n+1} \leq a_n$ for some $n \geq 1$, then

$$a_n \leq a_{n+2} \leq \sqrt{a_n^2 + 2a_{n+1}} < \sqrt{a_n^2 + 2a_n + 1} = a_n + 1,$$

hence $a_n = a_{n+2}$.

(b) If $a_n \leq a_{n+1}$ for some $n \geq 1$, then

$$a_n \leq a_{n+2} \leq \sqrt{a_n^2 + 2a_{n+1}} < \sqrt{a_{n+1}^2 + 2a_{n+1} + 1} = a_{n+1} + 1,$$

hence $a_n \leq a_{n+2} \leq a_{n+1}$.

Now let m be a positive integer such that $a_{m+2} > a_m$. By the observations above, we must have $a_m < a_{m+2} \leq a_{m+1}$. Thus the assertion (†) is true for $k = 1$. Assume that the assertion holds for some positive integer k. Using observation (a), we get $a_{m+2k+1} = a_{m+2k-1} = a_{m+1}$. Thus $a_{m+2k} \leq a_{m+2k+1}$, and then using observation (b), we get $a_{m+2k+2} \leq a_{m+2k+1} = a_{m+1}$, which proves the assertion (†).

3. Determine all positive integers k for which there exist a positive integer m and a set S of positive integers such that any integer $n > m$ can be written as a sum of distinct elements of S in exactly k ways.

Solution:

We claim that $k = 2^a$ for all $a \geq 0$.

Let $A = \{1, 2, 4, 8, \dots\}$ and $B = \mathbb{N} \backslash A$. For any set T, let $s(T)$ denote the sum of the elements of T. (If T is empty, we let $s(T) = 0$.)

We first show that any positive integer $k = 2^a$ satisfies the desired property. Let B' be a subset of B with a elements, and let $S = A \cup B'$. Recall that any nonnegative integer has a unique binary representation. Hence, for any integer $t > s(B')$ and any subset $B'' \subseteq B'$, the number $t - s(B'')$ can be written as a sum of distinct elements of A in a unique way. This means that t can be written as a sum of distinct elements of B' in exactly 2^a ways.

Next, assume that some positive integer k satisfies the desired property for a positive integer $m \geq 2$ and a set S. Clearly, S is infinite.

Lemma: For all sufficiently large $x \in S$, the smallest element of S larger than x is $2x$.

Proof of Lemma: Let $x \in S$ with $x > 3m$, and let $x < y < 2x$. We will show that $y \notin S$. Suppose first that $y > x + m$. Then $y - x$ can be written as a sum of distinct elements of S not including x in k ways. If $y \in S$, then y can be written as a sum of distinct elements of S in at least $k + 1$ ways, a contradiction. Suppose now that $y \leq x + m$. We consider $z \in (2x - m, 2x)$. Similarly as before, $z - x$ can be written as a sum of distinct elements of S not including x or y in k ways. If $y \in S$, then since $m < z - y < x$, $z - y$ can be written as a sum of distinct elements of S not including x or y. This means that z can be written as a sum of distinct elements of S in at least $k + 1$ ways, a contradiction.

We now show that $2x \in S$; assume for contradiction that this is not the case. Observe that $2x$ can be written as a sum of distinct elements of S including x in exactly $k - 1$ ways. This means that $2x$ can also be written as a sum of distinct elements of S not including x. If this sum includes any number less than $x - m$, then removing this number, we can write some number $y \in (x + m, 2x)$ as a sum of distinct elements of S not including x. Now if $y = y' + x$ where $y' \in (m, x)$ then y' can be written as

a sum of distinct elements of S including x in exactly k ways. Therefore y can be written as a sum of distinct elements of S in at least $k+1$ ways, a contradiction. Hence the sum only includes numbers in the range $[x-m, x)$. Clearly two numbers do not suffice. On the other hand, three such numbers sum to at least $3(x-m) > 2x$, a contradiction. □

From the Lemma, we have that $S = T \cup U$, where T is finite and $U = \{x, 2x, 4x, 8x, \dots\}$ for some positive integer x. Let y be any positive integer greater than $s(T)$. For any subset $T' \subseteq T$, if $y - s(T') \equiv 0 \pmod{x}$, then $y - s(T')$ can be written as a sum of distinct elements of U in a unique way; otherwise $y - s(T')$ cannot be written as a sum of distinct elements of U. Hence the number of ways to write y as a sum of distinct elements of S is equal to the number of subsets $T' \subseteq T$ such that $s(T') \equiv y \pmod{x}$. Since this holds for all y, for any $0 \le a \le x-1$ there are exactly k subsets $T' \subseteq T$ such that $s(T') \equiv a \pmod{x}$. This means that there are kx subsets of T in total. But the number of subsets of T is a power of 2, and therefore k is a power of 2, as claimed.

Solution 2. We give an alternative proof of the first half of the lemma in the Solution 1 above.

Let $s_1 < s_2 < \cdots$ be the elements of S. For any positive integer r, define $A_r(x) = \prod_{n=1}^{r}(1 + x^{s_n})$. For each n such that $m \le n < s_{r+1}$, all k ways of writing n as a sum of elements of S must only use $s_1, \dots, s_r$, so the coefficient of x^n in $A_r(x)$ is k. Similarly the number of ways of writing s_{r+1} as a sum of elements of S without using s_{r+1} is exactly $k-1$. Hence the coefficient of $x^{s_{r+1}}$ in $A_r(x)$ is $k-1$.

Fix a t such that $s_t > 2(m+1)$. Write

$$A_{t-1}(x) = u(x) + k\left(x^{m+1} + \cdots + x^{s_t - 1}\right) + x^{s_t} v(x)$$

for some $u(x), v(x)$ where $u(x)$ is of degree at most m.

Note that

$$A_{t+1}(x) = A_{t-1}(x) + x^{s_t} A_{t-1}(x) + x^{s_{t+1}} A_{t-1}(x) + x^{s_t + s_{t+1}} A_{t-1}(x).$$

If $s_{t+1} + m + 1 < 2s_t$, we can find the term $x^{s_{t+1}+m+1}$ in $x^{s_t} A_{t-1}(x)$ and in $x^{s_{t+1}} A_{t-1}(x)$. Hence the coefficient of $x^{s_{t+1}+m+1}$ in $A_{t+1}(x)$ is at least $2k$, which is impossible. So $s_{t+1} \ge 2s_t - (m+1) > s_t + m + 1$.

Now

$$A_t(x) = A_{t-1}(x) + x^{s_t} u(x) + k(x^{s_t+m+1} + \cdots x^{2s_t - 1}) + x^{2s_t} v(x).$$

Recall that the coefficent of $x^{s_{t+1}}$ in $A_t(x)$ is $k-1$. But if $s_t + m + 1 < s_{t+1} < s_{2t}$, then the coefficient of $x^{s_{t+1}}$ in $A_t(x)$ is at least k, which is a contradiction. Therefore $s_{t+1} \ge 2s_t$.

4. Let $\mathbb{Z}$ denote the set of all integers. Find all polynomials $P(x)$ with integer coefficients that satisfy the following property:

   For any infinite sequence $a_1, a_2, \dots$ of integers in which each integer in $\mathbb{Z}$ appears exactly once, there exist indices $i < j$ and an integer k such that $a_i + a_{i+1} + \cdots + a_j = P(k)$.

   Solution:

   Part 1: All polynomials with $\deg P = 1$ satisfy the given property.

   Suppose $P(x) = cx + d$, and assume without loss of generality that $c > d \ge 0$. Denote $s_i = a_1 + a_2 + \cdots + a_i \pmod{c}$. It suffices to show that there exist indices i and j such that $j - i \ge 2$ and $s_j - s_i \equiv d \pmod{c}$.

   Consider $c+1$ indices $e_1, e_2, \dots, e_{c+1} > 1$ such that $a_{e_l} \equiv d \pmod{c}$. By the pigeonhole principle, among the $n+1$ pairs $(s_{e_1 - 1}, s_{e_1}), (s_{e_2 - 1}, s_{e_2}), \dots, (s_{e_{n+1} - 1}, s_{e_{n+1}})$, some two are equal, say (s_{m-1}, s_m) and (s_{n-1}, s_n). We can then take $i = m-1$ and $j = n$.

   Part 2: All polynomials with $\deg P \ne 1$ do not satisfy the given property.

   Lemma: If $\deg P \ne 1$, then for any positive integers A, B, and C, there exists an integer y with $|y| > C$ such that no value in the range of P falls within the interval $[y - A, y + B]$.

   Proof of Lemma: The claim is immediate when P is constant or when $\deg P$ is even since P is bounded from below. Let $P(x) = a_n x^n + \cdots + a_1 x + a_0$ be of odd degree greater than 1, and assume without

loss of generality that $a_n > 0$. Since $P(x+1) - P(x) = a_n n x^{n-1} + \ldots$, and $n-1 > 0$, the gap between $P(x)$ and $P(x+1)$ grows arbitrarily for large x. The claim follows. $\square$

Suppose $\deg P \neq 1$. We will inductively construct a sequence $\{a_i\}$ such that for any indices $i < j$ and any integer k it holds that $a_i + a_{i+1} + \cdots + a_j \neq P(k)$. Suppose that we have constructed the sequence up to a_i, and m is an integer with smallest magnitude yet to appear in the sequence. We will add two more terms to the sequence. Take $a_{i+2} = m$. Consider all the new sums of at least two consecutive terms; each of them contains a_{i+1}. Hence all such sums are in the interval $[a_{i+1} - A, a_{i+1} + B]$ for fixed constants A, B. The lemma allows us to choose a_{i+1} so that all such sums avoid the range of P.

Alternate Solution for Part 1: Again, suppose $P(x) = cx + d$, and assume without loss of generality that $c > d \geq 0$. Let $S_i = \{a_j + a_{j+1} + \cdots + a_i \pmod c \mid j = 1, 2, \ldots, i\}$. Then $S_{i+1} = \{s_i + a_{i+1} \pmod c \mid s_i \in S_i\} \cup \{a_{i+1} \pmod c\}$. Hence $|S_{i+1}| = |S_i|$ or $S_{i+1} = |S_i| + 1$, with the former occuring exactly when $0 \in S_i$. Since $|S_i| \leq c$, the latter can only occur finitely many times, so there exists I such that $0 \in S_i$ for all $i \geq I$. Let $t > I$ be an index with $a_t \equiv d \pmod c$. Then we can find a sum of at least two consecutive terms ending at a_t and congruent to $d \pmod c$.

Alternate Construction when $P(x)$ is constant or of even degree
If $P(x)$ is of even degree, then P is bounded from below or from above. In case of P is constant or bounded from above, then there exists a positive integer c such that $P(x) < c$. Let $\{a_i\}$ be the sequence

$$0, 1, -1, 2, 3, -2, 4, 5, -3, \cdots$$

which is given by $a_{3n+1} = 2n, a_{3n+2} = 2n+1, a_{3n+3} = -(n+1)$ for all $n \geq 0$. Notice that for any $i < j$ we have $a_i + \cdots + a_j \geq 0$. Then for the sequence $\{b_n\}$ defined by $b_n = a_n + c$, clearly $b_i + \cdots + b_j \geq (a_i + \cdots + a_j) + 2c > c$ which is out side the range of $P(x)$.

Now if P is bounded from below, there exist a positive integer c such that $P(x) > -c$. In this case, take b_n to be $b_n = -a_n - c$. Then for all $i < j$ we have $b_i + \cdots + b_j \leq -(a_1 + \cdots a_n) - 2c < -c$ which is again out side the range of $P(x)$.

5. Let $n \geq 3$ be a fixed integer. The number 1 is written n times on a blackboard. Below the blackboard, there are two buckets that are initially empty. A *move* consists of erasing two of the numbers a and b, replacing them with the numbers 1 and $a + b$, then adding one stone to the first bucket and $\gcd(a, b)$ stones to the second bucket. After some finite number of moves, there are s stones in the first bucket and t stones in the second bucket, where s and t are positive integers. Find all possible values of the ratio $\frac{t}{s}$.

Solution:
The answer is the set of all rational numbers in the interval $[1, n-1)$. First, we show that no other numbers are possible. Clearly the ratio is at least 1, since for every move, at least one stone is added to the second bucket. Note that the number s of stones in the first bucket is always equal to $p - n$, where p is the sum of the numbers on the blackboard. We will assume that the numbers are written in a row, and whenever two numbers a and b are erased, $a + b$ is written in the place of the number on the right. Let $a_1, a_2, ..., a_n$ be the numbers on the blackboard from left to right, and let

$$q = 0 \cdot a_1 + 1 \cdot a_2 + \cdots + (n-1)a_n.$$

Since each number a_i is at least 1, we always have

$$q \leq (n-1)p - (1 + \cdots + (n-1)) = (n-1)p - \frac{n(n-1)}{2} = (n-1)s + \frac{n(n-1)}{2}.$$

Also, if a move changes a_i and a_j with $i < j$, then t changes by $\gcd(a_i, a_j) \leq a_i$ and q increases by

$$(j-1)a_i - (i-1)(a_i - 1) \geq ia_i - (i-1)(a_i - 1) \geq a_i.$$

Hence $q - t$ never decreases. We may assume without loss of generality that the first move involves the rightmost 1. Then immediately after this move, $q = 0 + 1 + \cdots + (n-2) + (n-1) \cdot 2 = \frac{(n+2)(n-1)}{2}$ and

$t = 1$. So after that move, we always have

$$\begin{aligned} t &\leq q + 1 - \frac{(n+2)(n-1)}{2} \\ &\leq (n-1)s + \frac{n(n-1)}{2} - \frac{(n+2)(n-1)}{2} + 1 \\ &= (n-1)s - (n-2) < (n-1)s. \end{aligned}$$

Hence, $\frac{t}{s} < n - 1$. So $\frac{t}{s}$ must be a rational number in $[1, n-1)$.

After a single move, we have $\frac{t}{s} = 1$, so it remains to prove that $\frac{t}{s}$ can be any rational number in $(1, n-1)$. We will now show by induction on n that for any positive integer a, it is possible to reach a situation where there are $n-1$ occurrences of 1 on the board and the number a^{n-1}, with t and s equal to $a^{n-2}(a-1)(n-1)$ and $a^{n-1} - 1$, respectively. For $n = 2$, this is clear as there is only one possible move at each step, so after $a - 1$ moves s and t will both be equal to $a - 1$. Now assume that the claim is true for $n - 1$, where $n > 2$. Call the algorithm which creates this situation using $n - 1$ numbers algorithm A. Then to reach the situation for size n, we apply algorithm A, to create the number a^{n-2} . Next, apply algorithm A again and then add the two large numbers, repeat until we get the number a^{n-1} . Then algorithm A was applied a times and the two larger numbers were added $a - 1$ times. Each time the two larger numbers are added, t increases by a^{n-2} and each time algorithm A is applied, t increases by $a^{n-3}(a-1)(n-2)$. Hence, the final value of t is

$$t = (a-1)a^{n-2} + a \cdot a^{n-3}(a-1)(n-2) = a^{n-2}(a-1)(n-1).$$

This completes the induction.

Now we can choose 1 and the large number b times for any positive integer b, and this will add b stones to each bucket. At this point we have

$$\frac{t}{s} = \frac{a^{n-2}(a-1)(n-1) + b}{a^{n-1} - 1 + b}.$$

So we just need to show that for any rational number $\frac{p}{q} \in (1, n-1)$, there exist positive integers a and b such that

$$\frac{p}{q} = \frac{a^{n-2}(a-1)(n-1) + b}{a^{n-1} - 1 + b}$$

Rearranging, we see that this happens if and only if

$$b = \frac{qa^{n-2}(a-1)(n-1) - p(a^{n-1} - 1)}{p - q}.$$

If we choose $a \equiv 1 \pmod{p-q}$, then this will be an integer, so we just need to check that the numerator is positive for sufficiently large a.

$$\begin{aligned} qa^{n-2}(a-1)(n-1) - p(a^{n-1} - 1) &> qa^{n-2}(a-1)(n-1) - pa^{n-1} \\ &= a^{n-2}\left(a(q(n-1) - p) - (n-1)\right), \end{aligned}$$

which is positive for sufficiently large a since $q(n-1) - p > 0$.

Alternative solution for the upper bound. Rather than starting with n occurrences of 1, we may start with infinitely many 1s, but we are restricted to having at most $n - 1$ numbers which are not equal to 1 on the board at any time. It is easy to see that this does not change the problem. Note also that we can ignore the 1 we write on the board each move, so the allowed move is to rub off two numbers and write their sum. We define the width and score of a number on the board as follows. Colour that number red, then reverse every move up to that point all the way back to the situation when the numbers are all 1s. Whenever a red number is split, colour the two replacement numbers

red. The width of the original number is equal to the maximum number of red integers greater than 1 which appear on the board at the same time. The score of the number is the number of stones which were removed from the second bucket during these splits. Then clearly the width of any number is at most $n-1$. Also, t is equal to the sum of the scores of the final numbers. We claim that if a number $p > 1$ has a width of at most w, then its score is at most $(p-1)w$. We will prove this by strong induction on p. If $p = 1$, then clearly p has a score of 0, so the claim is true. If $p > 1$, then p was formed by adding two smaller numbers a and b. Clearly a and b both have widths of at most w. Moreover, if a has a width of w, then at some point in the reversed process there will be w numbers in the set $\{2, 3, 4, ...\}$ that have split from a, and hence there can be no such numbers at this point which have split from b. Between this point and the final situation, there must always be at least one number in the set $\{2, 3, 4, ...\}$ that split from a, so the width of b is at most $w-1$. Therefore, a and b cannot both have widths of w, so without loss of generality, a has width at most w and b has width at most $w-1$. Then by the inductive hypothesis, a has score at most $(a-1)w$ and b has score at most $(b-1)(w-1)$. Hence, the score of p is at most

$$\begin{aligned}(a-1)w + (b-1)(w-1) + \gcd(a,b) &\leq (a-1)w + (b-1)(w-1) + b \\ &= (p-1)w + 1 - w \\ &\leq (p-1)w.\end{aligned}$$

This completes the induction.

Now, since each number p in the final configuration has width at most $(n-1)$, it has score less than $(n-1)(p-1)$. Hence the number t of stones in the second bucket is less than the sum over the values of $(n-1)(p-1)$, and s is equal to the sum of the the values of $(p-1)$. Therefore, $\frac{t}{s} < n-1$.

Solutions of APMO 2019

Problem 1. Let $\mathbb{Z}^+$ be the set of positive integers. Determine all functions $f : \mathbb{Z}^+ \to \mathbb{Z}^+$ such that $a^2 + f(a)f(b)$ is divisible by $f(a) + b$ for all positive integers a and b.

Answer: The answer is $f(n) = n$ for all positive integers n.

Clearly, $f(n) = n$ for all $n \in \mathbb{Z}^+$ satisfies the original relation. We show some possible approaches to prove that this is the only possible function.

Solution. First we perform the following substitutions on the original relation:

1. With $a = b = 1$, we find that $f(1) + 1 \mid f(1)^2 + 1$, which implies $f(1) = 1$.
2. With $a = 1$, we find that $b + 1 \mid f(b) + 1$. In particular, $b \leq f(b)$ for all $b \in \mathbb{Z}^+$.
3. With $b = 1$, we find that $f(a) + 1 \mid a^2 + f(a)$, and thus $f(a) + 1 \mid a^2 - 1$. In particular, $f(a) \leq a^2 - 2$ for all $a \geq 2$.

Now, let p be any odd prime. Substituting $a = p$ and $b = f(p)$ in the original relation, we find that $2f(p)|p^2 + f(p)f(f(p))$. Therefore, $f(p)|p^2$. Hence the possible values of $f(p)$ are $1, p$ and p^2. By (2) above, $f(p) \geq p$ and by (3) above $f(p) \leq p^2 - 2$. So $f(p) = p$ for all primes p.

Substituting $a = p$ into the original relation, we find that $b + p \mid p^2 + pf(b)$. However, since $(b + p)(f(b) + p - b) = p^2 - b^2 + bf(b) + pf(b)$, we have $b + p \mid bf(b) - b^2$. Thus, for any fixed b this holds for arbitrarily large primes p and therefore we must have $bf(b) - b^2 = 0$, or $f(b) = b$, as desired.

□

Solution 2: As above, we have relations (1)-(3). In (2) and (3), for $b = 2$ we have $3|f(2)+1$ and $f(2) + 1|3$. These imply $f(2) = 2$.

Now, using $a = 2$ we get $2 + b|4 + 2f(b)$. Let $f(b) = x$. We have

$$\begin{aligned} 1 + x &\equiv 0 \pmod{b+1} \\ 4 + 2x &\equiv 0 \pmod{b+2}. \end{aligned}$$

From the first equation $x \equiv b \pmod{b+1}$ so $x = b + (b + 1)t$ for some integer $t \geq 0$. Then

$$0 \equiv 4 + 2x \equiv 4 + 2(b + (b + 1)t) \equiv 4 + 2(-2 - t) \equiv -2t \pmod{b+2}.$$

Also $t \leq b - 2$ because $1 + x|b^2 - 1$ by (3).

If $b + 2$ is odd, then $t \equiv 0 \pmod{b+2}$. Then $t = 0$, which implies $f(b) = b$.

If $b + 2$ is even, then $t \equiv 0 \pmod{(b+2)/2}$. Then $t = 0$ or $t = (b + 2)/2$. But if $t \neq 0$, then by definition $(b + 4)/2 = (1 + t) = (x + 1)/(b + 1)$ and since $x + 1|b^2 - 1$, then $(b + 4)/2$ divides $b - 1$. Therefore $b + 4|10$ and the only possibility is $b = 6$. So for even b, $b \neq 6$ we have $f(b) = b$.

Finally, by (2) and (3), for $b = 6$ we have $7|f(6) + 1$ and $f(6) + 1|35$. This means $f(6) = 6$ or $f(6) = 34$. The later is discarded as, for $a = 5$, $b = 6$, we have by the original equation that $11|5(5 + f(6))$. Therefore $f(n) = n$ for every positive integer n.

□

Solution 3: We proceed by induction. As in Solution 1, we have $f(1) = 1$. Suppose that $f(n - 1) = n - 1$ for some integer $n \geq 2$.

With the substitution $a = n$ and $b = n - 1$ in the original relation we obtain that $f(n) + n - 1|n^2 + f(n)(n - 1)$. Since $f(n) + n - 1|(n - 1)(f(n) + n - 1)$, then $f(n) + n - 1|2n - 1$.

With the substitution $a = n - 1$ and $b = n$ in the original relation we obtain that $2n - 1|(n - 1)^2 + (n - 1)f(n) = (n - 1)(n - 1 + f(n))$. Since $(2n - 1, n - 1) = 1$, we deduce that $2n - 1|f(n) + n - 1$.

Therefore, $f(n) + n - 1 = 2n - 1$, which implies the desired $f(n) = n$.

□

Problem 2. Let m be a fixed positive integer. The infinite sequence $\{a_n\}_{n\geq 1}$ is defined in the following way: a_1 is a positive integer, and for every integer $n \geq 1$ we have

$$a_{n+1} = \begin{cases} a_n^2 + 2^m & \text{if } a_n < 2^m \\ a_n/2 & \text{if } a_n \geq 2^m. \end{cases}$$

For each m, determine all possible values of a_1 such that every term in the sequence is an integer.

Answer: The only value of m for which valid values of a_1 exist is $m = 2$. In that case, the only solutions are $a_1 = 2^\ell$ for $\ell \geq 1$.

Solution. Suppose that for integers m and a_1 all the terms of the sequence are integers. For each $i \geq 1$, write the ith term of the sequence as $a_i = b_i 2^{c_i}$ where b_i is the largest odd divisor of a_i (the "odd part" of a_i) and c_i is a nonnegative integer.

Lemma 1. The sequence $b_1, b_2, \ldots$ is bounded above by 2^m.

Proof. Suppose this is not the case and take an index i for which $b_i > 2^m$ and for which c_i is minimal. Since $a_i \geq b_i > 2^m$, we are in the second case of the recursion. Therefore, $a_{i+1} = a_i/2$ and thus $b_{i+1} = b_i > 2^m$ and $c_{i+1} = c_i - 1 < c_i$. This contradicts the minimality of c_i. □

Lemma 2. The sequence $b_1, b_2, \ldots$ is nondecreasing.

Proof. If $a_i \geq 2^m$, then $a_{i+1} = a_i/2$ and thus $b_{i+1} = b_i$. On the other hand, if $a_i < 2^m$, then

$$a_{i+1} = a_i^2 + 2^m = b_i^2 2^{2c_i} + 2^m,$$

and we have the following cases:

- If $2c_i > m$, then $a_{i+1} = 2^m(b_i^2 2^{2c_i - m} + 1)$, so $b_{i+1} = b_i^2 2^{2c_i - m} + 1 > b_i$.
- If $2c_i < m$, then $a_{i+1} = 2^{2c_i}(b_i^2 + 2^{m-2c_i})$, so $b_{i+1} = b_i^2 + 2^{m-2c_i} > b_i$.
- If $2c_i = m$, then $a_{i+1} = 2^{m+1} \cdot \frac{b_i^2+1}{2}$, so $b_{i+1} = (b_i^2 + 1)/2 \geq b_i$ since $b_i^2 + 1 \equiv 2 \pmod 4$.

□

By combining these two lemmas we obtain that the sequence $b_1, b_2, \ldots$ is eventually constant. Fix an index j such that $b_k = b_j$ for all $k \geq j$. Since a_n descends to $a_n/2$ whenever $a_n \geq 2^m$, there are infinitely many terms which are smaller than 2^m. Thus, we can choose an $i > j$ such that $a_i < 2^m$. From the proof of Lemma 2, $a_i < 2^m$ and $b_{i+1} = b_i$ can happen simultaneously only when $2c_i = m$ and $b_{i+1} = b_i = 1$. By Lemma 2, the sequence $b_1, b_2, \ldots$ is constantly 1 and thus $a_1, a_2, \ldots$ are all powers of two. Tracing the sequence starting from $a_i = 2^{c_i} = 2^{m/2} < 2^m$,

$$2^{m/2} \to 2^{m+1} \to 2^m \to 2^{m-1} \to 2^{2m-2} + 2^m.$$

Note that this last term is a power of two if and only if $2m - 2 = m$. This implies that m must be equal to 2. When $m = 2$ and $a_1 = 2^\ell$ for $\ell \geq 1$ the sequence eventually cycles through $2, 8, 4, 2, \ldots$. When $m = 2$ and $a_1 = 1$ the sequence fails as the first terms are $1, 5, 5/2$.

□

Solution 2: Let m be a positive integer and suppose that $\{a_n\}$ consists only of positive integers. Call a number *small* if it is smaller than 2^m and *large* otherwise. By the recursion,

after a small number we have a large one and after a large one we successively divide by 2 until we get a small one.

First, we note that $\{a_n\}$ is bounded. Indeed, a_1 turns into a small number after a finite number of steps. After this point, each small number is smaller than 2^m, so each large number is smaller than $2^{2m} + 2^m$. Now, since $\{a_n\}$ is bounded and consists only of positive integers, it is eventually periodic. We focus only on the cycle.

Any small number a_n in the cycle can be writen as $a/2$ for a large, so $a_n \geq 2^{m-1}$, then $a_{n+1} \geq 2^{2m-2} + 2^m = 2^{m-2}(4 + 2^m)$, so we have to divide a_{n+1} at least $m - 1$ times by 2 until we get a small number. This means that $a_{n+m} = (a_n^2 + 2^m)/2^{m-1}$, so $2^{m-1} | a_n^2$, and therefore $2^{\lceil (m-1)/2 \rceil} \mid a_n$ for any small number a_n in the cycle. On the other hand, $a_n \leq 2^m - 1$, so $a_{n+1} \leq 2^{2m} - 2^{m+1} + 1 + 2^m \leq 2^m(2^m - 1)$, so we have to divide a_{n+1} at most m times by two until we get a small number. This means that after a_n, the next small number is either $N = a_{m+n} = (a_n^2/2^{m-1}) + 2$ or $a_{m+n+1} = N/2$. In any case, $2^{\lceil (m-1)/2 \rceil}$ divides N.

If m is odd, then $x^2 \equiv -2 \pmod{2^{\lceil (m-1)/2 \rceil}}$ has a solution $x = a_n/2^{(m-1)/2}$. If $(m-1)/2 \geq 2 \iff m \geq 5$ then $x^2 \equiv -2 \pmod 4$, which has no solution. So if m is odd, then $m \leq 3$.

If m is even, then $2^{m-1} \mid a_n^2 \implies 2^{\lceil (m-1)/2 \rceil} \mid a_n \iff 2^{m/2} \mid a_n$. Then if $a_n = 2^{m/2}x$, $2x^2 \equiv -2 \pmod{2^{m/2}} \iff x^2 \equiv -1 \pmod{2^{(m/2)-1}}$, which is not possible for $m \geq 6$. So if m is even, then $m \leq 4$.

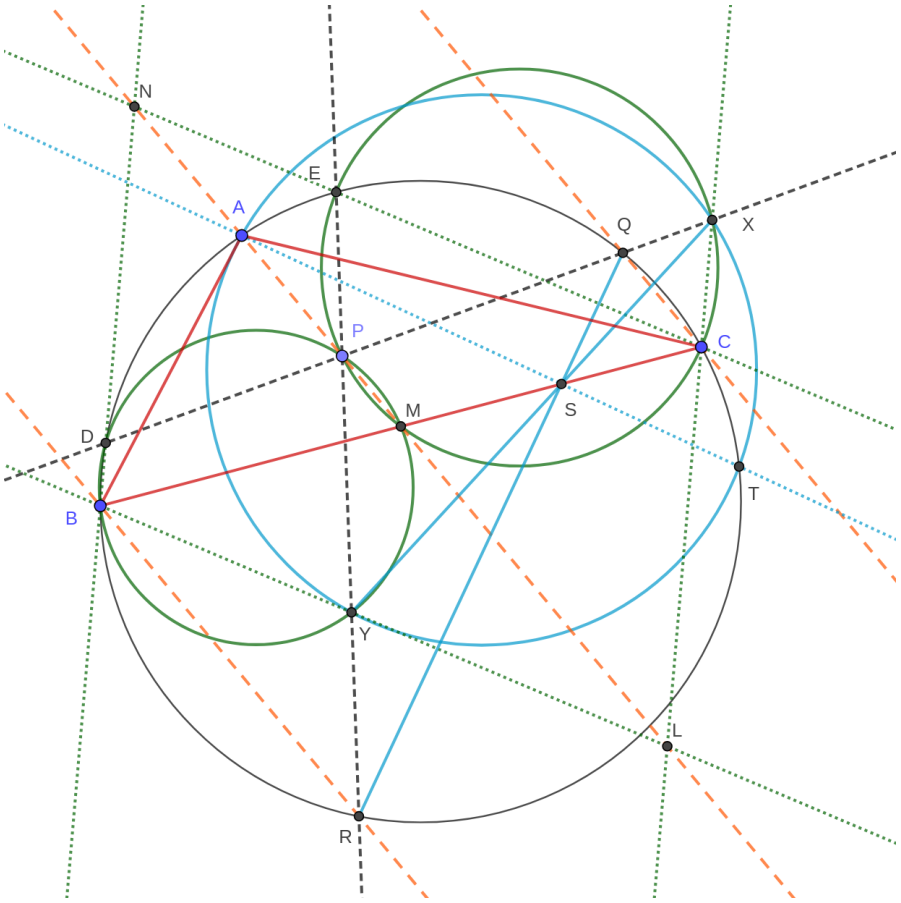
The cases $m = 1, 2, 3, 4$ are handed manually, checking the possible small numbers in the cycle, which have to be in the interval $[2^{m-1}, 2^m)$ and be divisible by $2^{\lceil (m-1)/2 \rceil}$:

- For $m = 1$, the only small number is 1, which leads to 5, then $5/2$.
- For $m = 2$, the only eligible small number is 2, which gives the cycle $(2, 8, 4)$. The only way to get to 2 is by dividing 4 by 2, so the starting numbers greater than 2 are all numbers that lead to 4, which are the powers of 2.
- For $m = 3$, the eligible small numbers are 4 and 6; we then obtain $4, 24, 12, 6, 44, 22, 11, 11/2$.
- For $m = 4$, the eligible small numbers are 8 and 12; we then obtain $8, 80, 40, 20, 10, \ldots$ or $12, 160, 80, 40, 20, 10, \ldots$, but in either case 10 is not an elegible small number.

□ **Problem 3**. Let ABC be a scalene triangle with circumcircle Γ. Let M be the midpoint of BC. A variable point P is selected in the line segment AM. The circumcircles of triangles BPM and CPM intersect Γ again at points D and E, respectively. The lines DP and EP intersect (a second time) the circumcircles to triangles CPM and BPM at X and Y, respectively. Prove that as P varies, the circumcircle of $\triangle AXY$ passes through a fixed point T distinct from A.

Solution. Let N be the radical center of the circumcircles of triangles ABC, BMP and CMP. The pairwise radical axes of these circles are BD, CE and PM, and hence they concur at N. Now, note that in directed angles:

$$\angle MCE = \angle MPE = \angle MPY = \angle MBY.$$



It follows that BY is parallel to CE, and analogously that CX is parallel to BD. Then, if L is the intersection of BY and CX, it follows that $BNCL$ is a parallelogram. Since $BM = MC$ we deduce that L is the reflection of N with respect to M, and therefore $L \in AM$. Using power of a point from L to the circumcircles of triangles BPM and CPM, we have

$$LY \cdot LB = LP \cdot LM = LX \cdot LC.$$

Hence, $BYXC$ is cyclic. Using the cyclic quadrilateral we find in directed angles:

$$\angle LXY = \angle LBC = \angle BCN = \angle NDE.$$

Since $CX \parallel BN$, it follows that $XY \parallel DE$.

Let Q and R be two points in Γ such that CQ, BR, and AM are all parallel. Then in directed angles:

$$\angle QDB = \angle QCB = \angle AMB = \angle PMB = \angle PDB.$$

Then D, P, Q are collinear. Analogously E, P, R are collinear. From here we get $\angle PRQ = \angle PDE = \angle PXY$, since XY and DE are parallel. Therefore $QRYX$ is cyclic. Let S be the radical center of the circumcircle of triangle ABC and the circles $BCYX$ and $QRYX$. This point lies in the lines BC, QR and XY because these are the radical axes of the circles. Let T be the second intersection of AS with Γ. By power of a point from S to the circumcircle of ABC and the circle $BCXY$ we have

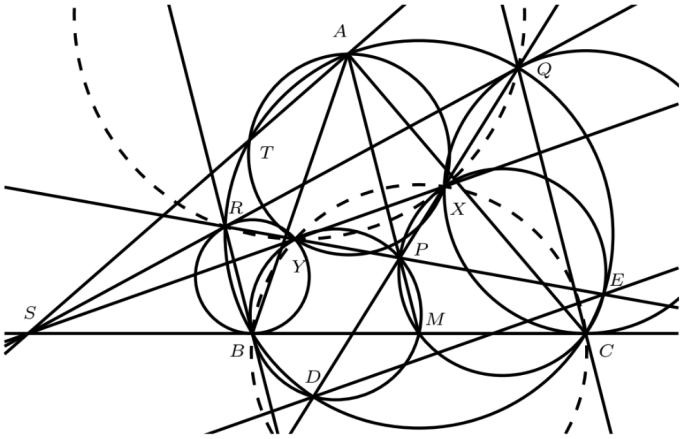
$$SX \cdot SY = SB \cdot SC = ST \cdot SA.$$

Therefore T is in the circumcircle of triangle AXY. Since Q and R are fixed regardless of the choice of P, then S is also fixed, since it is the intersection of QR and BC. This implies T is also fixed, and therefore, the circumcircle of triangle AXY goes through $T \neq A$ for any choice of P.

□

Now we show an alternative way to prove that $BCXY$ and $QRXT$ are cyclic.

Solution 2. Let the lines DP and EP meet the circumcircle of ABC again at Q and R, respectively. Then $\angle DQC \angle DBC = \angle DPM$, so $QC \parallel PM$. Similarly, $RB \parallel PM$.



Now, $\angle QCB = \angle PMB = \angle PXC = \angle(QX, CX)$, which is half of the arc QC in the circumcircle ω_C of QXC. So ω_C is tangent to BS; analogously, ω_B, the circumcicle of RYB, is also tangent to BC. Since $BR \parallel CQ$, the inscribed trapezoid $BRQC$ is isosceles, and by symmetry QR is also tangent to both circles, and the common perpendicular bisector of BR and CQ passes through the centers of ω_B and ω_C. Since $MB = MC$ and $PM \parallel BR \parallel CQ$, the line PM is the radical axis of ω_B and ω_C.

However, PM is also the radical axis of the circumcircles γ_B of PMB and γ_C of PMC. Let CX and PM meet at Z. Let $p(K, \omega)$ denote the power of a point K with respect to a circumference ω. We have

$$p(Z, \gamma_B) = p(Z, \gamma_C) = ZX \cdot ZC = p(Z, \omega_B) = p(Z, \omega_C).$$

Point Z is thus the radical center of $\gamma_B, \gamma_C, \omega_B, \omega_C$. Thus, the radical axes BY, CX, PM meet at Z. From here,

$$ZY \cdot ZB = ZC \cdot ZX \Rightarrow BCXY \text{ cyclic}$$
$$PY \cdot PR = PX \cdot PQ \Rightarrow QRXT \text{ cyclic}.$$

We may now finish as in Solution 1.

□

Problem 4. Consider a 2018×2019 board with integers in each unit square. Two unit squares are said to be neighbours if they share a common edge. In each turn, you choose some unit squares. Then for each chosen unit square the average of all its neighbours is calculated. Finally, after these calculations are done, the number in each chosen unit square is replaced by the corresponding average. Is it always possible to make the numbers in all squares become the same after finitely many turns?

Answer: No

Solution. Let n be a positive integer relatively prime to 2 and 3. We may study the whole process modulo n by replacing divisions by $2, 3, 4$ with multiplications by the corresponding inverses modulo n. If at some point the original process makes all the numbers equal, then the process modulo n will also have all the numbers equal. Our aim is to choose n and an initial configuration modulo n for which no process modulo n reaches a board with all numbers equal modulo n. We split this goal into two lemmas.

Lemma 1. There is a 2×3 board that stays constant modulo 5 and whose entries are not all equal.

Proof. Here is one such a board:

3	1	3
0	2	0

The fact that the board remains constant regardless of the choice of squares can be checked square by square.

□

Lemma 2. If there is an $r \times s$ board with $r \geq 2$, $s \geq 2$, that stays constant modulo 5, then there is also a $kr \times ls$ board with the same property.

Proof. We prove by a case by case analysis that repeateadly reflecting the $r \times s$ with respect to an edge preserves the property:

- If a cell had 4 neighbors, after reflections it still has the same neighbors.
- If a cell with a had 3 neighbors b, c, d, we have by hypothesis that $a \equiv 3^{-1}(b+c+d) \equiv 2(b+c+d) \pmod 5$. A reflection may add a as a neighbor of the cell and now
$$4^{-1}(a+b+c+d) \equiv 4(a+b+c+d) \equiv 4a+2a \equiv a \pmod 5$$
- If a cell with a had 2 neighbors b, c, we have by hypothesis that $a \equiv 2^{-1}(b+c) \equiv 3(b+c) \pmod 5$. If the reflections add one a as neighbor, now
$$3^{-1}(a+b+c) \equiv 2(3(b+c)+b+c) \equiv 8(b+c) \equiv 3(b+c) \equiv a \pmod 5$$
- If a cell with a had 2 neighbors b, c, we have by hypothesis that $a \equiv 2^{-1}(b+c) \pmod 5$. If the reflections add two a's as neighbors, now
$$4^{-1}(2a+b+c) \equiv (2^{-1}a + 2^{-1}a) \equiv a \pmod 5$$

In the three cases, any cell is still preserved modulo 5 after an operation. Hence we can fill in the $kr \times ls$ board by $k \times l$ copies by reflection. □

Since $2|2018$ and $3|2019$, we can get through reflections the following board:

3	1	3	3	1	3
0	2	0	0	2	0
0	2	0	0	2	0
3	1	3	3	1	3

... ...

By the lemmas above, the board is invariant modulo 5, so the answer is no.

□

Problem 5. Determine all the functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(x^2 + f(y)) = f(f(x)) + f(y^2) + 2f(xy)$$

for all real number x and y.

Answer: The possible functions are $f(x) = 0$ for all x and $f(x) = x^2$ for all x.

Solution. By substituting $x = y = 0$ in the given equation of the problem, we obtain that $f(0) = 0$. Also, by substituting $y = 0$, we get $f(x^2) = f(f(x))$ for any x.

Furthermore, by letting $y = 1$ and simplifying, we get

$$2f(x) = f(x^2 + f(1)) - f(x^2) - f(1),$$

from which it follows that $f(-x) = f(x)$ must hold for every x.

Suppose now that $f(a) = f(b)$ holds for some pair of numbers a, b. Then, by letting $y = a$ and $y = b$ in the given equation, comparing the two resulting identities and using the fact that $f(a^2) = f(f(a)) = f(f(b)) = f(b^2)$ also holds under the assumption, we get the fact that

$$f(a) = f(b) \Rightarrow f(ax) = f(bx) \quad \text{for any real number } x. \tag{1}$$

Consequently, if for some $a \neq 0$, $f(a) = 0$, then we see that, for any x, $f(x) = f(a \cdot \frac{x}{a}) = f(0 \cdot \frac{x}{a}) = f(0) = 0$, which gives a trivial solution to the problem.

In the sequel, we shall try to find a non-trivial solution for the problem. So, let us assume from now on that if $a \neq 0$ then $f(a) \neq 0$ must hold. We first note that since $f(f(x)) = f(x^2)$ for all x, the right-hand side of the given equation equals $f(x^2) + f(y^2) + 2f(xy)$, which is invariant if we interchange x and y. Therefore, we have

$$f(x^2) + f(y^2) + 2f(xy) = f(x^2 + f(y)) = f(y^2 + f(x)) \quad \text{for every pair } x, y. \tag{2}$$

Next, let us show that for any x, $f(x) \geq 0$ must hold. Suppose, on the contrary, $f(s) = -t^2$ holds for some pair s, t of non-zero real numbers. By setting $x = s, y = t$ in the right hand side of (2), we get $f(s^2 + f(t)) = f(t^2 + f(s)) = f(0) = 0$, so $f(t) = -s^2$. We also have $f(t^2) = f(-t^2) = f(f(s)) = f(s^2)$. By applying (2) with $x = \sqrt{s^2 + t^2}$ and $y = s$, we obtain

$$f(s^2 + t^2) + 2f(s \cdot \sqrt{s^2 + t^2}) = 0,$$

and similarly, by applying (2) with $x = \sqrt{s^2 + t^2}$ and $y = t$, we obtain

$$f(s^2 + t^2) + 2f(t \cdot \sqrt{s^2 + t^2}) = 0.$$

Consequently, we obtain

$$f(s \cdot \sqrt{s^2 + t^2}) = f(t \cdot \sqrt{s^2 + t^2}).$$

By applying (1) with $a = s\sqrt{s^2 + t^2}, b = t\sqrt{s^2 + t^2}$ and $x = 1/\sqrt{s^2 + t^2}$, we obtain $f(s) = f(t) = -s^2$, from which it follows that

$$0 = f(s^2 + f(s)) = f(s^2) + f(s^2) + 2f(s^2) = 4f(s^2),$$

a contradiction to the fact $s^2 > 0$. Thus we conclude that for all $x \neq 0$, $f(x) > 0$ must be satisfied.

Now, we show the following fact

$$k > 0, f(k) = 1 \Leftrightarrow k = 1. \tag{3}$$

Let $k > 0$ for which $f(k) = 1$. We have $f(k^2) = f(f(k)) = f(1)$, so by (1), $f(1/k) = f(k) = 1$, so we may assume $k \geq 1$. By applying (2) with $x = \sqrt{k^2 - 1}$ and $y = k$, and using $f(x) \geq 0$, we get

$$f(k^2 - 1 + f(k)) = f(k^2 - 1) + f(k^2) + 2f(k\sqrt{k^2 - 1}) \geq f(k^2 - 1) + f(k^2).$$

This simplifies to $0 \geq f(k^2 - 1) \geq 0$, so $k^2 - 1 = 0$ and thus $k = 1$.

Next we focus on showing $f(1) = 1$. If $f(1) = m \leq 1$, then we may proceed as above by setting $x = \sqrt{1 - m}$ and $y = 1$ to get $m = 1$. If $f(1) = m \geq 1$, now we note that $f(m) = f(f(1)) = f(1^2) = f(1) = m \leq m^2$. We may then proceed as above with $x = \sqrt{m^2 - m}$ and $y = 1$ to show $m^2 = m$ and thus $m = 1$.

We are now ready to finish. Let $x > 0$ and $m = f(x)$. Since $f(f(x)) = f(x^2)$, then $f(x^2) = f(m)$. But by (1), $f(m/x^2) = 1$. Therefore $m = x^2$. For $x < 0$, we have $f(x) = f(-x) = f(x^2)$ as well. Therefore, for all x, $f(x) = x^2$.

□

Solution 2 After proving that $f(x) > 0$ for $x \neq 0$ as in the previous solution, we may also proceed as follows. We claim that f is injective on the positive real numbers. Suppose that $a > b > 0$ satisfy $f(a) = f(b)$. Then by setting $x = 1/b$ in (1) we have $f(a/b) = f(1)$. Now, by induction on n and iteratively setting $x = a/b$ in (1) we get $f((a/b)^n) = 1$ for any positive integer n.

Now, let $m = f(1)$ and n be a positive integer such that $(a/b)^n > m$. By setting $x = \sqrt{(a/b)^n - m}$ and $y = 1$ in (2) we obtain that

$$f((a/b)^n - m + f(1)) = f((a/b)^n - m) + f(1^2) + 2f(\sqrt{(a/b)^n - m)}) \geq f((a/b)^n - m) + f(1).$$

Since $f((a/b)^n) = f(1)$, this last equation simplifies to $f((a/b)^n - m) \leq 0$ and thus $m = (a/b)^n$. But this is impossible since m is constant and $a/b > 1$. Thus, f is injective on the positive real numbers. Since $f(f(x)) = f(x^2)$, we obtain that $f(x) = x^2$ for any real value x.

□

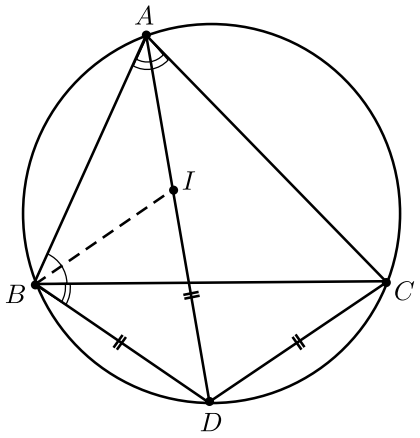
Solutions of APMO 2018

Problem 1. Let H be the orthocenter of the triangle ABC. Let M and N be the midpoints of the sides AB and AC, respectively. Assume that H lies inside the quadrilateral $BMNC$ and that the circumcircles of triangles BMH and CNH are tangent to each other. The line through H parallel to BC intersects the circumcircles of the triangles BMH and CNH in the points K and L, respectively. Let F be the intersection point of MK and NL and let J be the incenter of triangle MHN. Prove that $FJ = FA$.

Solution.

Lemma 1. In a triangle ABC, let D be the intersection of the interior angle bisector at A with the circumcircle of ABC, and let I be the incenter of $\triangle ABC$. Then

$$DI = DB = DC.$$



Proof.

$$\angle DBI = \frac{\angle BAC}{2} + \frac{\widehat{B}}{2} = \angle DIB \quad \Rightarrow \quad DI = DB.$$

Analogously $DI = DC$. □

We start solving the problem. First we state some position considerations. Since there is an arc of the circumcircle of BHM outside the triangle ABC, it must happen that K and N lie on opposite sides of AM. Similarly, L and M lie on opposite sides of AN. Also, K and L lie on the same side of MN, and opposite to A. Therefore, F lies inside the triangle AMN.

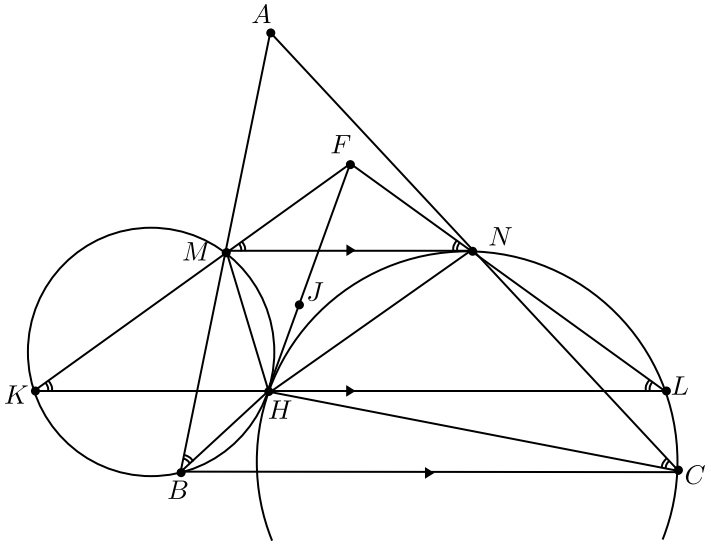
Now, since H is the orthocenter of $\triangle ABC$ and the circumcircles of BMH and CNH are tangent we have

$$\angle ABH = 90^\circ - \angle BAC = \angle ACH \quad \Rightarrow \quad \angle MHN = \angle MBH + \angle NCH = 180^\circ - 2\angle BAC. \quad (1)$$

So $\angle MBH = \angle MKH = \angle NCH = \angle NLH = 90^\circ - \angle BAC$ and, since $MN \| KL$, we have

$$\angle FMN = \angle FNM = 90^\circ - \angle BAC \quad \Rightarrow \quad \angle MFN = 2\angle BAC. \quad (2)$$

The relations (1) and (2) yield that the quadrilateral $MFNH$ is cyclic, with the vertices in this order around the circumference. Since $FM = FN$, $\angle MFN = 2\angle BAC$ and F is the correct side of MN we have that the point F is the circumcenter of triangle AMN, and thus $FA = FM = FN$.



Since the quadrilateral $MFNH$ is cyclic, $FM = FN$ and H lies on the correct side of MN, we have that H, J and F are collinear. According to Lemma 1, $FJ = FM = FN$. So $FJ = FA$.

□

Solution 2: According to Solution 1, we have $\angle MHN = 180^\circ - 2\angle BAC$ and since the point J is the incenter of $\triangle MHN$, we have $\angle MJN = 90^\circ + \frac{1}{2}\angle MHN = 180^\circ - \angle BAC$. So the quadrilateral $AMJN$ is cyclic.

According to Solution 1, the point F is the circumcenter of $\triangle AMN$. So $FJ = FA$.

□

Problem 2. Let $f(x)$ and $g(x)$ be given by

$$f(x) = \frac{1}{x} + \frac{1}{x-2} + \frac{1}{x-4} + \cdots + \frac{1}{x-2018}$$

and

$$g(x) = \frac{1}{x-1} + \frac{1}{x-3} + \frac{1}{x-5} + \cdots + \frac{1}{x-2017}\,.$$

Prove that

$$|f(x) - g(x)| > 2$$

for any non-integer real number x satisfying $0 < x < 2018$.

Solution 1 There are two cases: $2n-1 < x < 2n$ and $2n < x < 2n+1$. Note that $f(2018-x) = -f(x)$ and $g(2018-x) = -g(x)$, that is, a half turn about the point $(1009, 0)$ preserves the graphs of f and g. So it suffices to consider only the case $2n-1 < x < 2n$.

Let $d(x) = g(x) - f(x)$. We will show that $d(x) > 2$ whenever $2n-1 < x < 2n$ and $n \in \{1, 2, \ldots, 1009\}$.

For any non-integer x with $0 < x < 2018$, we have

$$d(x+2) - d(x) = \left(\frac{1}{x+1} - \frac{1}{x+2}\right) + \left(\frac{1}{x-2018} - \frac{1}{x-2017}\right) > 0 + 0 = 0.$$

Hence it suffices to prove $d(x) > 2$ for $1 < x < 2$. Since $x < 2$, it follows that $\frac{1}{x-2i-1} > \frac{1}{x-2i}$ for $i = 2, 3, \ldots, 1008$. We also have $\frac{1}{x-2018} < 0$. Hence it suffices to prove the following

for $1 < x < 2$.

$$\begin{aligned}&\frac{1}{x-1}+\frac{1}{x-3}-\frac{1}{x}-\frac{1}{x-2}>2\\ \Leftrightarrow&\left(\frac{1}{x-1}+\frac{1}{2-x}\right)+\left(\frac{1}{x-3}-\frac{1}{x}\right)>2\\ \Leftrightarrow&\frac{1}{(x-1)(2-x)}+\frac{3}{x(x-3)}>2.\end{aligned}$$

By the $GM-HM$ inequality (alternatively, by considering the maximum of the quadratic $(x-1)(2-x)$) we have

$$\frac{1}{x-1}\cdot\frac{1}{2-x}>\left(\frac{2}{(x-1)+(2-x)}\right)^2=4.$$

To find a lower bound for $\frac{3}{x(x-3)}$, note that $x(x-3)<0$ for $1<x<2$. So we seek an upper bound for $x(x-3)$. From the shape of the quadratic, this occurs at $x=1$ or $x=2$, both of which yield $\frac{3}{x(x-3)}>-\frac{3}{2}$.

It follows that $d(x)>4-\frac{3}{2}>2$, as desired. □

Solution 2

As in Solution 1, we may assume $2n-1<x<2n$ for some $1\le n\le 1009$. Let $d(x)=f(x)-g(x)$, and note that

$$d(x)=\frac{1}{x}+\sum_{m=1}^{1009}\frac{1}{(x-2m)(x-2m+1)}$$

We split the sum into three parts: the terms before $m=n$, after $m=n$, and the term $m=n$. The first two are

$$\begin{aligned}0&\le\sum_{m=1}^{n-1}\frac{1}{(x-2m)(x-2m+1)}\\ &\le\sum_{m=1}^{n-1}\frac{1}{(2n-1-2m)(2n-2m)}=\sum_{i=1}^{n-1}\frac{1}{(2i)(2i-1)}\le\sum_{i=1}^{1008}\frac{1}{2i-1}-\frac{1}{2i},\end{aligned}$$

$$\begin{aligned}0&\le\sum_{m=n+1}^{1009}\frac{1}{(2m-x)(2m-1-x)}\\ &\le\sum_{m=n+1}^{1009}\frac{1}{(2m-2n+1)(2m-2n)}=\sum_{i=1}^{1009-n}\frac{1}{(2i+1)(2i)}\le\sum_{i=1}^{1008}\frac{1}{2i}-\frac{1}{2i+1}.\end{aligned}$$

When we add the two sums the terms telescope and we are left with

$$0\le\sum_{1\le m\le 1009,m\ne n}\frac{1}{(x-2m)(x-2m+1)}\le 1-\frac{1}{2017}<1,$$

For the term $m=n$, we write

$$0<-(x-2n)(x-2n+1)=0.25-(x-2n+0.5)^2\le 0.25,$$

whence

$$-4 \geq \frac{1}{(x-2n)(x-2n+1)}.$$

Finally, $\frac{1}{x} < 1$ since $x > 2n-1 \geq 1$. Combining these we get

$$d(x) = \frac{1}{x} + \sum_{m=1}^{1009} \frac{1}{(x-2m)(x-2m+1)} < 1 + 1 - 4 < -2.$$

□

Solution 3

First notice that

$$f(x) - g(x) = \frac{1}{x} - \frac{1}{x-1} + \frac{1}{x-2} - \cdots - \frac{1}{x-2017} + \frac{1}{x-2018}.$$

As in Solution 1, we may deal only with the case $2n < x < 2n+1$. Then $x-2k+1$ and $x-2k$ never differ in sign for any integer k. Then

$$-\frac{1}{x-2k+1} + \frac{1}{x-2k} = \frac{1}{(x-2k+1)(x-2k)} > 0 \quad \text{for } k = 1, 2, \ldots, n-1, n+2, \ldots, 1009.$$

$$\frac{1}{x-2n} - \frac{1}{x-2n-1} = \frac{1}{(x-2n)(2n+1-x)} \geq \left(\frac{2}{x-2n+2n+1-x}\right)^2 = 4,$$

Therefore, summing all inequalities and collecting the remaining terms we find $f(x)-g(x) > 4 + \frac{1}{x-2} > 4-1 = 3$ for $0 < x < 1$ and, for $n > 0$,

$$\begin{aligned} f(x) - g(x) &> \frac{1}{x} - \frac{1}{x-2n+1} + 4 + \frac{1}{x-2n-2} \\ &= \frac{1}{x} - \frac{1}{x-2n+1} + 4 - \frac{1}{2n+2-x} \\ &> \frac{1}{x} - \frac{1}{2n-2n+1} + 4 - \frac{1}{2n+2-2n-1} \\ &= 2 + \frac{1}{x} > 2. \end{aligned}$$

□

Problem 3. A collection of n squares on the plane is called *tri-connected* if the following criteria are satisfied:

(i) All the squares are congruent.

(ii) If two squares have a point P in common, then P is a vertex of each of the squares.

(iii) Each square touches exactly three other squares.

How many positive integers n are there with $2018 \leq n \leq 3018$, such that there exists a collection of n squares that is tri-connected?

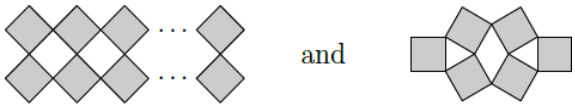
Answer: 501

Solution. We will prove that there is no tri-connected collection if n is odd, and that tri-connected collections exist for all even $n \geq 38$. Since there are 501 even numbers in the range from 2018 to 3018, this yields 501 as the answer.

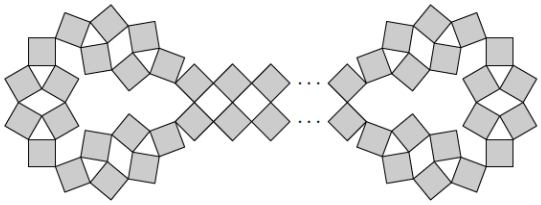
For any two different squares A and B, let us write $A \sim B$ to mean that square A touches square B. Since each square touches exactly three other squares, and there are n squares in total, the total number of instances of $A \sim B$ is $3n$. But $A \sim B$ if and only if $B \sim A$. Hence the total number of instances of $A \sim B$ is even. Thus $3n$ and hence also n is even.

We now construct tri-connected collections for each even n in the range. We show two

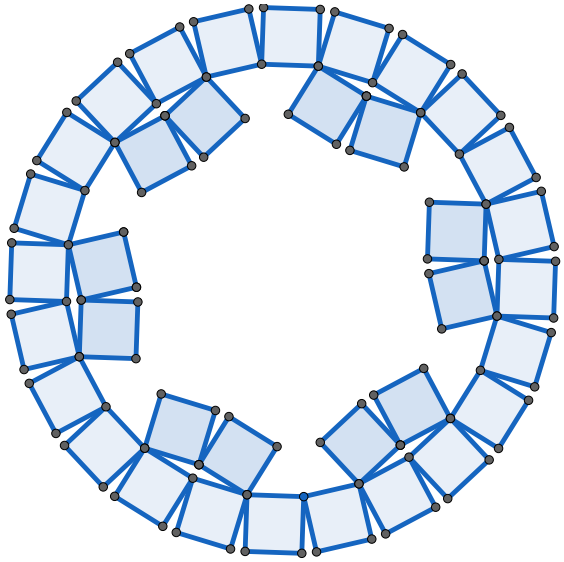
Construction 1 The idea is to use the following two configurations. Observe that in each configuration every square is related to three squares except for the leftmost and rightmost squares which are related to two squares. Note that the configuration on the left is of variable length. Also observe that multiple copies of the configuration on the right can be chained together to end around corners.



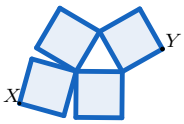
Putting the above two types of configurations together as in the following figure yields a tri-connected collection for every even $n \geq 38$.



Construction 2 Consider a regular $4n-$gon $A_1A_2\cdots A_{4n}$, and make $4n$ squares on the outside of the $4n-$gon with one side being on the $4n-$gon. Reflect squares sharing sides $A_{4m+2}A_{4m+3}, A_{4m+3}A_{4m+4}$ across line $A_{4m+2}A_{4m+4}$, for $0 \leq m \leq n-1$. This will produce a tri-connected set of $6n$ squares, as long as the squares inside the $4n-$gon do not intersect. When $n \geq 4$, this will be true. The picture for $n = 24$ is as follows:



To treat the other cases, consider the following gadget



Two squares touch 3 other squares, and the squares containing X, Y touch 2 other squares. Take the $4n-$gon from above, and break it into two along the line A_1A_{2n}, moving the two parts away from that line. Do so until the gaps can be exactly filled by inserting two copies of the above figure, so that the vertices X, Y touch the two vertices which used to be A_1 in one instance, and the two vertices which used to be A_{2n} in the other.
This gives us a valid configuration for $6n + 8$ squares, $n \geq 4$. Finally, if we had instead spread the two parts out more and inserted two copies of the above figure into each gap, we would get $6n + 16$ for $n \geq 4$, which finishes the proof for all even numbers at least 36.

Problem 4. Let ABC be an equilateral triangle. From the vertex A we draw a ray towards the interior of the triangle such that the ray reaches one of the sides of the triangle. When the ray reaches a side, it then bounces off following the *law of reflection*, that is, if it arrives with a directed angle α, it leaves with a directed angle $180° - \alpha$. After n bounces, the ray returns to A without ever landing on any of the other two vertices. Find all possible values of n.

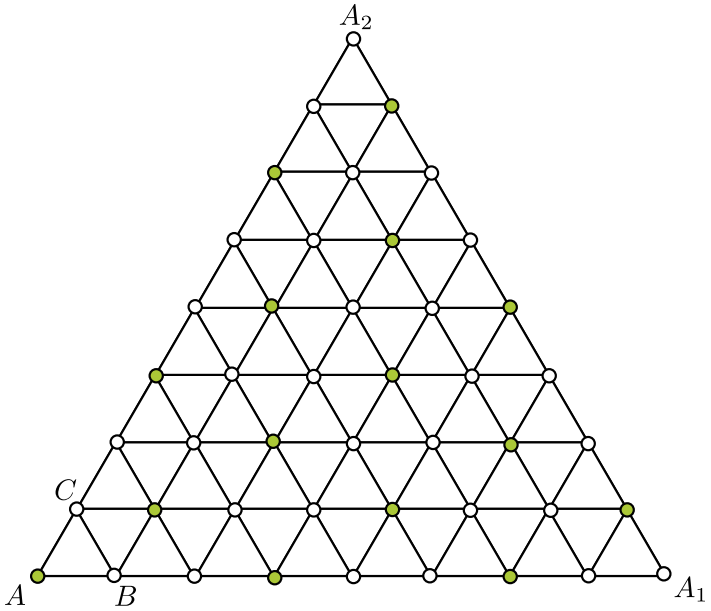
Answer: All $n \equiv 1, 5 \bmod 6$ with the exception of 5 and 17

Solution. Consider an equilateral triangle AA_1A_2 of side length m and triangulate it with unitary triangles. See the figure. To each of the vertices that remain after the triangulation we can assign a pair of coordinates (a, b) where a, b are non-negative integers, a is the number of edges we travel in the AA_1 direction and b is the number of edges we travel in the AA_2 direction to arrive to the vertex, (we have $A = (0, 0)$, $A_1 = (m, 0)$ and $A_2 = (0, m)$). The unitary triangle with vertex A will be our triangle ABC, ($B = (1, 0)$, $C = (0, 1)$). We can obtain every unitary triangle by starting with ABC and performing reflections with respect to a side (the vertex $(1, 1)$ is the reflection of A with respect to BC, the vertex $(0, 2)$ is the reflection of $B = (1, 0)$ with respect to the side formed by $C = (1, 0)$ and $(1, 1)$, and so on).

When we reflect a vertex (a, b) with respect to a side of one of the triangles, the congruence of $a-b$ is preserved modulo 3. Furthermore, an induction argument shows that any two vertices (a, b) and (a', b') with $a - b \equiv a' - b' \bmod 3$ can be obtained from each other by a series of such reflections. Therefore, the set of vertices V that result from the reflections of A will be those of the form (a, b) satisfying $a \equiv b \bmod 3$. See the green vertices in the figure.

Now, let U be the set of vertices u that satisfy that the line segment between u and A does not pass through any other vertex. A pair (a, b) is in U if and only if $\gcd(a, b) = 1$, since otherwise for $d = \gcd(a, b)$ we have that the vertex $(a/d, b/d)$ also lies on the line segment between u and A.

Observe that the rays that come out from A and eventually return to A are those that come out towards a vertex in $V \cap U$ (they would be in V to be able to come back to A and in U so that they do not reach a vertex beforehand). In the diagram, a ray toward one such vertex (a, b) will intersect exactly $(a - 1) + (b - 1) + (a + b - 1) = 2(a + b) - 3$ lines: $a - 1$ of them parallel to AB, $b - 1$ parallel to AC and $a + b - 1$ parallel to BC. Therefore, in the triangle ABC the ray will bounce $2(a + b) - 3$ times before returning to A. So we want to find all



$n = 2(a + b) - 3$ where $a \equiv b \bmod 3$ and $\gcd(a, b) = 1$.

If $a+b$ is a multiple of 3 then we cannot satisfy both conditions simultaneously, therefore n is not a multiple of 3. We also know that n is odd. Therefore $n \equiv 1, 5, 7, 11 \bmod 12$. Note that the pair $(1, 3k + 1)$ satisfies the conditions and we can create $n = 2(3k + 2) - 3 = 6k + 1$ for all $k \geq 0$ (this settles the question for $n \equiv 1, 7 \bmod 12$). For $n \equiv 5 \bmod 12$ consider the pair $(3k - 1, 3k + 5)$ when k is even or $(3k - 4, 3k + 8)$ when k is odd. This gives us all the integers of the form $12k + 5$ for $k \geq 2$. For $11 \bmod 12$, take the pairs $(3k - 1, 3k + 2)$ (with $k \geq 1$), which yield all positive integers of the form $12k - 1$.

Finally, to discard 5 and 17 note that the only pairs (a, b) that are solutions to $2(a+b)-3 = 5$ or $2(a + b) - 3 = 17$ with the same residue mod 3 in this range are the non-relatively prime pairs $(2, 2)$, $(2, 8)$ and $(5, 5)$.

□

Problem 5. Find all polynomials $P(x)$ with integer coefficients such that for all real numbers s and t, if $P(s)$ and $P(t)$ are both integers, then $P(st)$ is also an integer.

Answer: $P(x) = x^n + k, -x^n + k$ for n a non-negative integer and k an integer.

Solution 1: $P(x) = x^n + k, -x^n + k$ for n a non-negative integer and k an integer.

Notice that if $P(x)$ is a solution, then so is $P(x) + k$ and $-P(x) + k$ for any integer k, so we may assume that the leading coefficient of $P(x)$ is positive and that $P(0) = 0$, i.e., we can assume that $P(x) = \sum_{i=1}^{n} a_i x^i$ with $a_n > 0$. We are going to prove that $P(x) = x^n$ in this case.

Let p be a large prime such that $p > \sum_{i=1}^{n} |a_i|$. Because P has a positive leading coefficient and p is large enough, we can find $t \in \mathbb{R}$ such that $P(t) = p$. Denote the greatest common divisor of the polynomial $P(x) - p$ and $P(2x) - P(2t)$ as $f(x)$, and t is a root of it, so f is a non-constant polynomial. Notice that $P(2t)$ is an integer by using the hypothesis for $s = 2$ and t. Since $P(x) - p$ and $P(2x) - P(2t)$ are polynomials with integer coefficients, f can be chosen as a polynomial with rational coefficients.

In the following, we will prove that f is the same as $P(x) - p$ up to a constant multiplier. Say $P(x) - p = f(x)g(x)$, where f and g are non-constant polynomials. By Gauss's lemma, we can get f_1, g_1 with $P(x) - p = f_1(x)g_1(x)$ where f_1 is a scalar multiple of f and g_1 is a scalar multiple of g and one of f_1, g_1 has constant term ± 1 (this is because $-p = P(0) - p = f(0)g(0)$ with p prime). So $P(x) - p$ has at least one root r with absolute value not greater than 1 (using

Vieta, the product of the roots of the polynomial with constant term ± 1 is ± 1), but

$$|P(r) - p| = \left| \sum_{i=1}^{n} a_i r^i - p \right| > p - \sum_{i=1}^{n} |a_i| > 0,$$

hence we get a contradiction!

Therefore f is a constant multiple of $P(x) - p$, so $P(2x) - P(2t)$ is a constant multiple of $P(x) - p$ because they both have the same degree. By comparing leading coefficients we get that $P(2x) - P(2t) = 2^n(P(x) - p)$. Comparing the rest of the coefficients we get that $P(x) = a_n x^n$. If we let $a = b = (1/a_n)^{1/n}$, then $P(a) = P(b) = 1$, so $P(ab)$ must also be an integer. But $P(ab) = \frac{1}{a_n}$. Therefore $a_n = 1$ and the proof is complete.

□

Solution 2: Assume $P(x) = \sum_{i=0}^{n} a_i x^i$. Consider the following system of equations

$$\begin{aligned} a_0 &= P(0) \\ a_n t^n + a_{n-1} t^{n-1} + \cdots + a_0 &= P(t) \\ 2^n a_n t^n + 2^{n-1} a_{n-1} t^{n-1} + \cdots + a_0 &= P(2t) \\ &\vdots \\ n^n a_n t^n + n^{n-1} a_{n-1} t^{n-1} + \cdots + a_0 &= P(nt). \end{aligned}$$

viewing $a_k t^k$ as variables. Note that if $P(t)$ is an integer, then by the hypothesis all the terms on the right hand side of the equations are integers as well. By using Cramer's rule, we can get that $a_k t^k = D/M$, where D is an integer and M is the following determinant

$$\begin{vmatrix} 1 & 0 & 0 & \cdots & 0 \\ 1 & 1 & 1 & \cdots & 1 \\ 1 & 2 & 4 & \cdots & 2^n \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & n & n^2 & \cdots & n^n \end{vmatrix} \neq 0.$$

Thus, if we let r be the smallest positive index such that $a_r \neq 0$, we can express each $t \in \mathbb{R}$ with $P(t) \in \mathbb{Z}$ in the form $(\frac{m}{M'})^{1/r}$ for some integer m, and where $M' = M \times a_r$ is a constant.

We can choose L large enough such that $P|_{\mathbb{R}_{\geq L}}$ is injective, and for any larger N, the growth order of the number of values in the form $(\frac{m}{M'})^{1/r}$ is N^r, while the growth order of the number of integers in $[P(L), P(N)]$ is N^n, so $r = n$. Therefore $P(x)$ is of the form $a_n x^n + k$. The problem can be finished as in Solution 1.

□

Solutions of APMO 2017

Problem 1. We call a 5-tuple of integers *arrangeable* if its elements can be labeled a, b, c, d, e in some order so that $a-b+c-d+e=29$. Determine all 2017-tuples of integers $n_1, n_2, \ldots, n_{2017}$ such that if we place them in a circle in clockwise order, then any 5-tuple of numbers in consecutive positions on the circle is arrangeable.

Answer: $n_1 = \cdots = n_{2017} = 29$.

Solution. A valid 2017-tuple is $n_1 = \cdots = n_{2017} = 29$. We will show that it is the only solution.

We first replace each number n_i in the circle by $m_i := n_i - 29$. Since the condition $a-b+c-d+e=29$ can be rewritten as $(a-29)-(b-29)+(c-29)-(d-29)+(e-29)=0$, we have that any five consecutive replaced integers in the circle can be labeled a, b, c, d, e in such a way that $a-b+c-d+e=0$. We claim that this is possible only when all of the m_i's are 0 (and thus all of the original n_i's are 29).

We work with indexes modulo 2017. Notice that for every i, m_i and m_{i+5} have the same parity. Indeed, this follows from $m_i \equiv m_{i+1} + m_{i+2} + m_{i+3} + m_{i+4} \equiv m_{i+5} \pmod 2$. Since $\gcd(5, 2017) = 1$, this implies that all m_i's are of the same parity. Since $m_1+m_2+m_3+m_4+m_5$ is even, all m_i's must be even as well.

Suppose for the sake of contradiction that not all m_i's are zero. Then our condition still holds when we divide each number in the circle by 2. However, by performing repeated divisions, we eventually reach a point where some m_i is odd. This is a contradiction.

□

Problem 2. Let ABC be a triangle with $AB < AC$. Let D be the intersection point of the internal bisector of angle BAC and the circumcircle of ABC. Let Z be the intersection point of the perpendicular bisector of AC with the external bisector of angle $\angle BAC$. Prove that the midpoint of the segment AB lies on the circumcircle of triangle ADZ.

Solution 1. Let N be the midpoint of AC. Let M be the intersection point of the circumcircle of triangle ADZ and the segment AB. We will show that M is the midpoint of AB. To do this, let D' the reflection of D with respect to M. It suffices to show that $ADBD'$ is a parallelogram.

The internal and external bisectors of an angle in a triangle are perpendicular. This implies that ZD is a diameter of the circumcircle of AZD and thus $\angle ZMD = 90°$. This means that ZM is the perpendicular bisector of $D'D$ and thus $ZD' = ZD$. By construction, Z is in the perpendicular bisector of AC and thus $ZA = ZC$.

Now, let α be the angle $\angle BAD = \angle DAC$. In the cyclic quadrilateral $AZDM$ we get $\angle MZD = \angle MAD = \alpha$, and thus $\angle D'ZD = 2\alpha$. By angle chasing we get

$$\angle AZN = 90° - \angle ZAN = \angle DAC = \alpha,$$

which implies that $\angle AZC = 2\alpha$. Therefore,

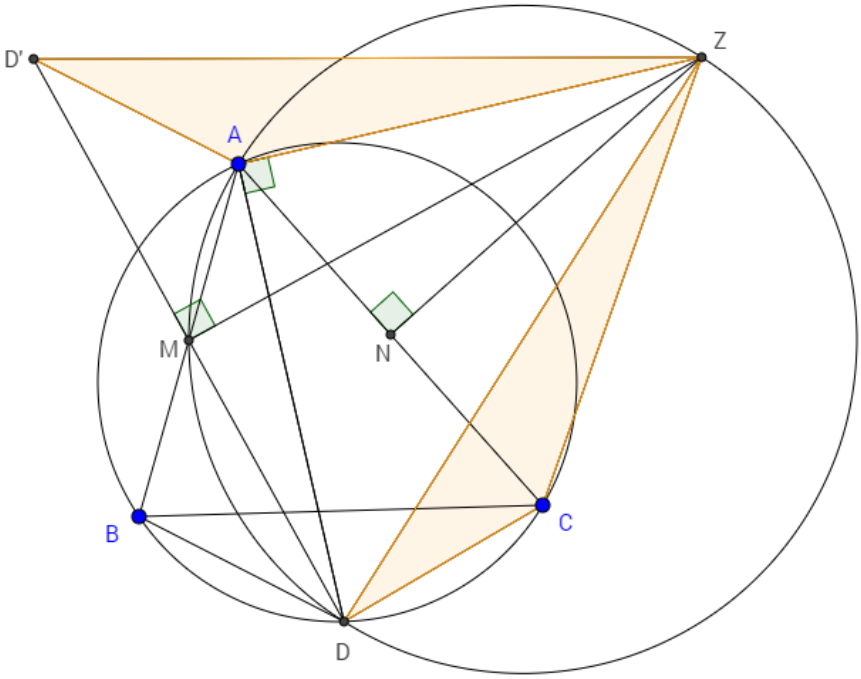
$$\angle D'ZA = \angle D'ZD - \angle AZD = 2\alpha - \angle AZD = \angle AZC - \angle AZD = \angle DZC.$$

Combining $\angle D'ZA = \angle DZC$, $ZD' = ZD$ and $ZA = ZC$, we obtain by the SAS criterion that the triangles $D'ZA$ and DZC are congruent. In particular, $D'A = DC$ and $\angle D'AZ = \angle DCZ$. From here $DB = DC = D'A$.

Finally, let $\beta = \angle ABC = \angle ADC$. We get the first of the following equalities by the sum of angles around point A and the second one by the sum of internal angles of quadrilateral $AZCD$

$$360^\circ = \angle D'AZ + \angle ZAD + \angle DAB + \angle BAD' = \angle D'AZ + 90^\circ + \alpha + \angle BAD'$$
$$360^\circ = \angle DCZ + \angle ZAD + \angle CZA + \angle ADC = \angle DCZ + 90^\circ + 2\alpha + \beta.$$

By canceling equal terms we conclude that $\angle BAD' = \alpha + \beta$. Also, $\angle ABD = \alpha + \beta$. Therefore, the segments $D'A$ and DB are parallel and have the same length. We conclude that $ADBD'$ is a parallelogram. As the diagonals of a parallelogram intersect at their midpoints, we obtain that M is the midpoint of AB as desired.



□

Variant of solution. The solution above is indirect in the sense that it assumes that M is in the circumcircle of AZD and then shows that M is the midpoint of AB. We point out that the same ideas in the solution can be used to give a direct solution. Here we present a sketch on how to proceed in this manner.

Now we know that M is the midpoint of the side AB. We construct the point D' in the same way. Now we have directly that $ADBD'$ is a parallelogram and thus $D'A = DB = DC$. By construction $ZA = ZC$. Also, the two sums of angles equal to 360° in the previous solution let us conclude that $\angle D'AZ = \angle DCZ$. Once again, we use (differently) the SAS criterion and obtain that the triangles $D'AZ$ and DCZ are congruent. Thus, $D'Z = DZ$.

We finish the problem by noting that ZM is a median of the isosceles triangle $D'ZD$, so it is also a perpendicular bisector. This shows that $\angle DMZ = 90^\circ = \angle DAZ$, and therefore M lies in the circumcircle of DAZ.

Solution 2. We proceed directly. As above, we name

$$\alpha = \angle DAC = \angle AZN = \angle CZN = \angle DCB.$$

Let L be the midpoint of the segment BC. Since M and N are midpoints of AB and AC, we have that $MN = CL$ and that the segment MN is parallel to BC. Thus, $\angle ANM = \angle ACB$.

Therefore, $\angle ZNM = \angle ACB + 90°$. Also, $\angle DCZ = \angle DCB + \angle ACB + \angle ZCA = \angle ACB + 90°$. We conclude that $\angle ZNM = \angle DCZ$.

Now, the triangles ZNC and CLD are similar since $\angle DCL = 90° = \angle CNZ$ and $\angle DCL = \alpha = \angle CZN$. We use this fact to obtain the following chain of equalities

$$\frac{CD}{MN} = \frac{CD}{CL} = \frac{CZ}{ZN}.$$

We combine the equality above with $\angle ZNM = \angle DCZ$ to conclude that the triangles ZNM and ZCD are similar. In particular, $\angle MZN = \angle DZC$ and $\frac{ZM}{ZN} = \frac{ZD}{ZC}$. Since we also have $\angle MZD = \angle MZN + \angle NZD = \angle DZC + \angle NZD = \angle NZC$, we conclude that the triangles MZD and NZC are similar. This yields that $\angle ZMD = 90°$ and therefore M is in the circumcircle of triangle DAZ.

□

Solution 3. Let m be the perpendicular bisector of AD; thus m passes through the center O of the circumcircle of triangle ABC. Since AD is the internal angle bisector of A and OM and ON are perpendicular to AB and AC respectively, we obtain that OM and ON form equal angles with AD. This implies that they are symmetric with respect to M.

Therefore, the line MO passes through the point Z' symmetrical to Z with respect to m. Since $\angle DAZ = 90°$, then also $\angle ADZ' = 90°$. Moreover, since $AZ = DZ'$, we have that $\angle AZZ' = \angle DZ'Z = 90°$, so $AZZ'D$ is a rectangle. Since $\angle AMZ' = \angle AMO = 90°$, we conclude that M is in the circle with diameter AZ', which is the circumcircle of the rectangle. Thus M lies on the circumcircle of the triangle ADZ.

Problem 3. Let $A(n)$ denote the number of sequences $a_1 \geq a_2 \geq \ldots \geq a_k$ of positive integers for which $a_1 + \cdots + a_k = n$ and each $a_i + 1$ is a power of two ($i = 1, 2, \ldots, k$). Let $B(n)$ denote the number of sequences $b_1 \geq b_2 \geq \ldots \geq b_m$ of positive integers for which $b_1 + \cdots + b_m = n$ and each inequality $b_j \geq 2b_{j+1}$ holds ($j = 1, 2, \ldots, m-1$).

Prove that $A(n) = B(n)$ for every positive integer n.

Solution. We say that a sequence $a_1 \geq a_2 \geq \ldots \geq a_k$ of positive integers has *type A* if $a_i + 1$ is a power of two for $i = 1, 2, \ldots, k$. We say that a sequence $b_1 \geq b_2 \geq \ldots \geq b_m$ of positive integer has *type B* if $b_j \geq 2b_{j+1}$ for $j = 1, 2, \ldots, m-1$.

Recall that the binary representation of a positive integer expresses it as a sum of distinct powers of two in a unique way. Furthermore, we have the following formula for every positive integer N

$$2^N - 1 = 2^{N-1} + 2^{N-2} + \cdots + 2^1 + 2^0.$$

Given a sequence $a_1 \geq a_2 \geq \ldots \geq a_k$ of type A, use the preceding formula to express each term as a sum of powers of two. Write these powers of two in left-aligned rows, in decreasing order of size. By construction, a_i is the sum of the numbers in the ith row. For example, we obtain the following array when we start with the type A sequence $15, 15, 7, 3, 3, 3, 1$.

8	4	2	1
8	4	2	1
4	2	1	
2	1		
2	1		
2	1		
1			
27	13	5	2

Define the sequence $b_1, b_2, \ldots, b_n$ by setting b_j to be the sum of the numbers in the jth column of the array. For example, we obtain the sequence $27, 13, 5, 2$ from the array above. We now show that this new sequence has type B. This is clear from the fact that each column in the array contains at least as many entries as the column to the right of it and that each number larger than 1 in the array is twice the number to the right of it. Furthermore, it is clear that $a_1 + a_2 + \cdots + a_k = b_1 + b_2 + \cdots + b_m$, since both are equal to the sum of all the entries in the array.

We now show that we can do this operation backwards. Suppose that we are given a type B sequence $b_1 \geq b_2 \geq \ldots \geq b_m$. We construct an array inductively as follows:

- We fill b_m left-aligned rows with the numbers $2^{m-1}, 2^{m-2}, \ldots, 2^1, 2^0$.
- Then we fill $b_{m-1} - 2b_m$ left aligned rows with the numbers $2^{m-2}, 2^{m-3}, \ldots, 2^1, 2^0$.
- Then we fill $b_{m-2} - 2b_{m-1}$ left aligned rows with the numbers $2^{m-3}, 2^{m-4}, \ldots, 2^1, 2^0$, and so on.
- In the last step we fill $b_1 - 2b_2$ left-aligned rows with the number 1.

For example, if we start with the type B sequence $27, 13, 5, 2$, we obtain once again the array above. We define the sequence $a_1, a_2, \ldots, a_k$ by setting a_i to be the sum of the numbers in the ith row of the array. By construction, this sequence has type A. Furthermore, it is clear that $a_1 + \cdots + a_k = b_1 + \cdots + b_m$, since once again both sums are equal to the sum of all the entries in the array.

We have defined an operation that starts with a sequence of type A, produces an array whose row sums are given by the sequence, and outputs a sequence of type B corresponding to the column sums. We have also defined an operation that starts with a sequence of type B, produces an array whose column sums are given by the sequence, and outputs a sequence of type A corresponding to the row sums. The arrays produced in both cases comprise left-aligned rows of the form $2^{N-1}, 2^{N-2}, \ldots, 2^1, 2^0$, with non-increasing lengths. Let us refer to arrays obeying these properties as *marvelous*.

To show that these two operations are inverses of each other, it then suffices to prove that marvelous arrays are uniquely defined by either their row sums or their column sums. The former is obviously true and the latter arises from the observation that each step in the above inductive algorithm was forced in order to create a marvelous array with the prescribed column sums.

Thus, we have produced a bijection between the sequences of type A with sum n and the sequences of type B with sum n. So we can conclude that $A(n) = B(n)$ for every positive integer n.

□

Remark The solution above provides a bijection between type A and type B sequences via an algorithm. There are alternative ways to provide such a bijection. For example, given the numbers $a_1 \geq \ldots \geq a_k$ we may define the b_i's as

$$b_j = \sum_i \left\lfloor \frac{a_i + 1}{2^j} \right\rfloor.$$

Conversely, given the numbers $b_1 \geq \ldots \geq b_m$, one may define the a_i's by taking, as in the solution, b_m numbers equal to $2^m - 1$, $b_{m-1} - 2b_m$ numbers equal to $2^{m-1} - 1, \ldots$, and $b_1 - 2b_2$ numbers equal to $2^1 - 1$. One now needs to verify that these maps are mutually inverse.

Problem 4. Call a rational number r *powerful* if r can be expressed in the form $\frac{p^k}{q}$ for some relatively prime positive integers p, q and some integer $k > 1$. Let a, b, c be positive

rational numbers such that $abc = 1$. Suppose there exist positive integers x, y, z such that $a^x + b^y + c^z$ is an integer. Prove that a, b, c are all *powerful*.

Solution. Let $a = \frac{a_1}{b_1}, b = \frac{a_2}{b_2}$, where $\gcd(a_1, b_1) = \gcd(a_2, b_2) = 1$. Then $c = \frac{b_1 b_2}{a_1 a_2}$. The condition that $a^x + b^y + c^z$ is an integer becomes

$$\frac{a_1^{x+z} a_2^z b_2^y + a_1^z a_2^{y+z} b_1^x + b_1^{x+z} b_2^{y+z}}{a_1^z a_2^z b_1^x b_2^y} \in \mathbb{Z},$$

which can be restated as

$$a_1^z a_2^z b_1^x b_2^y \mid a_1^{x+z} a_2^z b_2^y + a_1^z a_2^{y+z} b_1^x + b_1^{x+z} b_2^{y+z}. \tag{1}$$

In particular, a_1^z divides the right-hand side. Since it divides the first and second terms in the sum, we conclude that $a_1^z \mid b_1^{x+z} b_2^{y+z}$. Since $\gcd(a_1, b_1) = 1$, we have $a_1^z \mid b_2^{y+z}$.

Let p be a prime that divides a_1. Let $m, n \geq 1$ be integers such that $p^n \| a_1$ (i.e. $p^n | a_1$ but $p^{n+1} \nmid a_1$) and $p^m \| b_2$. The fact that $a_1^z \mid b_2^{y+z}$ implies $nz \leq m(y+z)$. Since $\gcd(a_1, b_1) = \gcd(a_2, b_2) = 1$, we have p does not divide b_1 and does not divide a_2. Thus

$$p^{nz} \| a_1^z a_2^{y+z} b_1^x \quad \text{and} \quad p^{m(y+z)} \| b_1^{x+z} b_2^{y+z}. \tag{2}$$

On the other hand, (1) implies that

$$p^{nz+my} \mid a_1^z a_2^{y+z} b_1^x + b_1^{x+z} b_2^{y+z}. \tag{3}$$

If $nz < m(y+z)$, then (2) gives $p^{nz} \| a_1^z a_2^{y+z} b_1^x + b_1^{x+z} b_2^{y+z}$, which contradicts (3). Thus $nz = m(y+z)$ so n is divisible by $k := \frac{y+z}{\gcd(z, y+z)} > 1$. Thus each exponent in the prime decomposition of a_1 must be divisible by k. Hence a_1 is a perfect k-power which means a is powerful. Similarly, b and c are also powerful.

□

Problem 5. Let n be a positive integer. A pair of n-tuples $(a_1, \ldots, a_n)$ and $(b_1, \ldots, b_n)$ with integer entries is called an *exquisite pair* if

$$|a_1 b_1 + \cdots + a_n b_n| \leq 1.$$

Determine the maximum number of distinct n-tuples with integer entries such that any two of them form an exquisite pair.

Answer: The maximum is $n^2 + n + 1$.

Solution. First, we construct an example with $n^2 + n + 1$ n-tuples, each two of them forming an exquisite pair. In the following list, $*$ represents any number of zeros as long as the total number of entries is n.

- $(*)$
- $(*, 1, *)$
- $(*, -1, *)$
- $(*, 1, *, 1, *)$
- $(*, 1, *, -1, *)$

For example, for $n = 2$ we have the tuples $(0, 0)$, $(0, 1)$, $(1, 0)$, $(0, -1)$, $(-1, 0)$, $(1, 1)$, $(1, -1)$.

The total number of such tuples is $1 + n + n + \binom{n}{2} + \binom{n}{2} = n^2 + n + 1$. For any two of them, at most two of the products $a_i b_i$ are non-zero. The only case in which two of them are non-zero is when we take a sequence $(*, 1, *, 1, *)$ and a sequence $(*, 1, *, -1, *)$ with zero entries in the same places. But in this case one $a_i b_i$ is 1 and the other -1. This shows that any two of these sequences form an exquisite pair.

Next, we claim that among any $n^2 + n + 2$ tuples, some two of them do not form an exquisite pair. We begin with lemma.

Lemma. Given $2n + 1$ distinct non-zero n-tuples of real numbers, some two of them $(a_1, \ldots, a_n)$ and $(b_1, \ldots, b_n)$ satisfy $a_1 b_1 + \cdots + a_n b_n > 0$.

Proof of Lemma. We proceed by induction. The statement is easy for $n = 1$ since for every three non-zero numbers there are two of them with the same sign. Assume that the statement is true for $n - 1$ and consider $2n + 1$ tuples with n entries. Since we are working with tuples of real numbers, we claim that we may assume that one of the tuples is $a = (0, 0, \ldots, 0, -1)$. Let us postpone the proof of this claim for the moment.

If one of the remaining tuples b has a negative last entry, then a and b satisfy the desired condition. So we may assume all the remaining tuples has a non-negative last entry. Now, from each tuple remove the last number. If two n-tuples b and c yield the same $(n - 1)$-tuple, then

$$b_1 c_1 + \cdots + b_{n-1} c_{n-1} + b_n c_n = b_1^2 + \cdots + b_{n-1}^2 + b_n c_n > 0,$$

and we are done. The remaining case is that all the n-tuples yield distinct $(n-1)$-tuples. Then at most one of them is the zero $(n-1)$-tuple, and thus we can use the inductive hypothesis on $2n - 1$ of them. So we find b and c for which

$$(b_1 c_1 + \cdots + b_{n-1} c_{n-1}) + b_n c_n > 0 + b_n c_n > 0.$$

The only thing that we are left to prove is that in the inductive step we may assume that one of the tuples is $a = (0, 0, \ldots, 0, -1)$. Fix one of the tuples $x = (x_1, \ldots, x_n)$. Set a real number φ for which $\tan \varphi = \frac{x_1}{x_2}$. Change each tuple $a = (a_1, a_2, \ldots, a_n)$ (including x), to the tuple

$$(a_1 \cos \varphi - a_2 \sin \varphi, a_1 \sin \varphi + a_2 \cos \varphi, a_3, a_4, \ldots, a_n).$$

A straightforward calculation shows that the first coordinate of the tuple x becomes 0, and that all the expressions of the form $a_1 b_1 + \cdots + a_n b_n$ are preserved. We may iterate this process until all the entries of x except for the last one are equal to 0. We finish by multiplying all the entries in all the tuples by a suitable constant that makes the last entry of x equal to -1. This preserves the sign of all the expressions of the form $a_1 b_1 + \cdots + a_n b_n$.

□

We proceed to the proof of our claim. Let A be a set of non-zero tuples among which any two form an exquisite pair. It suffices to prove that $|A| \leq n^2 + n$. We can write A as a disjoint union of subsets $A_1 \cup A_2 \cup \ldots \cup A_n$, where A_i is the set of tuples in A whose last non-zero entry appears in the ith position. We will show that $|A_i| \leq 2i$, which will finish our proof since $2 + 4 + \cdots + 2n = n^2 + n$.

Proceeding by contradiction, suppose that $|A_i| \geq 2i + 1$. If A_i has three or more tuples whose only non-zero entry is in the ith position, then for two of them this entry has the same sign. Since the tuples are different and their entries are integers, this yields two tuples for which $|\sum a_i b_i| \geq 2$, a contradiction. So there are at most two such tuples. We remove them from A_i.

Now, for each of the remaining tuples a, if it has a positive ith coordinate, we keep a as it is. If it has a negative ith coordinate, we replace it with the *opposite tuple* $-a$ with entries with opposite signs. This does not changes the exquisite pairs condition.

After making the necessary changes, we have two cases. The first case is that there are two tuples a and b that have the same first $i-1$ coordinates and thus

$$a_1b_1 + \cdots + a_{i-1}b_{i-1} = a_1^2 + \cdots + a_{i-1}^2 > 0,$$

and thus is at least 1 (the entries are integers). The second case is that no two tuples have the same first $i-1$ coordinates, but then by the Lemma we find two tuples a and b for which

$$a_1b_1 + \cdots + a_{i-1}b_{i-1} \geq 1.$$

In any case, we obtain

$$a_1b_1 + \cdots + a_{i-1}b_{i-1} + a_ib_i \geq 2.$$

This yields a final contradiction to the exquisite pair hypothesis.

□

Solutions of APMO 2016

Problem 1. We say that a triangle ABC is *great* if the following holds: for any point D on the side BC, if P and Q are the feet of the perpendiculars from D to the lines AB and AC, respectively, then the reflection of D in the line PQ lies on the circumcircle of the triangle ABC.

Prove that triangle ABC is great if and only if $\angle A = 90^\circ$ and $AB = AC$.

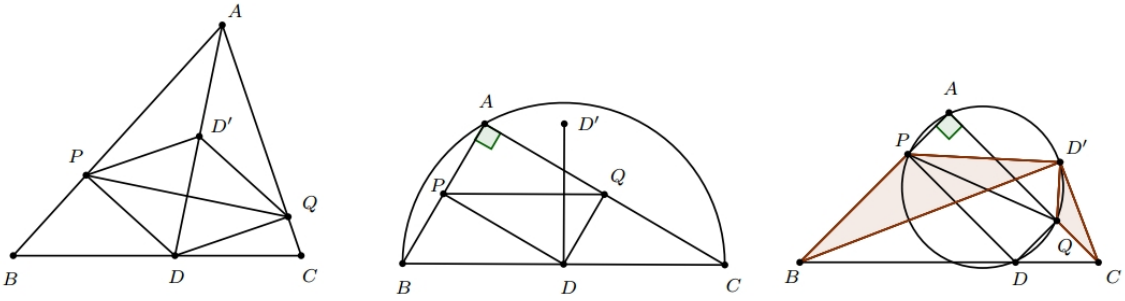
Solution. For every point D on the side BC, let D' be the reflection of D in the line PQ. We will first prove that if the triangle satisfies the condition then it is isosceles and right-angled at A.

Choose D to be the point where the angle bisector from A meets BC. Note that P and Q lie on the rays AB and AC respectively. Furthermore, P and Q are reflections of each other in the line AD, from which it follows that $PQ \perp AD$. Therefore, D' lies on the line AD and we may deduce that either $D' = A$ or D' is the second point of the angle bisector at A and the circumcircle of ABC. However, since $APDQ$ is a cyclic quadrilateral, the segment PQ intersects the segment AD. Therefore, D' lies on the ray DA and therefore $D' = A$. By angle chasing we obtain

$$\angle PD'Q = \angle PDQ = 180^\circ - \angle BAC,$$

and since $D' = A$ we also know $\angle PD'Q = \angle BAC$. This implies that $\angle BAC = 90^\circ$.

Now we choose D to be the midpoint of BC. Since $\angle BAC = 90^\circ$, we can deduce that DQP is the medial triangle of triangle ABC. Therefore, $PQ||BC$ from which it follows that $DD' \perp BC$. But the distance from D' to BC is equal to both the circumradius of triangle ABC and to the distance from A to BC. This can only happen if $A = D'$. This implies that ABC is isosceles and right-angled at A.



We will now prove that if ABC is isosceles and right-angled at A then the required property in the problem holds. Let D be any point on side BC. Then $D'P = DP$ and we also have $DP = BP$. Hence, $D'P = BP$ and similarly $D'Q = CQ$. Note that $APDQD'$ is cyclic with diameter PQ. Therefore, $\angle APD' = \angle AQD'$, from which we obtain $\angle BPD' = \angle CQD'$. So triangles $D'PB$ and $D'QC$ are similar. It follows that $\angle PD'Q = \angle PD'C + \angle CD'Q = \angle PD'C + \angle BD'P = \angle BD'C$ and $\frac{D'P}{D'Q} = \frac{D'B}{D'C}$. So we also obtain that triangles $D'PQ$ and $D'BC$ are similar. But since DPQ and $D'PQ$ are congruent, we may deduce that $\angle BD'C = \angle PD'Q = \angle PDQ = 90^\circ$. Therefore, D' lies on the circle with diameter BC, which is the circumcircle of triangle ABC.

□

Problem 2. A positive integer is called *fancy* if it can be expressed in the form

$$2^{a_1} + 2^{a_2} + \cdots + 2^{a_{100}},$$

where a_1, a_2, ..., a_{100} are non-negative integers that are not necessarily distinct.

Find the smallest positive integer n such that no multiple of n is a fancy number.

Answer: The answer is $n = 2^{101} - 1$.

Solution. Let k be any positive integer less than $2^{101} - 1$. Then k can be expressed in binary notation using at most 100 ones, and therefore there exists a positive integer r and non-negative integers a_1, a_2, ..., a_r such that $r \leq 100$ and $k = 2^{a_1} + \cdots + 2^{a_r}$. Notice that for a positive integer s we have:

$$\begin{aligned} 2^s k &= 2^{a_1+s} + 2^{a_2+s} + \cdots + 2^{a_{r-1}+s} + (1 + 1 + 2 + \cdots + 2^{s-1})2^{a_r} \\ &= 2^{a_1+s} + 2^{a_2+s} + \cdots + 2^{a_{r-1}+s} + 2^{a_r} + 2^{a_r} + \cdots + 2^{a_r+s-1}. \end{aligned}$$

This shows that k has a multiple that is a sum of $r + s$ powers of two. In particular, we may take $s = 100 - r \geq 0$, which shows that k has a multiple that is a fancy number.

We will now prove that no multiple of $n = 2^{101} - 1$ is a fancy number. In fact we will prove a stronger statement, namely, that no multiple of n can be expressed as the sum of at most 100 powers of 2.

For the sake of contradiction, suppose that there exists a positive integer c such that cn is the sum of at most 100 powers of 2. We may assume that c is the smallest such integer. By repeatedly merging equal powers of two in the representation of cn we may assume that

$$cn = 2^{a_1} + 2^{a_2} + \cdots + 2^{a_r}$$

where $r \leq 100$ and $a_1 < a_2 < \ldots < a_r$ are distinct non-negative integers. Consider the following two cases:

- If $a_r \geq 101$, then $2^{a_r} - 2^{a_r-101} = 2^{a_r-101}n$. It follows that $2^{a_1} + 2^{a_2} + \cdots + 2^{a_{r-1}} + 2^{a_r-101}$ would be a multiple of n that is smaller than cn. This contradicts the minimality of c.
- If $a_r \leq 100$, then $\{a_1, \ldots, a_r\}$ is a proper subset of $\{0, 1, \ldots, 100\}$. Then

  $$n \leq cn < 2^0 + 2^1 + \cdots + 2^{100} = n.$$

  This is also a contradiction.

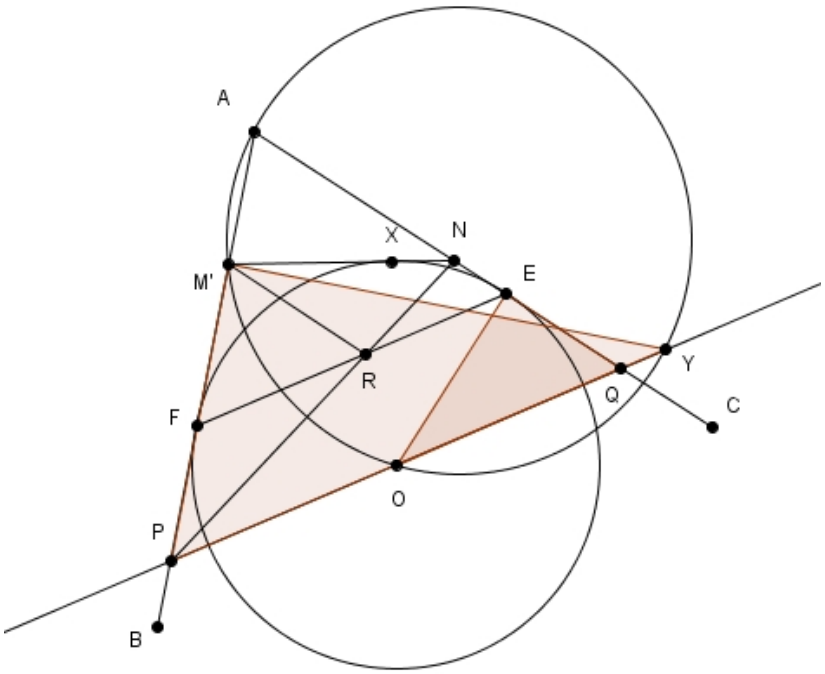
From these contradictions we conclude that it is impossible for cn to be the sum of at most 100 powers of 2. In particular, no multiple of n is a fancy number.

□

Problem 3. Let AB and AC be two distinct rays not lying on the same line, and let ω be a circle with center O that is tangent to ray AC at E and ray AB at F. Let R be a point on segment EF. The line through O parallel to EF intersects line AB at P. Let N be the intersection of lines PR and AC, and let M be the intersection of line AB and the line through R parallel to AC. Prove that line MN is tangent to ω.

Solution. We present two approaches. The first one introduces an auxiliary point and studies similarities in the figure. The second one reduces the problem to computations involving a particular exradius of a triangle. The second approach has two variants.

Solution 1.



Let the line through N tangent to ω at point $X \neq E$ intersect AB at point M'. It suffices to show that $M'R \parallel AC$, since this would yield $M' = M$.

Suppose that the line PO intersects AC at Q and the circumcircle of $AM'O$ at Y, respectively. Then

$$\angle AYM' = \angle AOM' = 90^\circ - \angle M'OP.$$

By angle chasing we have $\angle EOQ = \angle FOP = 90^\circ - \angle AOF = \angle M'AO = \angle M'YP$ and by symmetry $\angle EQO = \angle M'PY$. Therefore $\triangle M'YP \sim \triangle EOQ$.

On the other hand, we have

$$\angle M'OP = \angle M'OF + \angle FOP = \frac{1}{2}\left(\angle FOX + \angle FOP + \angle EOQ\right) =$$
$$= \frac{1}{2}\left(\frac{180^\circ - \angle XOE}{2}\right) = 90^\circ - \frac{\angle XOE}{2}.$$

Since we know that $\angle AYM'$ and $\angle M'OP$ are complementary this implies

$$\angle AYM' = \frac{\angle XOE}{2} = \angle NOE.$$

Therefore, $\angle AYM'$ and $\angle NOE$ are congruent angles, and this means that A and N are corresponding points in the similarity of triangles $\triangle M'YP$ and $\triangle EOQ$. It follows that

$$\frac{AM'}{M'P} = \frac{NE}{EQ} = \frac{NR}{RP}.$$

We conclude that $M'R \parallel AC$, as desired.

□

Solution 2a.

As in Solution 1, we introduce point M' and reduce the problem to proving $\frac{PR}{RN} = \frac{PM'}{M'A}$.

Menelaus theorem in triangle ANP with transversal line FRE yields

$$\frac{PR}{RN} \cdot \frac{NE}{EA} \cdot \frac{AF}{FP} = 1.$$

Since $AF = EA$, we have $\frac{FP}{NE} = \frac{PR}{RN}$, so that it suffices to prove

$$\frac{FP}{NE} = \frac{PM'}{M'A}. \tag{1}$$

This is a computation regarding the triangle $AM'N$ and its excircle opposite A. Indeed, setting $a = M'N$, $b = NA$, $c = M'A$, $s = \frac{a+b+c}{2}$, $x = s - a$, $y = s - b$ and $z = s - c$, then $AE = AF = s$, $M'F = z$ and $NE = y$. From $\triangle OFP \sim \triangle AFO$ we have $FP = \frac{r_a^2}{s}$, where $r_a = OF$ is the exradius opposite A. Combining the following two standard formulas for the area of a triangle

$$|AM'N|^2 = xyzs \quad \text{(Heron's formula)} \quad \text{and} \quad |AM'N| = r_a(s-a),$$

we have $r_a^2 = \frac{yzs}{x}$. Therefore, $FP = \frac{yz}{x}$. We can now write everything in (1) in terms of x, y, z. We conclude that we have to verify

$$\frac{\frac{yz}{x}}{y} = \frac{z + \frac{yz}{x}}{x + y},$$

which is easily seen to be true.

Note: Antoher approach using Menalaus theorem is to construct the tangent from M to create a point N' in AC and then prove, using the theorem, that P, R and N' are collinear. This also reduces to an algebraic identity.

Solution 2b.

As in Solution 1, we introduce point M'. Let the line through M' and parallel to AN intersect EF at R'. Let P' be the intersection of lines NR' and AM. It suffices to show that $P'O \parallel FE$, since this would yield $P = P'$, and then $R = R'$ and $M = M'$. Hence it is enough to prove that

$$\frac{AF}{FP'} = \frac{AD}{DO}, \tag{2}$$

where D is the intersection of AO and EF. Once again, this reduces to a computation regarding the triangle $AM'N$ and its excircle opposite A.

Let $u = P'F$ and x, y, z, s as in Solution 2a. Note that since $AE = AF$ and $M'R' \parallel AE$, we have $M'R' = M'F = z$. Since $M'R' \parallel AN$, we have $\frac{P'M'}{P'A} = \frac{M'R'}{NA}$, that is,

$$\frac{u+z}{u+x+y+z} = \frac{z}{x+z}.$$

From this last equation we obtain $u = \frac{yz}{x}$. Hence $\frac{AF}{FP'} = \frac{xs}{yz}$. Also, as in Solution 2a, we have $r_a^2 = \frac{yzs}{x}$.

Finally, using similar triangles ODF, FDA and OFA, and the above equalities, we have

$$\frac{AD}{DO} = \frac{AD}{DF} \cdot \frac{DF}{DO} = \frac{AF}{OF} \cdot \frac{AF}{OF} = \frac{s^2}{r_a^2} = \frac{s^2}{\frac{yzs}{x}} = \frac{xs}{yz} = \frac{AF}{FP'},$$

as required.

Problem 4. The country Dreamland consists of 2016 cities. The airline Starways wants to establish some one-way flights between pairs of cities in such a way that each city has exactly one flight out of it. Find the smallest positive integer k such that no matter how Starways establishes its flights, the cities can always be partitioned into k groups so that from any city it is not possible to reach another city in the same group by using at most 28 flights.

Answer: 57

Solution. The flights established by Starways yield a directed graph G on 2016 vertices in which each vertex has out-degree equal to 1.

We first show that we need at least 57 groups. For this, suppose that G has a directed cycle of length 57. Then, for any two cities in the cycle, one is reachable from the other using at most 28 flights. So no two cities in the cycle can belong to the same group. Hence, we need at least 57 groups.

We will now show that 57 groups are enough. Consider another auxiliary directed graph H in which the vertices are the cities of Dreamland and there is an arrow from city u to city v if u can be reached from v using at most 28 flights. Each city has out-degree at most 28. We will be done if we can split the cities of H in at most 57 groups such that there are no arrows between vertices of the same group. We prove the following stronger statement.

Lemma: Suppose we have a directed graph on $n \geq 1$ vertices such that each vertex has out-degree at most 28. Then the vertices can be partitioned into 57 groups in such a way that no vertices in the same group are connected by an arrow.

Proof: We apply induction. The result is clear for 1 vertex. Now suppose we have more than one vertex. Since the out-degree of each vertex is at most 28, there is a vertex, say v, with in-degree at most 28. If we remove the vertex v we obtain a graph with fewer vertices which still satifies the conditions, so by inductive hypothesis we may split it into at most 57 groups with no adjacent vertices in the same group. Since v has in-degree and out-degree at most 28, it has at most 56 neighboors in the original directed graph. Therefore, we may add v back and place it in a group in which it has no neighbors. This completes the inductive step.

□

Problem 5. Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that

$$(z+1)f(x+y) = f(xf(z)+y) + f(yf(z)+x), \tag{3}$$

for all positive real numbers x, y, z.

Answer: The only solution is $f(x) = x$ for all positive real numbers x.

Solution. The identity function $f(x) = x$ clearly satisfies the functional equation. Now, let f be a function satisfying the functional equation. Plugging $x = y = 1$ into (3) we get $2f(f(z)+1) = (z+1)(f(2))$ for all $z \in \mathbb{R}^+$. Hence, f is not bounded above.

Lemma. Let a, b, c be positive real numbers. If c is greater than 1, a/b and b/a, then the system of linear equations

$$cu + v = a \quad u + cv = b$$

has a positive real solution u, v.

Proof. The solution is

$$u = \frac{ca - b}{c^2 - 1} \quad v = \frac{cb - a}{c^2 - 1}.$$

The numbers u and v are positive if the conditions on c above are satisfied.

□

We will now prove that

$$f(a)+f(b)=f(c)+f(d) \quad \text{for all } a,b,c,d\in\mathbb{R}^+ \text{ with } a+b=c+d. \tag{4}$$

Consider $a,b,c,d\in\mathbb{R}^+$ such that $a+b=c+d$. Since f is not bounded above, we can choose a positive number e such that $f(e)$ is greater than 1, a/b, b/a, c/d and d/c. Using the above lemma, we can find $u,v,w,t\in\mathbb{R}^+$ satisfying

$$\begin{aligned} f(e)u+v=a, &\quad u+f(e)v=b\\ f(e)w+t=c, &\quad w+f(e)t=d. \end{aligned}$$

Note that $u+v=w+t$ since $(u+v)(f(e)+1)=a+b$ and $(w+t)(f(e)+1)=c+d$. Plugging $x=u$, $y=v$ and $z=e$ into (3) yields $f(a)+f(b)=(e+1)f(u+v)$. Similarly, we have $f(c)+f(d)=(e+1)f(w+t)$. The claim follows immediately.

We then have

$$yf(x)=f(xf(y)) \quad \text{for all } x,y\in\mathbb{R}^+ \tag{5}$$

since by (3) and (4),

$$(y+1)f(x)=f\left(\frac{x}{2}f(y)+\frac{x}{2}\right)+f\left(\frac{x}{2}f(y)+\frac{x}{2}\right)=f(xf(y))+f(x).$$

Now, let $a=f(1/f(1))$. Plugging $x=1$ and $y=1/f(1)$ into (5) yields $f(a)=1$. Hence $a=af(a)$ and $f(af(a))=f(a)=1$. Since $af(a)=f(af(a))$ by (5), we have $f(1)=a=1$. It follows from (5) that

$$f(f(y))=y \quad \text{for all } y\in\mathbb{R}^+. \tag{6}$$

Using (4) we have for all $x,y\in\mathbb{R}^+$ that

$$\begin{aligned} f(x+y)+f(1)=f(x)+f(y+1), &\quad \text{and}\\ f(y+1)+f(1)=f(y)+f(2). \end{aligned}$$

Therefore

$$f(x+y)=f(x)+f(y)+b \quad \text{for all } x,y\in\mathbb{R}^+, \tag{7}$$

where $b=f(2)-2f(1)=f(2)-2$. Using (5), (7) and (6), we get

$$4+2b=2f(2)=f(2f(2))=f(f(2)+f(2))=f(f(2))+f(f(2))+b=4+b.$$

This shows that $b=0$ and thus

$$f(x+y)=f(x)+f(y) \quad \text{for all } x,y\in\mathbb{R}^+.$$

In particular, f is strictly increasing.

We conclude as follows. Take any positive real number x. If $f(x)>x$, then $f(f(x))>f(x)>x=f(f(x))$, a contradiction. Similarly, it is not possible that $f(x)<x$. This shows that $f(x)=x$ for all positive real numbers x.

□

Solutions of APMO 2015

Problem 1. Let ABC be a triangle, and let D be a point on side BC. A line through D intersects side AB at X and ray AC at Y. The circumcircle of triangle BXD intersects the circumcircle ω of triangle ABC again at point $Z \neq B$. The lines ZD and ZY intersect ω again at V and W, respectively. Prove that $AB = VW$.

Solution. Suppose XY intersects ω at points P and Q, where Q lies between X and Y. We will show that V and W are the reflections of A and B with respect to the perpendicular bisector of PQ. From this, it follows that $AVWB$ is an isosceles trapezoid and hence $AB = VW$.

First, note that

$$\angle BZD = \angle AXY = \angle APQ + \angle BAP = \angle APQ + \angle BZP,$$

so $\angle APQ = \angle PZV = \angle PQV$, and hence V is the reflection of A with respect to the perpendicular bisector of PQ.

Now, suppose W' is the reflection of B with respect to the perpendicular bisector of PQ, and let Z' be the intersection of YW' and ω. It suffices to show that B, X, D, Z' are concyclic. Note that

$$\angle YDC = \angle PDB = \angle PCB + \angle QPC = \angle W'PQ + \angle QPC = \angle W'PC = \angle YZ'C.$$

So D, C, Y, Z' are concyclic. Next, $\angle BZ'D = \angle CZ'B - \angle CZ'D = 180^\circ - \angle BXD$ and due to the previous concyclicity we are done.

Alternative solution 1. Using cyclic quadrilaterals $BXDZ$ and $ABZV$ in turn, we have $\angle ZDY = \angle ZBA = \angle ZCY$. So $ZDCY$ is cyclic.

Using cyclic quadrilaterals $ABZC$ and $ZDCY$ in turn, we have $\angle AZB = \angle ACB = \angle WZV$ (or $180^\circ - \angle WZV$ if Z lies between W and C).

So $AB = VW$ because they subtend equal (or supplementary) angles in ω. □

Alternative solution 2. Using cyclic quadrilaterals $BXDZ$ and $ABZV$ in turn, we have $\angle ZDY = \angle ZBA = \angle ZCY$. So $ZDCY$ is cyclic.

Using cyclic quadrilaterals $BXDZ$ and $ABZV$ in turn, we have $\angle DXA = \angle VZB = 180^\circ - BAV$. So $XD \parallel AV$.

Using cyclic quadrilaterals $ZDCY$ and $BCWZ$ in turn, we have $\angle YDC = \angle YZC = \angle WBC$. So $XD \parallel BW$.

Hence $BW \parallel AV$ which implies that $AVWB$ is an isosceles trapezium with $AB = VW$. □

Problem 2. Let $S = \{2, 3, 4, \ldots\}$ denote the set of integers that are greater than or equal to 2. Does there exist a function $f : S \to S$ such that

$$f(a)f(b) = f(a^2b^2) \text{ for all } a, b \in S \text{ with } a \neq b?$$

Solution. We prove that there is no such function. For arbitrary elements a and b of S, choose an integer c that is greater than both of them. Since $bc > a$ and $c > b$, we have

$$f(a^4b^4c^4) = f(a^2)f(b^2c^2) = f(a^2)f(b)f(c).$$

Furthermore, since $ac > b$ and $c > a$, we have

$$f(a^4b^4c^4) = f(b^2)f(a^2c^2) = f(b^2)f(a)f(c).$$

Comparing these two equations, we find that for all elements a and b of S,

$$f(a^2)f(b) = f(b^2)f(a) \quad \Longrightarrow \quad \frac{f(a^2)}{f(a)} = \frac{f(b^2)}{f(b)}\,.$$

It follows that there exists a positive rational number k such that

$$f(a^2) = kf(a), \quad \text{for all } a \in S. \tag{1}$$

Substituting this into the functional equation yields

$$f(ab) = \frac{f(a)f(b)}{k}, \quad \text{for all } a, b \in S \text{ with } a \neq b. \tag{2}$$

Now combine the functional equation with equations (1) and (2) to obtain

$$f(a)f(a^2) = f(a^6) = \frac{f(a)f(a^5)}{k} = \frac{f(a)f(a)f(a^4)}{k^2} = \frac{f(a)f(a)f(a^2)}{k}, \quad \text{for all } a \in S.$$

It follows that $f(a) = k$ for all $a \in S$. Substituting $a = 2$ and $b = 3$ into the functional equation yields $k = 1$, however $1 \notin S$ and hence we have no solutions.

Problem 3. A sequence of real numbers $a_0, a_1, \ldots$ is said to be *good* if the following three conditions hold.

(i) The value of a_0 is a positive integer.

(ii) For each non-negative integer i we have $a_{i+1} = 2a_i + 1$ or $a_{i+1} = \dfrac{a_i}{a_i + 2}$.

(iii) There exists a positive integer k such that $a_k = 2014$.

Find the smallest positive integer n such that there exists a good sequence $a_0, a_1, \ldots$ of real numbers with the property that $a_n = 2014$.

Answer: 60.

Solution. Note that

$$a_{i+1} + 1 = 2(a_i + 1) \text{ or } a_{i+1} + 1 = \frac{a_i + a_i + 2}{a_i + 2} = \frac{2(a_i + 1)}{a_i + 2}.$$

Hence

$$\frac{1}{a_{i+1} + 1} = \frac{1}{2} \cdot \frac{1}{a_i + 1} \text{ or } \frac{1}{a_{i+1} + 1} = \frac{a_i + 2}{2(a_i + 1)} = \frac{1}{2} \cdot \frac{1}{a_i + 1} + \frac{1}{2}.$$

Therefore,

$$\frac{1}{a_k + 1} = \frac{1}{2^k} \cdot \frac{1}{a_0 + 1} + \sum_{i=1}^{k} \frac{\varepsilon_i}{2^{k-i+1}}, \tag{1}$$

where $\varepsilon_i = 0$ or 1. Multiplying both sides by $2^k(a_k + 1)$ and putting $a_k = 2014$, we get

$$2^k = \frac{2015}{a_0 + 1} + 2015 \cdot \left(\sum_{i=1}^{k} \varepsilon_i \cdot 2^{i-1} \right),$$

where $\varepsilon_i = 0$ or 1. Since $\gcd(2, 2015) = 1$, we have $a_0 + 1 = 2015$ and $a_0 = 2014$. Therefore,

$$2^k - 1 = 2015 \cdot \left(\sum_{i=1}^{k} \varepsilon_i \cdot 2^{i-1} \right),$$

where $\varepsilon_i = 0$ or 1. We now need to find the smallest k such that $2015 | 2^k - 1$. Since $2015 = 5 \cdot 13 \cdot 31$, from the Fermat little theorem we obtain $5 | 2^4 - 1$, $13 | 2^{12} - 1$ and $31 | 2^{30} - 1$. We also have $\operatorname{lcm}[4, 12, 30] = 60$, hence $5 | 2^{60} - 1$, $13 | 2^{60} - 1$ and $31 | 2^{60} - 1$, which gives $2015 | 2^{60} - 1$.

But $5 \nmid 2^{30} - 1$ and so $k = 60$ is the smallest positive integer such that $2015 | 2^k - 1$. To conclude, the smallest positive integer k such that $a_k = 2014$ is when $k = 60$.

Alternative solution 1. Clearly all members of the sequence are positive rational numbers. For each positive integer i, we have $a_i = \dfrac{a_{i+1} - 1}{2}$ or $a_i = \dfrac{2a_{i+1}}{1 - a_{i+1}}$. Since $a_i > 0$ we deduce that

$$a_i = \begin{cases} \dfrac{a_{i+1} - 1}{2} & \text{if } a_{i+1} > 1 \\ \dfrac{2a_{i+1}}{1 - a_{i+1}} & \text{if } a_{i+1} < 1. \end{cases}$$

Thus a_i is uniquely determined from a_{i+1}. Hence starting from $a_k = 2014$, we simply run the sequence backwards until we reach a positive integer. We compute as follows.

$$\frac{2014}{1}, \frac{2013}{2}, \frac{2011}{4}, \frac{2007}{8}, \frac{1999}{16}, \frac{1983}{32}, \frac{1951}{64}, \frac{1887}{128}, \frac{1759}{256}, \frac{1503}{512}, \frac{991}{1024}, \frac{1982}{33}, \frac{1949}{66}, \frac{1883}{132}, \frac{1751}{264}, \frac{1487}{528}, \frac{959}{1056}, \frac{1918}{97}, \frac{1821}{194}, \frac{1627}{388},$$

$$\frac{1239}{776}, \frac{463}{1552}, \frac{926}{1089}, \frac{1852}{163}, \frac{1689}{326}, \frac{1363}{652}, \frac{711}{1304}, \frac{1422}{593}, \frac{829}{1186}, \frac{1658}{357}, \frac{1301}{714}, \frac{587}{1428}, \frac{1174}{841}, \frac{333}{1682}, \frac{666}{1349}, \frac{1332}{683}, \frac{649}{1366}, \frac{1298}{717}, \frac{581}{1434}, \frac{1162}{853},$$

$$\frac{309}{1706}, \frac{618}{1397}, \frac{1236}{779}, \frac{457}{1558}, \frac{914}{1101}, \frac{1828}{187}, \frac{1641}{374}, \frac{1267}{748}, \frac{519}{1496}, \frac{1038}{977}, \frac{61}{1954}, \frac{122}{1893}, \frac{244}{1771}, \frac{488}{1527}, \frac{976}{1039}, \frac{1952}{63}, \frac{1889}{126}, \frac{1763}{252}, \frac{1511}{504}, \frac{1007}{1008}, \frac{2014}{1}.$$

There are 61 terms in the above list. Thus $k = 60$. □

Alternative solution 1 is quite computationally intensive. Calculating the first few terms indicates some patterns that are easy to prove. This is shown in the next solution.

Alternative solution 2. Start with $a_k = \dfrac{m_0}{n_0}$ where $m_0 = 2014$ and $n_0 = 1$ as in alternative solution 1. By inverting the sequence as in alternative solution 1, we have $a_{k-i} = \dfrac{m_i}{n_i}$ for $i \geq 0$ where

$$(m_{i+1}, n_{i+1}) = \begin{cases} (m_i - n_i, 2n_i) & \text{if } m_i > n_i \\ (2m_i, n_i - m_i) & \text{if } m_i < n_i. \end{cases}$$

Easy inductions show that $m_i + n_i = 2015$, $1 \leq m_i, n_i \leq 2014$ and $\gcd(m_i, n_i) = 1$ for $i \geq 0$. Since $a_0 \in \mathbb{N}^+$ and $\gcd(m_k, n_k) = 1$, we require $n_k = 1$. An easy induction shows that $(m_i, n_i) \equiv (-2^i, 2^i) \pmod{2015}$ for $i = 0, 1, \ldots, k$.

Thus $2^k \equiv 1 \pmod{2015}$. As in the official solution, the smallest such k is $k = 60$. This yields $n_k \equiv 1 \pmod{2015}$. But since $1 \leq n_k, m_k \leq 2014$, it follows that a_0 is an integer. □

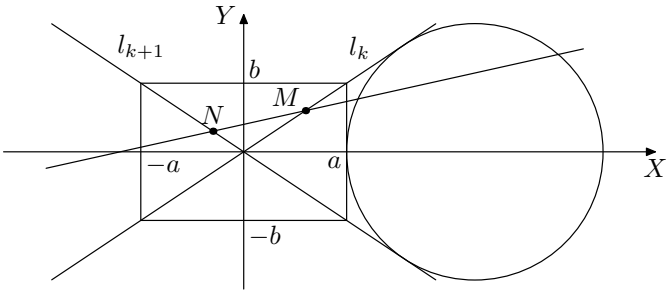
Problem 4. Let n be a positive integer. Consider $2n$ distinct lines on the plane, no two of which are parallel. Of the $2n$ lines, n are colored blue, the other n are colored red. Let $\mathcal{B}$ be the set of all points on the plane that lie on at least one blue line, and $\mathcal{R}$ the set of all points on the plane that lie on at least one red line. Prove that there exists a circle that intersects $\mathcal{B}$ in exactly $2n - 1$ points, and also intersects $\mathcal{R}$ in exactly $2n - 1$ points.

Solution. Consider a line ℓ on the plane and a point P on it such that ℓ is not parallel to any of the $2n$ lines. Rotate ℓ about P counterclockwise until it is parallel to one of the $2n$ lines. Take note of that line and keep rotating until all the $2n$ lines are met. The $2n$ lines are now ordered according to which line is met before or after. Say the lines are in order $\ell_1, \ldots, \ell_{2n}$. Clearly there must be $k \in \{1, \ldots, 2n - 1\}$ such that ℓ_k and ℓ_{k+1} are of different colors.

Now we set up a system of $X-$ and $Y-$ axes on the plane. Consider the two angular bisectors of ℓ_k and ℓ_{k+1}. If we rotate ℓ_{k+1} counterclockwise, the line will be parallel to one of the bisectors before the other. Let the bisector that is parallel to the rotation of ℓ_{k+1} first be the $X-$axis, and the other the $Y-$axis. From now on, we will be using the directed angle notation: for lines s and s', we define $\angle(s, s')$ to be a real number in $[0, \pi)$ denoting the angle in radians such that when s is rotated counterclockwise by $\angle(s, s')$ radian, it becomes parallel to s'. Using this

notation, we notice that there is no $i = 1, \ldots, 2n$ such that $\angle(X, l_i)$ is between $\angle(X, \ell_k)$ and $\angle(X, \ell_{k+1})$.

Because the $2n$ lines are distinct, the set S of all the intersections between ℓ_i and ℓ_j $(i \neq j)$ is a finite set of points. Consider a rectangle with two opposite vertices lying on ℓ_k and the other two lying on ℓ_{k+1}. With respect to the origin (the intersection of ℓ_k and ℓ_{k+1}), we can enlarge the rectangle as much as we want, while all the vertices remain on the lines. Thus, there is one of these rectangles R which contains all the points in S in its interior. Since each side of R is parallel to either X– or Y– axis, R is a part of the four lines $x = \pm a$, $y = \pm b$. where $a, b > 0$.



Consider the circle $\mathcal{C}$ tangent to the right of the $x = a$ side of the rectangle, and to both ℓ_k and ℓ_{k+1}. We claim that this circle intersects $\mathcal{B}$ in exactly $2n - 1$ points, and also intersects $\mathcal{R}$ in exactly $2n - 1$ points. Since $\mathcal{C}$ is tangent to both ℓ_k and ℓ_{k+1} and the two lines have different colors, it is enough to show that $\mathcal{C}$ intersects with each of the other $2n - 2$ lines in exactly 2 points. Note that no two lines intersect on the circle because all the intersections between lines are in S which is in the interior of R.

Consider any line L among these $2n-2$ lines. Let L intersect with ℓ_k and ℓ_{k+1} at the points M and N, respectively (M and N are not necessarily distinct). Notice that both M and N must be inside R. There are two cases:

(i) L intersects R on the $x = -a$ side once and another time on $x = a$ side;

(ii) L intersects $y = -b$ and $y = b$ sides.

However, if (ii) happens, $\angle(\ell_k, L)$ and $\angle(L, \ell_{k+1})$ would be both positive, and then $\angle(X, L)$ would be between $\angle(X, \ell_k)$ and $\angle(X, \ell_{k+1})$, a contradiction. Thus, only (i) can happen. Then L intersects $\mathcal{C}$ in exactly two points, and we are done.

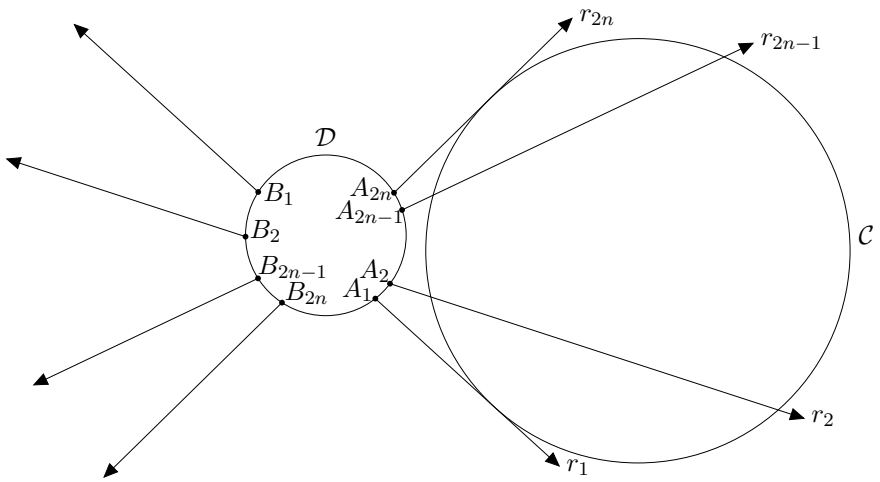
Alternative solution. By rotating the diagram we can ensure that no line is vertical. Let $\ell_1, \ell_2, \ldots, \ell_{2n}$ be the lines listed in order of increasing gradient. Then there is a k such that lines ℓ_k and ℓ_{k+1} are oppositely coloured. By rotating our coordinate system and cyclicly relabelling our lines we can ensure that $\ell_1, \ell_2, \ldots, \ell_{2n}$ are listed in order of increasing gradient, ℓ_1 and ℓ_{2n} are oppositely coloured, and no line is vertical.

Let $\mathcal{D}$ be a circle centred at the origin and of sufficiently large radius so that

- All intersection points of all pairs of lines lie strictly inside $\mathcal{D}$; and
- Each line ℓ_i intersects $\mathcal{D}$ in two points A_i and B_i say, such that A_i is on the right semicircle (the part of the circle in the positive x half plane) and B_i is on the left semicircle.

Note that the anticlockwise order of the points A_i, B_i around $\mathcal{D}$ is $A_1, A_2, \ldots, A_n, B_1, B_2, \ldots, B_n$.

(If A_{i+1} occurred before A_i then rays r_i and r_{i+1} (as defined below) would intersect outside $\mathcal{D}$.)



For each i, let r_i be the ray that is the part of the line ℓ_i starting from point A_i and that extends to the right. Let $\mathcal{C}$ be any circle tangent to r_1 and r_{2n}, that lies entirely to the right of $\mathcal{D}$. Then $\mathcal{C}$ intersects each of $r_2, r_3, \ldots, r_{2n-1}$ twice and is tangent to r_1 and r_{2n}. Thus $\mathcal{C}$ has the required properties. □

Problem 5. Determine all sequences $a_0, a_1, a_2, \ldots$ of positive integers with $a_0 \geq 2015$ such that for all integers $n \geq 1$:

(i) a_{n+2} is divisible by a_n;

(ii) $|s_{n+1} - (n+1)a_n| = 1$, where $s_{n+1} = a_{n+1} - a_n + a_{n-1} - \cdots + (-1)^{n+1}a_0$.

Answer: There are two families of answers:

(a) $a_n = c(n+2)!$ for all $n \geq 1$ and $a_0 = c+1$ for some integer $c \geq 2014$, and

(b) $a_n = c(n+2)!$ for all $n \geq 1$ and $a_0 = c-1$ for some integer $c \geq 2016$.

Solution. Let $\{a_n\}_{n=0}^{\infty}$ be a sequence of positive integers satisfying the given conditions. We can rewrite (ii) as $s_{n+1} = (n+1)a_n + h_n$, where $h_n \in \{-1, 1\}$. Substituting n with $n-1$ yields $s_n = na_{n-1} + h_{n-1}$, where $h_{n-1} \in \{-1, 1\}$. Note that $a_{n+1} = s_{n+1} + s_n$, therefore there exists $\delta_n \in \{-2, 0, 2\}$ such that

$$a_{n+1} = (n+1)a_n + na_{n-1} + \delta_n. \tag{1}$$

We also have $|s_2 - 2a_1| = 1$, which yields $a_0 = 3a_1 - a_2 \pm 1 \leq 3a_1$, and therefore $a_1 \geq \frac{a_0}{3} \geq 671$. Substituting $n = 2$ in (1), we find that $a_3 = 3a_2 + 2a_1 + \delta_2$. Since $a_1 | a_3$, we have $a_1 | 3a_2 + \delta_2$, and therefore $a_2 \geq 223$. Using (1), we obtain that $a_n \geq 223$ for all $n \geq 0$.

Lemma 1: For $n \geq 4$, we have $a_{n+2} = (n+1)(n+4)a_n$.

Proof. For $n \geq 3$ we have

$$a_n = na_{n-1} + (n-1)a_{n-2} + \delta_{n-1} > na_{n-1} + 3. \tag{2}$$

By applying (2) with n substituted by $n-1$ we have for $n \geq 4$,

$$a_n = na_{n-1} + (n-1)a_{n-2} + \delta_{n-1} < na_{n-1} + (a_{n-1} - 3) + \delta_{n-1} < (n+1)a_{n-1}. \tag{3}$$

Using (1) to write a_{n+2} in terms of a_n and a_{n-1} along with (2), we obtain that for $n \geq 3$,

$$\begin{aligned} a_{n+2} &= (n+3)(n+1)a_n + (n+2)na_{n-1} + (n+2)\delta_n + \delta_{n+1} \\ &< (n+3)(n+1)a_n + (n+2)na_{n-1} + 3(n+2) \\ &< (n^2+5n+5)a_n. \end{aligned}$$

Also for $n \geq 4$,

$$\begin{aligned} a_{n+2} &= (n+3)(n+1)a_n + (n+2)na_{n-1} + (n+2)\delta_n + \delta_{n+1} \\ &> (n+3)(n+1)a_n + na_n \\ &= (n^2+5n+3)a_n. \end{aligned}$$

Since $a_n | a_{n+2}$, we obtain that $a_{n+2} = (n^2+5n+4)a_n = (n+1)(n+4)a_n$, as desired. □

Lemma 2: For $n \geq 4$, we have $a_{n+1} = \dfrac{(n+1)(n+3)}{n+2}\, a_n$.

Proof. Using the recurrence $a_{n+3} = (n+3)a_{n+2} + (n+2)a_{n+1} + \delta_{n+2}$ and writing a_{n+3}, a_{n+2} in terms of a_{n+1}, a_n according to Lemma 1 we obtain

$$(n+2)(n+4)a_{n+1} = (n+3)(n+1)(n+4)a_n + \delta_{n+2}.$$

Hence $n+4 | \delta_{n+2}$, which yields $\delta_{n+2} = 0$ and $a_{n+1} = \dfrac{(n+1)(n+3)}{n+2}\, a_n$, as desired. □

Suppose there exists $n \geq 1$ such that $a_{n+1} \neq \dfrac{(n+1)(n+3)}{n+2}\, a_n$. By Lemma 2, there exist a greatest integer $1 \leq m \leq 3$ with this property. Then $a_{m+2} = \dfrac{(m+2)(m+4)}{m+3}\, a_{m+1}$. If $\delta_{m+1} = 0$, we have $a_{m+1} = \dfrac{(m+1)(m+3)}{m+2}\, a_m$, which contradicts our choice of m. Thus $\delta_{m+1} \neq 0$.

Clearly $m+3 | a_{m+1}$. Write $a_{m+1} = (m+3)k$ and $a_{m+2} = (m+2)(m+4)k$. Then $(m+1)a_m + \delta_{m+1} = a_{m+2} - (m+2)a_{m+1} = (m+2)k$. So, $a_m | (m+2)k - \delta_{m+1}$. But a_m also divides $a_{m+2} = (m+2)(m+4)k$. Combining the two divisibility conditions, we obtain $a_m | (m+4)\delta_{m+1}$. Since $\delta_{m+1} \neq 0$, we have $a_m | 2m+8 \leq 14$, which contradicts the previous result that $a_n \geq 223$ for all nonnegative integers n.

So, $a_{n+1} = \dfrac{(n+1)(n+3)}{n+2}\, a_n$ for $n \geq 1$. Substituting $n = 1$ yields $3 | a_1$. Letting $a_1 = 3c$, we have by induction that $a_n = n!(n+2)c$ for $n \geq 1$. Since $|s_2 - 2a_1| = 1$, we then get $a_0 = c \pm 1$, yielding the two families of solutions. By noting that $(n+2)n! = n! + (n+1)!$, we have $s_{n+1} = c(n+2)! + (-1)^n(c - a_0)$. Hence both families of solutions satisfy the given conditions.

Solutions of APMO 2014

Problem 1. For a positive integer m denote by $S(m)$ and $P(m)$ the sum and product, respectively, of the digits of m. Show that for each positive integer n, there exist positive integers $a_1, a_2, \ldots, a_n$ satisfying the following conditions:

$$S(a_1) < S(a_2) < \cdots < S(a_n) \text{ and } S(a_i) = P(a_{i+1}) \quad (i = 1, 2, \ldots, n).$$

(We let $a_{n+1} = a_1$.) (*Problem Committee of the Japan Mathematical Olympiad Foundation*)

Solution. Let k be a sufficiently large positive integer. Choose for each $i = 2, 3, \ldots, n$, a_i to be a positive integer among whose digits the number 2 appears exactly $k + i - 2$ times and the number 1 appears exactly $2^{k+i-1} - 2(k + i - 2)$ times, and nothing else. Then, we have $S(a_i) = 2^{k+i-1}$ and $P(a_i) = 2^{k+i-2}$ for each i, $2 \leq i \leq n$. Then, we let a_1 be a positive integer among whose digits the number 2 appears exactly $k + n - 1$ times and the number 1 appears exactly $2^k - 2(k + n - 1)$ times, and nothing else. Then, we see that a_1 satisfies $S(a_1) = 2^k$ and $P(a_1) = 2^{k+n-1}$. Such a choice of a_1 is possible if we take k to be large enough to satisfy $2^k > 2(k + n - 1)$ and we see that the numbers $a_1, \ldots, a_n$ chosen this way satisfy the given requirements.

Problem 2. Let $S = \{1, 2, \ldots, 2014\}$. For each non-empty subset $T \subseteq S$, one of its members is chosen as its *representative*. Find the number of ways to assign representatives to all non-empty subsets of S so that if a subset $D \subseteq S$ is a disjoint union of non-empty subsets $A, B, C \subseteq S$, then the representative of D is also the representative of at least one of A, B, C. (*Warut Suksompong, Thailand*)

Solution. *Answer:* $108 \cdot 2014!$.

For any subset X let $r(X)$ denotes the representative of X. Suppose that $x_1 = r(S)$. First, we prove the following fact:

$$\text{If } x_1 \in X \text{ and } X \subseteq S, \text{ then } x_1 = r(X).$$

If $|X| \leq 2012$, then we can write S as a disjoint union of X and two other subsets of S, which gives that $x_1 = r(X)$. If $|X| = 2013$, then let $y \in X$ and $y \neq x_1$. We can write X as a disjoint union of $\{x_1, y\}$ and two other subsets. We already proved that $r(\{x_1, y\}) = x_1$ (since $|\{x_1, y\}| = 2 < 2012$) and it follows that $y \neq r(X)$ for every $y \in X$ except x_1. We have proved the fact.

Note that this fact is true and can be proved similarly, if the ground set S would contain at least 5 elements.

There are 2014 ways to choose $x_1 = r(S)$ and for $x_1 \in X \subseteq S$ we have $r(X) = x_1$. Let $S_1 = S \setminus \{x_1\}$. Analogously, we can state that there are 2013 ways to choose $x_2 = r(S_1)$ and for $x_2 \in X \subseteq S_1$ we have $r(X) = x_2$. Proceeding similarly (or by induction), there are $2014 \cdot 2013 \cdots 5$ ways to choose $x_1, x_2, \ldots, x_{2010} \in S$ so that for all $i = 1, 2 \ldots, 2010$, $x_i = r(X)$ for each $X \subseteq S \setminus \{x_1, \ldots, x_{i-1}\}$ and $x_i \in X$.

We are now left with four elements $Y = \{y_1, y_2, y_3, y_4\}$. There are 4 ways to choose $r(Y)$. Suppose that $y_1 = r(Y)$. Then we clearly have $y_1 = r(\{y_1, y_2\}) = r(\{y_1, y_3\}) = r(\{y_1, y_4\})$. The only subsets whose representative has not been assigned yet are $\{y_1, y_2, y_3\}$, $\{y_1, y_2, y_4\}$, $\{y_1, y_3, y_4\}$, $\{y_2, y_3, y_4\}$, $\{y_2, y_3\}$, $\{y_2, y_4\}$, $\{y_3, y_4\}$. These subsets can be assigned in any way, hence giving $3^4 \cdot 2^3$ more choices.

In conclusion, the total number of assignments is $2014 \cdot 2013 \cdots 4 \cdot 3^4 \cdot 2^3 = 108 \cdot 2014!$.

Problem 3. Find all positive integers n such that for any integer k there exists an integer a for which $a^3 + a - k$ is divisible by n. (*Warut Suksompong, Thailand*)

Solution. *Answer:* All integers $n = 3^b$, where b is a nonnegative integer.

We are looking for integers n such that the set $A = \{a^3 + a \mid a \in \mathbf{Z}\}$ is a complete residue system by modulo n. Let us call this property by (*). It is not hard to see that $n = 1$ satisfies (*) and $n = 2$ does not.

If $a \equiv b \pmod{n}$, then $a^3 + a \equiv b^3 + b \pmod{n}$. So n satisfies (*) iff there are no $a, b \in \{0, \ldots, n-1\}$ with $a \neq b$ and $a^3 + a \equiv b^3 + b \pmod{n}$.

First, let us prove that 3^j satisfies (*) for all $j \geq 1$. Suppose that $a^3 + a \equiv b^3 + b \pmod{3^j}$ for $a \neq b$. Then $(a-b)(a^2 + ab + b^2 + 1) \equiv 0 \pmod{3^j}$. We can easily check mod 3 that $a^2 + ab + b^2 + 1$ is not divisible by 3.

Next note that if A is not a complete residue system modulo integer r, then it is also not a complete residue system modulo any multiple of r. Hence it remains to prove that any prime $p > 3$ does not satisfy (*).

If $p \equiv 1 \pmod{4}$, there exists b such that $b^2 \equiv -1 \pmod{p}$. We then take $a = 0$ to obtain the congruence $a^3 + a \equiv b^3 + b \pmod{p}$.

Suppose now that $p \equiv 3 \pmod{4}$. We will prove that there are integers $a, b \not\equiv 0 \pmod{p}$ such that $a^2 + ab + b^2 \equiv -1 \pmod{p}$. Note that we may suppose that $a \not\equiv b \pmod{p}$, since otherwise if $a \equiv b \pmod{p}$ satisfies $a^2 + ab + b^2 + 1 \equiv 0 \pmod{p}$, then $(2a)^2 + (2a)(-a) + a^2 + 1 \equiv 0 \pmod{p}$ and $2a \not\equiv -a \pmod{p}$. Letting c be the inverse of b modulo p (i.e. $bc \equiv 1 \pmod{p}$), the relation is equivalent to $(ac)^2 + ac + 1 \equiv -c^2 \pmod{p}$. Note that $-c^2$ can take on the values of all non-quadratic residues modulo p. If we can find an integer x such that $x^2 + x + 1$ is a non-quadratic residue modulo p, the values of a and c will follow immediately. Hence we focus on this latter task.

Note that if $x, y \in \{0, \ldots, p-1\} = B$, then $x^2 + x + 1 \equiv y^2 + y + 1 \pmod{p}$ iff p divides $x + y + 1$. We can deduce that $x^2 + x + 1$ takes on $(p+1)/2$ values as x varies in B. Since there are $(p-1)/2$ non-quadratic residues modulo p, the $(p+1)/2$ values that $x^2 + x + 1$ take on must be 0 and all the quadratic residues.

Let C be the set of quadratic residues modulo p and 0, and let $y \in C$. Suppose that $y \equiv z^2 \pmod{p}$ and let $z \equiv 2w + 1 \pmod{p}$ (we can always choose such w). Then $y + 3 \equiv 4(w^2 + w + 1) \pmod{p}$. From the previous paragraph, we know that $4(w^2 + w + 1) \in C$. This means that $y \in C \implies y + 3 \in C$. Unless $p = 3$, the relation implies that all elements of B are in C, a contradiction. This concludes the proof.

Problem 4. Let n and b be positive integers. We say n is *b-discerning* if there exists a set consisting of n different positive integers less than b that has no two different subsets U and V such that the sum of all elements in U equals the sum of all elements in V.

(a) Prove that 8 is a 100-discerning.

(b) Prove that 9 is not 100–discerning.

(*Senior Problems Committee of the Australian Mathematical Olympiad Committee*)

Solution.

(a) Take $S = \{3, 6, 12, 24, 48, 95, 96, 97\}$, i.e.

$$S = \{3 \cdot 2^k : 0 \leq k \leq 5\} \cup \{3 \cdot 2^5 - 1, 3 \cdot 2^5 + 1\}.$$

As k ranges between 0 to 5, the sums obtained from the numbers $3 \cdot 2^k$ are $3t$, where $1 \leq t \leq 63$. These are 63 numbers that are divisible by 3 and are at most $3 \cdot 63 = 189$.

Sums of elements of S are also the numbers $95 + 97 = 192$ and all the numbers that are sums of 192 and sums obtained from the numbers $3 \cdot 2^k$ with $0 \leq k \leq 5$. These are 64 numbers that are all divisible by 3 and at least equal to 192. In addition, sums of elements of S are the numbers 95 and all the numbers that are sums of 95 and sums obtained from the numbers $3 \cdot 2^k$ with $0 \leq k \leq 5$. These are 64 numbers that are all congruent to -1 mod 3.

Finally, sums of elements of S are the numbers 97 and all the numbers that are sums of 97 and sums obtained from the numbers $3 \cdot 2^k$ with $0 \leq k \leq 5$. These are 64 numbers that are all congruent to 1 mod 3.

Hence there are at least $63 + 64 + 64 + 64 = 255$ different sums from elements of S. On the other hand, S has $2^8 - 1 = 255$ non-empty subsets. Therefore S has no two different subsets with equal sums of elements. Therefore, 8 is 100-discerning.

(b) Suppose that 9 is 100-discerning. Then there is a set $S = \{s_1, \ldots, s_9\}$, $s_i < 100$ that has no two different subsets with equal sums of elements. Assume that $0 < s_1 < \cdots < s_9 < 100$.

Let X be the set of all subsets of S having at least 3 and at most 6 elements and let Y be the set of all subsets of S having exactly 2 or 3 or 4 elements greater than s_3.

The set X consists of

$$\binom{9}{3} + \binom{9}{4} + \binom{9}{5} + \binom{9}{6} = 84 + 126 + 126 + 84 = 420$$

subsets of S. The set in X with the largest sums of elements is $\{s_4, \ldots, s_9\}$ and the smallest sums is in $\{s_1, s_2, s_3\}$. Thus the sum of the elements of each of the 420 sets in X is at least $s_1 + s_2 + s_3$ and at most $s_4 + \cdots + s_9$, which is one of $(s_4 + \cdots + s_9) - (s_1 + s_2 + s_3) + 1$ integers. From the pigeonhole principle it follows that $(s_4 + \cdots + s_9) - (s_1 + s_2 + s_3) + 1 \geq 420$, i.e.,

$$(s_4 + \cdots + s_9) - (s_1 + s_2 + s_3) \geq 419. \tag{1}$$

Now let us calculate the number of subsets in Y. Observe that $\{s_4, \ldots, s_9\}$ has $\binom{6}{2}$ 2-element subsets, $\binom{6}{3}$ 3-element subsets and $\binom{6}{4}$ 4-element subsets, while $\{s_1, s_2, s_3\}$ has exactly 8 subsets. Hence the number of subsets of S in Y equals

$$8\left(\binom{6}{2} + \binom{6}{3} + \binom{6}{4}\right) = 8(15 + 20 + 15) = 400.$$

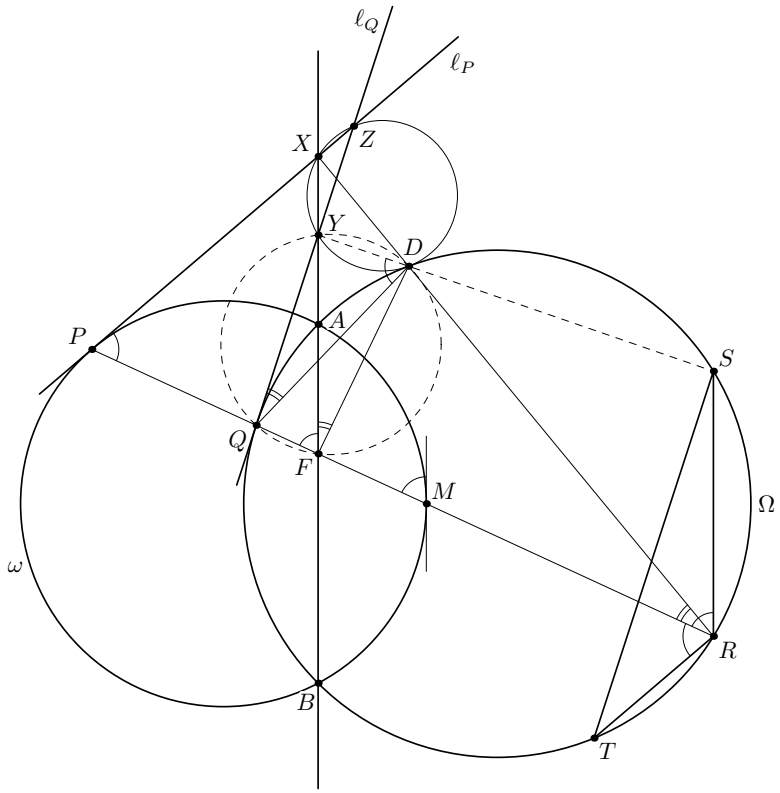
The set in Y with the largest sum of elements is $\{s_1, s_2, s_3, s_6, s_7, s_8, s_9\}$ and the smallest sum is in $\{s_4, s_5\}$. Again, by the pigeonhole principle it follows that $(s_1 + s_2 + s_3 + s_6 + s_7 + s_8 + s_9) - (s_4 + s_5) + 1 \geq 400$, i.e.,

$$(s_1 + s_2 + s_3 + s_6 + s_7 + s_8 + s_9) - (s_4 + s_5) \geq 399. \tag{2}$$

Adding (1) and (2) yields $2(s_6 + s_7 + s_8 + s_9) \geq 818$, so that $s_9 + 98 + 97 + 96 \geq s_9 + s_8 + s_7 + s_6 \geq 409$, i.e. $s_9 \geq 118$, a contradiction with $s_9 < 100$. Therefore, 9 is not 100-discerning.

Problem 5. Circles ω and Ω meet at points A and B. Let M be the midpoint of the arc AB of circle ω (M lies inside Ω). A chord MP of circle ω intersects Ω at Q (Q lies inside ω). Let ℓ_P be the tangent line to ω at P, and let ℓ_Q be the tangent line to Ω at Q. Prove that the circumcircle of the triangle formed by the lines ℓ_P, ℓ_Q, and AB is tangent to Ω. (*Ilya Bogdanov, Russia and Medeubek Kungozhin, Kazakhstan*)

Solution. Denote $X = AB \cap \ell_P$, $Y = AB \cap \ell_Q$, and $Z = \ell_P \cap \ell_Q$. Without loss of generality we have $AX < BX$. Let $F = MP \cap AB$.



Denote by R the second point of intersection of PQ and Ω; by S the point of Ω such that $SR \parallel AB$; and by T the point of Ω such that $RT \parallel \ell_P$. Since M is the midpoint of arc AB, the tangent ℓ_M at M to ω is parallel to AB, so $\angle(AB, PM) = \angle(PM, \ell_P)$. Therefore we have $\angle PRT = \angle MPX = \angle PFX = \angle PRS$. Thus the point Q is the midpoint of the arc TQS of Ω, hence $ST \parallel \ell_Q$. So the corresponding sides of the triangles RST and XYZ are parallel, and there exist a homothety h mapping RST to XYZ.

Let D be the second point of intersection of XR and Ω. We claim that D is the center of the homothety h; since $D \in \Omega$, this implies that the circumcircles of triangles RST and XYZ are tangent, as required. So, it remains to prove this claim. In order to do this, it suffices to show that $D \in SY$.

By $\angle PFX = \angle XPF$ we have $XF^2 = XP^2 = XA \cdot XB = XD \cdot XR$. Therefore, $\frac{XF}{XD} = \frac{XR}{XF}$, so the triangles XDF and XFR are similar, hence $\angle DFX = \angle XRF = \angle DRQ = \angle DQY$; thus the points D, Y, Q, and F are concyclic. It follows that $\angle YDQ = \angle YFQ = \angle SRQ = 180^\circ - \angle SDQ$ which means exactly that the points Y, D, and S are collinear, with D between S and Y.

Solutions of APMO 2013

Problem 1. Let ABC be an acute triangle with altitudes AD, BE and CF, and let O be the center of its circumcircle. Show that the segments OA, OF, OB, OD, OC, OE dissect the triangle ABC into three pairs of triangles that have equal areas.

Solution. Let M and N be midpoints of sides BC and AC, respectively. Notice that $\angle MOC = \frac{1}{2}\angle BOC = \angle EAB$, $\angle OMC = 90^\circ = \angle AEB$, so triangles OMC and AEB are similar and we get $\frac{OM}{AE} = \frac{OC}{AB}$. For triangles ONA and BDA we also have $\frac{ON}{BD} = \frac{OA}{BA}$. Then $\frac{OM}{AE} = \frac{ON}{BD}$ or $BD \cdot OM = AE \cdot ON$.

Denote by $S(\Phi)$ the area of the figure Φ. So, we see that $S(OBD) = \frac{1}{2}BD \cdot OM = \frac{1}{2}AE \cdot ON = S(OAE)$. Analogously, $S(OCD) = S(OAF)$ and $S(OCE) = S(OBF)$.

Alternative solution. Let R be the circumradius of triangle ABC, and as usual write A, B, C for angles $\angle CAB$, $\angle ABC$, $\angle BCA$ respectively, and a, b, c for sides BC, CA, AB respectively. Then the area of triangle OCD is

$$S(OCD) = \tfrac{1}{2} \cdot OC \cdot CD \cdot \sin(\angle OCD) = \tfrac{1}{2}R \cdot CD \cdot \sin(\angle OCD).$$

Now $CD = b\cos C$, and

$$\angle OCD = \frac{180^\circ - 2A}{2} = 90^\circ - A$$

(since triangle OBC is isosceles, and $\angle BOC = 2A$). So

$$S(OCD) = \tfrac{1}{2}Rb\cos C \sin(90^\circ - A) = \tfrac{1}{2}Rb\cos C\cos A.$$

A similar calculation gives

$$\begin{aligned} S(OAF) &= \tfrac{1}{2}OA \cdot AF \cdot \sin(\angle OAF) \\ &= \tfrac{1}{2}R \cdot (b\cos A)\sin(90^\circ - C) \\ &= \tfrac{1}{2}Rb\cos A\cos C, \end{aligned}$$

so OCD and OAF have the same area. In the same way we find that OBD and OAE have the same area, as do OCE and OBF.

Problem 2. Determine all positive integers n for which $\frac{n^2+1}{[\sqrt{n}]^2+2}$ is an integer. Here $[r]$ denotes the greatest integer less than or equal to r.

Solution. We will show that there are no positive integers n satisfying the condition of the problem.

Let $m = [\sqrt{n}]$ and $a = n - m^2$. We have $m \geq 1$ since $n \geq 1$. From $n^2+1 = (m^2+a)^2+1 \equiv (a-2)^2+1 \pmod{(m^2+2)}$, it follows that the condition of the problem is equivalent to the fact that $(a-2)^2+1$ is divisible by m^2+2. Since we have

$$0 < (a-2)^2 + 1 \leq \max\{2^2, (2m-2)^2\} + 1 \leq 4m^2 + 1 < 4(m^2+2),$$

we see that $(a-2)^2+1 = k(m^2+2)$ must hold with $k = 1, 2$ or 3. We will show that none of these can occur.

Case 1. When $k = 1$. We get $(a-2)^2 - m^2 = 1$, and this implies that $a-2 = \pm 1$, $m = 0$ must hold, but this contradicts with fact $m \geq 1$.

Case 2. When $k = 2$. We have $(a-2)^2+1 = 2(m^2+2)$ in this case, but any perfect square is congruent to $0, 1, 4 \bmod 8$, and therefore, we have $(a-2)^2+1 \equiv 1, 2, 5 \pmod 8$, while $2(m^2+2) \equiv 4, 6 \pmod 8$. Thus, this case cannot occur either.

Case 3. When $k = 3$. We have $(a-2)^2+1 = 3(m^2+2)$ in this case. Since any perfect square is congruent to 0 or 1 mod 3, we have $(a-2)^2+1 \equiv 1, 2 \pmod 3$, while $3(m^2+2) \equiv 0 \pmod 3$, which shows that this case cannot occur either.

Problem 3. For $2k$ real numbers $a_1, a_2, \ldots, a_k,\ b_1, b_2, \ldots, b_k$ define the sequence of numbers X_n by

$$X_n = \sum_{i=1}^{k} [a_i n + b_i] \quad (n = 1, 2, \ldots).$$

If the sequence X_n forms an arithmetic progression, show that $\sum_{i=1}^{k} a_i$ must be an integer. Here $[r]$ denotes the greatest integer less than or equal to r.

Solution. Let us write $A = \sum_{i=1}^{k} a_i$ and $B = \sum_{i=1}^{k} b_i$. Summing the corresponding terms of the following inequalities over i,

$$a_i n + b_i - 1 < [a_i n + b_i] \leq a_i n + b_i,$$

we obtain $An + B - k < X_n < An + B$. Now suppose that $\{X_n\}$ is an arithmetic progression with the common difference d, then we have $nd = X_{n+1} - X_1$ and $A + B - k < X_1 \leq A + B$ Combining with the inequalities obtained above, we get

$$A(n+1) + B - k < nd + X_1 < A(n+1) + B,$$

or

$$An - k \leq An + (A + B - X_1) - k < nd < An + (A + B - X_1) < An + k,$$

from which we conclude that $|A - d| < \frac{k}{n}$ must hold. Since this inequality holds for any positive integer n, we must have $A = d$. Since $\{X_n\}$ is a sequence of integers, d must be an integer also, and thus we conclude that A is also an integer.

Problem 4. Let a and b be positive integers, and let A and B be finite sets of integers satisfying:

(i) A and B are disjoint;

(ii) if an integer i belongs either to A or to B, then $i + a$ belongs to A or $i - b$ belongs to B.

Prove that $a|A| = b|B|$. (Here $|X|$ denotes the number of elements in the set X.)

Solution. Let $A^* = \{n - a : n \in A\}$ and $B^* = \{n + b : n \in B\}$. Then, by (ii), $A \cup B \subseteq A^* \cup B^*$ and by (i),

$$|A \cup B| \leq |A^* \cup B^*| \leq |A^*| + |B^*| = |A| + |B| = |A \cup B|. \tag{1}$$

Thus, $A \cup B = A^* \cup B^*$ and A^* and B^* have no element in common. For each finite set X of integers, let $\sum(X) = \sum_{x \in X} x$. Then

$$\begin{aligned} \sum(A) + \sum(B) &= \sum(A \cup B) \\ &= \sum(A^* \cup B^*) = \sum(A^*) + \sum(B^*) \\ &= \sum(A) - a|A| + \sum(B) + b|B|, \end{aligned} \tag{2}$$

which implies $a|A| = b|B|$.

Alternative solution. Let us construct a directed graph whose vertices are labelled by the members of $A \cup B$ and such that there is an edge from i to j iff $j \in A$ and $j = i + a$ or $j \in B$ and $j = i - b$. From (ii), each vertex has out-degree ≥ 1 and, from (i), each vertex has in-degree ≤ 1. Since the sum of the out-degrees equals the sum of the in-degrees, each vertex has in-degree and out-degree equal to 1. This is only possible if the graph is the union of disjoint cycles, say $G_1, G_2, \ldots, G_n$. Let $|A_k|$ be the number of elements of A in G_k and $|B_k|$ be the number of elements of B in G_k. The cycle G_k will involve increasing vertex labels by a a total of $|A_k|$ times and decreasing them by b a total of $|B_k|$ times. Since it is a cycle, we have $a|A_k| = b|B_k|$. Summing over all cycles gives the result.

Problem 5. Let $ABCD$ be a quadrilateral inscribed in a circle ω, and let P be a point on the extension of AC such that PB and PD are tangent to ω. The tangent at C intersects PD at Q and the line AD at R. Let E be the second point of intersection between AQ and ω. Prove that B, E, R are collinear.

Solution. To show B, E, R are collinear, it is equivalent to show the lines AD, BE, CQ are concurrent. Let CQ intersect AD at R and BE intersect AD at R'. We shall show $RD/RA = R'D/R'A$ so that $R = R'$.

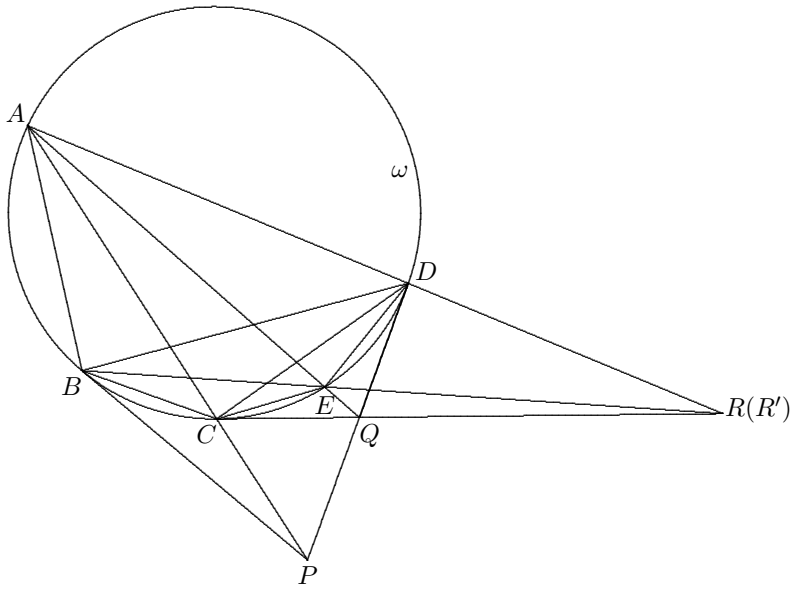
Since $\triangle PAD$ is similar to $\triangle PDC$ and $\triangle PAB$ is similar to $\triangle PBC$, we have $AD/DC = PA/PD = PA/PB = AB/BC$. Hence, $AB \cdot DC = BC \cdot AD$. By Ptolemy's theorem, $AB \cdot DC = BC \cdot AD = \frac{1}{2} CA \cdot DB$. Similarly $CA \cdot ED = CE \cdot AD = \frac{1}{2} AE \cdot DC$.

Thus

$$\frac{DB}{AB} = \frac{2DC}{CA}, \tag{3}$$

and

$$\frac{DC}{CA} = \frac{2ED}{AE}. \tag{4}$$



Since the triangles RDC and RCA are similar, we have $\frac{RD}{RC} = \frac{DC}{CA} = \frac{RC}{RA}$. Thus using (4)

$$\frac{RD}{RA} = \frac{RD \cdot RA}{RA^2} = \left(\frac{RC}{RA}\right)^2 = \left(\frac{DC}{CA}\right)^2 = \left(\frac{2ED}{AE}\right)^2. \tag{5}$$

Using the similar triangles ABR' and EDR', we have $R'D/R'B = ED/AB$. Using the similar triangles DBR' and EAR' we have $R'A/R'B = EA/DB$. Thus using (3) and (4),

$$\frac{R'D}{R'A} = \frac{ED \cdot DB}{EA \cdot AB} = \left(\frac{2ED}{AE}\right)^2. \tag{6}$$

It follows from (5) and (6) that $R = R'$.

SOLUTIONS FOR 2012 APMO PROBLEMS

Problem 1.

Solution: Let us denote by $\triangle XYZ$ the area of the triangle XYZ. Let $x = \triangle PAB$, $y = \triangle PBC$ and $z = \triangle PCA$.

From

$$y : z = \triangle BCP : \triangle ACP = BF : AF = \triangle BPF : \triangle APF = (x-1) : 1$$

follows that $z(x-1) = y$, which yields $(z+1)x = x+y+z$. Similarly, we get $(x+1)y = x+y+z$ and $(y+1)z = x+y+z$. Thus, we obtain $(x+1)y = (y+1)z = (z+1)x$.

We may assume without loss of generality that $x \leq y, z$. If we assume that $y > z$ holds, then we get $(y+1)z > (z+1)x$, which is a contradiction. Similarly, we see that $y < z$ leads to a contradiction $(x+1)y < (y+1)z$. Therefore, we must have $y = z$. Then, we also get from $(y+1)z = (z+1)x$ that $x = z$ must hold. We now obtain from $(x-1) : 1 = y : z = 1 : 1$ that $x = y = z = 2$ holds. Therefore, we conclude that the area of the triangle ABC equals $x+y+z = 6$.

Problem 2.

Solution: If we insert numbers as in the figure below ($0'$s are to be inserted in the remaining blank boxes), then we see that the condition of the problem is satisfied and the total number of all the numbers inserted is 5.

0	1	0	
1	1	1	
0	1	0	

We will show that the sum of all the numbers to be inserted in the boxes of the given grid cannot be more than 5 if the distribution of the numbers has to satisfy the requirement of the problem. Let $n = 2012$. Let us say that the row number (the column number) of a box in the given grid is i (j, respectively) if the box lies on the i-th row and the j-th column. For a pair of positive integers x and y, denote by $R(x, y)$ the sum of the numbers inserted in all of the boxes whose row number is greater than or equal to x and less than or equal to y (assign the value 0 if $x > y$).

First let a be the largest integer satisfying $1 \leq a \leq n$ and $R(1, a-1) \leq 1$, and then choose the smallest integer c satisfying $a \leq c \leq n$ and $R(c+1, n) \leq 1$. It is possible to choose such a pair a, c since $R(1, 0) = 0$ and $R(n+1, n) = 0$. If $a < c$, then we have $a < n$ and so, by the maximality of a, we must have $R(1, a) > 1$, while from the minimality of c, we must have $R(a+1, n) > 1$. Then by splitting the grid into 2 rectangles by means of the horizontal line bordering the a-th row and the $a+1$-th row, we get the splitting contradicting the requirement of the problem. Thus, we must have $a = c$.

Similarly, if for any pair of integers x, y we define $C(x, y)$ to be the sum of the numbers inserted in all of the boxes whose column number is greater than or equal to x and less than or equal to y ($C(x, y) = 0$ if $x > y$), then we get a number b for which

$$C(1, b-1) \leq 1, \quad C(b+1, n) \leq 1, \quad 1 \leq b \leq n.$$

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If we let r be the number inserted in the box whose row number is a and the column number is b, then since $r \leq 1$, we conclude that the sum of the numbers inserted into all of the boxes is

$$\leq R(1, a-1) + R(a+1, n) + C(1, b-1) + C(b+1, n) + r \leq 5.$$

Problem 3.

Solution

For integers a, b and a positive integer m, let us write $a \equiv b \pmod{m}$ if $a - b$ is divisible by m. Since $\frac{n^p+1}{p^n+1}$ must be a positive integer, we see that $p^n \leq n^p$ must hold. This means that if $p = 2$, then $2^n \leq n^2$ must hold. As it is easy to show by induction that $2^n > n^2$ holds if $n \geq 5$, we conclude that if $p = 2$, then $n \leq 4$ must be satisfied. And we can check that $(p, n) = (2, 2), (2, 4)$ satisfy the condition of the problem, while $(2, 3)$ does not.

Next, we consider the case where $p \geq 3$.

Suppose s is an integer satisfying $s \geq p$. If $s^p \leq p^s$ for such an s, then we have

$$\begin{aligned}(s+1)^p &= s^p\left(1+\frac{1}{s}\right)^p \leq p^s\left(1+\frac{1}{p}\right)^p \\ &= p^s \sum_{r=0}^{p} {}_pC_r \frac{1}{p^r} < p^s \sum_{r=0}^{p} \frac{1}{r!} \\ &\leq p^s\left(1+\sum_{r=1}^{p} \frac{1}{2^{r-1}}\right) \\ &< p^s(1+2) \leq p^{s+1}\end{aligned}$$

Thus we have $(s+1)^p < p^{s+1}$, and by induction on n, we can conclude that if $n > p$, then $n^p < p^n$. This implies that we must have $n \leq p$ in order to satisfy our requirement $p^n \leq n^p$.

We note that since $p^n + 1$ is even, so is $n^p + 1$, which, in turn implies that n must be odd and therefore, $p^n + 1$ is divisible by $p + 1$, and $n^p + 1$ is also divisible by $p + 1$. Thus we have $n^p \equiv -1 \pmod{(p+1)}$, and therefore, $n^{2p} \equiv 1 \pmod{(p+1)}$.

Now, let e be the smallest positive integer for which $n^e \equiv 1 \pmod{(p+1)}$. Then, we can write $2p = ex + y$, where x, y are non-negative integers and $0 \leq y < e$, and we have

$$1 \equiv n^{2p} = (n^e)^x \cdot n^y \equiv n^y \pmod{(p+1)},$$

which implies, because of the minimality of e, that $y = 0$ must hold. This means that $2p$ is an integral multiple of e, and therefore, e must equal one of the numbers $1, 2, p, 2p$.

Now, if $e = 1, p$, then we get $n^p \equiv 1 \pmod{(p+1)}$, which contradicts the fact that p is an odd prime. Since n and $p + 1$ are relatively prime, we have by Euler's Theorem that $n^{\varphi(p+1)} \equiv 1 \pmod{(p+1)}$, where $\varphi(m)$ denotes the number of integers $j (1 \leq j \leq m)$ which are relatively prime with m. From $\varphi(p+1) < p+1 < 2p$ and the minimality of e, we can then conclude that $e = 2$ must hold.

From $n^2 \equiv 1 \pmod{(p+1)}$, we get

$$-1 \equiv n^p = n^{(2 \cdot \frac{p-1}{2}+1)} \equiv n \pmod{(p+1)},$$

which implies that $p+1$ divides $n+1$. Therefore, we must have $p \leq n$, which, together with the fact $n \leq p$, show that $p = n$ must hold.

It is clear that the pair (p, p) for any prime $p \geq 3$ satisfies the condition of the problem, and thus, we conclude that the pairs (p, n) which satisfy the condition of the problem must be of the form $(2, 4)$ and (p, p) with any prime p.

Alternate Solution. Let us consider the case where $p \geq 3$. As we saw in the preceding solution, n must be odd if the pair (p, n) satisfy the condition of the problem. Now, let q be a prime factor of $p+1$. Then, since $p+1$ divides p^n+1, q must be a prime factor of p^n+1 and of n^p+1 as well. Suppose $q \geq 3$. Then, from $n^p \equiv -1 \pmod{q}$, it follows that $n^{2p} \equiv 1 \pmod{q}$ holds. If we let e be the smallest positive integer satisfying $n^e \equiv 1 \pmod{q}$, then by using the same argument as we used in the preceding solution, we can conclude that e must equal one of the numbers $1, 2, p, 2p$. If $e = 1, p$, then we get $n^p \equiv 1 \pmod{q}$, which contradicts the assumption $q \geq 3$. Since n is not a multiple of q, by Fermat's Little Theorem we get $n^{q-1} \equiv 1 \pmod{q}$, and therefore, we get by the minimality of e that $e = 2$ must hold. From $n^2 \equiv 1 \pmod{q}$, we also get

$$n^p = n^{(2 \cdot \frac{p-1}{2}+1)} \equiv n \pmod{q},$$

and since $n^p \equiv -1 \pmod{q}$, we have $n \equiv -1 \pmod{q}$ as well.

Now, if $q = 2$ then since n is odd, we have $n \equiv -1 \pmod{q}$ as well. Thus, we conclude that for an arbitrary prime factor q of $p+1$, $n \equiv -1 \pmod{q}$ must hold.

Suppose, for a prime q, q^k for some positive integer k is a factor of $p+1$. Then q^k must be a factor of n^p+1 as well. But since

$$n^p + 1 = (n+1)(n^{p-1} - n^{p-2} + \cdots - n + 1) \quad \text{and}$$

$$n^{p-1} - n^{p-2} + \cdots - n + 1 \equiv (-1)^{p-1} - (-1)^{p-2} + \cdots - (-1) + 1 \not\equiv 0 \pmod{q},$$

we see that q^k must divide $n+1$. By applying the argument above for each prime factor q of $p+1$, we can then conclude that $n+1$ must be divisible by $p+1$, and as we did in the preceding proof, we can conclude that $n = p$ must hold.

Problem 4.

Solution: If $AB = AC$, then we get $BF = CF$ and the conclusion of the problem is clearly satisfied. So, we assume that $AB \neq AC$ in the sequel.

Due to symmetry, we may suppose without loss of generality that $AB > AC$. Let K be the point on the circle Γ such that AK is a diameter of this circle. Then, we get

$$\angle BCK = \angle ACK - \angle ACB = 90^\circ - \angle ACB = \angle CBH$$

and

$$\angle CBK = \angle ABK - \angle ABC = 90^\circ - \angle ABC = \angle BCH,$$

from which we conclude that the triangles BCK and CBH are congruent. Therefore, the quadrilateral $BKCH$ is a parallelogram, and its diagonal HK passes through the center M of the other diagonal BC. Therefore, the 3 points H, M, K lie on the same straight line, and we have $\angle AEM = \angle AEK = 90^\circ$.

From $\angle AED = 90^\circ = \angle ADM$, we see that the 4 points A, E, D, M lie on the circumference of the same circle, from which we obtain $\angle AMB = \angle AED = \angle AEF = \angle ACF$. Putting this fact together with the fact that $\angle ABM = \angle AFC$, we conclude that the triangles ABM and AFC are similar, and we get $\frac{AM}{BM} = \frac{AC}{FC}$. By a similar argument, we get that the triangles ACM and AFB are similar, and

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therefore, that $\frac{AM}{CM} = \frac{AB}{FB}$ holds. Noting that $BM = CM$, we also get $\frac{AC}{FC} = \frac{AB}{FB}$, from which we can conclude that $\frac{BF}{CF} = \frac{AB}{AC}$, proving the assertion of the problem.

Problem 5.

Solution: Let us note first that if $i \neq j$, then since $a_i a_j \leq \frac{a_i^2+a_j^2}{2}$, we have

$$n - a_i a_j \geq n - \frac{a_i^2 + a_j^2}{2} \geq n - \frac{n}{2} = \frac{n}{2} > 0.$$

If we set $b_i = |a_i|$ $(i = 1, 2, \ldots, n)$, then we get $b_1^2 + b_2^2 + \cdots + b_n^2 = n$ and $\frac{1}{n-a_i a_j} \leq \frac{1}{n-b_i b_j}$, which shows that it is enough to prove the assertion of the problem in the case where all of $a_1, a_2, \cdots, a_n$ are non-negative. Hence, we assume from now on that $a_1, a_2, \cdots, a_n$ are all non-negative.

By multiplying by n the both sides of the desired inequality we get the inequality:

$$\sum_{1 \leq i < j \leq n} \frac{n}{n - a_i a_j} \leq \frac{n^2}{2}$$

and since $\frac{n}{n-a_i a_j} = 1 + \frac{a_i a_j}{n-a_i a_j}$, we obtain from the inequality above by subtracting $\frac{n(n-1)}{2}$ from both sides the following inequality:

(i) $$\sum_{1 \leq i < j \leq n} \frac{a_i a_j}{n - a_i a_j} \leq \frac{n}{2}$$

We will show that this inequality (i) holds.

If for some i the equality $a_i^2 = n$ is valid, then $a_j = 0$ must hold for all $j \neq i$ and the inequality (i) is trivially satisfied. So, we assume from now on that $a_i^2 < n$ is valid for each i.

Let us assume that $i \neq j$ from now on. Since $0 \leq a_i a_j \leq \left(\frac{a_i + a_j}{2}\right)^2 \leq \frac{a_i^2+a_j^2}{2}$ holds, we have

(ii) $$\frac{a_i a_j}{n - a_i a_j} \leq \frac{a_i a_j}{n - \frac{a_i^2+a_j^2}{2}} \leq \frac{\left(\frac{a_i+a_j}{2}\right)^2}{n - \frac{a_i^2+a_j^2}{2}} = \frac{1}{2} \cdot \frac{(a_i + a_j)^2}{(n - a_i^2) + (n - a_j^2)}.$$

Since $n - a_i^2 > 0$, $n - a_j^2 > 0$, we also get from the Cauchy-Schwarz inequality that

$$\left(\frac{a_j^2}{n - a_i^2} + \frac{a_i^2}{n - a_j^2}\right)((n - a_i^2) + (n - a_j^2)) \geq (a_i + a_j)^2,$$

from which it follows that

(iii) $$\frac{(a_i + a_j)^2}{(n - a_i^2) + (n - a_j^2)} \leq \left(\frac{a_j^2}{n - a_i^2} + \frac{a_i^2}{n - a_j^2}\right)$$

holds. Combining the inequalities (ii) and (iii), we get

$$\begin{aligned}\sum_{1\le i<j\le n}\frac{a_ia_j}{n-a_ia_j} &\le \frac{1}{2}\sum_{1\le i<j\le n}\left(\frac{a_j^2}{n-a_i^2}+\frac{a_i^2}{n-a_j^2}\right)\\ &= \frac{1}{2}\sum_{i\ne j}\frac{a_j^2}{n-a_i^2}\\ &= \frac{1}{2}\sum_{i=1}^{n}\frac{n-a_i^2}{n-a_i^2}\\ &= \frac{n}{2},\end{aligned}$$

which establishes the desired inequality (i).

SOLUTIONS FOR 2011 APMO PROBLEMS

Problem 1.

Solution: Suppose all of the 3 numbers a^2+b+c, b^2+c+a and c^2+a+b are perfect squares. Then from the fact that a^2+b+c is a perfect square bigger than a^2 it follows that $a^2+b+c \geq (a+1)^2$, and therefore, $b+c \geq 2a+1$. Similarly we obtain $c+a \geq 2b+1$ and $a+b \geq 2c+1$.

Adding the corresponding sides of the preceding 3 inequalities, we obtain $2(a+b+c) \geq 2(a+b+c)+3$, a contradiction. This proves that it is impossible to have all the 3 given numbers to be perfect squares.

Alternate Solution: Since the given conditions of the problem are symmetric in a, b, c, we may assume that $a \geq b \geq c$ holds. From the assumption that a^2+b+c is a perfect square, we can deduce as in the solution above the inequality $b+c \geq 2a+1$. But then we have

$$2a \geq b+c \geq 2a+1,$$

a contradiction, which proves the assertion of the problem.

Problem 2.

Solution: We will show that 36° is the desired answer for the problem.

First, we observe that if the given 5 points form a regular pentagon, then the minimum of the angles formed by any triple among the five vertices is 36°, and therefore, the answer we seek must be bigger than or equal to 36°.

Next, we show that for any configuration of 5 points satisfying the condition of the problem, there must exist an angle smaller than or equal to 36° formed by a triple chosen from the given 5 points. For this purpose, let us start with any 5 points, say A_1, A_2, A_3, A_4, A_5, on the plane satisfying the condition of the problem, and consider the smallest convex subset, call it Γ, in the plane containing all of the 5 points. Since this convex subset Γ must be either a triangle or a quadrilateral or a pentagon, it must have an interior angle with 108° or less. We may assume without loss of generality that this angle is $\angle A_1A_2A_3$. By the definition of Γ it is clear that the remaining 2 points A_4 and A_5 lie in the interior of the angular region determined by $\angle A_1A_2A_3$, and therefore, there must be an angle smaller than or equal to $\frac{1}{3} \cdot 108^\circ = 36^\circ$, which is formed by a triple chosen from the given 5 points, and this proves that 36° is the desired maximum.

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Problem 3.

Solution: Since $\angle B_1BB_2 = 90^\circ$, the circle having B_1B_2 as its diameter goes through the points B, B_1, B_2. From $B_1A : B_1C = B_2A : B_2C = BA : BC$, it follows that this circle is the Apolonius circle with the ratio of the distances from the points A and C being $BA : BC$. Since the point P lies on this circle, we have

$$PA : PC = BA : BC = \sin C : \sin A,$$

from which it follows that $PA \sin A = PC \sin C$. Similarly, we have $PA \sin A = PB \sin B$, and therefore, $PA \sin A = PB \sin B = PC \sin C$.

Let us denote by D, E, F the foot of the perpendicular line drawn from P to the line segment BC, CA and AB, respectively. Since the points E, F lie on a circle having PA as its diameter, we have by the law of sines $EF = PA \sin A$. Similarly, we have $FD = PB \sin B$ and $DE = PC \sin C$. Consequently, we conclude that DEF is an equilateral triangle. Furthermore, we have $\angle CPE = \angle CDE$, since the quadrilateral $CDPE$ is cyclic. Similarly, we have $\angle FPB = \angle FDB$. Putting these together, we get

$$\begin{aligned}\angle BPC &= 360^\circ - (\angle CPE + \angle FPB + \angle EPF) \\ &= 360^\circ - \{(\angle CDE + \angle FDB) + (180^\circ - \angle FAE)\} \\ &= 360^\circ - (120^\circ + 150^\circ) = 90^\circ,\end{aligned}$$

which proves the assertion of the problem.

Alternate Solution: Let O be the midpoint of the line segment B_1B_2. Then the points B and P lie on the circle with center at O and going through the point B_1. From

$$\angle OBC = \angle OBB_1 - \angle CBB_1 = \angle OB_1B - \angle B_1BA = \angle BAC$$

it follows that the triangles OCD and OBA are similar, and therefore we have that $OC \cdot OA = OB^2 = OP^2$. Thus we conclude that the triangles OCP and OPA are similar, and therefore, we have $\angle OPC = \angle PAC$. Using this fact, we obtain

$$\begin{aligned}\angle PBC - \angle PBA &= (\angle B_1BC + \angle PBB_1) - (\angle ABB_1 - \angle PBB_1) \\ &= 2\angle PBB_1 = \angle POB_1 = \angle PCA - \angle OPC \\ &= \angle PCA - \angle PAC,\end{aligned}$$

from which we conclude that $\angle PAC + \angle PBC = \angle PBA + \angle PCA$. Similarly, we get $\angle PAB + \angle PCB = \angle PBA + \angle PCA$. Putting these facts together and taking into account the fact that

$$(\angle PAC + \angle PBC) + (\angle PAB + \angle PCB) + (\angle PBA + \angle PCA) = 180^\circ,$$

we conclude that $\angle PBA + \angle PCA = 60^\circ$, and finally that

$$\angle BPC = (\angle PBA + \angle PAB) + (\angle PCA + \angle PAC) = \angle BAC + (\angle PBA + \angle PCA) = 90^\circ,$$

proving the assertion of the problem.

Problem 4.

Solution: We will show that the desired maximum value for m is $n(n-1)$.

First, let us show that $m \leq n(n-1)$ always holds for any sequence $P_0, P_1, \cdots, P_{m+1}$ satisfying the conditions of the problem.

Call a point a **turning point** if it coincides with P_i for some i with $1 \leq i \leq m$. Let us say also that 2 points $\{P, Q\}$ are **adjacent** if $\{P, Q\} = \{P_{i-1}, P_i\}$ for some i with $1 \leq i \leq m$, and **vertically adjacent** if, in addition, PQ is parallel to the y-axis.

Any turning point is vertically adjacent to exactly one other turning point. Therefore, the set of all turning points is partitioned into a set of pairs of points using the relation of "vertical adjacency". Thus we can conclude that if we fix $k \in \{1, 2, \cdots, n\}$, the number of turning points having the x-coordinate k must be even, and hence it is less than or equal to $n-1$. Therefore, altogether there are less than or equal to $n(n-1)$ turning points, and this shows that $m \leq n(n-1)$ must be satisfied.

It remains now to show that for any positive odd number n one can choose a sequence for which $m = n(n-1)$. We will show this by using the mathematical induction on n. For $n = 1$, this is clear. For $n = 3$, choose

$$P_0 = (0, 1), \qquad P_1 = (1, 1), \qquad P_2 = (1, 2), \qquad P_3 = (2, 2),$$

$$P_4 = (2, 1), \qquad P_5 = (3, 1), \qquad P_6 = (3, 3), \qquad P_7 = (4, 3).$$

It is easy to see that these points satisfy the requirements (See fig. 1 below).

figure 1

Let n be an odd integer ≥ 5, and suppose there exists a sequence satisfying the desired conditions for $n-4$. Then, it is possible to construct a sequence which gives a configuration indicated in the following diagram (fig. 2), where the configuration inside of the dotted square is given by the induction hypothesis:

figure 2

By the induction hypothesis, there are exactly $(n-4)(n-5)$ turning points for the configuration inside of the dotted square in the figure 2 above, and all of the lattice points in the figure 2 lying outside of the dotted square except for the 4 points $(n, 2)$, $(n-1, n-2)$, $(2, 3)$, $(1, n-1)$ are turning points. Therefore, the total

number of turning points in this configuration is

$$(n-4)(n-5)+(n^2-(n-4)^2-4)=n(n-1),$$

showing that for this n there exists a sequence satisfying the desired properties, and thus completing the induction process.

Problem 5.

Solution: By substituting $x=1$ and $y=1$ into the given identity we obtain $f(f(1))=f(1)$. Next, by substituting $x=1$ and $y=f(1)$ into the given identity and using $f(f(1))=f(1)$, we get $f(1)^2=f(1)$, from which we conclude that either $f(1)=0$ or $f(1)=1$. But if $f(1)=1$, then substituting $y=1$ into the given identity, we get $f(x)=x$ for all x, which contradicts the condition (1). Therefore, we must have $f(1)=0$.

By substituting $x=1$ into the given identity and using the fact $f(1)=0$, we then obtain $f(f(y))=2f(y)$ for all y. This means that if a number t belongs to the range of the function f, then so does $2t$, and by induction we can conclude that for any non-negative integer n, 2^nt belongs to the range of f if t does. Now suppose that there exists a real number a for which $f(a)>0$, then for any non-negative integer n $2^nf(a)$ must belong to the range of f, which leads to a contradiction to the condition (1). Thus we conclude that $f(x)\leq 0$ for any real number x.

By substituting $\frac{x}{2}$ for x and $f(y)$ for y in the given identity and using the fact that $f(f(y))=2f(y)$, we obtain

$$f(xf(y))+f(y)f\left(\frac{x}{2}\right)=xf(y)+f\left(\frac{x}{2}f(y)\right),$$

from which it follows that $xf(y)-f(xf(y))=f(y)f\left(\frac{x}{2}\right)-f\left(\frac{x}{2}f(y)\right)\geq 0$, since the values of f are non-positive. Combining this with the given identity, we conclude that $yf(x)\geq f(xy)$. When $x>0$, by letting y to be $\dfrac{1}{x}$ and using the fact that $f(1)=0$, we get $f(x)\geq 0$. Since $f(x)\leq 0$ for any real number x, we conclude that $f(x)=0$ for any positive real number x. We also have $f(0)=f(f(1))=2f(1)=0$.

If f is identically 0, i.e., $f(x)=0$ for all x, then clearly, this f satisfies the given identity. If f satisfies the given identity but not identically 0, then there exists a $b<0$ for which $f(b)<0$. If we set $c=f(b)$, then we have $f(c)=f(f(b))=2f(b)=2c$. For any negative real number x, we have $cx>0$ so that $f(cx)=f(2cx)=0$, and by substituting $y=c$ into the given identity, we get

$$f(2cx)+cf(x)=2cx+f(cx),$$

from which it follows that $f(x)=2x$ for any negative real x.

We therefore conclude that if f satisfies the given identity and is not identically 0, then f is of the form $f(x)=\begin{cases}0 & \text{if } x\geq 0\\ 2x & \text{if } x<0.\end{cases}$ Finally, let us show that the function f of the form shown above does satisfy the conditions of the problem. Clearly, it satisfies the condition (1). We can check that f satisfies the condition (2) as well by separating into the following 4 cases depending on whether x,y are non-negative or negative.

- when both x and y are non-negative, both sides of the given identity are 0.
- when x is non-negative and y is negative, we have $xy\leq 0$ and both sides of the given identity are $4xy$.

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- when x is negative and y is non-negative, we have $xy \leq 0$ and both sides of the given identity are $2xy$.
- when both x and y are negative, we have $xy > 0$ and both sides of the given identity are $2xy$.

Summarizing the arguments above, we conclude that the functions f satisfying the conditions of the problem are

$$f(x) = 0 \qquad \text{and} \qquad f(x) = \begin{cases} 0 & \text{if } x \geq 0 \\ 2x & \text{if } x < 0. \end{cases}$$

SOLUTIONS FOR 2010 APMO PROBLEMS

Problem 1. Let ABC be a triangle with $\angle BAC \neq 90^\circ$. Let O be the circumcenter of the triangle ABC and let Γ be the circumcircle of the triangle BOC. Suppose that Γ intersects the line segment AB at P different from B, and the line segment AC at Q different from C. Let ON be a diameter of the circle Γ. Prove that the quadrilateral $APNQ$ is a parallelogram.

Solution: From the assumption that the circle Γ intersects both of the line segments AB and AC, it follows that the 4 points N, C, Q, O are located on Γ in the order of N, C, Q, O or in the order of N, C, O, Q. The following argument for the proof of the assertion of the problem is valid in either case. Since $\angle NQC$ and $\angle NOC$ are subtended by the same arc $\overset{\frown}{NC}$ of Γ at the points Q and O, respectively, on Γ, we have $\angle NQC = \angle NOC$. We also have $\angle BOC = 2\angle BAC$, since $\angle BOC$ and $\angle BAC$ are subtended by the same arc $\overset{\frown}{BC}$ of the circum-circle of the triangle ABC at the center O of the circle and at the point A on the circle, respectively. From $OB = OC$ and the fact that ON is a diameter of Γ, it follows that the triangles OBN and OCN are congruent, and therefore we obtain $2\angle NOC = \angle BOC$. Consequently, we have $\angle NQC = \frac{1}{2}\angle BOC = \angle BAC$, which shows that the 2 lines AP, QN are parallel.

In the same manner, we can show that the 2 lines AQ, PN are also parallel. Thus, the quadrilateral $APNQ$ is a parallelogram.

Problem 2. For a positive integer k, call an integer a *pure k-th power* if it can be represented as m^k for some integer m. Show that for every positive integer n there exist n distinct positive integers such that their sum is a pure 2009-th power, and their product is a pure 2010-th power.

Solution: For the sake of simplicity, let us set $k = 2009$.

First of all, choose n distinct positive integers $b_1, \cdots, b_n$ suitably so that their product is a pure $k+1$-th power (for example, let $b_i = i^{k+1}$ for $i = 1, \cdots, n$). Then we have $b_1 \cdots b_n = t^{k+1}$ for some positive integer t. Set $b_1 + \cdots + b_n = s$.

Now we set $a_i = b_i s^{k^2-1}$ for $i = 1, \cdots, n$, and show that $a_1, \cdots, a_n$ satisfy the required conditions. Since $b_1, \cdots, b_n$ are distinct positive integers, it is clear that so are $a_1, \cdots, a_n$. From

$$a_1 + \cdots + a_n = s^{k^2-1}(b_1 + \cdots + b_n) = s^{k^2} = (s^k)^{2009},$$
$$a_1 \cdots a_n = (s^{k^2-1})^n b_1 \cdots b_n = (s^{k^2-1})^n t^{k+1} = (s^{(k-1)n} t)^{2010}$$

we can see that $a_1, \cdots, a_n$ satisfy the conditions on the sum and the product as well. This ends the proof of the assertion.

Remark: We can find the appropriate exponent $k^2 - 1$ needed for the construction of the a_i's by solving the simultaneous congruence relations: $x \equiv 0 \pmod{k+1}$, $x \equiv -1 \pmod{k}$.

Problem 3. Let n be a positive integer. n people take part in a certain party. For any pair of the participants, either the two are acquainted with each other or they are not. What is the maximum possible number of the pairs for which the two are not acquainted but have a common acquaintance among the participants?

Solution: When 1 participant, say the person A, is mutually acquainted with each of the remaining $n-1$ participants, and if there are no other acquaintance relationships among the participants, then for any pair of participants not involving A, the two are not mutual acquaintances, but they have a common acquaintance, namely A, so any such pair satisfies the requirement. Thus, the number desired in this case is $\frac{(n-1)(n-2)}{2} = \frac{n^2-3n+2}{2}$.

Let us show that $\frac{n^2-3n+2}{2}$ is the maximum possible number of the pairs satisfying the requirement of the problem. First, let us observe that in the process of trying to find the maximum possible number of such pairs, if we split the participants into two non-empty subsets T and S which are disjoint, we may assume that there is a pair consisting of one person chosen from T and the other chosen from S who are mutual acquaintances. This is so, since if there are no such pair for some splitting T and S, then among the pairs consisting of one person chosen from T and the other chosen from S, there is no pair for which the two have a common acquaintance among participants, and therefore, if we arbitrarily choose a person $A \in T$ and $B \in S$ and declare that A and B are mutual acquaintances, the number of the pairs satisfying the requirement of the problem does not decrease.

Let us now call a set of participants a *group* if it satisfies the following 2 conditions:

• One can connect any person in the set with any other person in the set by tracing a chain of mutually acquainted pairs. More precisely, for any pair of people A, B in the set there exists a sequence of people $A_0, A_1, \cdots, A_n$ for which $A_0 = A$, $A_n = B$ and, for each $i : 0 \leq i \leq n-1$, A_i and A_{i+1} are mutual acquaintances.

• No person in this set can be connected with a person not belonging to this set by tracing a chain of mutually acquainted pairs.

In view of the discussions made above, we may assume that the set of all the participants to the party forms a group of n people. Let us next consider the following lemma.

Lemma. In a group of n people, there are at least $n-1$ pairs of mutual acquaintances.

Proof: If you choose a mutually acquainted pair in a group and declare the two in the pair are not mutually acquainted, then either the group stays the same or splits into 2 groups. This means that by changing the status of a mutually acquainted pair in a group to that of a non-acquainted pair, one can increase the number of groups at most by 1. Now if in a group of n people you change the status of all of the mutually acquainted pairs to that of non-acquainted pairs, then obviously, the number of groups increases from 1 to n. Therefore, there must be at least $n-1$ pairs of mutually acquainted pairs in a group consisting of n people. □

The lemma implies that there are at most $\frac{n(n-1)}{2} - (n-1) = \frac{n^2-3n+2}{2}$ pairs satisfying the condition of the problem. Thus the desired maximum number of pairs satisfying the requirement of the problem is $\frac{n^2-3n+2}{2}$.

Remark: One can give a somewhat different proof by separating into 2 cases depending on whether there are at least $n-1$ mutually acquainted pairs, or at most $n-2$ such pairs. In the former case, one can argue in the same way as the proof above, while in the latter case, the Lemma above implies that there would be 2 or more groups to start with, but then, in view of the comment made before the definition of a group above, these groups can be combined to form one group, thereby one can reduce the argument to the former case.

Alternate Solution 1: The construction of an example for the case for which the number $\frac{n^2-3n+2}{2}$ appears, and the argument for the case where there is only 1 group would be the same as in the preceding proof.

Suppose, then, n participants are separated into $k(k \geq 2)$ groups, and the number of people in each group is given by $a_i, i = 1, \cdots, k$. In such a case, the number of pairs for which paired people are not mutually acquainted but have a common acquaintance is at most $\sum_{i=1}^{k} {}_{a_i}C_2$, where we set ${}_1C_2 = 0$ for convenience. Since ${}_aC_2 + {}_bC_2 \leq {}_{a+b}C_2$ holds for any pair of positive integers a, b, we have $\sum_{i=1}^{k} {}_{a_i}C_2 \leq {}_{a_1}C_2 + {}_{n-a_1}C_2$. From

$$ {}_{a_1}C_2 + {}_{n-a_1}C_2 = a_1^2 - na_1 + \frac{n^2-n}{2} = \left(a_1 - \frac{n}{2}\right)^2 + \frac{n^2-2n}{4} $$

it follows that ${}_{a_1}C_2 + {}_{n-a_1}C_2$ takes its maximum value when $a_1 = 1, n-1$. Therefore, we have $\sum_{i=1}^{k} {}_{a_i}C_2 \leq {}_{n-1}C_2$, which shows that in the case where the number of groups are 2 or more, the number of the pairs for which paired people are not mutually acquainted but have a common acquaintance is at most ${}_{n-1}C_2 = \frac{n^2-3n+2}{2}$, and hence the desired maximum number of the pairs satisfying the requirement is $\frac{n^2-3n+2}{2}$.

Alternate Solution 2: Construction of an example would be the same as the preceding proof.

For a participant, say A, call another participant, say B, a *familiar face* if A and B are not mutually acquainted but they have a common acquaintance among the participants, and in this case call the pair A, B a familiar pair.

Suppose there is a participant P who is mutually acquainted with d participants. Denote by S the set of these d participants, and by T the set of participants different from P and not belonging to the set S. Suppose there are e pairs formed by a person in S and a person in T who are mutually acquainted.

Then the number of participants who are familiar faces to P is at most e. The number of pairs formed by two people belonging to the set S and are mutually acquainted is at most ${}_dC_2$. The number of familiar pairs formed by two people belonging to the set T is at most ${}_{n-d-1}C_2$. Since there are e pairs formed by a person in the set S and a person in the set T who are mutually acquainted (and so the pairs are not familiar pairs), we have at most $d(n-1-d)-e$ familiar pairs formed by a person chosen from S and a person chosen from T. Putting these together we conclude that there are at most $e + {}_dC_2 + {}_{n-1-d}C_2 + d(n-1-d) - e$ familiar pairs. Since

$$ e + {}_dC_2 + {}_{n-1-d}C_2 + d(n-1-d) - e = \frac{n^2-3n+2}{2}, $$

the number we seek is at most $\frac{n^2-3n+2}{2}$, and hence this is the desired solution to the problem.

Problem 4. Let ABC be an acute triangle satisfying the condition $AB > BC$ and $AC > BC$. Denote by O and H the circumcenter and the orthocenter, respectively, of the triangle ABC. Suppose that the circumcircle of the triangle AHC intersects the line AB at M different from A, and that the circumcircle of the triangle AHB intersects the line AC at N different from A. Prove that the circumcenter of the triangle MNH lies on the line OH.

Solution: In the sequel, we denote $\angle BAC = \alpha, \angle CBA = \beta, \angle ACB = \gamma$. Let O' be the circumcenter of the triangle MNH. The lengths of line segments starting from the point H will be treated as signed quantities.

Let us denote by M', N' the point of intersection of CH, BH, respectively, with the circumcircle of the triangle ABC (distinct from C, B, respectively.) From the fact that 4 points A, M, H, C lie on the same circle, we see that $\angle MHM' = \alpha$ holds. Furthermore, $\angle BM'C, \angle BN'C$ and α are all subtended by the same arc $\overset{\frown}{BC}$ of the circumcircle of the triangle ABC at points on the circle, and therefore, we have $\angle BM'C = \alpha$, and $\angle BN'C = \alpha$ as well. We also have $\angle ABH = \angle ACN'$ as they are subtended by the same arc $\overset{\frown}{AN'}$ of the circumcircle of the triangle ABC at points on the circle. Since $HM' \perp BM, HN' \perp AC$, we conclude that

$$\angle M'HB = 90^\circ - \angle ABH = 90^\circ - \angle ACN' = \alpha$$

is valid as well. Putting these facts together, we obtain the fact that the quadrilateral $HBM'M$ is a rhombus. In a similar manner, we can conclude that the quadrilateral $HCN'N$ is also a rhombus. Since both of these rhombuses are made up of 4 right triangles with an angle of magnitude α, we also see that these rhombuses are similar.

Let us denote by P, Q the feet of the perpendicular lines on HM and HN, respectively, drawn from the point O'. Since O' is the circumcenter of the triangle MNH, P, Q are respectively, the midpoints of the line segments HM, HN. Furthermore, if we denote by R, S the feet of the perpendicular lines on HM and HN, respectively, drawn from the point O, then since O is the circumcenter of both the triangle $M'BC$ and the triangle $N'BC$, we see that R is the intersection point of HM and the perpendicular bisector of BM', and S is the intersection point of HN and the perpendicular bisector of CN'.

We note that the similarity map ϕ between the rhombuses $HBM'M$ and $HCN'N$ carries the perpendicular bisector of BM' onto the perpendicular bisector of CN', and straight line HM onto the straight line HN, and hence ϕ maps R onto S, and P onto Q. Therefore, we get $HP : HR = HQ : HS$. If we now denote by X, Y the intersection points of the line HO' with the line through R and perpendicular to HP, and with the line through S and perpendicular to HQ, respectively, then we get

$$HO' : HX = HP : HR = HQ : HS = HO' : HY$$

so that we must have $HX = HY$, and therefore, $X = Y$. But it is obvious that the point of intersection of the line through R and perpendicular to HP with the line through S and perpendicular to HQ must be O, and therefore, we conclude that $X = Y = O$ and that the points H, O', O are collinear.

Alternate Solution: Deduction of the fact that both of the quadrilaterals $HBM'M$ and $HCN'N$ are rhombuses is carried out in the same way as in the preceding proof.

We then see that the point M is located in a symmetric position with the point B with respect to the line CH, we conclude that we have $\angle CMB = \beta$. Similarly, we have $\angle CNB = \gamma$. If we now put $x = \angle AHO'$, then we get

$$\angle O' = \beta - \alpha - x, \;\; \angle MNH = 90^\circ - \beta - \alpha + x,$$

from which it follows that

$$\angle ANM = 180^\circ - \angle MNH - (90^\circ - \alpha) = \beta - x.$$

Similarly, we get

$$\angle NMA = \gamma + x.$$

Using the laws of sines, we then get

$$\frac{\sin(\gamma+x)}{\sin(\beta-x)} = \frac{AN}{AM} = \frac{AC}{AM}\cdot\frac{AB}{AC}\cdot\frac{AN}{AB}$$
$$= \frac{\sin\beta}{\sin(\beta-\alpha)}\cdot\frac{\sin\gamma}{\sin\beta}\cdot\frac{\sin(\gamma-\alpha)}{\sin\gamma} = \frac{\sin(\gamma-\alpha)}{\sin(\beta-\alpha)}.$$

On the other hand, if we let $y = \angle AHO$, we then get

$$\angle OHB = 180^\circ - \gamma - y,\ \ \angle CHO = 180^\circ - \beta + y,$$

and since

$$\angle HBO = \gamma - \alpha,\ \ \angle OCH = \beta - \alpha,$$

using the laws of sines and observing that $OB = OC$, we get

$$\frac{\sin(\gamma-\alpha)}{\sin(\beta-\alpha)} = \frac{\sin\angle HBO}{\sin\angle OCH} = \frac{\sin(180^\circ-\gamma-y)\cdot\frac{OH}{OB}}{\sin(180^\circ-\beta+y)\cdot\frac{OH}{OC}}$$
$$= \frac{\sin(180^\circ-\gamma-y)}{\sin(180^\circ-\beta+y)} = \frac{\sin(\gamma+y)}{\sin(\beta-y)}.$$

We then get $\sin(\gamma+x)\sin(\beta-y) = \sin(\beta-x)\sin(\gamma+y)$. Expanding both sides of the last identity by using the addition formula for the sine function and after factoring and using again the addition formula we obtain that $\sin(x-y)\sin(\beta+\gamma) = 0$. This implies that $x-y$ must be an integral multiple of 180°, and hence we conclude that H, O, O' are collinear.

Problem 5. Find all functions f from the set $\mathbf{R}$ of real numbers into $\mathbf{R}$ which satisfy for all $x, y, z \in \mathbf{R}$ the identity

$$f(f(x)+f(y)+f(z)) = f(f(x)-f(y)) + f(2xy+f(z)) + 2f(xz-yz).$$

Solution: It is clear that if f is a constant function which satisfies the given equation, then the constant must be 0. Conversely, $f(x) = 0$ clearly satisfies the given equation, so, the identically 0 function is a solution. In the sequel, we consider the case where f is not a constant function.

Let $t \in \mathbf{R}$ and substitute $(x, y, z) = (t, 0, 0)$ and $(x, y, z) = (0, t, 0)$ into the given functional equation. Then, we obtain, respectively,

$$f(f(t)+2f(0)) = f(f(t)-f(0)) + f(f(0)) + 2f(0),$$
$$f(f(t)+2f(0)) = f(f(0)-f(t)) + f(f(0)) + 2f(0),$$

from which we conclude that $f(f(t)-f(0)) = f(f(0)-f(t))$ holds for all $t \in \mathbf{R}$. Now, suppose for some pair u_1, u_2, $f(u_1) = f(u_2)$ is satisfied. Then by substituting $(x, y, z) = (s, 0, u_1)$ and $(x, y, z) = (s, 0, u_2)$ into the functional equation and comparing the resulting identities, we can easily conclude that

$$f(su_1) = f(su_2) \qquad (*)$$

holds for all $s \in \mathbf{R}$. Since f is not a constant function there exists an s_0 such that $f(s_0)-f(0) \neq 0$. If we put $u_1 = f(s_0) - f(0), u_2 = -u_1$, then $f(u_1) = f(u_2)$, so we have by $(*)$

$$f(su_1) = f(su_2) = f(-su_1)$$

for all $s \in \mathbf{R}$. Since $u_1 \neq 0$, we conclude that

$$f(x) = f(-x)$$

holds for all $x \in \mathbf{R}$.

Next, if $f(u) = f(0)$ for some $u \neq 0$, then by $(*)$, we have $f(su) = f(s0) = f(0)$ for all s, which implies that f is a constant function, contradicting our assumption. Therefore, we must have $f(s) \neq f(0)$ whenever $s \neq 0$.

We will now show that if $f(x) = f(y)$ holds, then either $x = y$ or $x = -y$ must hold. Suppose on the contrary that $f(x_0) = f(y_0)$ holds for some pair of non-zero numbers x_0, y_0 for which $x_0 \neq y_0, x_0 \neq -y_0$. Since $f(-y_0) = f(y_0)$, we may assume, by replacing y_0 by $-y_0$ if necessary, that x_0 and y_0 have the same sign. In view of $(*)$, we see that $f(sx_0) = f(sy_0)$ holds for all s, and therefore, there exists some $r > 0, r \neq 1$ such that

$$f(x) = f(rx)$$

holds for all x. Replacing x by rx and y by ry in the given functional equation, we obtain

$$f(f(rx) + f(ry) + f(z)) = f(f(rx) - f(ry)) + f(2r^2xy + f(z)) + 2f(r(x-y)z) \qquad \text{(i)},$$

and replacing x by r^2x in the functional equation, we get

$$f(f(r^2x) + f(y) + f(z)) = f(f(r^2x) - f(y)) + f(2r^2xy + f(z)) + 2f((r^2x - y)z) \quad \text{(ii)}.$$

Since $f(rx) = f(x)$ holds for all $x \in \mathbf{R}$, we see that except for the last term on the right-hand side, all the corresponding terms appearing in the identities (i) and (ii) above are equal, and hence we conclude that

$$f(r(x-y)z) = f((r^2x - y)z)) \qquad \text{(iii)}$$

must hold for arbitrary choice of $x, y, z \in \mathbf{R}$. For arbitrarily fixed pair $u, v \in \mathbf{R}$, substitute $(x, y, z) = (\frac{v-u}{r^2-1}, \frac{v-r^2u}{r^2-1}, 1)$ into the identity (iii). Then we obtain $f(v) = f(ru) = f(u)$, since $x - y = u, r^2x - y = v, z = 1$. But this implies that the function f is a constant, contradicting our assumption. Thus we conclude that if $f(x) = f(y)$ then either $x = y$ or $x = -y$ must hold.

By substituting $z = 0$ in the functional equation, we get

$$f(f(x) + f(y) + f(0)) = f(f(x) - f(y) + f(0)) = f((f(x) - f(y)) + f(2xy + f(0)) + 2f(0).$$

Changing y to $-y$ in the identity above and using the fact that $f(y) = f(-y)$, we see that all the terms except the second term on the right-hand side in the identity above remain the same. Thus we conclude that $f(2xy + f(0)) = f(-2xy + f(0))$, from which we get either $2xy + f(0) = -2xy + f(0)$ or $2xy + f(0) = 2xy - f(0)$ for all $x, y \in \mathbf{R}$. The first of these alternatives says that $4xy = 0$, which is impossible if $xy \neq 0$. Therefore the second alternative must be valid and we get that $f(0) = 0$.

Finally, let us show that if f satisfies the given functional equation and is not a constant function, then $f(x) = x^2$. Let $x = y$ in the functional equation, then since $f(0) = 0$, we get

$$f(2f(x) + f(z)) = f(2x^2 + f(z)),$$

from which we conclude that either $2f(x) + f(z) = 2x^2 + f(z)$ or $2f(x) + f(z) = -2x^2 - f(z)$ must hold. Suppose there exists x_0 for which $f(x_0) \neq x_0^2$, then from the second alternative, we see that $f(z) = -f(x_0) - x_0^2$ must hold for all z, which means that f must be a constant function, contrary to our assumption. Therefore, the first alternative above must hold, and we have $f(x) = x^2$ for all x, establishing our claim.

It is easy to check that $f(x) = x^2$ does satisfy the given functional equation, so we conclude that $f(x) = 0$ and $f(x) = x^2$ are the only functions that satisfy the requirement.

XXI Asian Pacific Mathematics Olympiad

March, 2009

Problem 1. Consider the following operation on positive real numbers written on a blackboard:

> Choose a number r written on the blackboard, erase that number, and then write a pair of positive real numbers a and b satisfying the condition $2r^2 = ab$ on the board.

Assume that you start out with just one positive real number r on the blackboard, and apply this operation $k^2 - 1$ times to end up with k^2 positive real numbers, not necessarily distinct. Show that there exists a number on the board which does not exceed kr.

(Solution) Using AM-GM inequality, we obtain

$$\frac{1}{r^2} = \frac{2}{ab} = \frac{2ab}{a^2b^2} \leq \frac{a^2+b^2}{a^2b^2} \leq \frac{1}{a^2} + \frac{1}{b^2}. \qquad (*)$$

Consequently, if we let S_ℓ be the sum of the squares of the reciprocals of the numbers written on the board after ℓ operations, then S_ℓ increases as ℓ increases, that is,

$$S_0 \leq S_1 \leq \cdots \leq S_{k^2-1}. \qquad (**)$$

Therefore if we let s be the smallest real number written on the board after $k^2 - 1$ operations, then $\frac{1}{s^2} \geq \frac{1}{t^2}$ for any number t among k^2 numbers on the board and hence

$$k^2 \times \frac{1}{s^2} \geq S_{k^2-1} \geq S_0 = \frac{1}{r^2},$$

which implies that $s \leq kr$ as desired. □

Remark. The nature of the problem does not change at all if the numbers on the board are restricted to be positive integers. But that may mislead some contestants to think the problem is a number theoretic problem rather than a combinatorial problem.

Problem 2. Let a_1, a_2, a_3, a_4, a_5 be real numbers satisfying the following equations:

$$\frac{a_1}{k^2+1} + \frac{a_2}{k^2+2} + \frac{a_3}{k^2+3} + \frac{a_4}{k^2+4} + \frac{a_5}{k^2+5} = \frac{1}{k^2} \quad \text{for } k = 1, 2, 3, 4, 5.$$

Find the value of $\frac{a_1}{37} + \frac{a_2}{38} + \frac{a_3}{39} + \frac{a_4}{40} + \frac{a_5}{41}$. (Express the value in a single fraction.)

(Solution) Let $R(x) := \frac{a_1}{x^2+1} + \frac{a_2}{x^2+2} + \frac{a_3}{x^2+3} + \frac{a_4}{x^2+4} + \frac{a_5}{x^2+5}$. Then $R(\pm 1) = 1$, $R(\pm 2) = \frac{1}{4}$, $R(\pm 3) = \frac{1}{9}$, $R(\pm 4) = \frac{1}{16}$, $R(\pm 5) = \frac{1}{25}$ and $R(6)$ is the value to be found. Let's put $P(x) := (x^2+1)(x^2+2)(x^2+3)(x^2+4)(x^2+5)$ and $Q(x) := R(x)P(x)$. Then for $k = \pm 1, \pm 2, \pm 3, \pm 4, \pm 5$, we get $Q(k) = R(k)P(k) = \frac{P(k)}{k^2}$, that is, $P(k) - k^2 Q(k) = 0$. Since $P(x) - x^2 Q(x)$ is a polynomial of degree 10 with roots $\pm 1, \pm 2, \pm 3, \pm 4, \pm 5$, we get

$$P(x) - x^2 Q(x) = A(x^2-1)(x^2-4)(x^2-9)(x^2-16)(x^2-25). \qquad (*)$$

Putting $x = 0$, we get $A = \frac{P(0)}{(-1)(-4)(-9)(-16)(-25)} = -\frac{1}{120}$. Finally, dividing both sides of $(*)$ by $P(x)$ yields

$$1 - x^2 R(x) = 1 - x^2 \frac{Q(x)}{P(x)} = -\frac{1}{120} \cdot \frac{(x^2-1)(x^2-4)(x^2-9)(x^2-16)(x^2-25)}{(x^2+1)(x^2+2)(x^2+3)(x^2+4)(x^2+5)}$$

and hence that

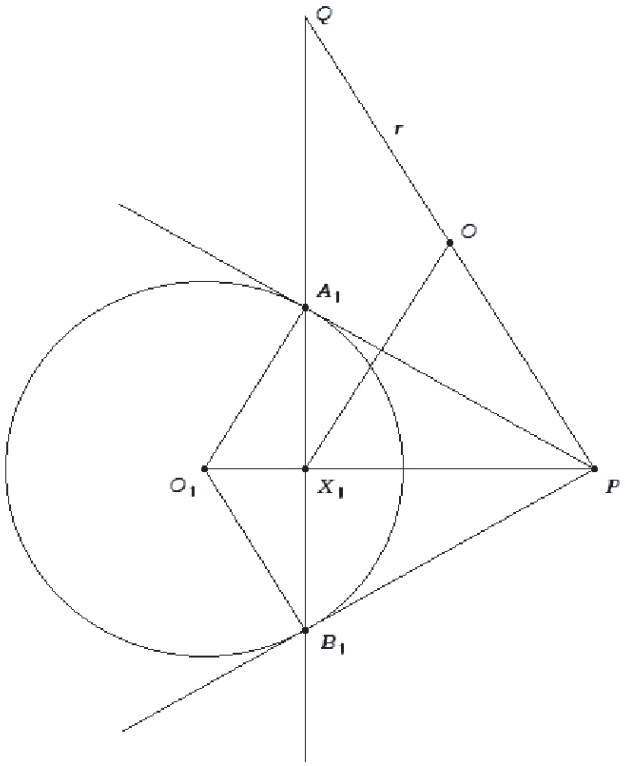
$$1 - 36R(6) = -\frac{35 \times 32 \times 27 \times 20 \times 11}{120 \times 37 \times 38 \times 39 \times 40 \times 41} = -\frac{3 \times 7 \times 11}{13 \times 19 \times 37 \times 41} = -\frac{231}{374699},$$

which implies $R(6) = \frac{187465}{6744582}$.

Remark. We can get $a_1 = \frac{1105}{72}, a_2 = -\frac{2673}{40}, a_3 = \frac{1862}{15}, a_4 = -\frac{1885}{18}, a_5 = \frac{1323}{40}$ by solving the given system of linear equations, which is extremely messy and takes a lot of time.

Problem 3. Let three circles $\Gamma_1, \Gamma_2, \Gamma_3$, which are non-overlapping and mutually external, be given in the plane. For each point P in the plane, outside the three circles, construct six points $A_1, B_1, A_2, B_2, A_3, B_3$ as follows: For each $i = 1, 2, 3$, A_i, B_i are distinct points on the circle Γ_i such that the lines PA_i and PB_i are both tangents to Γ_i. Call the point P *exceptional* if, from the construction, three lines A_1B_1, A_2B_2, A_3B_3 are concurrent. Show that every exceptional point of the plane, if exists, lies on the same circle.

(Solution) Let O_i be the center and r_i the radius of circle Γ_i for each $i = 1, 2, 3$. Let P be an exceptional point, and let the three corresponding lines A_1B_1, A_2B_2,A_3B_3 concur at Q. Construct the circle with diameter PQ. Call the circle Γ, its center O and its radius r. We now claim that all exceptional points lie on Γ.



Let PO_1 intersect A_1B_1 in X_1. As $PO_1 \perp A_1B_1$, we see that X_1 lies on Γ. As PA_1 is a tangent to Γ_1, triangle PA_1O_1 is right-angled and similar to triangle $A_1X_1O_1$. It follows that

$$\frac{O_1X_1}{O_1A_1} = \frac{O_1A_1}{O_1P}, \quad \text{i.e.,} \quad O_1X_1 \cdot O_1P = {O_1A_1}^2 = {r_1}^2.$$

On the other hand, $O_1X_1 \cdot O_1P$ is also the power of O_1 with respect to Γ, so that

$$r_1^2 = O_1X_1 \cdot O_1P = (O_1O - r)(O_1O + r) = O_1O^2 - r^2, \qquad (*)$$

and hence

$$r^2 = OO_1^2 - r_1^2 = (OO_1 - r_1)(OO_1 + r_1).$$

Thus, r^2 is the power of O with respect to Γ_1. By the same token, r^2 is also the power of O with respect to Γ_2 and Γ_3. Hence O must be the radical center of the three given circles. Since r, as the square root of the power of O with respect to the three given circles, does not depend on P, it follows that all exceptional points lie on Γ. □

Remark. In the event of the radical point being at infinity (and hence the three radical axes being parallel), there are no exceptional points in the plane, which is consistent with the statement of the problem.

Problem 4. Prove that for any positive integer k, there exists an arithmetic sequence

$$\frac{a_1}{b_1}, \quad \frac{a_2}{b_2}, \quad \cdots, \quad \frac{a_k}{b_k}$$

of rational numbers, where a_i, b_i are relatively prime positive integers for each $i = 1, 2, \ldots, k$, such that the positive integers $a_1, b_1, a_2, b_2, \ldots, a_k, b_k$ are all distinct.

(Solution) For $k = 1$, there is nothing to prove. Henceforth assume $k \geq 2$.
Let $p_1, p_2, \ldots, p_k$ be k distinct primes such that

$$k < p_k < \cdots < p_2 < p_1$$

and let $N = p_1 p_2 \cdots p_k$. By Chinese Remainder Theorem, there exists a positive integer x satisfying

$$x \equiv -i \pmod{p_i}$$

for all $i = 1, 2, \ldots, k$ and $x > N^2$. Consider the following sequence:

$$\frac{x+1}{N}, \quad \frac{x+2}{N}, \quad , \ldots, \quad \frac{x+k}{N}.$$

This sequence is obviously an arithmetic sequence of positive rational numbers of length k. For each $i = 1, 2, \ldots, k$, the numerator $x + i$ is divisible by p_i but not by p_j for $j \neq i$, for otherwise p_j divides $|i - j|$, which is not possible because $p_j > k > |i - j|$. Let

$$a_i := \frac{x+i}{p_i}, \quad b_i := \frac{N}{p_i} \qquad \text{for all } \ i = 1, 2, \ldots, k.$$

Then

$$\frac{x+i}{N} = \frac{a_i}{b_i}, \quad \gcd(a_i, b_i) = 1 \qquad \text{for all } \ i = 1, 2, \ldots, k,$$

and all b_i's are distinct from each other. Moreover, $x > N^2$ implies

$$a_i = \frac{x+i}{p_i} > \frac{N^2}{p_i} > N > \frac{N}{p_j} = b_j \qquad \text{for all } \ i, j = 1, 2, \ldots, k$$

and hence all a_i's are distinct from b_i's. It only remains to show that all a_i's are distinct from each other. This follows from

$$a_j = \frac{x+j}{p_j} > \frac{x+i}{p_j} > \frac{x+i}{p_i} = a_i \qquad \text{for all } \ i < j$$

by our choice of $p_1, p_2, \ldots, p_k$. Thus, the arithmetic sequence

$$\frac{a_1}{b_1}, \quad \frac{a_2}{b_2}, \quad \cdots, \quad \frac{a_k}{b_k}$$

of positive rational numbers satisfies the conditions of the problem. □

Remark. Here is a much easier solution :

For any positive integer $k \geq 2$, consider the sequence

$$\frac{(k!)^2+1}{k!}, \frac{(k!)^2+2}{k!}, \ldots, \frac{(k!)^2+k}{k!}.$$

Note that $\gcd(k!, (k!)^2+i) = i$ for all $i = 1, 2, \ldots, k$. So, taking

$$a_i := \frac{(k!)^2+i}{i}, \quad b_i := \frac{k!}{i} \qquad \text{for all } \ i = 1, 2, \ldots, k,$$

we have $\gcd(a_i, b_i) = 1$ and

$$a_i = \frac{(k!)^2+i}{i} > a_j = \frac{(k!)^2+j}{j} > b_i = \frac{k!}{i} > b_j = \frac{k!}{j}$$

for any $1 \leq i < j \leq k$. Therefore this sequence satisfies every condition given in the problem.

Problem 5. Larry and Rob are two robots travelling in one car from Argovia to Zillis. Both robots have control over the steering and steer according to the following algorithm: Larry makes a 90° left turn after every ℓ kilometer driving from start; Rob makes a 90° right turn after every r kilometer driving from start, where ℓ and r are relatively prime positive integers. In the event of both turns occurring simultaneously, the car will keep going without changing direction. Assume that the ground is flat and the car can move in any direction.

Let the car start from Argovia facing towards Zillis. For which choices of the pair (ℓ, r) is the car guaranteed to reach Zillis, regardless of how far it is from Argovia?

(Solution) Let Zillis be d kilometers away from Argovia, where d is a positive real number. For simplicity, we will position Argovia at $(0,0)$ and Zillis at $(d,0)$, so that the car starts out facing east. We will investigate how the car moves around in the period of travelling the first ℓr kilometers, the second ℓr kilometers, ..., and so on. We call each period of travelling ℓr kilometers a *section*. It is clear that the car will have identical behavior in every section except the direction of the car at the beginning.

`Case 1:` $\ell - r \equiv 2 \pmod 4$. After the first section, the car has made $\ell - 1$ right turns and $r - 1$ left turns, which is a net of $2(\equiv \ell - r \pmod 4)$ right turns. Let the displacement vector for the first section be (x, y). Since the car has rotated 180°, the displacement vector for the second section will be $(-x, -y)$, which will take the car back to $(0,0)$ facing east again. We now have our original situation, and the car has certainly never travelled further than ℓr kilometers from Argovia. So, the car cannot reach Zillis if it is further apart from Argovia.

`Case 2:` $\ell - r \equiv 1 \pmod 4$. After the first section, the car has made a net of 1 right turn. Let the displacement vector for the first section again be (x, y). This time the car has rotated 90° clockwise. We can see that the displacements for the second, third and fourth section will be $(y, -x)$, $(-x, -y)$ and $(-y, x)$, respectively, so after four sections the car is back at $(0,0)$ facing east. Since the car has certainly never travelled further than $2\ell r$ kilometers from Argovia, the car cannot reach Zillis if it is further apart from Argovia.

`Case 3:` $\ell - r \equiv 3 \pmod 4$. An argument similar to that in `Case 2` (switching the roles of left and right) shows that the car cannot reach Zillis if it is further apart from Argovia.

`Case 4:` $\ell \equiv r \pmod 4$. The car makes a net turn of 0° after each section, so it must be facing east. We are going to show that, after traversing the first section, the car will be at $(1,0)$. It will be useful to interpret the Cartesian plane as the complex plane, i.e. writing $x + iy$ for (x, y), where $i = \sqrt{-1}$. We will denote the k-th kilometer of movement by m_{k-1},

which takes values from the set $\{1, i, -1, -i\}$, depending on the direction. We then just have to show that

$$\sum_{k=0}^{\ell r-1} m_k = 1,$$

which implies that the car will get to Zillis no matter how far it is apart from Argovia.

`Case 4a:` $\underline{\ell \equiv r \equiv 1 \pmod 4}$. First note that for $k = 0, 1, \ldots, \ell r - 1$,

$$m_k = i^{\lfloor k/\ell \rfloor}(-i)^{\lfloor k/r \rfloor}$$

since $\lfloor k/\ell \rfloor$ and $\lfloor k/r \rfloor$ are the exact numbers of left and right turns before the $(k+1)$st kilometer, respectively. Let $a_k (\equiv k \pmod \ell)$ and $b_k (\equiv k \pmod r)$ be the remainders of k when divided by ℓ and r, respectively. Then, since

$$a_k = k - \left\lfloor \frac{k}{\ell} \right\rfloor \ell \equiv k - \left\lfloor \frac{k}{\ell} \right\rfloor \pmod 4 \quad \text{and} \quad b_k = k - \left\lfloor \frac{k}{r} \right\rfloor r \equiv k - \left\lfloor \frac{k}{r} \right\rfloor \pmod 4,$$

we have $\lfloor k/\ell \rfloor \equiv k - a_k \pmod 4$ and $\lfloor k/r \rfloor \equiv k - b_k \pmod 4$. We therefore have

$$m_k = i^{k-a_k}(-i)^{k-b_k} = (-i^2)^k i^{-a_k}(-i)^{-b_k} = (-i)^{a_k}\, i^{b_k}\,.$$

As ℓ and r are relatively prime, by Chinese Remainder Theorem, there is a bijection between pairs $(a_k, b_k) = (k \pmod \ell, k \pmod r)$ and the numbers $k = 0, 1, 2, \ldots, \ell r - 1$. Hence

$$\sum_{k=0}^{\ell r-1} m_k = \sum_{k=0}^{\ell r-1} (-i)^{a_k}\, i^{b_k} = \left(\sum_{k=0}^{\ell-1} (-i)^{a_k} \right) \left(\sum_{k=0}^{r-1} i^{b_k} \right) = 1 \times 1 = 1$$

as required because $\ell \equiv r \equiv 1 \pmod 4$.

`Case 4b:` $\underline{\ell \equiv r \equiv 3 \pmod 4}$. In this case, we get

$$m_k = i^{a_k}(-i)^{b_k},$$

where $a_k (\equiv k \pmod \ell)$ and $b_k (\equiv k \pmod r)$ for $k = 0, 1, \ldots, \ell r - 1$. Then we can proceed analogously to `Case 4a` to obtain

$$\sum_{k=0}^{\ell r-1} m_k = \sum_{k=0}^{\ell r-1} (-i)^{a_k}\, i^{b_k} = \left(\sum_{k=0}^{\ell-1} (-i)^{a_k} \right) \left(\sum_{k=0}^{r-1} i^{b_k} \right) = i \times (-i) = 1$$

as required because $\ell \equiv r \equiv 3 \pmod 4$.

Now clearly the car traverses through all points between $(0, 0)$ and $(1, 0)$ during the first section and, in fact, covers all points between $(n-1, 0)$ and $(n, 0)$ during the n-th section. Hence it will eventually reach $(d, 0)$ for any positive d.

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To summarize: (ℓ, r) satisfies the required conditions if and only if

$$\ell \equiv r \equiv 1 \pmod 4 \quad \text{or} \quad \ell \equiv r \equiv 3 \pmod 4. \qquad \square$$

Remark. In case $\gcd(\ell, r) = d \neq 1$, the answer is:

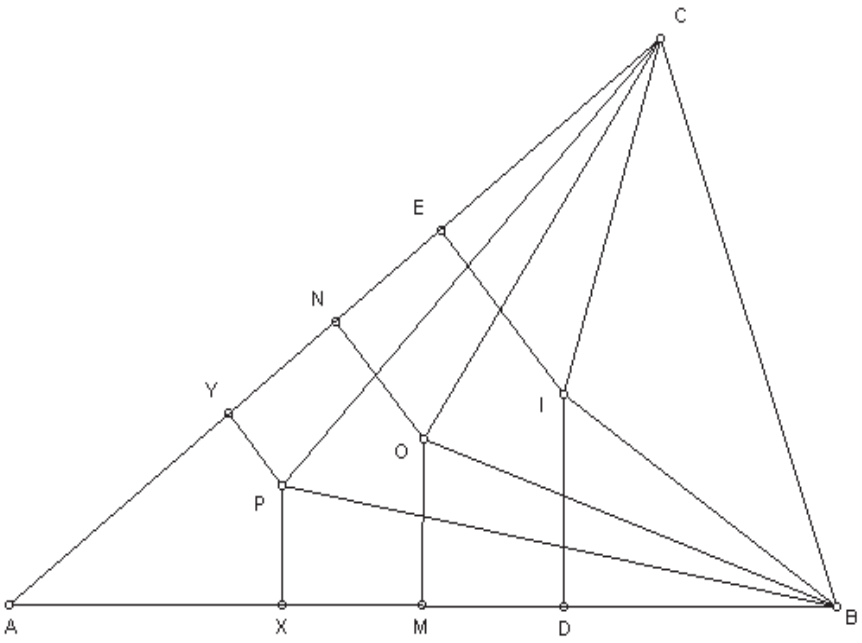
$$\frac{\ell}{d} \equiv \frac{r}{d} \equiv 1 \pmod 4 \quad \text{or} \quad \frac{\ell}{d} \equiv \frac{r}{d} \equiv 3 \pmod 4.$$

XX Asian Pacific Mathematics Olympiad

March, 2008

Problem 1. Let ABC be a triangle with $\angle A < 60°$. Let X and Y be the points on the sides AB and AC, respectively, such that $CA + AX = CB + BX$ and $BA + AY = BC + CY$. Let P be the point in the plane such that the lines PX and PY are perpendicular to AB and AC, respectively. Prove that $\angle BPC < 120°$.

(Solution) Let I be the incenter of $\triangle ABC$, and let the feet of the perpendiculars from I to AB and to AC be D and E, respectively. (Without loss of generality, we may assume that AC is the longest side. Then X lies on the line segment AD. Although P may or may not lie inside $\triangle ABC$, the proof below works for both cases. Note that P is on the line perpendicular to AB passing through X.) Let O be the midpoint of IP, and let the feet of the perpendiculars from O to AB and to AC be M and N, respectively. Then M and N are the midpoints of DX and EY, respectively.



The conditions on the points X and Y yield the equations

$$AX = \frac{AB + BC - CA}{2} \qquad \text{and} \qquad AY = \frac{BC + CA - AB}{2}.$$

From $AD = AE = \dfrac{CA + AB - BC}{2}$, we obtain

$$BD = AB - AD = AB - \frac{CA + AB - BC}{2} = \frac{AB + BC - CA}{2} = AX.$$

Since M is the midpoint of DX, it follows that M is the midpoint of AB. Similarly, N is the midpoint of AC. Therefore, the perpendicular bisectors of AB and AC meet at O, that is, O is the circumcenter of $\triangle ABC$. Since $\angle BAC < 60^\circ$, O lies on the same side of BC as the point A and

$$\angle BOC = 2\angle BAC.$$

We can compute $\angle BIC$ as follows:

$$\begin{aligned}\angle BIC &= 180^\circ - \angle IBC - \angle ICB = 180^\circ - \frac{1}{2}\angle ABC - \frac{1}{2}\angle ACB \\ &= 180^\circ - \frac{1}{2}(\angle ABC + \angle ACB) = 180^\circ - \frac{1}{2}(180^\circ - \angle BAC) = 90^\circ + \frac{1}{2}\angle BAC\end{aligned}$$

It follows from $\angle BAC < 60^\circ$ that

$$2\angle BAC < 90^\circ + \frac{1}{2}\angle BAC, \quad \text{i.e.,} \quad \angle BOC < \angle BIC.$$

From this it follows that I lies inside the circumcircle of the isosceles triangle BOC because O and I lie on the same side of BC. However, as O is the midpoint of IP, P must lie outside the circumcircle of triangle BOC and on the same side of BC as O. Therefore

$$\angle BPC < \angle BOC = 2\angle BAC < 120^\circ.$$

Remark. If one assumes that $\angle A$ is smaller than the other two, then it is clear that the line PX (or the line perpendicular to AB at X if $P = X$) runs through the excenter I_C of the excircle tangent to the side AB. Since $2\angle ACI_C = \angle ACB$ and $BC < AC$, we have $2\angle PCB > \angle C$. Similarly, $2\angle PBC > \angle B$. Therefore,

$$\angle BPC = 180^\circ - (\angle PBC + \angle PCB) < 180^\circ - \left(\frac{\angle B + \angle C}{2}\right) = 90 + \frac{\angle A}{2} < 120^\circ.$$

In this way, a special case of the problem can be easily proved.

Problem 2. Students in a class form groups each of which contains exactly three members such that any two distinct groups have at most one member in common. Prove that, when the class size is 46, there is a set of 10 students in which no group is properly contained.

(Solution) We let C be the set of all 46 students in the class and let

$$s := \max\{\, |S| \,:\, S \subseteq C \text{ such that } S \text{ contains no group properly} \,\}.$$

Then it suffices to prove that $s \geq 10$. (If $|S| = s > 10$, we may choose a subset of S consisting of 10 students.)

Suppose that $s \leq 9$ and let S be a set of size s in which no group is properly contained. Take any student, say v, from outside S. Because of the maximality of s, there should be a group containing the student v and two other students in S. The number of ways to choose two students from S is

$$\binom{s}{2} \leq \binom{9}{2} = 36.$$

On the other hand, there are at least $37 = 46 - 9$ students outside of S. Thus, among those 37 students outside, there is at least one student, say u, who does not belong to any group containing two students in S and one outside. This is because no two distinct groups have two members in common. But then, S can be enlarged by including u, which is a contradiction.

Remark. One may choose a subset S of C that contains no group properly. Then, assuming $|S| < 10$, prove that there is a student outside S, say u, who does not belong to any group containing two students in S. After enlarging S by including u, prove that the enlarged S still contains no group properly.

Problem 3. Let Γ be the circumcircle of a triangle ABC. A circle passing through points A and C meets the sides BC and BA at D and E, respectively. The lines AD and CE meet Γ again at G and H, respectively. The tangent lines of Γ at A and C meet the line DE at L and M, respectively. Prove that the lines LH and MG meet at Γ.

(Solution) Let MG meet Γ at P. Since $\angle MCD = \angle CAE$ and $\angle MDC = \angle CAE$, we have $MC = MD$. Thus

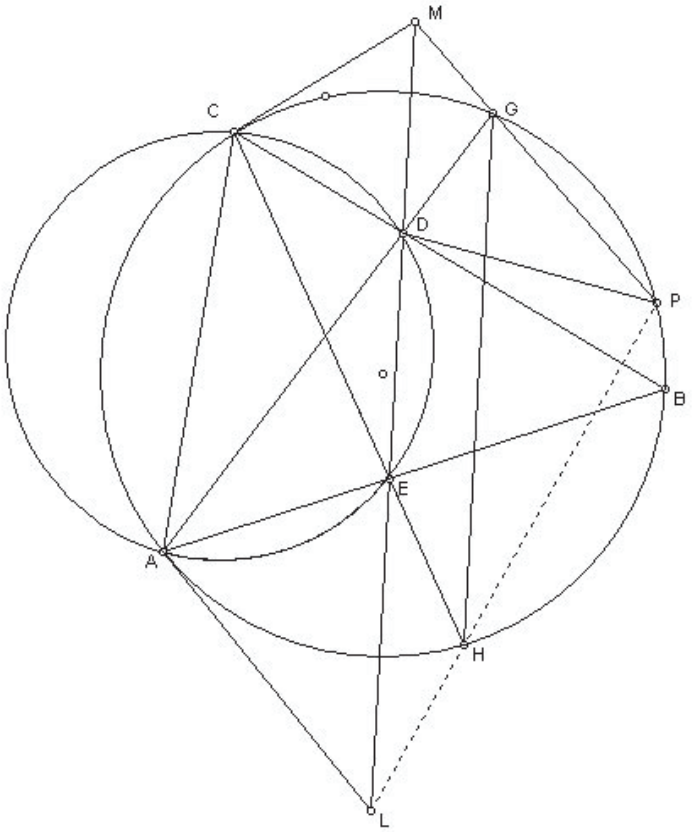
$$MD^2 = MC^2 = MG \cdot MP$$

and hence MD is tangent to the circumcircle of $\triangle DGP$. Therefore $\angle DGP = \angle EDP$.

Let Γ' be the circumcircle of $\triangle BDE$. If $B = P$, then, since $\angle BGD = \angle BDE$, the tangent lines of Γ' and Γ at B should coincide, that is Γ' is tangent to Γ from inside. Let $B \neq P$. If P lies in the same side of the line BC as G, then we have

$$\angle EDP + \angle ABP = 180^\circ$$

because $\angle DGP + \angle ABP = 180^\circ$. That is, the quadrilateral $BPDE$ is cyclic, and hence P is on the intersection of Γ' with Γ.



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Otherwise,

$$\angle EDP = \angle DGP = \angle AGP = \angle ABP = \angle EBP.$$

Therefore the quadrilateral $PBDE$ is cyclic, and hence P again is on the intersection of Γ' with Γ.

Similarly, if LH meets Γ at Q, we either have $Q = B$, in which case Γ' is tangent to Γ from inside, or $Q \neq B$. In the latter case, Q is on the intersection of Γ' with Γ. In either case, we have $P = Q$.

Problem 4. Consider the function $f : \mathbb{N}_0 \to \mathbb{N}_0$, where $\mathbb{N}_0$ is the set of all non-negative integers, defined by the following conditions:

$$\text{(i) } f(0) = 0, \text{ (ii) } f(2n) = 2f(n) \text{ and (iii) } f(2n+1) = n + 2f(n) \text{ for all } n \geq 0.$$

(a) Determine the three sets $L := \{\, n \mid f(n) < f(n+1) \,\}$, $E := \{\, n \mid f(n) = f(n+1) \,\}$, and $G := \{\, n \mid f(n) > f(n+1) \,\}$.

(b) For each $k \geq 0$, find a formula for $a_k := \max\{f(n) : 0 \leq n \leq 2^k\}$ in terms of k.

(Solution) (a) Let

$$L_1 := \{2k : k > 0\}, \quad E_1 := \{0\} \cup \{4k+1 : k \geq 0\}, \quad \text{and} \quad G_1 := \{4k+3 : k \geq 0\}.$$

We will show that $L_1 = L, E_1 = E$, and $G_1 = G$. It suffices to verify that $L_1 \subseteq E$, $E_1 \subseteq E$, and $G_1 \subseteq G$ because L_1, E_1, and G_1 are mutually disjoint and $L_1 \cup E_1 \cup G_1 = \mathbb{N}_0$.

Firstly, if $k > 0$, then $f(2k) - f(2k+1) = -k < 0$ and therefore $L_1 \subseteq L$.

Secondly, $f(0) = 0$ and

$$\begin{aligned} f(4k+1) &= 2k + 2f(2k) = 2k + 4f(k) \\ f(4k+2) &= 2f(2k+1) = 2(k + 2f(k)) = 2k + 4f(k) \end{aligned}$$

for all $k \geq 0$. Thus, $E_1 \subseteq E$.

Lastly, in order to prove $G_1 \subset G$, we claim that $f(n+1) - f(n) \leq n$ for all n. (In fact, one can prove a stronger inequality: $f(n+1) - f(n) \leq n/2$.) This is clearly true for even n from the definition since for $n = 2t$,

$$f(2t+1) - f(2t) = t \leq n.$$

If $n = 2t+1$ is odd, then (assuming inductively that the result holds for all nonnegative $m < n$), we have

$$\begin{aligned} f(n+1) - f(n) &= f(2t+2) - f(2t+1) = 2f(t+1) - t - 2f(t) \\ &= 2(f(t+1) - f(t)) - t \leq 2t - t = t < n. \end{aligned}$$

For all $k \geq 0$,

$$\begin{aligned} &f(4k+4) - f(4k+3) = f(2(2k+2)) - f(2(2k+1)+1) \\ &= 4f(k+1) - (2k+1+2f(2k+1)) = 4f(k+1) - (2k+1+2k+4f(k)) \\ &= 4(f(k+1) - f(k)) - (4k+1) \leq 4k - (4k+1) < 0. \end{aligned}$$

This proves $G_1 \subseteq G$.

(b) Note that $a_0 = a_1 = f(1) = 0$. Let $k \geq 2$ and let $N_k = \{0, 1, 2, \ldots, 2^k\}$. First we claim that the maximum a_k occurs at the largest number in $G \cap N_k$, that is, $a_k = f(2^k - 1)$. We use mathematical induction on k to prove the claim. Note that $a_2 = f(3) = f(2^2 - 1)$.

Now let $k \geq 3$. For every even number $2t$ with $2^{k-1} + 1 < 2t \leq 2^k$,

$$f(2t) = 2f(t) \leq 2a_{k-1} = 2f(2^{k-1} - 1) \tag{$\dagger$}$$

by induction hypothesis. For every odd number $2t+1$ with $2^{k-1} + 1 \leq 2t + 1 < 2^k$,

$$\begin{aligned} f(2t+1) &= t + 2f(t) \leq 2^{k-1} - 1 + 2f(t) \\ &\leq 2^{k-1} - 1 + 2a_{k-1} = 2^{k-1} - 1 + 2f(2^{k-1} - 1) \end{aligned} \tag{$\ddagger$}$$

again by induction hypothesis. Combining (†), (‡) and

$$f(2^k-1) = f(2(2^{k-1}-1)+1) = 2^{k-1}-1+2f(2^{k-1}-1),$$

we may conclude that $a_k = f(2^k-1)$ as desired.

Furthermore, we obtain

$$a_k = 2a_{k-1} + 2^{k-1} - 1$$

for all $k \geq 3$. Note that this recursive formula for a_k also holds for $k \geq 0, 1$ and 2. Unwinding this recursive formula, we finally get

$$\begin{aligned} a_k &= 2a_{k-1} + 2^{k-1} - 1 = 2(2a_{k-2} + 2^{k-2} - 1) + 2^{k-1} - 1 \\ &= 2^2 a_{k-2} + 2 \cdot 2^{k-1} - 2 - 1 = 2^2(2a_{k-3} + 2^{k-3} - 1) + 2 \cdot 2^{k-1} - 2 - 1 \\ &= 2^3 a_{k-3} + 3 \cdot 2^{k-1} - 2^2 - 2 - 1 \\ &\vdots \\ &= 2^k a_0 + k2^{k-1} - 2^{k-1} - 2^{k-2} - ... - 2 - 1 \\ &= k2^{k-1} - 2^k + 1 \quad \text{for all } k \geq 0. \end{aligned}$$

Problem 5. Let a, b, c be integers satisfying $0 < a < c - 1$ and $1 < b < c$. For each k, $0 \le k \le a$, let r_k, $0 \le r_k < c$, be the remainder of kb when divided by c. Prove that the two sets $\{r_0, r_1, r_2, \ldots, r_a\}$ and $\{0, 1, 2, \ldots, a\}$ are different.

(Solution) Suppose that two sets are equal. Then $\gcd(b, c) = 1$ and the polynomial

$$f(x) := (1 + x^b + x^{2b} + \cdots + x^{ab}) - (1 + x + x^2 + \cdots + x^{a-1} + x^a)$$

is divisible by $x^c - 1$. (This is because: $m = n + cq \implies x^m - x^n = x^{n+cq} - x^n = x^n(x^{cq} - 1)$ and $(x^{cq} - 1) = (x^c - 1)((x^c)^{q-1} + (x^c)^{q-2} + \cdots + 1)$.) From

$$f(x) = \frac{x^{(a+1)b} - 1}{x^b - 1} - \frac{x^{a+1} - 1}{x - 1} = \frac{F(x)}{(x-1)(x^b - 1)} ,$$

where $F(x) = x^{ab+b+1} + x^b + x^{a+1} - x^{ab+b} - x^{a+b+1} - x$, we have

$$F(x) \equiv 0 \pmod{x^c - 1} .$$

Since $x^c \equiv 1 \pmod{x^c - 1}$, we may conclude that

$$\{ab + b + 1, b, a + 1\} \equiv \{ab + b, a + b + 1, 1\} \pmod{c}. \qquad (\dagger)$$

Thus,

$$b \equiv \ ab + b,\ a + b + 1 \ \text{ or } \ 1 \pmod{c}.$$

But neither $b \equiv 1 \pmod{c}$ nor $b \equiv a + b + 1 \pmod{c}$ are possible by the given conditions. Therefore, $b \equiv ab + b \pmod{c}$. But this is also impossible because $\gcd(b, c) = 1$.

XIX Asian Pacific Mathematics Olympiad

Problem 1. Let S be a set of 9 distinct integers all of whose prime factors are at most 3. Prove that S contains 3 distinct integers such that their product is a perfect cube.

Solution. Without loss of generality, we may assume that S contains only positive integers. Let

$$S = \{2^{a_i}3^{b_i} \mid a_i, b_i \in \mathbb{Z},\ a_i, b_i \geq 0,\ 1 \leq i \leq 9\}.$$

It suffices to show that there are $1 \leq i_1, i_2, i_3 \leq 9$ such that

$$a_{i_1} + a_{i_2} + a_{i_3} \equiv b_{i_1} + b_{i_2} + b_{i_3} \equiv 0 \pmod 3. \qquad (\dagger)$$

For $n = 2^a3^b \in S$, let's call ($a \pmod 3$, $b \pmod 3$) the *type* of n. Then there are 9 possible types:

$$(0,0), (0,1), (0,2), (1,0), (1,1), (1,2), (2,0), (2,1), (2,2).$$

Let $N(i,j)$ be the number of integers in S of type (i,j). We obtain 3 distinct integers whose product is a perfect cube when

(1) $N(i,j) \geq 3$ for some i, j, or

(2) $N(i,0)N(i,1)N(i,2) \neq 0$ for some $i = 0, 1, 2$, or

(3) $N(0,j)N(1,j)N(2,j) \neq 0$ for some $j = 0, 1, 2$, or

(4) $N(i_1,j_1)N(i_2,j_2)N(i_3,j_3) \neq 0$, where $\{i_1, i_2, i_3\} = \{j_1, j_2, j_3\} = \{0, 1, 2\}$.

Assume that none of the conditions (1)$\sim$(3) holds. Since $N(i,j) \leq 2$ for all (i,j), there are at least five $N(i,j)$'s that are nonzero. Furthermore, among those nonzero $N(i,j)$'s, no three have the same i nor the same j. Using these facts, one may easily conclude that the condition (4) should hold. (For example, if one places each nonzero $N(i,j)$ in the (i,j)-th box of a regular 3×3 array of boxes whose rows and columns are indexed by 0,1 and 2, then one can always find three boxes, occupied by at least one nonzero $N(i,j)$, whose rows and columns are all distinct. This implies (4).)

Second solution. Up to (†), we do the same as above and get 9 possible types:

$$(a \pmod 3, b \pmod 3) = (0,0), (0,1), (0,2), (1,0), (1,1), (1,2), (2,0), (2,1), (2,2)$$

for $n = 2^a 3^b \in S$.

Note that (i) among any 5 integers, there exist 3 whose sum is 0 (mod 3), and that (ii) if $i, j, k \in \{0,1,2\}$, then $i+j+k \equiv 0 \pmod 3$ if and only if $i = j = k$ or $\{i, j, k\} = \{0, 1, 2\}$.

Let's define

T: the set of types of the integers in S;
$N(i)$: the number of integers in S of the type $(i, \cdot)$;
$M(i)$: the number of integers $j \in \{0,1,2\}$ such that $(i,j) \in T$.

If $N(i) \geq 5$ for some i, the result follows from (i). Otherwise, for some permutation (i,j,k) of $(0,1,2)$,

$$N(i) \geq 3, \quad N(j) \geq 3, \quad N(k) \geq 1.$$

If $M(i)$ or $M(j)$ is 1 or 3, the result follows from (ii). Otherwise $M(i) = M(j) = 2$. Then either

$$(i,x), (i,y), (j,x), (j,y) \in T \quad \text{or} \quad (i,x), (i,y), (j,x), (j,z) \in T$$

for some permutation (x,y,z) of $(0,1,2)$. Since $N(k) \geq 1$, at least one of $(k,x), (k,y)$ and (k,z) contained in T. Therefore, in any case, the result follows from (ii). (For example, if $(k,y) \in T$, then take $(i,y), (j,y), (k,y)$ or $(i,x), (j,z), (k,y)$ from T.)

Problem 2. Let ABC be an acute angled triangle with $\angle BAC = 60^\circ$ and $AB > AC$. Let I be the incenter, and H the orthocenter of the triangle ABC. Prove that

$$2\angle AHI = 3\angle ABC.$$

Solution. Let D be the intersection point of the lines AH and BC. Let K be the intersection point of the circumcircle O of the triangle ABC and the line AH. Let the line through I perpendicular to BC meet BC and the minor arc BC of the circumcircle O at E and N, respectively. We have

$$\angle BIC = 180^\circ - (\angle IBC + \angle ICB) = 180^\circ - \frac{1}{2}(\angle ABC + \angle ACB) = 90^\circ + \frac{1}{2}\angle BAC = 120^\circ$$

and also $\angle BNC = 180^\circ - \angle BAC = 120^\circ = \angle BIC$. Since $IN \perp BC$, the quadrilateral $BICN$ is a kite and thus $IE = EN$.

Now, since H is the orthocenter of the triangle ABC, $HD = DK$. Also because $ED \perp IN$ and $ED \perp HK$, we conclude that $IHKN$ is an isosceles trapezoid with $IH = NK$.

Hence

$$\angle AHI = 180^\circ - \angle IHK = 180^\circ - \angle AKN = \angle ABN.$$

Since $IE = EN$ and $BE \perp IN$, the triangles IBE and NBE are congruent. Therefore

$$\angle NBE = \angle IBE = \angle IBC = \angle IBA = \frac{1}{2}\angle ABC$$

and thus

$$\angle AHI = \angle ABN = \frac{3}{2}\angle ABC.$$

Second solution. Let P, Q and R be the intersection points of BH, CH and AH with AC, AB and BC, respectively. Then we have $\angle IBH = \angle ICH$. Indeed,

$$\angle IBH = \angle ABP - \angle ABI = 30^\circ - \frac{1}{2}\angle ABC$$

and

$$\angle ICH = \angle ACI - \angle ACH = \frac{1}{2}\angle ACB - 30^\circ = 30^\circ - \frac{1}{2}\angle ABC,$$

because $\angle ABH = \angle ACH = 30^\circ$ and $\angle ACB + \angle ABC = 120^\circ$. (Note that $\angle ABP > \angle ABI$ and $\angle ACI > \angle ACH$ because AB is the longest side of the triangle ABC under the given conditions.) Therefore $BIHC$ is a cyclic quadrilateral and thus

$$\angle BHI = \angle BCI = \frac{1}{2}\angle ACB.$$

On the other hand,

$$\angle BHR = 90^\circ - \angle HBR = 90^\circ - (\angle ABC - \angle ABH) = 120^\circ - \angle ABC.$$

Therefore,

$$\begin{aligned}\angle AHI &= 180^\circ - \angle BHI - \angle BHR = 60^\circ - \frac{1}{2}\angle ACB + \angle ABC \\ &= 60^\circ - \frac{1}{2}(120^\circ - \angle ABC) + \angle ABC = \frac{3}{2}\angle ABC.\end{aligned}$$

Problem 3. Consider n disks $C_1, C_2, \ldots, C_n$ in a plane such that for each $1 \leq i < n$, the center of C_i is on the circumference of C_{i+1}, and the center of C_n is on the circumference of C_1. Define the *score* of such an arrangement of n disks to be the number of pairs (i, j) for which C_i properly contains C_j. Determine the maximum possible score.

Solution. The answer is $(n-1)(n-2)/2$.

Let's call a set of n disks satisfying the given conditions an *n-configuration*. For an n-configuration $\mathcal{C} = \{C_1, \ldots, C_n\}$, let $S_{\mathcal{C}} = \{(i, j) \mid C_i \text{ properly contains } C_j\}$. So, the score of an n-configuration $\mathcal{C}$ is $|S_{\mathcal{C}}|$.

We'll show that (i) there is an n-configuration $\mathcal{C}$ for which $|S_{\mathcal{C}}| = (n-1)(n-2)/2$, and that (ii) $|S_{\mathcal{C}}| \leq (n-1)(n-2)/2$ for any n-configuration $\mathcal{C}$.

Let C_1 be any disk. Then for $i = 2, \ldots, n-1$, take C_i inside C_{i-1} so that the circumference of C_i contains the center of C_{i-1}. Finally, let C_n be a disk whose center is on the circumference of C_1 and whose circumference contains the center of C_{n-1}. This gives $S_{\mathcal{C}} = \{(i, j) \mid 1 \leq i < j \leq n-1\}$ of size $(n-1)(n-2)/2$, which proves (i).

For any n-configuration $\mathcal{C}$, $S_{\mathcal{C}}$ must satisfy the following properties:

(1) $(i, i) \notin S_{\mathcal{C}}$,

(2) $(i+1, i) \notin S_{\mathcal{C}}$, $(1, n) \notin S_{\mathcal{C}}$,

(3) if $(i, j), (j, k) \in S_{\mathcal{C}}$, then $(i, k) \in S_{\mathcal{C}}$,

(4) if $(i, j) \in S_{\mathcal{C}}$, then $(j, i) \notin S_{\mathcal{C}}$.

Now we show that a set G of ordered pairs of integers between 1 and n, satisfying the conditions (1)$\sim$(4), can have no more than $(n-1)(n-2)/2$ elements. Suppose that there exists a set G that satisfies the conditions (1)$\sim$(4), and has more than $(n-1)(n-2)/2$ elements. Let n be the least positive integer with which there exists such a set G. Note that G must have $(i, i+1)$ for some $1 \leq i \leq n$ or $(n, 1)$, since otherwise G can have at most

$$\binom{n}{2} - n = \frac{n(n-3)}{2} < \frac{(n-1)(n-2)}{2}$$

elements. Without loss of generality we may assume that $(n, 1) \in G$. Then $(1, n-1) \notin G$, since otherwise the condition (3) yields $(n, n-1) \in G$ contradicting the condition (2). Now let $G' = \{(i, j) \in G \mid 1 \leq i, j \leq n-1\}$, then G' satisfies the conditions (1)$\sim$(4), with $n-1$.

We now claim that $|G - G'| \leq n-2$:

Suppose that $|G - G'| > n-2$, then $|G - G'| = n-1$ and hence for each $1 \leq i \leq n-1$, either (i, n) or (n, i) must be in G. We already know that $(n, 1) \in G$ and $(n-1, n) \in G$ (because $(n, n-1) \notin G$) and this implies that $(n, n-2) \notin G$ and $(n-2, n) \in G$. If we keep doing this process, we obtain $(1, n) \in G$, which is a contradiction.

Since $|G - G'| \leq n - 2$, we obtain

$$|G'| \geq \frac{(n-1)(n-2)}{2} - (n-2) = \frac{(n-2)(n-3)}{2}.$$

This, however, contradicts the minimality of n, and hence proves (ii).

Problem 4. Let x, y and z be positive real numbers such that $\sqrt{x}+\sqrt{y}+\sqrt{z}=1$. Prove that

$$\frac{x^2+yz}{\sqrt{2x^2(y+z)}}+\frac{y^2+zx}{\sqrt{2y^2(z+x)}}+\frac{z^2+xy}{\sqrt{2z^2(x+y)}}\geq 1.$$

Solution. We first note that

$$\begin{aligned}\frac{x^2+yz}{\sqrt{2x^2(y+z)}} &= \frac{x^2-x(y+z)+yz}{\sqrt{2x^2(y+z)}}+\frac{x(y+z)}{\sqrt{2x^2(y+z)}}\\ &= \frac{(x-y)(x-z)}{\sqrt{2x^2(y+z)}}+\sqrt{\frac{y+z}{2}}\\ &\geq \frac{(x-y)(x-z)}{\sqrt{2x^2(y+z)}}+\frac{\sqrt{y}+\sqrt{z}}{2}. \qquad (1)\end{aligned}$$

Similarly, we have

$$\frac{y^2+zx}{\sqrt{2y^2(z+x)}}\geq \frac{(y-z)(y-x)}{\sqrt{2y^2(z+x)}}+\frac{\sqrt{z}+\sqrt{x}}{2}, \qquad (2)$$

$$\frac{z^2+xy}{\sqrt{2z^2(x+y)}}\geq \frac{(z-x)(z-y)}{\sqrt{2z^2(x+y)}}+\frac{\sqrt{x}+\sqrt{y}}{2}. \qquad (3)$$

We now add (1)∼(3) to get

$$\begin{aligned}&\frac{x^2+yz}{\sqrt{2x^2(y+z)}}+\frac{y^2+zx}{\sqrt{2y^2(z+x)}}+\frac{z^2+xy}{\sqrt{2z^2(x+y)}}\\ &\geq \frac{(x-y)(x-z)}{\sqrt{2x^2(y+z)}}+\frac{(y-z)(y-x)}{\sqrt{2y^2(z+x)}}+\frac{(z-x)(z-y)}{\sqrt{2z^2(x+y)}}+\sqrt{x}+\sqrt{y}+\sqrt{z}\\ &= \frac{(x-y)(x-z)}{\sqrt{2x^2(y+z)}}+\frac{(y-z)(y-x)}{\sqrt{2y^2(z+x)}}+\frac{(z-x)(z-y)}{\sqrt{2z^2(x+y)}}+1.\end{aligned}$$

Thus, it suffices to show that

$$\frac{(x-y)(x-z)}{\sqrt{2x^2(y+z)}}+\frac{(y-z)(y-x)}{\sqrt{2y^2(z+x)}}+\frac{(z-x)(z-y)}{\sqrt{2z^2(x+y)}}\geq 0. \qquad (4)$$

Now, assume without loss of generality, that $x\geq y\geq z$. Then we have

$$\frac{(x-y)(x-z)}{\sqrt{2x^2(y+z)}}\geq 0$$

and

$$\frac{(z-x)(z-y)}{\sqrt{2z^2(x+y)}}+\frac{(y-z)(y-x)}{\sqrt{2y^2(z+x)}}=\frac{(y-z)(x-z)}{\sqrt{2z^2(x+y)}}-\frac{(y-z)(x-y)}{\sqrt{2y^2(z+x)}}$$
$$\geq\frac{(y-z)(x-y)}{\sqrt{2z^2(x+y)}}-\frac{(y-z)(x-y)}{\sqrt{2y^2(z+x)}}=(y-z)(x-y)\left(\frac{1}{\sqrt{2z^2(x+y)}}-\frac{1}{\sqrt{2y^2(z+x)}}\right).$$

The last quantity is non-negative due to the fact that

$$y^2(z+x)=y^2z+y^2x\geq yz^2+z^2x=z^2(x+y).$$

This completes the proof.

Second solution. By Cauchy-Schwarz inequality,

$$\left(\frac{x^2}{\sqrt{2x^2(y+z)}}+\frac{y^2}{\sqrt{2y^2(z+x)}}+\frac{z^2}{\sqrt{2z^2(x+y)}}\right)\tag{5}$$
$$\times\left(\sqrt{2(y+z)}+\sqrt{2(z+x)}+\sqrt{2(x+y)}\right)\geq(\sqrt{x}+\sqrt{y}+\sqrt{z})^2=1,$$

and

$$\left(\frac{yz}{\sqrt{2x^2(y+z)}}+\frac{zx}{\sqrt{2y^2(z+x)}}+\frac{xy}{\sqrt{2z^2(x+y)}}\right)\tag{6}$$
$$\times\left(\sqrt{2(y+z)}+\sqrt{2(z+x)}+\sqrt{2(x+y)}\right)\geq\left(\sqrt{\frac{yz}{x}}+\sqrt{\frac{zx}{y}}+\sqrt{\frac{xy}{z}}\right)^2.$$

We now combine (5) and (6) to find

$$\left(\frac{x^2+yz}{\sqrt{2x^2(y+z)}}+\frac{y^2+zx}{\sqrt{2y^2(z+x)}}+\frac{z^2+xy}{\sqrt{2z^2(x+y)}}\right)$$
$$\times\left(\sqrt{2(x+y)}+\sqrt{2(y+z)}+\sqrt{2(z+x)}\right)$$
$$\geq 1+\left(\sqrt{\frac{yz}{x}}+\sqrt{\frac{zx}{y}}+\sqrt{\frac{xy}{z}}\right)^2\geq 2\left(\sqrt{\frac{yz}{x}}+\sqrt{\frac{zx}{y}}+\sqrt{\frac{xy}{z}}\right).$$

Thus, it suffices to show that

$$2\left(\sqrt{\frac{yz}{x}}+\sqrt{\frac{zx}{y}}+\sqrt{\frac{xy}{z}}\right)\geq\sqrt{2(y+z)}+\sqrt{2(z+x)}+\sqrt{2(x+y)}\,.\tag{7}$$

Consider the following inequality using AM-GM inequality

$$\left[\sqrt{\frac{yz}{x}}+\left(\frac{1}{2}\sqrt{\frac{zx}{y}}+\frac{1}{2}\sqrt{\frac{xy}{z}}\right)\right]^2\geq 4\sqrt{\frac{yz}{x}}\left(\frac{1}{2}\sqrt{\frac{zx}{y}}+\frac{1}{2}\sqrt{\frac{xy}{z}}\right)=2(y+z),$$

or equivalently

$$\sqrt{\frac{yz}{x}}+\left(\frac{1}{2}\sqrt{\frac{zx}{y}}+\frac{1}{2}\sqrt{\frac{xy}{z}}\right)\geq\sqrt{2(y+z)}\,.$$

Similarly, we have

$$\sqrt{\frac{zx}{y}}+\left(\frac{1}{2}\sqrt{\frac{xy}{z}}+\frac{1}{2}\sqrt{\frac{yz}{x}}\right)\geq\sqrt{2(z+x)}\,,$$
$$\sqrt{\frac{xy}{z}}+\left(\frac{1}{2}\sqrt{\frac{yz}{x}}+\frac{1}{2}\sqrt{\frac{zx}{y}}\right)\geq\sqrt{2(x+y)}\,.$$

Adding the last three inequalities, we get

$$2\left(\sqrt{\frac{yz}{x}}+\sqrt{\frac{zx}{y}}+\sqrt{\frac{xy}{z}}\right)\geq\sqrt{2(y+z)}+\sqrt{2(z+x)}+\sqrt{2(x+y)}\,.$$

This completes the proof.

Problem 5. A regular (5×5)-array of lights is defective, so that toggling the switch for one light causes each adjacent light in the same row and in the same column as well as the light itself to change state, from on to off, or from off to on. Initially all the lights are switched off. After a certain number of toggles, exactly one light is switched on. Find all the possible positions of this light.

Solution. We assign the following *first labels* to the 25 positions of the lights:

1	1	0	1	1
0	0	0	0	0
1	1	0	1	1
0	0	0	0	0
1	1	0	1	1

For each on-off combination of lights in the array, define its *first value* to be the sum of the first labels of those positions at which the lights are switched on. It is easy to check that toggling any switch always leads to an on-off combination of lights whose first value has the same parity(the remainder when divided by 2) as that of the previous on-off combination.

The 90° rotation of the first labels gives us another labels (let us call it the *second labels*) which also makes the parity of the *second value*(the sum of the second labels of those positions at which the lights are switched on) invariant under toggling.

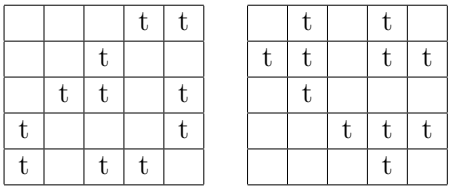
1	0	1	0	1
1	0	1	0	1
0	0	0	0	0
1	0	1	0	1
1	0	1	0	1

Since the parity of the first and the second values of the initial status is 0, after certain number of toggles the parity must remain unchanged with respect to the first labels and the second labels as well. Therefore, if exactly one light is on after some number of toggles, the label of that position must be 0 with respect to both labels. Hence according to the above pictures, the possible positions are the ones marked with $*_i$'s in the following picture:

	$*_2$		$*_1$	
		$*_0$		
	$*_3$		$*_4$	

Now we demonstrate that all five positions are possible :

Toggling the positions checked by t (the order of toggling is irrelevant) in the first picture makes the center($*_0$) the only position with light on and the second picture makes the position $*_1$ the only position with light on. The other $*_i$'s can be obtained by rotating the second picture appropriately.



Problem 1. Let n be a positive integer. Find the largest nonnegative real number $f(n)$ (depending on n) with the following property: whenever $a_1, a_2, \ldots, a_n$ are real numbers such that $a_1 + a_2 + \cdots + a_n$ is an integer, there exists some i such that $| a_i - \frac{1}{2} | \geq f(n)$.

(Solution) The answer is

$$f(n) = \begin{cases} 0 & \text{if } n \text{ is even,} \\ \frac{1}{2n} & \text{if } n \text{ is odd.} \end{cases}$$

First, assume that n is even. If $a_i = \frac{1}{2}$ for all i, then the sum $a_1 + a_2 + \cdots + a_n$ is an integer. Since $| a_i - \frac{1}{2} | = 0$ for all i, we may conclude $f(n) = 0$ for any even n.

Now assume that n is odd. Suppose that $| a_i - \frac{1}{2} | < \frac{1}{2n}$ for all $1 \leq i \leq n$. Then, since $\sum_{i=1}^{n} a_i$ is an integer,

$$\frac{1}{2} \leq \left| \sum_{i=1}^{n} a_i - \frac{n}{2} \right| \leq \sum_{i=1}^{n} \left| a_i - \frac{1}{2} \right| < \frac{1}{2n} \cdot n = \frac{1}{2},$$

a contradiction. Thus $|a_i - \frac{1}{2}| \geq \frac{1}{2n}$ for some i, as required. On the other hand, putting $n = 2m + 1$ and $a_i = \frac{m}{2m+1}$ for all i gives $\sum a_i = m$, while

$$\left| a_i - \frac{1}{2} \right| = \frac{1}{2} - \frac{m}{2m+1} = \frac{1}{2(2m+1)} = \frac{1}{2n}$$

for all i. Therefore, $f(n) = \frac{1}{2n}$ is the best possible for any odd n.

Problem 2. Prove that every positive integer can be written as a finite sum of distinct integral powers of the golden mean $\tau = \frac{1+\sqrt{5}}{2}$. Here, an integral power of τ is of the form τ^i, where i is an integer (not necessarily positive).

(Solution) We will prove this statement by induction using the equality

$$\tau^2 = \tau + 1.$$

If $n = 1$, then $1 = \tau^0$. Suppose that $n - 1$ can be written as a finite sum of integral powers of τ, say

$$n - 1 = \sum_{i=-k}^{k} a_i \tau^i, \tag{1}$$

where $a_i \in \{0, 1\}$ and $n \geq 2$. We will write (1) as

$$n - 1 = a_k \cdots a_1 a_0 . a_{-1} a_{-2} \cdots a_{-k}. \tag{2}$$

For example,

$$1 = 1.0 = 0.11 = 0.1011 = 0.101011.$$

Firstly, we will prove that we may assume that in (2) we have $a_i a_{i+1} = 0$ for all i with $-k \le i \le k-1$. Indeed, if we have several occurrences of 11, then we take the leftmost such occurrence. Since we may assume that it is preceded by a 0, we can replace 011 with 100 using the identity $\tau^{i+1} + \tau^i = \tau^{i+2}$. By doing so repeatedly, if necessary, we will eliminate all occurrences of two 1's standing together. Now we have the representation

$$n - 1 = \sum_{i=-K}^{K} b_i \tau^i, \tag{3}$$

where $b_i \in \{0, 1\}$ and $b_i b_{i+1} = 0$.

If $b_0 = 0$ in (3), then we just add $1 = \tau^0$ to both sides of (3) and we are done.

Suppose now that there is 1 in the unit position of (3), that is $b_0 = 1$. If there are two 0's to the right of it, i.e.

$$n - 1 = \cdots 1.00 \cdots,$$

then we can replace 1.00 with 0.11 because $1 = \tau^{-1} + \tau^{-2}$, and we are done because we obtain 0 in the unit position. Thus we may assume that

$$n - 1 = \cdots 1.010 \cdots.$$

Again, if we have $n - 1 = \cdots 1.0100 \cdots$, we may rewrite it as

$$n - 1 = \cdots 1.0100 \cdots = \cdots 1.0011 \cdots = \cdots 0.1111 \cdots$$

and obtain 0 in the unit position. Therefore, we may assume that

$$n - 1 = \cdots 1.01010 \cdots.$$

Since the number of 1's is finite, eventually we will obtain an occurrence of 100 at the end, i.e.

$$n - 1 = \cdots 1.01010 \cdots 100.$$

Then we can shift all 1's to the right to obtain 0 in the unit position, i.e.

$$n - 1 = \cdots 0.11 \cdots 11,$$

and we are done.

Problem 3. Let $p \ge 5$ be a prime and let r be the number of ways of placing p checkers on a $p \times p$ checkerboard so that not all checkers are in the same row (but they may all be in the same column). Show that r is divisible by p^5. Here, we assume that all the checkers are identical.

(Solution) Note that $r = \binom{p^2}{p} - p$. Hence, it suffices to show that

$$(p^2 - 1)(p^2 - 2) \cdots (p^2 - (p-1)) - (p-1)! \equiv 0 \pmod{p^4}. \tag{1}$$

Now, let

$$f(x) := (x-1)(x-2)\cdots(x-(p-1)) = x^{p-1} + s_{p-2}x^{p-2} + \cdots + s_1 x + s_0. \qquad (2)$$

Then the congruence equation (1) is same as $f(p^2) - s_0 \equiv 0 \pmod{p^4}$. Therefore, it suffices to show that $s_1 p^2 \equiv 0 \pmod{p^4}$ or $s_1 \equiv 0 \pmod{p^2}$.

Since $a^{p-1} \equiv 1 \pmod{p}$ for all $1 \le a \le p-1$, we can factor

$$x^{p-1} - 1 \equiv (x-1)(x-2)\cdots(x-(p-1)) \pmod{p}. \qquad (3)$$

Comparing the coefficients of the left hand side of (3) with those of the right hand side of (2), we obtain $p \,|\, s_i$ for all $1 \le i \le p-2$ and $s_0 \equiv -1 \pmod{p}$. On the other hand, plugging p for x in (2), we get

$$f(p) = (p-1)! = s_0 = p^{p-1} + s_{p-2}p^{p-2} + \cdots + s_1 p + s_0,$$

which implies

$$p^{p-1} + s_{p-2}p^{p-2} + \cdots + s_2 p^2 = -s_1 p.$$

Since $p \ge 5$, $p \,|\, s_2$ and hence $s_1 \equiv 0 \pmod{p^2}$ as desired.

Problem 4. Let A, B be two distinct points on a given circle O and let P be the midpoint of the line segment AB. Let O_1 be the circle tangent to the line AB at P and tangent to the circle O. Let ℓ be the tangent line, different from the line AB, to O_1 passing through A. Let C be the intersection point, different from A, of ℓ and O. Let Q be the midpoint of the line segment BC and O_2 be the circle tangent to the line BC at Q and tangent to the line segment AC. Prove that the circle O_2 is tangent to the circle O.

(Solution) Let S be the tangent point of the circles O and O_1 and let T be the intersection point, different from S, of the circle O and the line SP. Let X be the tangent point of ℓ to O_1 and let M be the midpoint of the line segment XP. Since $\angle TBP = \angle ASP$, the triangle TBP is similar to the triangle ASP. Therefore,

$$\frac{PT}{PB} = \frac{PA}{PS}.$$

Since the line ℓ is tangent to the circle O_1 at X, we have

$$\angle SPX = 90^\circ - \angle XSP = 90^\circ - \angle APM = \angle PAM$$

which implies that the triangle PAM is similar to the triangle SPX. Consequently,

$$\frac{XS}{XP} = \frac{MP}{MA} = \frac{XP}{2MA} \quad \text{and} \quad \frac{XP}{PS} = \frac{MA}{AP}.$$

From this and the above observation follows

$$\frac{XS}{XP}\cdot\frac{PT}{PB}=\frac{XP}{2MA}\cdot\frac{PA}{PS}=\frac{XP}{2MA}\cdot\frac{MA}{XP}=\frac{1}{2}\,. \tag{1}$$

Let A' be the intersection point of the circle O and the perpendicular bisector of the chord BC such that A, A' are on the same side of the line BC, and N be the intersection point of the lines $A'Q$ and CT. Since

$$\angle NCQ=\angle TCB=\angle TCA=\angle TBA=\angle TBP$$

and

$$\angle CA'Q=\frac{\angle CAB}{2}=\frac{\angle XAP}{2}=\angle PAM=\angle SPX,$$

the triangle NCQ is similar to the triangle TBP and the triangle $CA'Q$ is similar to the triangle SPX. Therefore

$$\frac{QN}{QC}=\frac{PT}{PB}\quad\text{and}\quad\frac{QC}{QA'}=\frac{XS}{XP}\,.$$

and hence $QA'=2QN$ by (1). This implies that N is the midpoint of the line segment QA'. Let the circle O_2 touch the line segment AC at Y. Since

$$\angle ACN=\angle ACT=\angle BCT=\angle QCN$$

and $|\,CY|=|\,CQ|$, the triangles YCN and QCN are congruent and hence $NY\perp AC$ and $NY=NQ=NA'$. Therefore, N is the center of the circle O_2, which completes the proof.

Remark: Analytic solutions are possible: For example, one can prove for a triangle ABC inscribed in a circle O that $AB=k(2+2t), AC=k(1+2t), BC=k(1+4t)$ for some positive numbers k,t if and only if there exists a circle O_1 such that O_1 is tangent to the side AB at its midpoint, the side AC and the circle O.

One obtains $AB=k'(1+4t'), AC=k'(1+2t'), BC=k'(2+2t')$ by substituting $t=1/4t'$ and $k=2k't'$. So, there exists a circle O_2 such that O_2 is tangent to the side BC at its midpoint, the side AC and the circle O.

In the above, $t=\tan^2\alpha$ and $k=\frac{4R\tan\alpha}{(1+\tan^2\alpha)(1+4\tan^2\alpha)}$, where R is the radius of O and $\angle A=2\alpha$. Furthermore, $t'=\tan^2\gamma$ and $k'=\frac{4R\tan\gamma}{(1+\tan^2\gamma)(1+4\tan^2\gamma)}$, where $\angle C=2\gamma$. Observe that $\sqrt{tt'}=\tan\alpha\cdot\tan\gamma=\frac{XS}{XP}\cdot\frac{PT}{PB}=\frac{1}{2}$, which implies $tt'=\frac{1}{4}$. It is now routine easy to check that $k=2k't'$.

Problem 5. In a circus, there are n clowns who dress and paint themselves up using a selection of 12 distinct colours. Each clown is required to use at least five different colours. One day, the ringmaster of the circus orders that no two clowns have exactly the same set

of colours and no more than 20 clowns may use any one particular colour. Find the largest number n of clowns so as to make the ringmaster's order possible.

(Solution) Let C be the set of n clowns. Label the colours $1, 2, 3, \ldots, 12$. For each $i = 1, 2, \ldots, 12$, let E_i denote the set of clowns who use colour i. For each subset S of $\{1, 2, \ldots, 12\}$, let E_S be the set of clowns who use exactly those colours in S. Since $S \neq S'$ implies $E_S \cap E_{S'} = \emptyset$, we have

$$\sum_S |E_S| = |C| = n,$$

where S runs over all subsets of $\{1, 2, \ldots, 12\}$. Now for each i,

$$E_S \subseteq E_i \quad \text{if and only if} \quad i \in S,$$

and hence

$$|E_i| = \sum_{i \in S} |E_S|.$$

By assumption, we know that $|E_i| \leq 20$ and that if $E_S \neq \emptyset$, then $|S| \geq 5$. From this we obtain

$$20 \times 12 \geq \sum_{i=1}^{12} |E_i| = \sum_{i=1}^{12} \left(\sum_{i \in S} |E_S| \right) \geq 5 \sum_S |E_S| = 5n.$$

Therefore $n \leq 48$.

Now, define a sequence $\{c_i\}_{i=1}^{52}$ of colours in the following way:

$$\begin{array}{l} 1\ 2\ 3\ 4 \mid 5\ 6\ 7\ 8 \mid 9\ 10\ 11\ 12 \mid \\ 4\ 1\ 2\ 3 \mid 8\ 5\ 6\ 7 \mid 12\ 9\ 10\ 11 \mid \\ 3\ 4\ 1\ 2 \mid 7\ 8\ 5\ 6 \mid 11\ 12\ 9\ 10 \mid \\ 2\ 3\ 4\ 1 \mid 6\ 7\ 8\ 5 \mid 10\ 11\ 12\ 9 \mid 1\ 2\ 3\ 4 \end{array}$$

The first row lists $c_1, \ldots, c_{12}$ in order, the second row lists $c_{13}, \ldots, c_{24}$ in order, the third row lists $c_{25}, \ldots, c_{36}$ in order, and finally the last row lists $c_{37}, \ldots, c_{52}$ in order. For each j, $1 \leq j \leq 48$, assign colours $c_j, c_{j+1}, c_{j+2}, c_{j+3}, c_{j+4}$ to the j-th clown. It is easy to check that this assignment satisfies all conditions given above. So, 48 is the largest for n.

Remark: The fact that $n \leq 48$ can be obtained in a much simpler observation that

$$5n \leq 12 \times 20 = 240.$$

There are many other ways of constructing 48 distinct sets consisting of 5 colours. For example, consider the sets

$$\begin{array}{cccc} \{1,2,3,4,5,6\}, & \{3,4,5,6,7,8\}, & \{5,6,7,8,9,10\}, & \{7,8,9,10,11,12\}, \\ \{9,10,11,12,1,2\}, & \{11,12,1,2,3,4\}, & \{1,2,5,6,9,10\}, & \{3,4,7,8,11,12\}. \end{array}$$

Each of the above 8 sets has 6 distinct subsets consisting of exactly 5 colours. It is easy to check that the 48 subsets obtained in this manner are all distinct.

XVII APMO - March, 2005

Problems and Solutions

Problem 1. Prove that for every irrational real number a, there are irrational real numbers b and b' so that $a+b$ and ab' are both rational while ab and $a+b'$ are both irrational.

(Solution) Let a be an irrational number. If a^2 is irrational, we let $b=-a$. Then, $a+b=0$ is rational and $ab=-a^2$ is irrational. If a^2 is rational, we let $b=a^2-a$. Then, $a+b=a^2$ is rational and $ab=a^2(a-1)$. Since

$$a=\frac{ab}{a^2}+1$$

is irrational, so is ab.

Now, we let $b'=\frac{1}{a}$ or $b'=\frac{2}{a}$. Then $ab'=1$ or 2, which is rational. Note that

$$a+b'=\frac{a^2+1}{a} \quad \text{or} \quad a+b'=\frac{a^2+2}{a}.$$

Since,

$$\frac{a^2+2}{a}-\frac{a^2+1}{a}=\frac{1}{a},$$

at least one of them is irrational.

Problem 2. Let a, b and c be positive real numbers such that $abc = 8$. Prove that

$$\frac{a^2}{\sqrt{(1+a^3)(1+b^3)}} + \frac{b^2}{\sqrt{(1+b^3)(1+c^3)}} + \frac{c^2}{\sqrt{(1+c^3)(1+a^3)}} \;\geq\; \frac{4}{3}\,.$$

(Solution) Observe that

$$\frac{1}{\sqrt{1+x^3}} \geq \frac{2}{2+x^2}\,. \tag{1}$$

In fact, this is equivalent to $(2+x^2)^2 \geq 4(1+x^3)$, or $x^2(x-2)^2 \geq 0$. Notice that equality holds in (1) if and only if $x = 2$.

We substitute x by a, b, c in (1), respectively, to find

$$\begin{aligned}&\frac{a^2}{\sqrt{(1+a^3)(1+b^3)}} + \frac{b^2}{\sqrt{(1+b^3)(1+c^3)}} + \frac{c^2}{\sqrt{(1+c^3)(1+a^3)}}\\ &\geq \frac{4a^2}{(2+a^2)(2+b^2)} + \frac{4b^2}{(2+b^2)(2+c^2)} + \frac{4c^2}{(2+c^2)(2+a^2)}\,.\end{aligned} \tag{2}$$

We combine the terms on the right hand side of (2) to obtain

$$\text{Left hand side of (2)} \geq \frac{2S(a,b,c)}{36+S(a,b,c)} = \frac{2}{1+36/S(a,b,c)}\,, \tag{3}$$

where $S(a,b,c) := 2(a^2+b^2+c^2) + (ab)^2 + (bc)^2 + (ca)^2$. By AM-GM inequality, we have

$$\begin{aligned}a^2+b^2+c^2 &\geq 3\sqrt[3]{(abc)^2} = 12\,,\\ (ab)^2+(bc)^2+(ca)^2 &\geq 3\sqrt[3]{(abc)^4} = 48\,.\end{aligned}$$

Note that the equalities holds if and only if $a = b = c = 2$. The above inequalities yield

$$S(a,b,c) = 2(a^2+b^2+c^2) + (ab)^2 + (bc)^2 + (ca)^2 \geq 72\,. \tag{4}$$

Therefore

$$\frac{2}{1+36/S(a,b,c)} \geq \frac{2}{1+36/72} = \frac{4}{3}\,, \tag{5}$$

which is the required inequality.

Problem 3. Prove that there exists a triangle which can be cut into 2005 congruent triangles.

(Solution) Suppose that one side of a triangle has length n. Then it can be cut into n^2 congruent triangles which are similar to the original one and whose corresponding sides to the side of length n have lengths 1.

Since $2005 = 5 \times 401$ where 5 and 401 are primes and both primes are of the type $4k + 1$, it is representable as a sum of two integer squares. Indeed, it is easy to see that

$$\begin{aligned} 2005 &= 5 \times 401 = (2^2 + 1)(20^2 + 1) \\ &= 40^2 + 20^2 + 2^2 + 1 \\ &= (40 - 1)^2 + 2 \times 40 + 20^2 + 2^2 \\ &= 39^2 + 22^2. \end{aligned}$$

Let ABC be a right-angled triangle with the legs AB and BC having lengths 39 and 22, respectively. We draw the altitude BK, which divides ABC into two similar triangles. Now we divide ABK into 39^2 congruent triangles as described above and BCK into 22^2 congruent triangles. Since ABK is similar to BKC, all 2005 triangles will be congruent.

Problem 4. In a small town, there are $n \times n$ houses indexed by (i, j) for $1 \leq i, j \leq n$ with $(1, 1)$ being the house at the top left corner, where i and j are the row and column indices, respectively. At time 0, a fire breaks out at the house indexed by $(1, c)$, where $c \leq \frac{n}{2}$. During each subsequent time interval $[t, t+1]$, the fire fighters defend a house which is not yet on fire while the fire spreads to all undefended *neighbors* of each house which was on fire at time t. Once a house is defended, it remains so all the time. The process ends when the fire can no longer spread. At most how many houses can be saved by the fire fighters? A house indexed by (i, j) is a *neighbor* of a house indexed by (k, ℓ) if $|i - k| + |j - \ell| = 1$.

(Solution) At most $n^2 + c^2 - nc - c$ houses can be saved. This can be achieved under the following order of defending:

$$\begin{array}{c}(2, c), (2, c+1); (3, c-1), (3, c+2); (4, c-2), (4, c+3); \ldots \\ (c+1, 1), (c+1, 2c); (c+1, 2c+1), \ldots, (c+1, n).\end{array} \tag{6}$$

Under this strategy, there are

2 columns (column numbers $c, c+1$) at which $n-1$ houses are saved
2 columns (column numbers $c-1, c+2$) at which $n-2$ houses are saved
$\ldots$
2 columns (column numbers $1, 2c$) at which $n-c$ houses are saved
$n-2c$ columns (column numbers $n-2c+1, \ldots, n$) at which $n-c$ houses are saved

Adding all these we obtain:

$$2[(n-1) + (n-2) + \cdots + (n-c)] + (n-2c)(n-c) = n^2 + c^2 - cn - c. \tag{7}$$

We say that a house indexed by (i, j) is at level t if $|i-1| + |j-c| = t$. Let $d(t)$ be the number of houses at level t defended by time t, and $p(t)$ be the number of houses at levels greater than t defended by time t. It is clear that

$$p(t) + \sum_{i=1}^{t} d(i) \leq t \text{ and } p(t+1) + d(t+1) \leq p(t) + 1.$$

Let $s(t)$ be the number of houses at level t which are not burning at time t. We prove that

$$s(t) \leq t - p(t) \leq t$$

for $1 \leq t \leq n-1$ by induction. It is obvious when $t = 1$. Assume that it is true for $t = k$. The union of the neighbors of any $k - p(k) + 1$ houses at level $k+1$ contains at least $k - p(k) + 1$ vertices at level k. Since $s(k) \leq k - p(k)$, one of these houses at level k is burning. Therefore, at most $k - p(k)$ houses at level $k+1$ have no neighbor burning. Hence we have

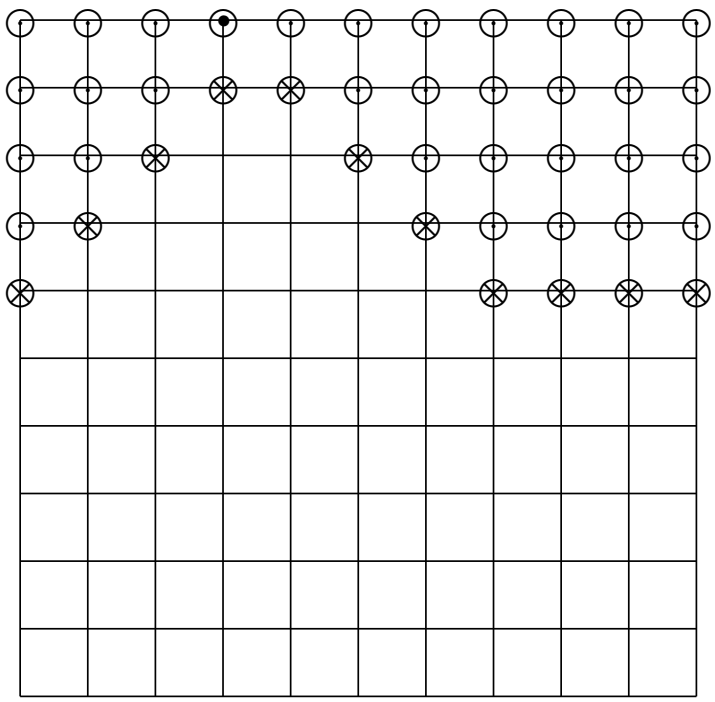
$$\begin{aligned} s(k+1) &\leq k - p(k) + d(k+1) \\ &= (k+1) - (p(k) + 1 - d(k+1)) \\ &\leq (k+1) - p(k+1). \end{aligned}$$

We now prove that the strategy given above is optimal. Since

$$\sum_{t=1}^{n-1} s(t) \leq \binom{n}{2},$$

the maximum number of houses at levels less than or equal to $n-1$, that can be saved under any strategy is at most $\binom{n}{2}$, which is realized by the strategy above. Moreover, at levels bigger than $n-1$, every house is saved under the strategy above.

The following is an example when $n = 11$ and $c = 4$. The houses with $\bigcirc$ mark are burned. The houses with $\bigotimes$ mark are blocked ones and hence those and the houses below them are saved.



Problem 5. In a triangle ABC, points M and N are on sides AB and AC, respectively, such that $MB = BC = CN$. Let R and r denote the circumradius and the inradius of the triangle ABC, respectively. Express the ratio MN/BC in terms of R and r.

(Solution) Let ω, O and I be the circumcircle, the circumcenter and the incenter of ABC, respectively. Let D be the point of intersection of the line BI and the circle ω such that $D \neq B$. Then D is the midpoint of the arc AC. Hence $OD \perp CN$ and $OD = R$.

We first show that triangles MNC and IOD are similar. Because $BC = BM$, the line BI (the bisector of $\angle MBC$) is perpendicular to the line CM. Because $OD \perp CN$ and $ID \perp MC$, it follows that

$$\angle ODI = \angle NCM \tag{8}$$

Let $\angle ABC = 2\beta$. In the triangle BCM, we have

$$\frac{CM}{NC} = \frac{CM}{BC} = 2\sin\beta \tag{9}$$

Since $\angle DIC = \angle DCI$, we have $ID = CD = AD$. Let E be the point of intersection of the line DO and the circle ω such that $E \neq D$. Then DE is a diameter of ω and $\angle DEC = \angle DBC = \beta$. Thus we have

$$\frac{DI}{OD} = \frac{CD}{OD} = \frac{2R\sin\beta}{R} = 2\sin\beta. \tag{10}$$

Combining equations (8), (9), and (10) shows that triangles MNC and IOD are similar. It follows that

$$\frac{MN}{BC} = \frac{MN}{NC} = \frac{IO}{OD} = \frac{IO}{R}\,. \tag{11}$$

The well-known Euler's formula states that

$$OI^2 = R^2 - 2Rr. \tag{12}$$

Therefore,

$$\frac{MN}{BC} = \sqrt{1 - \frac{2r}{R}}\,. \tag{13}$$

(Alternative Solution) Let a (resp., b, c) be the length of BC (resp., AC, AB). Let α (resp., β, γ) denote the angle $\angle BAC$ (resp., $\angle ABC$, $\angle ACB$). By introducing coordinates $B = (0,0)$, $C = (a,0)$, it is immediate that the coordinates of M and N are

$$M = (a\cos\beta, a\sin\beta), \quad N = (a - a\cos\gamma, a\sin\gamma), \tag{14}$$

respectively. Therefore,

$$\begin{aligned}(MN/BC)^2 &= [(a - a\cos\gamma - a\cos\beta)^2 + (a\sin\gamma - a\sin\beta)^2]/a^2\\ &= (1 - \cos\gamma - \cos\beta)^2 + (\sin\gamma - \sin\beta)^2\\ &= 3 - 2\cos\gamma - 2\cos\beta + 2(\cos\gamma\cos\beta - \sin\gamma\sin\beta)\\ &= 3 - 2\cos\gamma - 2\cos\beta + 2\cos(\gamma + \beta)\\ &= 3 - 2\cos\gamma - 2\cos\beta - 2\cos\alpha\\ &= 3 - 2(\cos\gamma + \cos\beta + \cos\alpha).\end{aligned} \tag{15}$$

Now we claim

$$\cos\gamma + \cos\beta + \cos\alpha = \frac{r}{R} + 1. \tag{16}$$

From

$$\begin{aligned}a &= b\cos\gamma + c\cos\beta\\ b &= c\cos\alpha + a\cos\gamma\\ c &= a\cos\beta + b\cos\alpha\end{aligned} \tag{17}$$

we get

$$a(1 + \cos\alpha) + b(1 + \cos\beta) + c(1 + \cos\gamma) = (a + b + c)(\cos\alpha + \cos\beta + \cos\gamma). \tag{18}$$

Thus

$$\begin{aligned}&\cos\alpha + \cos\beta + \cos\gamma\\ &= \frac{1}{a+b+c}\left(a(1 + \cos\alpha) + b(1 + \cos\beta) + c(1 + \cos\gamma)\right)\\ &= \frac{1}{a+b+c}\left(a\left(1 + \frac{b^2 + c^2 - a^2}{2bc}\right) + b\left(1 + \frac{a^2 + c^2 - b^2}{2ac}\right) + c\left(1 + \frac{a^2 + b^2 - c^2}{2ab}\right)\right)\\ &= \frac{1}{a+b+c}\left(a + b + c + \frac{a^2(b^2 + c^2 - a^2) + b^2(a^2 + c^2 - b^2) + c^2(a^2 + b^2 - c^2)}{2abc}\right)\\ &= 1 + \frac{2a^2b^2 + 2b^2c^2 + 2c^2a^2 - a^4 - b^4 - c^4}{2abc(a + b + c)}.\end{aligned} \tag{19}$$

On the other hand, from $R = \dfrac{a}{2\sin\alpha}$ it follows that

$$\begin{aligned}R^2 &= \frac{a^2}{4(1 - \cos^2\alpha)} = \frac{a^2}{4\left(1 - \left(\dfrac{b^2 + c^2 - a^2}{2bc}\right)^2\right)}\\ &= \frac{a^2b^2c^2}{2a^2b^2 + 2b^2c^2 + 2c^2a^2 - a^4 - b^4 - c^4}.\end{aligned} \tag{20}$$

Also from $\frac{1}{2}(a+b+c)r = \frac{1}{2}bc\sin\alpha$, it follows that

$$\begin{aligned} r^2 &= \frac{b^2c^2(1-\cos^2\alpha)}{(a+b+c)^2} = \frac{b^2c^2\left(1-\left(\frac{b^2+c^2-a^2}{2bc}\right)^2\right)}{(a+b+c)^2} \\ &= \frac{2a^2b^2+2b^2c^2+2c^2a^2-a^4-b^4-c^4}{4(a+b+c)^2}. \end{aligned} \tag{21}$$

Combining (19), (20) and (21), we get (16) as desired.

Finally, by (15) and (16) we have

$$\frac{MN}{BC} = \sqrt{1-\frac{2r}{R}}. \tag{22}$$

Another proof of (16) from R.A. Johnson's "Advanced Euclidean Geometry"[1]:

Construct the perpendicular bisectors OD, OE, OF, where D, E, F are the midpoints of BC, CA, AB, respectively. By Ptolemy's Theorem applied to the cyclic quadrilateral $OEAF$, we get

$$\frac{a}{2}\cdot R = \frac{b}{2}\cdot OF + \frac{c}{2}\cdot OE.$$

Similarly

$$\frac{b}{2}\cdot R = \frac{c}{2}\cdot OD + \frac{a}{2}\cdot OF, \quad \frac{c}{2}\cdot R = \frac{a}{2}\cdot OE + \frac{b}{2}\cdot OD.$$

Adding, we get

$$sR = OD\cdot\frac{b+c}{2} + OE\cdot\frac{c+a}{2} + OF\cdot\frac{a+b}{2}, \tag{23}$$

where s is the semiperimeter. But also, the area of triangle OBC is $OD\cdot\frac{a}{2}$, and adding similar formulas for the areas of triangles OCA and OAB gives

$$rs = \triangle ABC = OD\cdot\frac{a}{2} + OE\cdot\frac{b}{2} + OF\cdot\frac{c}{2} \tag{24}$$

Adding (23) and (24) gives $s(R+r) = s(OD+OE+OF)$, or

$$OD+OE+OF = R+r.$$

Since $OD = R\cos A$ *etc.*, (16) follows.

[1]This proof was introduced to the coordinating country by Professor Bill Sands of Canada.

APMO 2004 – Problems and Solutions

Problem 1
Determine all finite nonempty sets S of positive integers satisfying

$$\frac{i+j}{(i,j)} \quad \text{is an element of } S \text{ for all } i,j \text{ in } S,$$

where (i,j) is the greatest common divisor of i and j.

Answer: $S = \{2\}$.

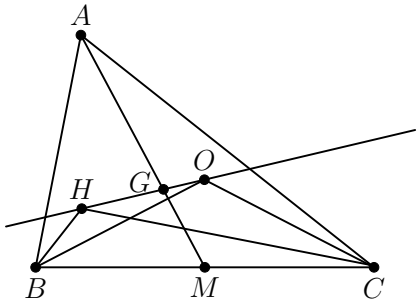
Solution
Let $k \in S$. Then $\frac{k+k}{(k,k)} = 2$ is in S as well.
Suppose for the sake of contradiction that there is an odd number in S, and let k be the largest such odd number. Since $(k,2) = 1$, $\frac{k+2}{(k,2)} = k+2 > k$ is in S as well, a contradiction. Hence S has no odd numbers.
Now suppose that $\ell > 2$ is the second smallest number in S. Then ℓ is even and $\frac{\ell+2}{(\ell,2)} = \frac{\ell}{2} + 1$ is in S. Since $\ell > 2 \implies \frac{\ell}{2} + 1 > 2$, $\frac{\ell}{2} + 1 \geq \ell \iff \ell \leq 2$, a contradiction again.
Therefore S can only contain 2, and $S = \{2\}$ is the only solution.

Problem 2

Let O be the circumcentre and H the orthocentre of an acute triangle ABC. Prove that the area of one of the triangles AOH, BOH and COH is equal to the sum of the areas of the other two.

Solution 1

Suppose, without loss of generality, that B and C lies in the same side of line OH. Such line is the *Euler line* of ABC, so the centroid G lies in this line.



Let M be the midpoint of BC. Then the distance between M and the line OH is the average of the distances from B and C to OH, and the sum of the areas of triangles BOH and COH is

$$[BOH] + [COH] = \frac{OH \cdot d(B, OH)}{2} + \frac{OH \cdot d(C, OH)}{2} = \frac{OH \cdot 2d(M, OH)}{2}.$$

Since $AG = 2GM$, $d(A, OH) = 2d(M, OH)$. Hence

$$[BOH] + [COH] = \frac{OH \cdot d(A, OH)}{2} = [AOH],$$

and the result follows.

Solution 2

One can use barycentric coordinates: it is well known that

$$A = (1 : 0 : 0), \quad B = (0 : 1 : 0), \quad C = (0 : 0 : 1),$$
$$O = (\sin 2A : \sin 2B : \sin 2C) \quad \text{and} \quad H = (\tan A : \tan B : \tan C).$$

Then the (signed) area of AOH is proportional to

$$\begin{vmatrix} 1 & 0 & 0 \\ \sin 2A & \sin 2B & \sin 2C \\ \tan A & \tan B & \tan C \end{vmatrix}$$

Adding all three expressions we find that the sum of the signed sums of the areas is a constant times

$$\begin{vmatrix} 1 & 0 & 0 \\ \sin 2A & \sin 2B & \sin 2C \\ \tan A & \tan B & \tan C \end{vmatrix} + \begin{vmatrix} 0 & 1 & 0 \\ \sin 2A & \sin 2B & \sin 2C \\ \tan A & \tan B & \tan C \end{vmatrix} + \begin{vmatrix} 0 & 0 & 1 \\ \sin 2A & \sin 2B & \sin 2C \\ \tan A & \tan B & \tan C \end{vmatrix}.$$

By multilinearity of the determinant, this sum equals

$$\begin{vmatrix} 1 & 1 & 1 \\ \sin 2A & \sin 2B & \sin 2C \\ \tan A & \tan B & \tan C \end{vmatrix},$$

which contains, in its rows, the coordinates of the centroid, the circumcenter, and the orthocenter. Since these three points lie in the Euler line of ABC, the signed sum of the areas is 0, which means that one of the areas of AOH, BOH, COH is the sum of the other two areas.

Comment: Both solutions can be adapted to prove a stronger result: if the centroid G of triangle ABC belongs to line XY then one of the areas of triangles AXY, BXY, and CXY is equal to the sum of the other two.

Problem 3

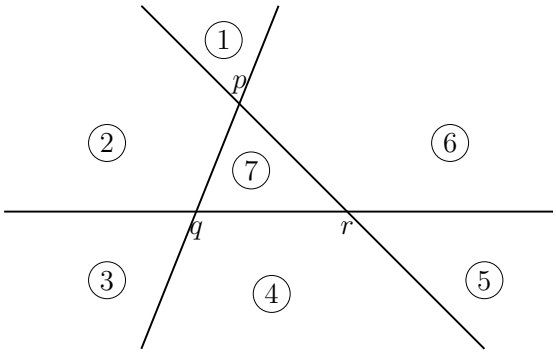
Let a set S of 2004 points in the plane be given, no three of which are collinear. Let $\mathcal{L}$ denote the set of all lines (extended indefinitely in both directions) determined by pairs of points from the set. Show that it is possible to colour the points of S with at most two colours, such that for any points p, q of S, the number of lines in $\mathcal{L}$ which separate p from q is odd if and only if p and q have the same colour.

Note: A line ℓ separates two points p and q if p and q lie on opposite sides of ℓ with neither point on ℓ.

Solution

Choose any point p from S and color it, say, blue. Let $n(q, r)$ be the number of lines from $\mathcal{L}$ that separates q and r. Then color any other point q blue if $n(p, q)$ is odd and red if $n(p, q)$ is even.

Now it remains to show that q and r have the same color if and only if $n(q, r)$ is odd for all $q \neq p$ and $r \neq p$, which is equivalent to proving that $n(p, q) + n(p, r) + n(q, r)$ is always odd. For this purpose, consider the seven numbered regions defined by lines pq, pr, and qr:



Any line that do not pass through any of points p, q, r meets the sides pq, qr, pr of triangle pqr in an even number of points (two sides or no sides), so these lines do not affect the parity of $n(p, q) + n(p, r) + n(q, r)$. Hence the only lines that need to be considered are the ones that pass through one of vertices p, q, r and cuts the opposite side in the triangle pqr.

Let n_i be the number of points in region i, p, q, and r excluded, as depicted in the diagram. Then the lines through p that separate q and r are the lines passing through p and points from regions 1, 4, and 7. The same applies for p, q and regions 2, 5, and 7; and p, r and regions 3, 6, and 7. Therefore

$$\begin{aligned} n(p, q) + n(q, r) + n(p, r) &\equiv (n_2 + n_5 + n_7) + (n_1 + n_4 + n_7) + (n_3 + n_6 + n_7) \\ &\equiv n_1 + n_2 + n_3 + n_4 + n_5 + n_6 + n_7 = 2004 - 3 \equiv 1 \pmod 2, \end{aligned}$$

and the result follows.

Comment: The problem statement is also true if 2004 is replaced by any even number and is not true if 2004 is replaced by any odd number greater than 1.

Problem 4

For a real number x, let $\lfloor x \rfloor$ stand for the largest integer that is less than or equal to x. Prove that

$$\left\lfloor \frac{(n-1)!}{n(n+1)} \right\rfloor$$

is even for every positive integer n.

Solution

Consider four cases:

- $n \leq 5$. Then $\left\lfloor \frac{(n-1)!}{n(n+1)} \right\rfloor = 0$ is an even number.
- *n and $n+1$ are both composite (in particular, $n \geq 8$).* Then $n = ab$ and $n+1 = cd$ for $a, b, c, d \geq 2$. Moreover, since n and $n+1$ are coprime, a, b, c, d are all distinct and smaller than n, and one can choose a, b, c, d such that exactly one of these four numbers is even. Hence $\frac{(n-1)!}{n(n+1)}$ is an integer. As $n \geq 8 > 6$, $(n-1)!$ has at least three even factors, so $\frac{(n-1)!}{n(n+1)}$ is an even integer.
- *$n \geq 7$ is an odd prime.* By Wilson's theorem, $(n-1)! \equiv -1 \pmod{n}$, that is, $\frac{(n-1)!+1}{n}$ is an integer, as $\frac{(n-1)!+n+1}{n} = \frac{(n-1)!+1}{n} + 1$ is. As before, $\frac{(n-1)!}{n+1}$ is an even integer; therefore $\frac{(n-1)!+n+1}{n+1} = \frac{(n-1)!}{n+1} + 1$ is an odd integer.

  Also, n and $n+1$ are coprime and n divides the odd integer $\frac{(n-1)!+n+1}{n+1}$, so $\frac{(n-1)!+n+1}{n(n+1)}$ is also an odd integer. Then

  $$\left\lfloor \frac{(n-1)!}{n(n+1)} \right\rfloor = \frac{(n-1)!+n+1}{n(n+1)} - 1$$

  is even.
- *$n+1 \geq 7$ is an odd prime.* Again, since n is composite, $\frac{(n-1)!}{n}$ is an even integer, and $\frac{(n-1)!+n}{n}$ is an odd integer. By Wilson's theorem, $n! \equiv -1 \pmod{n+1} \iff (n-1)! \equiv 1 \pmod{n+1}$. This means that $n+1$ divides $(n-1)!+n$, and since n and $n+1$ are coprime, $n+1$ also divides $\frac{(n-1)!+n}{n}$. Then $\frac{(n-1)!+n}{n(n+1)}$ is also an odd integer and

  $$\left\lfloor \frac{(n-1)!}{n(n+1)} \right\rfloor = \frac{(n-1)!+n}{n(n+1)} - 1$$

  is even.

Problem 5
Prove that

$$(a^2+2)(b^2+2)(c^2+2) \geq 9(ab+bc+ca)$$

for all real numbers $a, b, c > 0$.

Solution 1
Let $p = a+b+c$, $q = ab+bc+ca$, and $r = abc$. The inequality simplifies to

$$a^2b^2c^2 + 2(a^2b^2+b^2c^2+c^2a^2) + 4(a^2+b^2+c^2) + 8 - 9(ab+bc+ca) \geq 0.$$

Since $a^2b^2+b^2c^2+c^2a^2 = q^2-2pr$ and $a^2+b^2+c^2 = p^2-2q$,

$$r^2 + 2q^2 - 4pr + 4p^2 - 8q + 8 - 9q \geq 0,$$

which simplifies to

$$r^2 + 2q^2 + 4p^2 - 17q - 4pr + 8 \geq 0. \qquad (I)$$

Bearing in mind that equality occurs for $a = b = c = 1$, which means that, for instance, $p = 3r$, one can rewrite (I) as

$$\left(r - \frac{p}{3}\right)^2 - \frac{10}{3}pr + \frac{35}{9}p^2 + 2q^2 - 17q + 8 \geq 0. \qquad (II)$$

Since $(ab-bc)^2 + (bc-ca)^2 + (ca-ab)^2 \geq 0$ is equivalent to $q^2 \geq 3pr$, rewrite (II) as

$$\left(r - \frac{p}{3}\right)^2 + \frac{10}{9}(q^2-3pr) + \frac{35}{9}p^2 + \frac{8}{9}q^2 - 17q + 8 \geq 0. \qquad (III)$$

Finally, $a = b = c = 1$ implies $q = 3$; then rewrite (III) as

$$\left(r - \frac{p}{3}\right)^2 + \frac{10}{9}(q^2-3pr) + \frac{35}{9}(p^2-3q) + \frac{8}{9}(q-3)^2 \geq 0.$$

This final inequality is true because $q^2 \geq 3pr$ and $p^2 - 3q = \frac{1}{2}[(a-b)^2+(b-c)^2+(c-a)^2] \geq 0$.

Solution 2
We prove the stronger inequality

$$(a^2+2)(b^2+2)(c^2+2) \geq 3(a+b+c)^2, \qquad (*)$$

which implies the proposed inequality because $(a+b+c)^2 \geq 3(ab+bc+ca)$ is equivalent to $(a-b)^2+(b-c)^2+(c-a)^2 \geq 0$, which is immediate.
The inequality $(*)$ is equivalent to

$$\big((b^2+2)(c^2+2)-3\big)a^2 - 6(b+c)a + 2(b^2+2)(c^2+2) - 3(b+c)^2 \geq 0.$$

Seeing this inequality as a quadratic inequality in a with positive leading coefficient $(b^2+2)(c^2+2) - 3 = b^2c^2 + 2b^2 + 2c^2 + 1$, it suffices to prove that its discriminant is non-positive, which is equivalent to

$$(3(b+c))^2 - \big((b^2+2)(c^2+2)-3\big)\big(2(b^2+2)(c^2+2) - 3(b+c)^2\big) \leq 0.$$

This simplifies to

$$-2(b^2+2)(c^2+2) + 3(b+c)^2 + 6 \leq 0. \qquad (**)$$

Now we look $(**)$ as a quadratic inequality in b with negative leading coefficient $-2c^2-1$:

$$(-2c^2-1)b^2 + 6cb - c^2 - 2 \leq 0.$$

If suffices to show that the discriminant of $(**)$ is non-positive, which is equivalent to

$$9c^2 - (2c^2+1)(c^2+2) \leq 0.$$

It simplifies to $-2(c^2-1)^2 \leq 0$, which is true. The equality occurs for $c^2 = 1$, that is, $c = 1$, for which $b = \frac{6c}{2(2c^2+1)} = 1$, and $a = \frac{6(b+c)}{2((b^2+2)(c^2+2)-3)} = 1$.

Solution 3

Let A, B, C angles in $(0, \pi/2)$ such that $a = \sqrt{2}\tan A$, $b = \sqrt{2}\tan B$, and $c = \sqrt{2}\tan C$. Then the inequality is equivalent to

$$4\sec^2 A \sec^2 B \sec^2 C \geq 9(\tan A \tan B + \tan B \tan C + \tan C \tan A).$$

Substituting $\sec x = \frac{1}{\cos x}$ for $x \in \{A, B, C\}$ and clearing denominators, the inequality is equivalent to

$$\cos A \cos B \cos C(\sin A \sin B \cos C + \cos A \sin B \sin C + \sin A \cos B \sin C) \leq \frac{4}{9}.$$

Since

$$\begin{aligned} &\cos(A + B + C) = \cos A \cos(B + C) - \sin A \sin(B + C) \\ = \ &\cos A \cos B \cos C - \cos A \sin B \sin C - \sin A \cos B \sin C - \sin A \sin B \cos C, \end{aligned}$$

we rewrite our inequality as

$$\cos A \cos B \cos C(\cos A \cos B \cos C - \cos(A + B + C)) \leq \frac{4}{9}.$$

The cosine function is concave down on $(0, \pi/2)$. Therefore, if $\theta = \frac{A+B+C}{3}$, by the AM-GM inequality and Jensen's inequality,

$$\cos A \cos B \cos C \leq \left(\frac{\cos A + \cos B + \cos C}{3}\right)^3 \leq \cos^3 \frac{A + B + C}{3} = \cos^3 \theta.$$

Therefore, since $\cos A \cos B \cos C - \cos(A + B + C) = \sin A \sin B \cos C + \cos A \sin B \sin C + \sin A \cos B \sin C > 0$, and recalling that $\cos 3\theta = 4\cos^3 \theta - 3\cos \theta$,

$$\cos A \cos B \cos C(\cos A \cos B \cos C - \cos(A+B+C)) \leq \cos^3 \theta(\cos^3 \theta - \cos 3\theta) = 3\cos^4 \theta(1 - \cos^2 \theta).$$

Finally, by AM-GM (notice that $1 - \cos^2 \theta = \sin^2 \theta > 0$),

$$3\cos^4 \theta(1 - \cos^2 \theta) = \frac{3}{2}\cos^2 \theta \cdot \cos^2 \theta(2 - 2\cos^2 \theta) \leq \frac{3}{2}\left(\frac{\cos^2 \theta + \cos^2 \theta + (2 - 2\cos^2 \theta)}{3}\right)^3 = \frac{4}{9},$$

and the result follows.

XV APMO: Solutions and Marking Schemes

1. Let a, b, c, d, e, f be real numbers such that the polynomial

$$p(x) = x^8 - 4x^7 + 7x^6 + ax^5 + bx^4 + cx^3 + dx^2 + ex + f$$

factorises into eight linear factors $x - x_i$, with $x_i > 0$ for $i = 1, 2, \ldots, 8$. Determine all possible values of f.

Solution.

From

$$x^8 - 4x^7 + 7x^6 + ax^5 + bx^4 + cx^3 + dx^2 + ex + f = (x - x_1)(x - x_2)\ldots(x - x_8)$$

we have

$$\sum_{i=1}^{8} x_i = 4 \qquad \text{and} \qquad \sum x_i x_j = 7,$$

where the second sum is over all pairs (i, j) of integers where $1 \le i < j \le 8$. Since this sum can also be written

$$\frac{1}{2}\left[\left(\sum_{i=1}^{8} x_i\right)^2 - \sum_{i=1}^{8} x_i^2\right],$$

we get

$$14 = \left(\sum_{i=1}^{8} x_i\right)^2 - \sum_{i=1}^{8} x_i^2 = 16 - \sum_{i=1}^{8} x_i^2,$$

so

$$\sum_{i=1}^{8} x_i^2 = 2 \qquad \text{while} \qquad \sum_{i=1}^{8} x_i = 4. \qquad \textbf{[3 marks]} \tag{1}$$

Now

$$\sum_{i=1}^{8} (2x_i - 1)^2 = 4\sum_{i=1}^{8} x_i^2 - 4\sum_{i=1}^{8} x_i + 8 = 4(2) - 4(4) + 8 = 0,$$

which forces $x_i = 1/2$ for all i. **[3 marks]** Therefore

$$f = \prod_{i=1}^{8} x_i = \left(\frac{1}{2}\right)^8 = \frac{1}{256}. \qquad \textbf{[1 mark]}$$

Alternate solution: After obtaining (1) **[3 marks]**, use Cauchy's inequality to get

$$16 = (x_1 \cdot 1 + x_2 \cdot 1 + \cdots + x_8 \cdot 1)^2 \le (x_1^2 + x_2^2 + \cdots + x_8^2)(1^2 + 1^2 + \cdots + 1^2) = 8 \cdot 2 = 16\ ;$$

or the power mean inequality to get

$$\frac{1}{2} = \frac{1}{8}\sum_{i=1}^{8} x_i \le \left(\frac{1}{8}\sum_{i=1}^{8} x_i^2\right)^{1/2} = \frac{1}{2}. \qquad \textbf{[2 marks]}$$

Either way, equality must hold, which can only happen if all the terms x_i are equal, that is, if $x_i = 1/2$ for all i. **[1 mark]** Thus $f = 1/256$ as above. **[1 mark]**

2. Suppose $ABCD$ is a square piece of cardboard with side length a. On a plane are two parallel lines ℓ_1 and ℓ_2, which are also a units apart. The square $ABCD$ is placed on the plane so that sides AB and AD intersect ℓ_1 at E and F respectively. Also, sides CB and CD intersect ℓ_2 at G and H respectively. Let the perimeters of $\triangle AEF$ and $\triangle CGH$ be m_1 and m_2 respectively. Prove that no matter how the square was placed, $m_1 + m_2$ remains constant.

Solution 1.

Let EH intersect FG at O. The distance from G to line FD and line EF are both a. So FG bisects $\angle EFD$. Similarly, EH bisects $\angle BEF$. So O is an excentre of $\triangle AEF$. Similarly, O is an excentre of $\triangle CGH$. **[2 marks]** Construct these excircles with centre O. Let M, N, P, Q be on sides AB, BC, CD, DA respectively, where these excircles touch the square. Then $OM \perp AB$, $ON \perp BC$, $OP \perp CD$, and $OQ \perp DA$. Since $AB \parallel CD$ and $AD \parallel BC$, M, O, P are collinear and N, O, Q are collinear. Now $MP = NQ = a$. **[2 marks]** Using the fact that the two tangents from a point to a circle have the same length, we get $EF = EM + FQ$ and $GH = GN + HP$. **[1 mark]** Then

$$m_1 = AE + AF + EF = AE + AF + (EM + FQ) = AM + AQ = OQ + OM$$

and

$$m_2 = CG + CH + GH = CG + CH + (GN + HP) = CN + CP = OP + ON. \qquad \textbf{[1 mark]}$$

Therefore

$$m_1 + m_2 = (OQ + OM) + (OP + ON) = MP + NQ = 2a. \qquad \textbf{[1 mark]}$$

Solution 2.

Extend AB to I and DC to J so that $AE = BI = CJ$. Let ℓ_2 intersect IJ at M, and let K lie on IJ so that $GK \perp IJ$. Then, since $AE = GK$, $\triangle AEF$ and $\triangle KGM$ are congruent. **[1 mark]** Thus, since $GK = CJ$ and $GC = KJ$,

$$m_1 + m_2 = \text{perimeter}(KGM) + \text{perimeter}(CGH) = \text{perimeter}(HMJ). \qquad \textbf{[2 marks]}$$

Let L lie on CD so that $EL \perp CD$. Then a circle with centre E and radius a will touch DC at L, IJ at I, and the interior of HM at some point N, so

$$\text{perimeter}(HMJ) = JH + (HN + NM) + JM = (JH + HL) + (MI + JM) = JL + IJ = a + a = 2a.$$

[4 marks] Thus $m_1 + m_2 = 2a$.

Solution 3.

Without loss of generality, assume the square has side $a = 1$. Let θ be the acute angle between ℓ_1 (or ℓ_2) and the sides AB and CD of the square. Then, letting $EF = x$ and $GH = y$, we have

$$EA = x\cos\theta, \quad AF = x\sin\theta, \qquad CH = y\cos\theta, \quad CG = y\sin\theta.$$

Thus

$$m_1 + m_2 = (x + y)(\sin\theta + \cos\theta + 1). \qquad \textbf{[2 marks]} \qquad (1)$$

Draw lines parallel to ℓ_1, ℓ_2 through A and C respectively. The distance between these lines is $\sin\theta + \cos\theta$ **[1 mark]**, as can be seen by drawing a mutual perpendicular to these lines through B, say. Also, the altitudes from A to EF and from C to GH have lengths $x\sin\theta\cos\theta$ and $y\sin\theta\cos\theta$ respectively **[1 mark]**. Therefore the distance between ℓ_1 and ℓ_2 must be

$$(\sin\theta + \cos\theta) - x\sin\theta\cos\theta - y\sin\theta\cos\theta.$$

But we are given that this distance is $a = 1$, so

$$(x + y)\sin\theta\cos\theta + 1 = \sin\theta + \cos\theta,$$

or

$$x + y = \frac{\sin\theta + \cos\theta - 1}{\sin\theta\cos\theta}. \qquad \textbf{[1 mark]}$$

Therefore, by (1),

$$\begin{aligned} m_1 + m_2 &= \frac{(\sin\theta + \cos\theta - 1)(\sin\theta + \cos\theta + 1)}{\sin\theta\cos\theta} \\ &= \frac{(\sin^2\theta + \cos^2\theta + 2\sin\theta\cos\theta) - 1}{\sin\theta\cos\theta} \\ &= \frac{1 + 2\sin\theta\cos\theta - 1}{\sin\theta\cos\theta} = 2. \qquad \textbf{[2 marks]} \end{aligned}$$

3. Let $k \geq 14$ be an integer, and let p_k be the largest prime number which is strictly less than k. You may assume that $p_k \geq 3k/4$. Let n be a composite integer. Prove:

(a) if $n = 2p_k$, then n does not divide $(n-k)!$;

(b) if $n > 2p_k$, then n divides $(n-k)!$.

Solution.

(a) Note that $n - k = 2p_k - k < 2p_k - p_k = p_k$, so $p_k \nmid (n-k)!$, so $2p_k \nmid (n-k)!$. **[1 mark]**

(b) Note that $n > 2p_k \geq 3k/2$ implies $k < 2n/3$, so $n - k > n/3$. So if we can find integers $a, b \geq 3$ such that $n = ab$ and $a \neq b$, then both a and b will appear separately in the product $(n-k)! = 1 \times 2 \times \cdots \times (n-k)$, which means $n|(n-k)!$. Observe that $k \geq 14$ implies $p_k \geq 13$, so that $n > 2p_k \geq 26$.

If $n = 2^\alpha$ for some integer $\alpha \geq 5$, then take $a = 2^2$, $b = 2^{\alpha-2}$. **[1 mark]** Otherwise, since $n \geq 26 > 16$, we can take a to be an odd prime factor of n and $b = n/a$ **[1 mark]**, unless $b < 3$ or $b = a$.

Case (i): $b < 3$. Since n is composite, this means $b = 2$, so that $2a = n > 2p_k$. As a is a prime number and p_k is the largest prime number which is strictly less than k, it follows that $a \geq k$. From $n - k = 2a - k \geq 2a - a = a > 2$ we see that $n = 2a$ divides into $(n-k)!$. **[2 marks]**

Case (ii): $b = a$. Then $n = a^2$ and $a > 6$ since $n \geq 26$. Thus $n - k > n/3 = a^2/3 > 2a$, so that both a and $2a$ appear among $\{1, 2, \ldots, n-k\}$. Hence $n = a^2$ divides into $(n-k)!$. **[2 marks]**

4. Let a, b, c be the sides of a triangle, with $a + b + c = 1$, and let $n \geq 2$ be an integer. Show that

$$\sqrt[n]{a^n + b^n} + \sqrt[n]{b^n + c^n} + \sqrt[n]{c^n + a^n} < 1 + \frac{\sqrt[n]{2}}{2}.$$

Solution.

Without loss of generality, assume $a \leq b \leq c$. As $a + b > c$, we have

$$\frac{\sqrt[n]{2}}{2} = \frac{\sqrt[n]{2}}{2}(a+b+c) > \frac{\sqrt[n]{2}}{2}(c+c) = \sqrt[n]{2c^n} \geq \sqrt[n]{b^n + c^n}. \qquad \textbf{[2 marks]} \tag{1}$$

As $a \leq c$ and $n \geq 2$, we have

$$\begin{aligned}
(c^n + a^n) - \left(c + \frac{a}{2}\right)^n &= a^n - \sum_{k=1}^{n} \binom{n}{k} c^{n-k} \left(\frac{a}{2}\right)^k \\
&\leq \left[1 - \sum_{k=1}^{n} \binom{n}{k} \left(\frac{1}{2}\right)^k\right] a^n \qquad (\text{since } c^{n-k} \geq a^{n-k}) \\
&= \left[\left(1 - \frac{n}{2}\right) - \sum_{k=2}^{n} \binom{n}{k} \left(\frac{1}{2}\right)^k\right] a^n < 0.
\end{aligned}$$

Thus

$$\sqrt[n]{c^n + a^n} < c + \frac{a}{2}. \qquad \textbf{[3 marks]} \tag{2}$$

Likewise

$$\sqrt[n]{b^n + a^n} < b + \frac{a}{2}. \qquad \textbf{[1 mark]} \tag{3}$$

Adding (1), (2) and (3), we get

$$\sqrt[n]{a^n + b^n} + \sqrt[n]{b^n + c^n} + \sqrt[n]{c^n + a^n} < \frac{\sqrt[n]{2}}{2} + c + \frac{a}{2} + b + \frac{a}{2} = 1 + \frac{\sqrt[n]{2}}{2}. \qquad \textbf{[1 mark]}$$

5. Given two positive integers m and n, find the smallest positive integer k such that among any k people, either there are $2m$ of them who form m pairs of mutually acquainted people or there are $2n$ of them forming n pairs of mutually unacquainted people.

Solution.

Let the smallest positive integer k satisfying the condition of the problem be denoted $r(m,n)$. We shall show that

$$r(m,n) = 2(m+n) - \min\{m,n\} - 1.$$

Observe that, by symmetry, $r(m,n) = r(n,m)$. Therefore it suffices to consider the case where $m \geq n$, and to prove that

$$r(m,n) = 2m + n - 1. \qquad \textbf{[1 mark]} \tag{1}$$

First we prove that

$$r(m,n) \geq 2m + n - 1$$

by an example. Call a group of k people, every two of whom are mutually acquainted, a *k-clique*. Consider a set of $2m+n-2$ people consisting of a $(2m-1)$-clique together with an additional $n-1$ people none of whom know anyone else. (Call such people *isolated*.) Then there are not $2m$ people forming m mutually acquainted pairs, and there also are not $2n$ people forming n mutually unacquainted pairs. Thus $r(m,n) \geq (2m-1)+(n-1)+1 = 2m+n-1$ by the definition of $r(m,n)$. **[1 mark]**

To establish (1), we need to prove that $r(m,n) \leq 2m+n-1$. To do this, we now show that

$$r(m,n) \leq r(m-1,n-1) + 3 \qquad \text{for all } m \geq n \geq 2. \tag{2}$$

Let G be a group of $t = r(m-1,n-1)+3$ people. Notice that

$$t \geq 2(m-1) + (n-1) - 1 + 3 = 2m + n - 1 \geq 2m \geq 2n.$$

If G is a t-clique, then G contains $2m$ people forming m mutually acquainted pairs, and if G has only isolated people, then G contains $2n$ people forming n mutually unacquainted pairs. Otherwise, there are three people in G, say a, b and c, such that a, b are acquainted but a, c are not. Now consider the group A obtained by removing a, b and c from G. A has $t-3 = r(m-1,n-1)$ people, so by the definition of $r(m-1,n-1)$, A either contains $2(m-1)$ people forming $m-1$ mutually acquainted pairs, or else contains $2(n-1)$ people forming $n-1$ mutually unacquainted pairs. In the former case, we add the acquainted pair a, b to A to form m mutually acquainted pairs in G. In the latter case, we add the unacquainted pair a, c to A to form n mutually unacquainted pairs in G. This proves (2). **[3 marks]**

Trivially, $r(s,1) = 2s$ for all s **[1 mark]**, so $r(m,n) \leq 2m+n-1$ holds whenever $n = 1$. Proceeding by induction on n, by (2) we obtain

$$r(m,n) \leq r(m-1,n-1) + 3 \leq 2(m-1) + (n-1) - 1 + 3 = 2m + n - 1,$$

which completes the proof. **[1 mark]**

Note. Give an additional 1 mark to any student who gets at most 5 marks by the above marking scheme, but in addition gives a valid argument that $r(2,2) = 5$.

XIV APMO: Solutions and Marking Schemes

1. Let $a_1, a_2, a_3, \ldots, a_n$ be a sequence of non-negative integers, where n is a positive integer. Let

$$A_n = \frac{a_1 + a_2 + \cdots + a_n}{n}.$$

Prove that

$$a_1! a_2! \ldots a_n! \geq (\lfloor A_n \rfloor !)^n,$$

where $\lfloor A_n \rfloor$ is the greatest integer less than or equal to A_n, and $a! = 1 \times 2 \times \cdots \times a$ for $a \geq 1$ (and $0! = 1$). When does equality hold?

Solution 1.

Assume without loss of generality that $a_1 \geq a_2 \geq \cdots \geq a_n \geq 0$, and let $s = \lfloor A_n \rfloor$. Let k be any (fixed) index for which $a_k \geq s \geq a_{k+1}$.

Our inequality is equivalent to proving that

$$\frac{a_1!}{s!} \cdot \frac{a_2!}{s!} \cdot \ldots \cdot \frac{a_k!}{s!} \geq \frac{s!}{a_{k+1}!} \cdot \frac{s!}{a_{k+2}!} \cdot \ldots \cdot \frac{s!}{a_n!}. \tag{1}$$

Now for $i = 1, 2, \ldots, k$, $a_i!/s!$ is the product of $a_i - s$ factors. For example, $9!/5! = 9 \cdot 8 \cdot 7 \cdot 6$. The left side of inequality (1) therefore is the product of $A = a_1 + a_2 + \cdots + a_k - ks$ factors, all of which are greater than s. Similarly, the right side of (1) is the product of $B = (n-k)s - (a_{k+1} + a_{k+2} + \cdots + a_n)$ factors, all of which are at most s. Since $\sum_{i=1}^{n} a_i = nA_n \geq ns$, $A \geq B$. This proves the inequality. **[5 marks to here.]**

Equality in (1) holds if and only if either:

(i) $A = B = 0$, that is, both sides of (1) are the empty product, which occurs if and only if $a_1 = a_2 = \cdots = a_n$; or

(ii) $a_1 = 1$ and $s = 0$, that is, the only factors on either side of (1) are 1's, which occurs if and only if $a_i \in \{0, 1\}$ for all i. **[2 marks for both (i) and (ii), no marks for (i) only.]**

Solution 2.

Assume without loss of generality that $0 \leq a_1 \leq a_2 \leq \cdots \leq a_n$. Let $d = a_n - a_1$ and $m = |\{i : a_i = a_1\}|$. Our proof is by induction on d.

We first do the case $d = a_n - a_1 = 0$ or 1 separately. Then $a_1 = a_2 = \cdots = a_m = a$ and $a_{m+1} = \cdots = a_n = a + 1$ for some $1 \leq m \leq n$ and $a \geq 0$. In this case we have $\lfloor A_n \rfloor = a$, so the inequality to be proven is just $a_1! a_2! \ldots a_n! \geq (a!)^n$, which is obvious. Equality holds if and only if either $m = n$, that is, $a_1 = a_2 = \cdots = a_n = a$; or if $a = 0$, that is, $a_1 = \cdots = a_m = 0$ and $a_{m+1} = \cdots = a_n = 1$. **[2 marks to here.]**

So assume that $d = a_n - a_1 \geq 2$ and that the inequality holds for all sequences with smaller values of d, or with the same value of d and smaller values of m. Then the sequence

$$a_1 + 1,\ a_2,\ a_3,\ \ldots,\ a_{n-1},\ a_n - 1,$$

though not necessarily in non-decreasing order any more, does have either a smaller value of d, or the same value of d and a smaller value of m, but in any case has the same value of A_n. Thus, by induction and since $a_n > a_1 + 1$,

$$\begin{aligned} a_1!a_2!\ldots a_n! &= (a_1+1)!a_2!\ldots a_{n-1}!(a_n-1)!\cdot\frac{a_n}{a_1+1} \\ &\geq (\lfloor A_n\rfloor!)^n\cdot\frac{a_n}{a_1+1} \\ &> (\lfloor A_n\rfloor!)^n, \end{aligned}$$

which completes the proof. Equality cannot hold in this case.

2. Find all positive integers a and b such that

$$\frac{a^2+b}{b^2-a} \quad \text{and} \quad \frac{b^2+a}{a^2-b}$$

are both integers.

Solution.

By the symmetry of the problem, we may suppose that $a \leq b$. Notice that $b^2 - a \geq 0$, so that if $\dfrac{a^2+b}{b^2-a}$ is a positive integer, then $a^2 + b \geq b^2 - a$. Rearranging this inequality and factorizing, we find that $(a+b)(a-b+1) \geq 0$. Since $a, b > 0$, we must have $a \geq b - 1$. [**3 marks to here.**] We therefore have two cases:

Case 1: $a = b$. Substituting, we have

$$\frac{a^2+a}{a^2-a} = \frac{a+1}{a-1} = 1 + \frac{2}{a-1},$$

which is an integer if and only if $(a-1)|2$. As $a > 0$, the only possible values are $a - 1 = 1$ or 2. Hence, $(a, b) = (2, 2)$ or $(3, 3)$. [**1 mark.**]

Case 2: $a = b - 1$. Substituting, we have

$$\frac{b^2+a}{a^2-b} = \frac{(a+1)^2+a}{a^2-(a+1)} = \frac{a^2+3a+1}{a^2-a-1} = 1 + \frac{4a+2}{a^2-a-1}.$$

Once again, notice that $4a + 2 > 0$, and hence, for $\dfrac{4a+2}{a^2-a-1}$ to be an integer, we must have $4a + 2 \geq a^2 - a - 1$, that is, $a^2 - 5a - 3 \leq 0$. Hence, since a is an integer, we can bound a by $1 \leq a \leq 5$. Checking all the ordered pairs $(a, b) = (1, 2), (2, 3), \ldots, (5, 6)$, we find that only $(1, 2)$ and $(2, 3)$ satisfy the given conditions. [**3 marks.**]

Thus, the ordered pairs that work are

$$(2, 2), (3, 3), (1, 2), (2, 3), (2, 1), (3, 2),$$

where the last two pairs follow by symmetry. [**2 marks if these solutions are found without proof that there are no others.**]

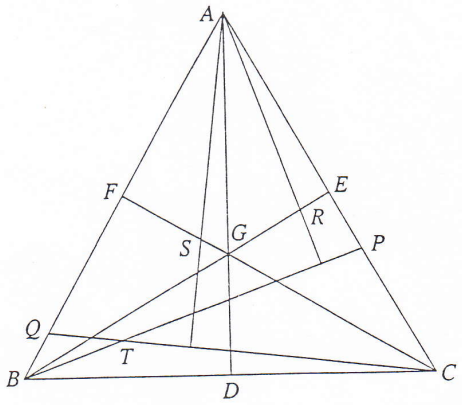
3. Let ABC be an equilateral triangle. Let P be a point on the side AC and Q be a point on the side AB so that both triangles ABP and ACQ are acute. Let R be the orthocentre of triangle ABP and S be the orthocentre of triangle ACQ. Let T be the point common to the segments BP and CQ. Find all possible values of $\angle CBP$ and $\angle BCQ$ such that triangle TRS is equilateral.

Solution.
We are going to show that this can only happen when

$$\angle CBP = \angle BCQ = 15^\circ.$$

Lemma. If $\angle CBP > \angle BCQ$, then $RT > ST$.

Proof. Let AD, BE and CF be the altitudes of triangle ABC concurrent at its centre G. Then P lies on CE, Q lies on BF, and thus T lies in triangle BDG.



Note that

$$\angle FAS = \angle FCQ = 30^\circ - \angle BCQ > 30^\circ - \angle CBP = \angle EBP = \angle EAR.$$

Since $AF = AE$, we have $FS > ER$ so that

$$GS = GF - FS < GE - ER = GR.$$

Let T_x be the projection of T onto BC and T_y be the projection of T onto AD, and similarly for R and S. We have

$$R_xT_x = DR_x + DT_x > |DS_x - DT_x| = S_xT_x$$

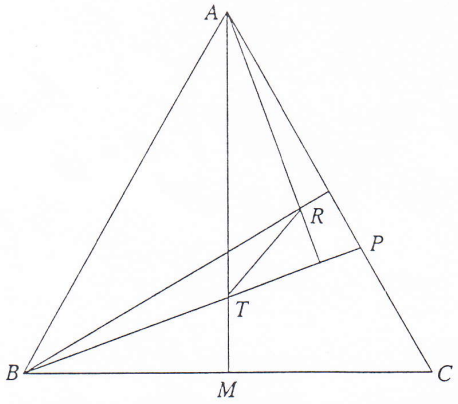
and

$$R_yT_y = GR_y + GT_y > GS_y + GT_y = S_yT_y.$$

It follows that $RT > ST$. □

[1 mark for stating the Lemma, 3 marks for proving it.]

Thus, if $\triangle TRS$ is equilateral, we must have $\angle CBP = \angle BCQ$.



It is clear from the symmetry of the figure that $TR = TS$, so $\triangle TRS$ is equilateral if and only if $\angle RTA = 30°$. Now, as BR is an altitude of the triangle ABC, $\angle RBA = 30°$. So $\triangle TRS$ is equilateral if and only if $RTBA$ is a cyclic quadrilateral. Therefore, $\triangle TRS$ is equilateral if and only if $\angle TBR = \angle TAR$. But

$$\begin{aligned} 90° &= \angle TBA + \angle BAR \\ &= (\angle TBR + \angle RBA) + (\angle BAT + \angle TAR) \\ &= (\angle TBR + 30°) + (30° + \angle TAR) \end{aligned}$$

and so

$$30° = \angle TAR + \angle TBR.$$

But these angles must be equal, so $\angle TAR = \angle TBR = 15°$. Therefore $\angle CBP = \angle BCQ = 15°$. **[3 marks for finishing the proof with the assumption that $\angle CBP = \angle BCQ$.]**

4. Let x, y, z be positive numbers such that

$$\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 1.$$

Show that

$$\sqrt{x+yz} + \sqrt{y+zx} + \sqrt{z+xy} \geq \sqrt{xyz} + \sqrt{x} + \sqrt{y} + \sqrt{z}.$$

Solution 1.

$$\begin{aligned} \sum_{\text{cyclic}} \sqrt{x+yz} &= \sqrt{xyz} \sum_{\text{cyclic}} \sqrt{\frac{1}{x} + \frac{1}{yz}} \\ &= \sqrt{xyz} \sum_{\text{cyclic}} \sqrt{\frac{1}{x}\left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z}\right) + \frac{1}{yz}} \qquad \textbf{[1 mark.]} \\ &= \sqrt{xyz} \sum_{\text{cyclic}} \sqrt{\left(\frac{1}{x} + \frac{1}{y}\right)\left(\frac{1}{x} + \frac{1}{z}\right)} \qquad \textbf{[1 mark.]} \end{aligned}$$

$$= \sqrt{xyz}\sum_{\text{cyclic}}\sqrt{\left(\frac{1}{x}+\frac{1}{\sqrt{yz}}\right)^2+\frac{(\sqrt{y}-\sqrt{z})^2}{xyz}} \qquad \text{[2 marks.]}$$

$$\geq \sqrt{xyz}\sum_{\text{cyclic}}\left(\frac{1}{x}+\frac{1}{\sqrt{yz}}\right) \qquad \text{[1 mark.]}$$

$$= \sqrt{xyz}\left(1+\sum_{\text{cyclic}}\frac{1}{\sqrt{yz}}\right) \qquad \text{[1 mark.]}$$

$$= \sqrt{xyz}+\sum_{\text{cyclic}}\sqrt{x}. \qquad \text{[1 mark.]}$$

Note. It is easy to check that equality holds if and only if $x = y = z = 3$.

Solution 2.
Squaring both sides of the given inequality, we obtain

$$\sum_{\text{cyclic}} x+\sum_{\text{cyclic}} yz+2\sum_{\text{cyclic}}\sqrt{x+yz}\sqrt{y+zx}$$

$$\geq xyz+2\sqrt{xyz}\sum_{\text{cyclic}}\sqrt{x}+\sum_{\text{cyclic}} x+2\sum_{\text{cyclic}}\sqrt{xy}. \qquad \text{[1 mark.]}$$

It follows from the given condition $\frac{1}{x}+\frac{1}{y}+\frac{1}{z}=1$ that $xyz=\sum_{\text{cyclic}} xy$. Therefore, the given inequality is equivalent to

$$\sum_{\text{cyclic}}\sqrt{x+yz}\sqrt{y+zx}\geq\sqrt{xyz}\sum_{\text{cyclic}}\sqrt{x}+\sum_{\text{cyclic}}\sqrt{xy}. \qquad \text{[2 marks.]}$$

Using the Cauchy–Schwarz inequality [or just $x^2+y^2\geq 2xy$], we see that

$$(x+yz)(y+zx)\geq(\sqrt{xy}+\sqrt{xyz^2})^2, \qquad \text{[1 mark.]}$$

or

$$\sqrt{x+yz}\sqrt{y+zx}\geq\sqrt{xy}+\sqrt{z}\sqrt{xyz}. \qquad \text{[1 mark.]}$$

Taking the cyclic sum of this inequality over x, y and z, we get the desired inequality. [2 marks.]

Solution 3.
This is another way of presenting the idea in the first solution.
Using the condition $\frac{1}{x}+\frac{1}{y}+\frac{1}{z}=1$ and the AM-GM inequality, we have

$$x+yz-\left(\sqrt{\frac{yz}{x}}+\sqrt{x}\right)^2 = yz\left(1-\frac{1}{x}\right)-2\sqrt{yz}$$

$$= yz\left(\frac{1}{y}+\frac{1}{z}\right)-2\sqrt{yz}=y+z-2\sqrt{yz}\geq 0,$$

which gives

$$\sqrt{x+yz}\geq\sqrt{\frac{yz}{x}}+\sqrt{x}. \qquad \text{[3 marks.]}$$

Similarly, we have

$$\sqrt{y+zx} \geq \sqrt{\frac{zx}{y}} + \sqrt{y} \quad \text{and} \quad \sqrt{z+xy} \geq \sqrt{\frac{xy}{z}} + \sqrt{z}.$$

Addition yields

$$\sqrt{x+yz} + \sqrt{y+zx} + \sqrt{z+xy} \geq \sqrt{\frac{yz}{x}} + \sqrt{\frac{zx}{y}} + \sqrt{\frac{xy}{z}} + \sqrt{x} + \sqrt{y} + \sqrt{z}.$$

[2 marks.] Using the condition $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 1$ again, we have

$$\sqrt{\frac{yz}{x}} + \sqrt{\frac{zx}{y}} + \sqrt{\frac{xy}{z}} = \sqrt{xyz}\left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z}\right) = \sqrt{xyz}, \qquad \textbf{[1 mark.]}$$

and thus

$$\sqrt{x+yz} + \sqrt{y+zx} + \sqrt{z+xy} \geq \sqrt{xyz} + \sqrt{x} + \sqrt{y} + \sqrt{z}. \qquad \textbf{[1 mark.]}$$

Solution 4.
This is also another way of presenting the idea in the first solution.
We make the substitution $a = \frac{1}{x}$, $b = \frac{1}{y}$, $c = \frac{1}{z}$. Then it is enough to show that

$$\sqrt{\frac{1}{a} + \frac{1}{bc}} + \sqrt{\frac{1}{b} + \frac{1}{ca}} + \sqrt{\frac{1}{c} + \frac{1}{ab}} \geq \sqrt{\frac{1}{abc}} + \sqrt{\frac{1}{a}} + \sqrt{\frac{1}{b}} + \sqrt{\frac{1}{c}},$$

where $a + b + c = 1$. Multiplying this inequality by $\sqrt{abc}$, we find that it can be written

$$\sqrt{a+bc} + \sqrt{b+ca} + \sqrt{c+ab} \geq 1 + \sqrt{bc} + \sqrt{ca} + \sqrt{ab}. \qquad \textbf{[1 mark.]}$$

This is equivalent to

$$\sqrt{a(a+b+c)+bc} + \sqrt{b(a+b+c)+ca} + \sqrt{c(a+b+c)+ab}$$
$$\geq a + b + c + \sqrt{bc} + \sqrt{ca} + \sqrt{ab}, \qquad \textbf{[1 mark.]}$$

which in turn is equivalent to

$$\sqrt{(a+b)(a+c)} + \sqrt{(b+c)(b+a)} + \sqrt{(c+a)(c+b)} \geq a + b + c + \sqrt{bc} + \sqrt{ca} + \sqrt{ab}.$$

[1 mark.] (This is a *homogeneous* version of the original inequality.) By the Cauchy–Schwarz inequality (or since $b + c \geq 2\sqrt{bc}$), we have

$$[(\sqrt{a})^2 + (\sqrt{b})^2][(\sqrt{a})^2 + (\sqrt{c})^2] \geq (\sqrt{a}\sqrt{a} + \sqrt{b}\sqrt{c})^2$$

or

$$\sqrt{(a+b)(a+c)} \geq a + \sqrt{bc}. \qquad \textbf{[2 marks.]}$$

Taking the cyclic sum of this inequality over a, b, c, we get the desired inequality. **[2 marks.]**

5. Let $\mathbf{R}$ denote the set of all real numbers. Find all functions f from $\mathbf{R}$ to $\mathbf{R}$ satisfying:
(i) there are only finitely many s in $\mathbf{R}$ such that $f(s) = 0$, and
(ii) $f(x^4 + y) = x^3 f(x) + f(f(y))$ for all x, y in $\mathbf{R}$.

Solution 1.
The only such function is the identity function on $\mathbf{R}$.
Setting $(x, y) = (1, 0)$ in the given functional equation (ii), we have $f(f(0)) = 0$. Setting $x = 0$ in (ii), we find

$$f(y) = f(f(y)) \tag{1}$$

[1 mark.] and thus $f(0) = f(f(0)) = 0$ **[1 mark.]**. It follows from (ii) that $f(x^4 + y) = x^3 f(x) + f(y)$ for all $x, y \in \mathbf{R}$. Set $y = 0$ to obtain

$$f(x^4) = x^3 f(x) \tag{2}$$

for all $x \in \mathbf{R}$, and so

$$f(x^4 + y) = f(x^4) + f(y) \tag{3}$$

for all $x, y \in \mathbf{R}$. The functional equation (3) suggests that f is *additive*, that is, $f(a + b) = f(a) + f(b)$ for all $a, b \in \mathbf{R}$. **[1 mark.]** We now show this.
First assume that $a \geq 0$ and $b \in \mathbf{R}$. It follows from (3) that

$$f(a + b) = f((a^{1/4})^4 + b) = f((a^{1/4})^4) + f(b) = f(a) + f(b).$$

We next note that f is an *odd* function, since from (2)

$$f(-x) = \frac{f(x^4)}{(-x)^3} = \frac{f(x^4)}{-x^3} = -f(x), \quad x \neq 0.$$

Since f is odd, we have that, for $a < 0$ and $b \in \mathbf{R}$,

$$\begin{aligned} f(a + b) &= -f((-a) + (-b)) = -(f(-a) + f(-b)) \\ &= -(-f(a) - f(b)) = f(a) + f(b). \end{aligned}$$

Therefore, we conclude that $f(a + b) = f(a) + f(b)$ for all $a, b \in \mathbf{R}$. **[2 marks.]**
We now show that $\{s \in \mathbf{R} | f(s) = 0\} = \{0\}$. Recall that $f(0) = 0$. Assume that there is a nonzero $h \in \mathbf{R}$ such that $f(h) = 0$. Then, using the fact that f is additive, we inductively have $f(nh) = 0$ or $nh \in \{s \in \mathbf{R} | f(s) = 0\}$ for all $n \in \mathbf{N}$. However, this is a contradiction to the given condition (i). **[1 mark.]**
It's now easy to check that f is *one-to-one*. Assume that $f(a) = f(b)$ for some $a, b \in \mathbf{R}$. Then, we have $f(b) = f(a) = f(a - b) + f(b)$ or $f(a - b) = 0$. This implies that $a - b \in \{s \in \mathbf{R} | f(s) = 0\} = \{0\}$ or $a = b$, as desired. From (1) and the fact that f is one-to-one, we deduce that $f(x) = x$ for all $x \in \mathbf{R}$. **[1 mark.]** This completes the proof.

Solution 2.
Again, the only such function is the identity function on $\mathbf{R}$.
As in Solution 1, we first show that $f(f(y)) = f(y)$, $f(0) = 0$, and $f(x^4) = x^3 f(x)$. **[2 marks.]** From the latter follows

$$f(x) = 0 \Longrightarrow f(x^4) = 0,$$

and from condition (i) we get that $f(x) = 0$ only possibly for $x \in \{0, 1, -1\}$. [**1 mark.**]

Next we prove

$$f(a) = b \Longrightarrow f\left(\sqrt[4]{|a-b|}\right) = 0.$$

This is clear if $a = b$. If $a > b$ then

$$\begin{aligned} f(a) &= f((a-b)+b) = (a-b)^{3/4} f(\sqrt[4]{a-b}) + f(f(b)) \\ &= (a-b)^{3/4} f(\sqrt[4]{a-b}) + f(b) \\ &= (a-b)^{3/4} f(\sqrt[4]{a-b}) + f(f(a)) \\ &= (a-b)^{3/4} f(\sqrt[4]{a-b}) + f(a), \end{aligned}$$

so $(a-b)^{3/4} f(\sqrt[4]{a-b}) = 0$ which means $f\left(\sqrt[4]{|a-b|}\right) = 0$. If $a < b$ we get similarly

$$\begin{aligned} f(b) &= f((b-a)+a) = (b-a)^{3/4} f(\sqrt[4]{b-a}) + f(f(a)) \\ &= (b-a)^{3/4} f(\sqrt[4]{b-a}) + f(b), \end{aligned}$$

and again $f\left(\sqrt[4]{|a-b|}\right) = 0$. [**2 marks.**]

Thus $f(a) = b \Longrightarrow |a-b| \in \{0, 1\}$. Suppose that $f(x) = x + b$ for some x, where $|b| = 1$. Then from $f(x^4) = x^3 f(x)$ and $f(x^4) = x^4 + a$ for some $|a| \leq 1$ we get $x^3 = a/b$, so $|x| \leq 1$. Thus $f(x) = x$ for all x except possibly $x = \pm 1$. [**1 mark.**] But for example,

$$f(1) = f(2^4 - 15) = 2^3 f(2) + f(f(-15)) = 2^3 \cdot 2 - 15 = 1$$

and

$$f(-1) = f(2^4 - 17) = 2^3 f(2) + f(f(-17)) = 2^3 \cdot 2 - 17 = -1.$$

[**1 mark.**] This finishes the proof.

XI APMO - SOLUTIONS AND MARKING SCHEMES

Problem 1.

First solution.

Let us prove that $n = S(n) + 9T(n)$ proceding by induction over the number of digits of n.

1 POINT.

If n has one digit then the result is trivial. Suppose that $n = S(n) + 9T(n)$ is true for any integer n of k digits. Now any number m of $k+1$ digits can be writen as $m = 10n + a$ where n is a number of k digits. Obviously,

$$T(m) = n + T(n) \quad \text{and} \quad (m) = S(n) + a.$$

2 POINTS (1 POINT for each of the last equalities).

Therefore

$$\begin{aligned} m - S(m) &= 10n + a - S(n) - a \\ &= 10n - S(n) \\ &= (n - S(n)) + 9n \\ &= 9T(n) + 9n \\ &= 9T(m), \end{aligned}$$

as required.

4 POINTS.

Second solution.

Let $n = \overline{a_k a_{k-1} \ldots a_1 a_0} = 10^k a_k + 10^{k-1} a_{k-1} + \ldots + 10a_1 + a_0$, where a_k, a_{k-1}, $\ldots$, a_1, a_0 are digits. The stumps of n are

$$\begin{aligned} \overline{a_k a_{k-1} \ldots a_1} &= 10^{k-1} a_k + 10^{k-2} a_{k-1} + \ldots + 10a_2 + a_1 \\ \overline{a_k a_{k-1} \ldots a_2} &= 10^{k-2} a_k + 10^{k-3} a_{k-1} + \ldots + a_2 \\ &\vdots \\ \overline{a_k a_{k-1}} &= 10a_k + a_{k-1} \\ a_k &= a_k. \end{aligned}$$

1 POINT.

Since $10^{m-1} + 10^{m-2} + \ldots + 10 + 1 = \frac{10^m - 1}{9}$ then the sum of all stumps of n is

$$T(n) = \frac{10^k - 1}{9} a_k + \frac{10^{k-1} - 1}{9} a_{k-1} + \ldots + \frac{10 - 1}{9} a_1,$$

3 POINTS.

and hence

$$\begin{aligned} 9T(n) &= 10^k a_k + 10^{k-1} a_{k-1} + \ldots + 10a_1 + a_0 - (a_k + a_{k-1} + \ldots + a_1 + a_0) \\ &= n - S(n). \end{aligned}$$

Consequently $n = S(n) + 9T(n)$.

3 POINTS.

Third Solution.

Let $k(n)$ be the number of zeros at the end of the decimal representation of n or, which is the same, the largest power of 10 which divides n. The the following two observations are straightforward:

1. $S(n) = S(n-1) - (9k(n) - 1)$,
2. $k(1) + k(2) + \ldots + k(n) = T(n)$.

4 POINTS (2 POINTS for each of these equalities).

Then summing the following equalities up

$$\begin{aligned} S(1) &= S(0) - (9k(1) - 1), \\ S(2) &= S(1) - (9k(2) - 1), \\ &\vdots \\ S(n-1) &= S(n-2) - (9k(n-1) - 1), \\ S(n) &= S(n-1) - (9k(n-1) - 1) \end{aligned}$$

we get $S(n) = n - (k(1) + k(2) + \ldots + k(n)) = n - 9T(n)$.

3 POINT for concluding.

Problem 2.

First Solution.

This is equivalent to find the largest positive integer solution of the equation

$$\left\lfloor \frac{N}{3} \right\rfloor = \left\lfloor \frac{N}{5} \right\rfloor + \left\lfloor \frac{N}{7} \right\rfloor - \left\lfloor \frac{N}{35} \right\rfloor,$$

or

$$\left\lfloor \frac{N}{3} \right\rfloor + \left\lfloor \frac{N}{35} \right\rfloor = \left\lfloor \frac{N}{5} \right\rfloor + \left\lfloor \frac{N}{7} \right\rfloor. \tag{1}$$

1 POINT for equality (1).

For N to be a solution of (1) it is necessary that

$$\frac{N-2}{3} + \frac{N-34}{35} \leq \frac{N}{5} + \frac{N}{7},$$

which simplifies to $N \leq 86$.

1 POINTS for finding that $N \leq 86$.

However, if $N \geq 70$ then because $N \leq 86$, (1) implies that

$$\frac{N-2}{3} + \frac{N-16}{35} \leq \frac{N}{5} + \frac{N}{7}$$

which simplifies to $N \leq 59$, contradicting $N \geq 70$. it follows that N must be at most 69.

4 POINTS for finding that $N \leq 69$.

Checking (1) for $N \leq 69$ we find that

when	$N = 69,$	(1) is	$23 + 1 = 13 + 9,$	false
when	$N = 68, 67, 66,$	(1) is	$22 + 1 = 13 + 9,$	false
when	$N = 65,$	(1) is	$21 + 1 = 13 + 9,$	**true**.

Thus the answer is $N = 65$.

1 POINT for concluding.

Second solution.

This is equivalent to find the largest positive integer solution of the equation

$$\left\lfloor \frac{N}{3} \right\rfloor = \left\lfloor \frac{N}{5} \right\rfloor + \left\lfloor \frac{N}{7} \right\rfloor - \left\lfloor \frac{N}{35} \right\rfloor, \tag{1}$$

1 POINT for equality (1).

Let $N = 35k + r$ $(0 \leq r < 35)$ be a solution of (1). Then (1) can be writen as

$$\left\lfloor \frac{35k + r}{3} \right\rfloor = 11k + \left\lfloor \frac{r}{5} \right\rfloor + \left\lfloor \frac{r}{7} \right\rfloor.$$

1 POINT.

Now $\frac{35k+r-2}{3} \leq \left\lfloor \frac{35k+r}{3} \right\rfloor$, $\lfloor \frac{r}{5} \rfloor \leq \frac{r}{5}$ and $\lfloor \frac{r}{7} \rfloor \leq \frac{r}{7}$. Therefore

$$\frac{35k+r-2}{3} \leq 11k + \frac{r}{5} + \frac{r}{7} = \frac{12r}{35},$$

which implies that $70k \leq r + 70 < 35 + 70 = 105$. Then $k \leq 1$ or equivalently $N \leq 69$.

4 POINTS for finding that $N \leq 69$.

As in the first solution, checking (1) for $N \leq 69$ we find the answer $N = 65$.

1 POINT for concluding.

Remark.
2 POINTS can be given for finding a good upper bound for example $N \leq 100$ (in the first solution we found $n \leq 86$). 6 POINTS for proving that $N \leq 69$.

1 POINT can be given for the correct answer.

Problem 3.

First Solution.

It is easy to see that the intersection $S \cap T$ has $2n$ sides only if the sides of $S \cap T$ alternate: blue, red, blue, red, etc.

1 POINT.

Denote the vertices of $S \cap T$ clockwise as $C_1D_1C_2D_2 \ldots C_nD_n$ so that the sides C_1D_1, $C_2D_2, \ldots C_nD_n$ are blue. Denote the vertices of S by A_1, $A_2, \ldots,$ A_n and the vertices of T by B_1, $B_2, \ldots,$ B_n so that $C_1D_1 \subset B_1B_2, \ldots, C_nD_n \subset B_nB_1$ and $D_nC_1 \subset A_1A_2, \ldots, D_{n-1}C_n \subset A_nA_1$.

It is easy to check that all the triangles $D_nB_1C_1$, $D_1B_2C_2, \ldots, D_{n-1}B_nC_n$ and $C_1A_2D_1$, $C_2A_3D_2, \ldots, C_nA_1D_n$ are similar.

1 POINT.

Therefore,

$$\frac{D_nC_1}{D_nB_1 + B_1C_1} = \frac{D_1C_2}{D_1B_2 + B_2C_2} = \ldots = \frac{D_{n-1}C_n}{D_{n-1}B_n + B_nC_n} =$$
$$= \frac{C_1D_1}{C_1A_2 + A_2D_1} = \frac{C_1D_2}{C_2A_3 + A_3D_2} = \ldots = \frac{C_nD_n}{C_nA_1 + A_1D_n}.$$

1 POINT.

Hence,

$$\frac{D_nC_1 + D_1C_2 + \ldots + D_{n-1}C_n}{D_nB_1 + B_1C_1 + D_1B_2 + B_2C_2 + \ldots + D_{n-1}B_n + B_nC_n} =$$

$$= \frac{C_1D_1 + C_1D_2 + \ldots + C_nD_n}{C_1A_2 + A_2D_1 + C_2A_3 + A_3D_2 + \ldots + C_nA_1 + A_1D_n}. \quad (1)$$

Let $x = D_nC_1 + D_1C_2 + \ldots + D_{n-1}C_n$ and $y = C_1D_1 + C_1D_2 + \ldots + C_nD_n$ then x is the sum of the blue sides of $S \cap T$ and y is the sum of the red sides. If a is the length of a side of S (or T), then the equality (1) can be written in the following form

$$\frac{x}{na - y} = \frac{y}{na - x}.$$

It follows that

$$\begin{aligned} nax - x^2 &= nay - y^2 \\ na(x - y) &= (x + y)(x - y) \\ (na - x - y)(x - y) &= 0. \end{aligned}$$

Since the perimeter $x + y$ of $S \cap T$ is strictly less than the perimeter na of S or T, $na - x - y > 0$. We obtain $x - y = 0$ and $x = y$ q.e.d.

4 POINTS for concluding.

Second Solution.

As in the first solution, $S \cap T$ has $2n$ sides only if the sides of $S \cap T$ alternate.

1 POINT.

Label the vertices of the red n-gon $R_1, R_2, \ldots R_n$ and the vertices of the blue n-gon $B_1, B_2, \ldots, B_n$. Place the n-gons so that the vertices are in the following clockwise order: $B_1, R_1, B_2, R_2, \ldots, B_n, R_n$. Each of these vertices together with the opposite side determines a triangle and all these triangles are similar.

1 POINT.

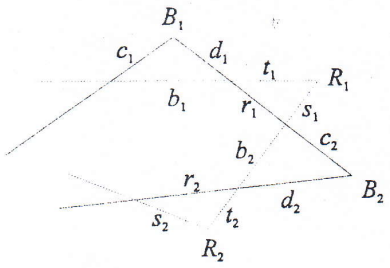
For each $i = 1, \ldots, n$, we let the lengths of the sides of the triangle determined by B_i be b_i, c_i, d_i in the clockwise order where b_i is the side opposite B_i such that $b_i/b_1 = c_i/c_1 = d_i/d_1 = p_i$. We also let the lengths of the sides of the triangle determined by R_i be r_i, s_i, t_i in the counter clockwise order such that $r_i/b_1 = s_i/c_1 = t_i/d_1 = q_i$.

1 POINT.

Then we want to prove that $b_1 + \cdots + b_n = r_1 + \cdots + r_n$ or $b_1(p_1 + \cdots + p_n) = b_1(q_1 + \cdots + q_n)$, where $p_1 = 1$, or $p = q$ where $p = (p_1 + \cdots + p_n), q = (q_1 + \cdots + q_n)$. The perimeter of the blue n-gon is $c_1 + d_1 + r_1 + \cdots + c_n + d_n + r_n = pc_1 + pd_1 + qb_1$. Likewise the perimeter of the red n-gon is $pb_1 + qc_1 + qd_1$. Equating the two we have $p(c_1 + d_1 - b_1) = q(c_1 + d_1 - b_1)$, which implies $p = q$ as

required.

4 POINTS for concluding.



Problem 4.

Answer. All the polynomials of degree 1 with rational coefficients.

First Solution.

Note that a polynomial that satisfies the conditions of the problem takes rational values for rational numbers and irrational values for irrational numbers. Let $f(x)$ be a polynomial of degree n such that $f(r) \in Q$ for every $r \in Q$. For distinct rational numbers $r_0,\ r_1, \ldots, r_n$, where $n = \deg f(x)$ let us define the polynomial

$$\begin{aligned} g(x) \;=\; & c_0(x-r_1)(x-r_2)\ldots(x-r_n) + c_1(x-r_0)(x-r_2)\ldots(x-r_n) + \ldots \\ & + c_i(x-r_0)\ldots(x-r_{i-1})(x-r_{i+1})\ldots(x-r_n) + \ldots \\ & + c_n(x-r_0)(x-r_1)\ldots(x-r_{n-1}), \end{aligned} \tag{1}$$

where $c_0,\ c_1, \ldots,\ c_n$ are real numbers.

Suppose that $g(r_i) = f(r_i)$, $i = 0, 1, \ldots, n$. Since $g(r_i) = c_i(r_i - r_0)\ldots(r_i - r_{i-1})(r_i - r_{i+1})\ldots(r_i - r_n)$ then

$$c_i = \frac{g(r_i)}{(r_i - r_0)\ldots(r_i - r_{i-1})(r_i - r_{i+1})} = \frac{f(r_i)}{(r_i - r_0)\ldots(r_i - r_{i-1})(r_i - r_{i+1})}. \tag{2}$$

Clearly c_i is a rational number for $i = 0, 1, \ldots, n$ and therefore the coefficients of $g(x)$ are rational. The polynomial $g(x)$ defined in (1) with coefficients $c_0, c_1, \ldots, c_n$ satisfying (2) coincides with $f(x)$ in $n+1$ points and both polynomials f and g have degree n. It follows that for every real x, $f(x) = g(x)$. Therefore the coefficients of $f(x)$ are rational.

Thus if a polynomial satisfies the conditions, all its coefficients are rational.

1 POINT for proving that the polinomial has rational coefficients.

It is easy to see that all the polynomials of first degree with rational coefficients satisfy the conditions of the problem and polynomials of degree 0 do not satisfy it. Let us prove that no other polynomials exist.

Suppose that $f(x) = a_0 + a_1x + \ldots + a_nx^n$ is a polynomial with rational coefficients and degree $n \geq 2$ that satisfies the conditions of the problem. We may assume that the coefficients of $f(x)$ are integers, because the sets of solutions of equations $f(x) = r$ and $af(x) = ar$, where a is an integer, coincide. Moreover let us denote $g(x) = a_n^{n-1} f(\frac{x}{a_n})$. $g(x)$ is a polynomial with integer coefficients whose leading coefficient is 1. The equation $f(x) = r$ has an irrational root if and only if $g(x) = a_n^{n-1}r$ has an irrational root. Therefore, we may assume WLOG that $f(x)$ has integer coefficients and $a_n = 1$.

1 POINT more for proving that it is suficient to consider polinomials with integer and leading coefficient equal to 1.

Let r be a sufficiently large prime, such that

$$r > \max\{f(1) - f(0), x_1, x_2, \ldots, x_k\},$$

where $\{x_1, x_2, \ldots, x_k\}$ denote the set of all real roots of $f(x) - f(0) - x = 0$. Putting $q = r + f(0) \in Z$, and considering the equality

$$f(x) - q = f(x) - f(0) - r$$

we then have

$$f(1) - q = f(1) - f(0) - r < 0.$$

On the other hand, by the choice of r, we have

$$f(1) - q = f(r) - f(0) - r > 0.$$

It follows from the intermediate value theorem that there is at least one real root p of $f(x) - q = 0$ between 1 and r. Notice that from the criterion theorem for rational roots, the possible positive rational roots of the equation $f(x) - q = f(x) - f(0) - r = 0$ are 1 and r. Thus p must be irrational.

5 POINTS for concluding.

Second Solution.
As in the first solution we may assume WLOG that $f(x)$ has integer coefficients and the leading coefficient is $a_n = 1$.

2 POINTS.

Observing the graph of $f(x)$, it is easy to see that there exists a sufficiently great integer r such that $f(x) = r$ has one possitive root x_0 and for $x \geq x_0$ the derivative $f'(x)$ is greater than 1. The equation $f(x) = r + 1$ has also one positive root $x_1 > x_0$. Since $a_n = 1$, rational roots x_0 and x_1 must be integers. Then $x_1 - x_0 \geq 1$ and

$$\frac{f(x_1) - f(x_0)}{x_1 - x_0} \leq 1.$$

It follows that $f'(z) < 1$ for $z \in [x_0, x_1]$. This contradiction proves that $f(x) = r$ necessarily has an irrational root for at least one integer r.

5 POINTS for concluding.

Remark.
No points can be given just for the answer.

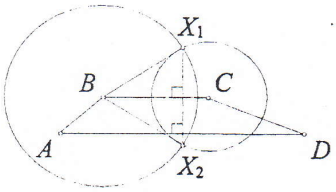
Problem 5.

First Solution.

One of the sides AX_i or BX_i is equal to CD, thus X_i is on one of the circles of radius CD and center A or B. In the same way X_i is on one of circles of radius AB with center C or D. The intersection of these four circles has no more than 8 points so that $n \leq 8$.

1 POINT for finding that $n \leq 8$.

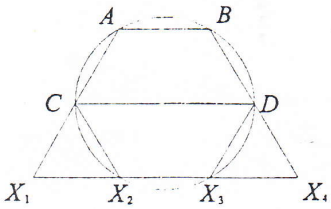
Suppose that circle S_B with center B and radius CD intersects circle S_C with center C and radius AB in two points X_1 and X_2 which satisfy the conditions of the problem. Then in triangles ABX_1 and CDX_1 we have $BX_1 = CD$ and $CX_1 = AB$. Since these triangles are congruent then $AX_1 = DX_1$, therefore X_1 and X_2 are on the perpendicular bisector of AD. On the other hand X_1X_2 is perpendicular to segment BC. Then $BC \| AD$ and AB and CD are the diagonals or nonparallel sides of a trapezoid.



Suppose that $AB < CD$. Then $BX_1 = CD > AB = CX_1$. It follows that the distance from A to the perpendicular bisector of BC must be less than the distance from D to this line otherwise we obtain a contradiction to the condition $AB < CD$. Then for any point X in the perpendicular bisector of BC we have $AX < DX$ and it is not possible to have $AX = CD$, $DX = AB$. Thus if the circle with center A and radius CD intersects the circle with center D and radius AB, then the points of intersection do not satisfy the condition of congruence. Therefore if the points of intersection of S_B with S_C satisfy the condition of congruence, then the points of intersection of S_A with S_D do not. Thus no more than half of the 8 points of intersection of these circles can satisfy the condition of congruence, i.e. $n \leq 4$.

4 POINTS for proving that $n \leq 4$.

If $n = 4$ we have the following example of a regular hexagon.

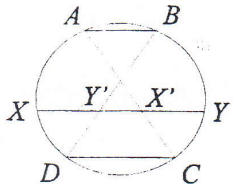


2 POINTS for proving that $n \geq 4$.

Second Solution.

The greatest possible value of n is 4. First we will prove that $n \geq 4$ and finally we will prove that $n \leq 4$.

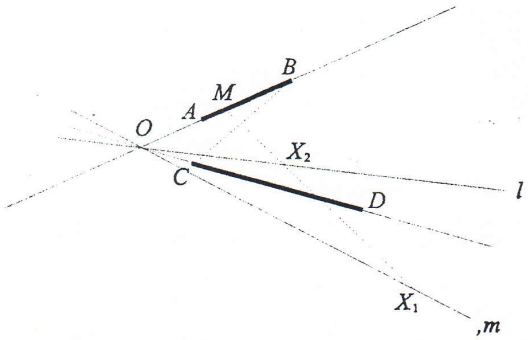
- To prove that $n \geq 4$ it is enough to show a configuration of 8 points that satisfies the conditions of the problem. Let us choose 6 points A, B, Y, C, D, X on a circle such that triangle AYD is equilateral and B, C, X are points on arcs AY, YD, DA respectively, such that arcs AB, YC, DX are equal and less than $60°$. Then it is easy to see that AB, CD and XY are parallel and $\angle DXY = \angle CYX = 60°$.



If XY intersects AC and BD at X', Y', respectively, then $X, Y, X'Y'$ satisfy the conditions of the problem.

2 POINTS for proving that $n \geq 4$.

- To show that $n \leq 4$ let us consider the following figure.



Suppose that $AB < CD$. The trace of point K such that $(ABK) = (CDK)$ is two lines l and m through the point of intersection O of AB with CD. (If $AB \| CD$ then the trace of K is formed by two parallel lines.)

If X is a point such that $\triangle ABX \cong \triangle CDX$ then X must be on one or more of the perpendicular bisectors of the segments AC, AD, BC, BD. Since X lies on l or m, on each of the perpendicular bisectors of AC, AD, BC, BD there can be no more than 2 intersection points, i.e. $n \leq 8$.

1 POINT for proving that $n \leq 8$.

We will prove that at most one point on each perpendicular bisector satisfies the conditions of the problem.

WLOG Suposse that X_1 and X_2 are points on the perpendicular bisector of BC such that $\triangle ABX_1 \cong \triangle CDX_1$ and $\triangle ABX_2 \cong \triangle CDX_2$ with $X_1 \in l$ and $X_2 \in m$.

Let us prove that $AX_1 = CD$. Since $AB \neq CD$ one of the segments AX_1 or BX_1 is equal to CD. If $BX_1 = CD$ then $CX_1 = CD$ by construction, and $\triangle CDX_1$ has two sides equal to CD, thus $AX_1 = CD$.

In the same way we have $DX_1 = AB$

The same argument can be used if we consider the point X_2. In this case we conclude that $AX_1 = AX_2$ and $DX_1 = DX_2$, then AD and X_1X_2 are perpendicular. But BC and X_1X_2 perpendicular and therefore $AD \| BC$. We are considering the case when AD and BC are diagonals of a cuadrilateral, therefore they are not parallel. This is a contradiction.

If we consider the perpendicular bisector of BD, the same argument allows us to conclude that AC and BD are parallel. Therefore a point X on l or m (i.e $(XAB) = (XCD)$) satisfies $(XOB) = (XOD)$ because triangles OAC and OBD are similar. Thus, the points of intersection of AC with l and BD with l are midpoints of the respective segments. Then $X_1X_2 \subset l$ and, since l is the perpendicular bisector and median, triangle ABD is isosceles with $OB = OC$. So, $AB = CD$, which is a contradiction. Therefore $n \leq 4$.

4 POINT for proving that $n \leq 4$.

APMO 2000 – Problems and Solutions

Problem 1
Compute the sum $S = \sum_{i=0}^{101} \frac{x_i^3}{1-3x_i+3x_i^2}$ for $x_i = \frac{i}{101}$.

Answer: $S = 51$.

Solution
Since $x_{101-i} = \frac{101-i}{101} = 1 - \frac{i}{101} = 1 - x_i$ and

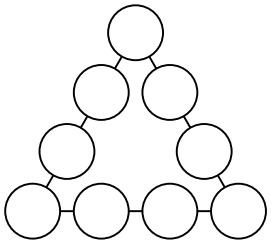
$$1 - 3x_i + 3x_i^2 = (1 - 3x_i + 3x_i^2 - x_i^3) + x_i^3 = (1 - x_i)^3 + x_i^3 = x_{101-i}^3 + x_i^3,$$

we have, by replacing i by $101 - i$ in the second sum,

$$2S = S + S = \sum_{i=0}^{101} \frac{x_i^3}{x_{101-i}^3 + x_i^3} + \sum_{i=0}^{101} \frac{x_{101-i}^3}{x_i^3 + x_{101-i}^3} = \sum_{i=0}^{101} \frac{x_i^3 + x_{101-i}^3}{x_{101-i}^3 + x_i^3} = 102,$$

so $S = 51$.

Problem 2
Given the following arrangement of circles:



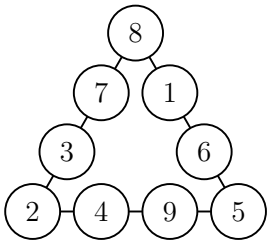
Each of the numbers $1, 2, \ldots, 9$ is to be written into one of these circles, so that each circle contains exactly one of these numbers and

(i) the sums of the four numbers on each side of the triangle are equal;

(ii) the sums of squares of the four numbers on each side of the triangle are equal.

Find all ways in which this can be done.

Answer: The only solutions are



and the ones generated by permuting the vertices, adjusting sides and exchanging the two middle numbers on each side.

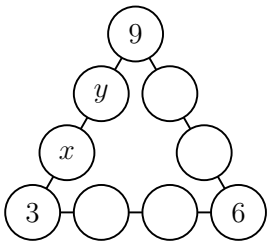
Solution
Let a, b, and c be the numbers in the vertices of the triangular arrangement. Let s be the sum of the numbers on each side and t be the sum of the squares of the numbers on each side. Summing the numbers (or their squares) on the three sides repeats each once the numbers on the vertices (or their squares):

$$3s = a + b + c + (1 + 2 + \cdots + 9) = a + b + c + 45$$
$$3t = a^2 + b^2 + c^2 + (1^2 + 2^2 + \cdots + 9^2) = a^2 + b^2 + c^2 + 285$$

At any rate, $a + b + c$ and $a^2 + b^2 + c^2$ are both multiples of 3. Since $x^2 \equiv 0, 1 \pmod 3$, either a, b, c are all multiples of 3 or none is a multiple of 3. If two of them are $1, 2 \bmod 3$ then $a + b + c \equiv 0 \pmod 3$ implies that the other should be a multiple of 3, which is not possible. Thus a, b, c are all congruent modulo 3, that is,

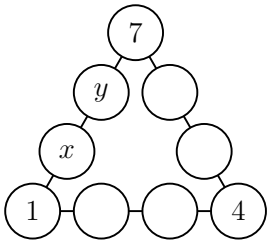
$$\{a, b, c\} = \{3, 6, 9\}, \quad \{1, 4, 7\}, \quad \text{or} \quad \{2, 5, 8\}.$$

Case 1: $\{a,b,c\} = \{3,6,9\}$. Then $3t = 3^2 + 6^2 + 9^2 + 285 \iff t = 137$.



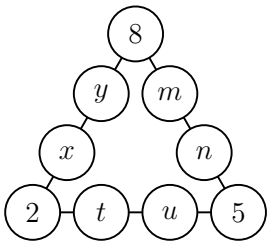
In this case $x^2+y^2+3^2+9^2=137 \iff x^2+y^2=47$. However, 47 cannot be written as the sum of two squares. One can check manually, or realize that $47 \equiv 3 \pmod 4$, and since $x^2, y^2 \equiv 0, 1 \pmod 4$, $x^2+y^2 \equiv 0, 1, 2 \pmod 4$ cannot be 47.
Hence there are no solutions in this case.

Case 2: $\{a,b,c\} = \{1,4,7\}$. Then $3t = 1^2+4^2+7^2+285 \iff t = 117$.



In this case $x^2+y^2+1^2+7^2=117 \iff x^2+y^2=67 \equiv 3 \pmod 4$, and as in the previous case there are no solutions.

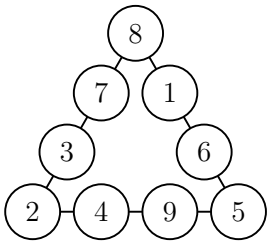
Case 3: $\{a,b,c\} = \{2,5,8\}$. Then $3t = 2^2+5^2+8^2+285 \iff t = 126$.



Then

$$\begin{cases} x^2+y^2+2^2+8^2=126 \\ t^2+u^2+2^2+5^2=126 \\ m^2+n^2+5^2+8^2=126 \end{cases} \iff \begin{cases} x^2+y^2=58 \\ t^2+u^2=97 \\ m^2+n^2=37 \end{cases}$$

The only solutions to $t^2+u^2=97$ and $m^2+n^2=37$ are $\{t,u\}=\{4,9\}$ and $\{m,n\}=\{1,6\}$, respectively (again, one can check manually.) Then $\{x,y\}=\{3,7\}$, and the solutions are



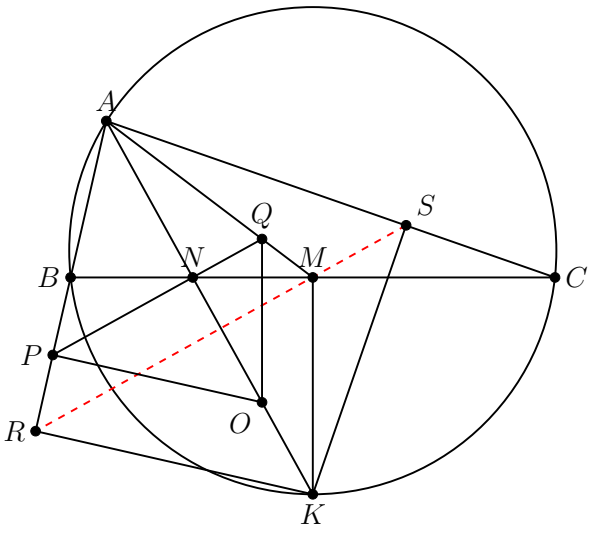
and the ones generated by permuting the vertices, adjusting sides and exchanging the two middle numbers on each side. There are $3! \cdot 2^3 = 48$ such solutions.

Problem 3

Let ABC be a triangle. Let M and N be the points in which the median and angle bisector, respectively, at A meet the side BC. Let Q and P be the points in which the perpendicular at N to NA meets MA and BA, respectively, and O be the point in which the perpendicular at P to BA meets AN produced. Prove that QO is perpendicular to BC.

Solution 1

Let AN meet the circumcircle of ABC at point K, the midpoint of arc BC that does not contain A.



The orthogonal projection of K onto side BC is M. Let R and S be the orthogonal projections of K onto lines AB and AC, respectively. Points R, M, and S lie in the *Simson line* of K with respect to ABC. Since K is in the bisector of $\angle BAC$, $ARKS$ is a kite, and the Simson line RMS is perpendicular to AN, and therefore parallel to PQ.

Now consider the homothety with center A that takes O to K. Since $OP \perp AB$ and $KR \perp AB$, OP and KR are parallel, which means that P is taken to R. Finally, line PQ is parallel to line RS, so line PQ is taken to line RS by the homothety. Then Q is taken to M, and since O is taken to K, line OQ is taken to line MK. We are done now: this means that OQ is parallel to MK, which is perpendicular to BC (it is its perpendicular bisector, as $MB = MC$ and $KB = KC$.)

Solution 2

Consider a cartesian plane with $A = (0, 0)$ as the origin and the bisector AN as x-axis. Thus AB has equation $y = mx$ and AC has equation $y = -mx$. Let $B = (b, mb)$ and $C = (c, -mc)$. By symmetry, the problem is immediate if $AB = AC$, that is, if $b = c$. Suppose that $b \neq c$ from now on. Line BC has slope $\frac{mb-(-mc)}{b-c} = \frac{m(b+c)}{b-c}$. Let $N = (n, 0)$.

Point M is the midpoint $\left(\frac{b+c}{2}, \frac{mb-mc}{2}\right)$ of BC, so AM has slope $\frac{m(b-c)}{b+c}$.

The line through N that is perpendicular to the x-axis AN is $x = n$. Therefore

$$P = (n, mn) \quad \text{and} \quad Q = \left(n, \frac{m(b-c)n}{b+c}\right).$$

In the right triangle APO, with altitude AN, $AN \cdot AO = AP^2$. Thus

$$n \cdot AO = (0-n)^2 + (0-mn)^2 \iff AO = n(m^2+1) \implies O = (n(m^2+1), 0).$$

Finally, the slope of OQ is

$$\frac{\frac{m(b-c)n}{b+c} - 0}{n - n(m^2+1)} = -\frac{b-c}{(b+c)m}.$$

Since the product of the slopes of OQ and BC is

$$-\frac{b-c}{(b+c)m}\cdot\frac{m(b+c)}{b-c}=-1,$$

OQ and BC are perpendicular, and we are done.

Comment: The second solution shows that N can be any point in the bisector of $\angle A$. In fact, if we move N in the bisector and construct O, P and Q accordingly, then all lines OQ obtained are parallel: just consider a homothety with center A and variable ratios.

Problem 4
Let n, k be given positive integers with $n > k$. Prove that

$$\frac{1}{n+1} \cdot \frac{n^n}{k^k(n-k)^{n-k}} < \frac{n!}{k!\,(n-k)!} < \frac{n^n}{k^k(n-k)^{n-k}}.$$

Solution
The inequality is equivalent to

$$\frac{n^n}{n+1} < \binom{n}{k} k^k(n-k)^{n-k} < n^n,$$

which suggests investigating the binomial expansion of

$$n^n = ((n-k)+k)^n = \sum_{i=0}^{n} \binom{n}{i} (n-k)^{n-i}k^i.$$

The $(k+1)$th term T_{k+1} of the expansion is $\binom{n}{k}k^k(n-k)^{n-k}$, and all terms in the expansion are positive, which implies the right inequality.
Now, for $1 \le i \le n$,

$$\frac{T_{i+1}}{T_i} = \frac{\binom{n}{i}(n-k)^{n-i}k^i}{\binom{n}{i-1}(n-k)^{n-i+1}k^{i-1}} = \frac{(n-i+1)k}{i(n-k)},$$

and

$$\frac{T_{i+1}}{T_i} > 1 \iff (n-i+1)k > i(n-k) \iff i < k + \frac{k}{n} \iff i \le k.$$

This means that

$$T_1 < T_2 < \cdots < T_{k+1} > T_{k+2} > \cdots > T_{n+1},$$

that is, $T_{k+1} = \binom{n}{k}k^k(n-k)^{n-k}$ is the largest term in the expansion. The maximum term is greater that the average, which is the sum n^n divided by the quantity $n+1$, therefore

$$\binom{n}{k} k^k(n-k)^{n-k} > \frac{n^n}{n+1},$$

as required.

Comment: If we divide further by n^n one finds

$$\frac{1}{n+1} < \binom{n}{k} \left(\frac{k}{n}\right)^k \left(1 - \frac{k}{n}\right)^{n-k} < 1.$$

The middle term is the probability $P(X = k)$ of k successes in a binomial distribution with n trials and success probability $p = \frac{k}{n}$. The right inequality is immediate from the fact that $P(X = k)$ is not the only possible event in this distribution, and the left inequality comes from the fact that the mode of the binomial distribution are given by $\lfloor (n+1)p \rfloor = \lfloor (n+1)\frac{k}{n} \rfloor = k$ and $\lceil (n+1)p - 1 \rceil = k$. However, the proof of this fact is identical to the above solution.

Problem 5

Given a permutation $(a_0, a_1, \ldots, a_n)$ of the sequence $0, 1, \ldots, n$. A transposition of a_i with a_j is called *legal* if $a_i = 0$ for $i > 0$, and $a_{i-1} + 1 = a_j$. The permutation $(a_0, a_1, \ldots, a_n)$ is called *regular* if after a number of legal transpositions it becomes $(1, 2, \ldots, n, 0)$. For which numbers n is the permutation $(1, n, n-1, \ldots, 3, 2, 0)$ regular?

Answer: $n = 2$ and $n = 2^k - 1$, k positive integer.

Solution

A legal transposition consists of looking at the number immediately before 0 and exchanging 0 and its successor; therefore, we can perform at most one legal transposition to any permutation, and a legal transposition is not possible only and if only 0 is preceded by n.

If $n = 1$ or $n = 2$ there is nothing to do, so $n = 1 = 2^1 - 1$ and $n = 2$ are solutions. Suppose that $n > 3$ in the following.

Call a *pass* a maximal sequence of legal transpositions that move 0 to the left. We first illustrate what happens in the case $n = 15$, which is large enough to visualize what is going on. The first pass is

$$(1, 15, 14, 13, 12, 11, 10, 9, 8, 7, 6, 5, 4, \mathbf{3}, 2, 0)$$
$$(1, 15, 14, 13, 12, 11, 10, 9, 8, 7, 6, \mathbf{5}, 4, 0, 2, 3)$$
$$(1, 15, 14, 13, 12, 11, 10, 9, 8, \mathbf{7}, 6, 0, 4, 5, 2, 3)$$
$$(1, 15, 14, 13, 12, 11, 10, \mathbf{9}, 8, 0, 6, 7, 4, 5, 2, 3)$$
$$(1, 15, 14, 13, 12, \mathbf{11}, 10, 0, 8, 9, 6, 7, 4, 5, 2, 3)$$
$$(1, 15, 14, \mathbf{13}, 12, 0, 10, 11, 8, 9, 6, 7, 4, 5, 2, 3)$$
$$(1, \mathbf{15}, 14, 0, 12, 13, 10, 11, 8, 9, 6, 7, 4, 5, 2, 3)$$
$$(1, 0, 14, 15, 12, 13, 10, 11, 8, 9, 6, 7, 4, 5, 2, 3)$$

After exchanging 0 and 2, the second pass is

$$(1, 2, 14, 15, 12, 13, 10, 11, 8, 9, \mathbf{6}, 7, 4, 5, 0, 3)$$
$$(1, 2, 14, 15, 12, 13, \mathbf{10}, 11, 8, 9, 0, 7, 4, 5, 6, 3)$$
$$(1, 2, \mathbf{14}, 15, 12, 13, 0, 11, 8, 9, 10, 7, 4, 5, 6, 3)$$
$$(1, 2, 0, 15, 12, 13, 14, 11, 8, 9, 10, 7, 4, 5, 6, 3)$$

After exchanging 0 and 3, the third pass is

$$(1, 2, 3, 15, 12, 13, 14, 11, 8, 9, 10, \mathbf{7}, 4, 5, 6, 0)$$
$$(1, 2, 3, 15, 12, 13, 14, \mathbf{11}, 8, 9, 10, 0, 4, 5, 6, 7)$$
$$(1, 2, 3, \mathbf{15}, 12, 13, 14, 0, 8, 9, 10, 11, 4, 5, 6, 7)$$
$$(1, 2, 3, 0, 12, 13, 14, 15, 8, 9, 10, 11, 4, 5, 6, 7)$$

After exchanging 0 and 4, the fourth pass is

$$(1, 2, 3, 4, \mathbf{12}, 13, 14, 15, 8, 9, 10, 11, 0, 5, 6, 7)$$
$$(1, 2, 3, 4, 0, 13, 14, 15, 8, 9, 10, 11, 12, 5, 6, 7)$$

And then one can successively perform the operations to eventually find

$$(1, 2, 3, 4, 5, 6, 7, 0, 8, 9, 10, 11, 12, 13, 14, 15)$$

after which 0 will move one unit to the right with each transposition, and $n = 15$ is a solution. The general case follows.

Case 1: $n > 2$ even: After the first pass, in which 0 is transposed successively with $3, 5, \ldots, n-1$, after which 0 is right after n, and no other legal transposition can be performed. So n is not a solution in this case.
Case 2: $n = 2^k - 1$: Denote $N = n+1$, $R = 2^r$, $[a : b] = (a, a+1, a+2, \ldots, b)$, and concatenation by a comma. Let P_r be the permutation

$$[1 : R-1], (0), [N-R : N-1], [N-2R : N-R-1], \ldots, [2R : 3R-1], [R : 2R-1]$$

P_r is formed by the blocks $[1 : R-1], (0)$, and other $2^{k-r} - 1$ blocks of size $R = 2^r$ with consecutive numbers, beginning with tR and finishing with $(t+1)R-1$, in decreasing order of t. Also define P_0 as the initial permutation.
Then it can be verified that P_{r+1} is obtained from P_r after a number of legal transpositions: it can be easily verified that P_0 leads to P_1, as 0 is transposed successively with $3, 5, \ldots, n-1$, shifting cyclically all numbers with even indices; this is P_1.
Starting from P_r, $r > 0$, 0 is successively transposed with $R, 3R, \ldots, N-R$. The numbers $0, N-R, N-3R, \ldots, 3R, R$ are cyclically shifted. This means that R precedes 0, and the blocks become

$$[1 : R], (0), [N-R+1 : N-1], [N-2R : N-R], [N-3R+1 : N-2R-1], \ldots,$$
$$[3R+1 : 4R-1], [2R : 3R], [R+1 : 2R-1]$$

Note that the first block and the ones on even positions greater than 2 have one more number and the other blocks have one less number.
Now $0, N-R+1, N-3R+1, \ldots, 3R+1, R+1$ are shifted. Note that, for every ith block, i odd greater than 1, the first number is cyclically shifted, and the blocks become

$$[1 : R+1], (0), [N-R+2 : N-1], [N-2R : N-R+1], [N-3R+2 : N-2R-1], \ldots,$$
$$[3R+1 : 4R-1], [2R : 3R+1], [R+2 : 2R-1]$$

The same phenomenom happened: the first block and the ones on even positions greater than 2 have one more number and the other blocks have one less number. This pattern continues: $0, N-R+u, N-3R+u, \ldots, R+u$ are shifted, $u = 0, 1, 2, \ldots, R-1$, the first block and the ones on even positions greater than 2 have one more number and the other blocks have one less number, until they vanish. We finish with

$$[1 : 2R-1], (0), [N-2R : N-1], \ldots, [2R : 4R-1],$$

which is precisely P_{r+1}.
Since $P_k = [1 : N-1], (0)$, $n = 2^k - 1$ is a solution.
Case 3: n is odd, but is not of the form $2^k - 1$. Write $n+1$ as $n+1 = 2^a(2b+1)$, $b \geq 1$, and define $P_0, \ldots, P_a$ as in the previous case. Since 2^a divides $N = n+1$, the same rules apply, and we obtain P_a:

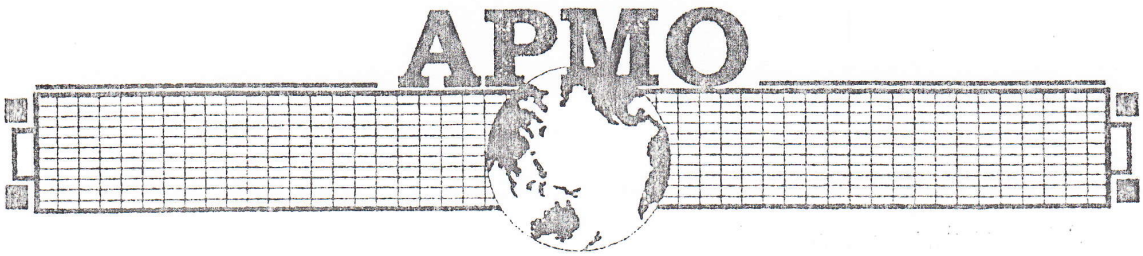
$$[1 : 2^a - 1], (0), [N-2^a : N-1], [N-2^{a+1} : N-2^a-1], \ldots, [2^{a+1} : 3 \cdot 2^a - 1], [2^a : 2^{a+1} - 1].$$

But then 0 is transposed with $2^a, 3 \cdot 2^a, \ldots, (2b-1) \cdot 2^a = N - 2^{a+1}$, after which 0 is put immediately after $N - 1 = n$, and cannot be transposed again. Therefore, n is not a solution.
All cases were studied, and so we are done.

Comment: The general problem of finding the number of regular permutations for any n seems to be difficult. A computer finds the first few values

$$1, 2, 5, 14, 47, 189, 891, 4815, 29547,$$

which is not catalogued at oeis.org.



XI APMO - SOLUTIONS AND MARKING SCHEMES

Problem 1. Find the smallest positive integer n with the following property : There does not exist an arithmetic progression of 1999 terms of real numbers containing exactly n integers.

Solution and Marking Scheme:

We first note that the integer terms of any arithmetic progression are "equally spaced", because if the ith term a_i and the $(i+j)$th term a_{i+j} of an arithmetic progression are both integers, then so is the $(i+2j)$th term $a_{i+2j} = a_{i+j} + (a_{i+j} - a_i)$.

1 POINT for realizing that the integers must be "equally spaced".

Thus, by scaling and translation, we can assume that the integer terms of the arithmetic progression are $1, 2, \cdots, n$ and we need only to consider arithmetic progression of the form

$$1, 1+\frac{1}{k}, 1+\frac{2}{k}, \cdots, 1+\frac{k-1}{k}, 2, 2+\frac{1}{k}, \cdots, n-1, \cdots, n-1+\frac{k-1}{k}, n$$

This has $kn - k + 1$ terms of which exactly n are integers. Moreover we can add up to $k-1$ terms on either end and get another arithmetic progression without changing the number of integer terms.

2 POINTS for noticing that the maximal sequence has an equal number of terms on either side of the integers appearing in the sequence (this includes the 1 POINT above). In other words, 2 POINTS for the scaled and translated form of the progression including the $k-1$ terms on either side.

Thus there are arithmetic progressions with n integers whose length is any integer lying in the interval $[kn - k + 1, kn + k - 1]$, where k is any positive integer. Thus we want to find the smallest $n > 0$ so that, if k is the largest integer satisfying $kn + k - 1 \leq 1998$, then $(k+1)n - (k+1) + 1 \geq 2000$.

4 POINTS for clarifying the nature of the number n in this way, which includes counting the terms of the maximal and minimal sequences containing n integers and bounding them accordingly (this includes the 2 POINTS above).

That is, putting $k = \lfloor 1999/(n+1) \rfloor$, we want the smallest integer n so that

$$\left\lfloor \frac{1999}{n+1} \right\rfloor (n-1) + n \geq 2000.$$

This inequality does not hold if

$$\frac{1999}{n+1}\cdot(n-1)+n<2000.$$

2 POINTS for setting up an inequality for n.

This simplifies to $n^2<3999$, that is, $n\le 63$. Now we check integers from $n=64$ on:

$$\text{for } n=64,\ \left\lfloor\tfrac{1999}{65}\right\rfloor\cdot 63+64=30\cdot 63+64=1954<2000\ ;$$

$$\text{for } n=65,\ \left\lfloor\tfrac{1999}{66}\right\rfloor\cdot 64+65=30\cdot 64+65=1985<2000\ ;$$

$$\text{for } n=66,\ \left\lfloor\tfrac{1999}{67}\right\rfloor\cdot 65+66=29\cdot 65+66=1951<2000\ ;$$

$$\text{for } n=67,\ \left\lfloor\tfrac{1999}{68}\right\rfloor\cdot 66+67=29\cdot 66+67=1981<2000\ ;$$

$$\text{for } n=68,\ \left\lfloor\tfrac{1999}{69}\right\rfloor\cdot 67+68=28\cdot 67+68=1944<2000\ ;$$

$$\text{for } n=69,\ \left\lfloor\tfrac{1999}{70}\right\rfloor\cdot 68+69=28\cdot 68+69=1973<2000\ ;$$

$$\text{for } n=70,\ \left\lfloor\tfrac{1999}{71}\right\rfloor\cdot 69+70=28\cdot 69+70=2002\ge 2000\ .$$

Thus the answer is $n=70$.

1 POINT for checking these numbers and finding that $n=70$.

Problem 2. Let $a_1,a_2,\cdots$ be a sequence of real numbers satisfying $a_{i+j}\le a_i+a_j$ for all $i,j=1,2,\cdots$. Prove that

$$a_1+\frac{a_2}{2}+\frac{a_3}{3}+\cdots+\frac{a_n}{n}\ge a_n$$

for each positive integer n.

Solution and Marking Scheme:

Letting $b_i=a_i/i$, $(i=1,2,\cdots)$, we prove that

$$b_1+\cdots+b_n\ge a_n\quad(n=1,2,\cdots)$$

by induction on n. For $n=1$, $b_1=a_1\ge a_1$, and the induction starts. Assume that

$$b_1+\cdots+b_k\ge a_k$$

for all $k=1,2,\cdots,n-1$. It suffices to prove that $b_1+\cdots+b_n\ge a_n$ or equivalently that

$$nb_1 + \cdots + nb_{n-1} \geq (n-1)a_n.$$

3 POINTS for separating a_n from $b_1, \cdots, b_{n-1}$.

$$\begin{aligned} nb_1 + \cdots + nb_{n-1} &= (n-1)b_1 + (n-2)b_2 + \cdots + b_{n-1} + b_1 + 2b_2 + \cdots + (n-1)b_{n-1} \\ &= b_1 + (b_1 + b_2) + \cdots + (b_1 + b_2 + \cdots + b_{n-1}) + (a_1 + a_2 + \cdots + a_{n-1}) \\ &\geq 2(a_1 + a_2 + \cdots + a_{n-1}) = \sum_{i=1}^{n-1} (a_i + a_{n-i}) \geq (n-1)a_n. \end{aligned}$$

3 POINTS for the first inequality and 1 POINT for the rest.

Problem 3. Let Γ_1 and Γ_2 be two circles intersecting at P and Q. The common tangent, closer to P, of Γ_1 and Γ_2 touches Γ_1 at A and Γ_2 at B. The tangent of Γ_1 at P meets Γ_2 at C, which is different from P and the extension of AP meets BC at R. Prove that the circumcircle of triangle PQR is tangent to BP and BR.

Solution and Marking Scheme:

Let $\alpha = \angle PAB$, $\beta = \angle ABP$ y $\gamma = \angle QAP$. Then, since PC is tangent to Γ_1, we have $\angle QPC = \angle QBC = \gamma$. Thus A, B, R, Q are concyclic.

3 POINTS for proving that A, B, R, Q are concyclic.

Since AB is a common tangent to Γ_1 and Γ_2 then $\angle AQP = \alpha$ and $\angle PQB = \angle PCB = \beta$. Therefore, since A, B, R, Q are concyclic, $\angle ARB = \angle AQB = \alpha + \beta$ and $\angle BQR = \alpha$. Thus $\angle PQR = \angle PQB + \angle BQR = \alpha + \beta$.

2 POINTS for proving that $\angle PQR = \angle PRB = \alpha + \beta$.

Since $\angle BPR$ is an exterior angle of triangle ABP, $\angle BPR = \alpha + \beta$. We have

$$\angle PQR = \angle BPR = \angle BRP$$

1 POINT for proving $\angle BPR = \alpha + \beta$.

So circumcircle of PQR is tangent to BP and BR.

1 POINT for concluding.

Remark. *2 POINTS can be given for proving that $\angle PRB = \angle RPB$ and 1 more POINT for attempting to prove (unsuccessfully) that $\angle PRB = \angle RPB = \angle PQR$.*

Problem 4. Determine all pairs (a, b) of integers with the property that the numbers $a^2 + 4b$ and $b^2 + 4a$ are both perfect squares.

First Solution and Marking Scheme:

Without loss of generality, assume that $|b| \leq |a|$. If $b = 0$, then a must be a perfect square. So $(a = k^2, b = 0)$ for each $k \in Z$ is a solution.

Now we consider the case $b \neq 0$. Because $a^2 + 4b$ is a perfect square, the quadratic equation

$$x^2 + ax - b = 0 \qquad (*)$$

has two non-zero integral roots x_1, x_2.

2 POINTS for noticing that this equation has integral roots.

Then $x_1 + x_2 = -a$ and $x_1 x_2 = -b$, and from this it follows that

$$\frac{1}{|x_1|} + \frac{1}{|x_2|} \geq \left|\frac{1}{x_1} + \frac{1}{x_2}\right| = \frac{|a|}{|b|} \geq 1.$$

Hence there is at least one root, say x_1, such that $|x_1| \leq 2$.

3 POINTS for finding that $|x_1| \leq 2$.

There are the following possibilities.

(1) $x_1 = 2$. Substituting $x_1 = 2$ into $(*)$ we get $b = 2a + 4$. So we have $b^2 + 4a = (2a + 4)^2 + 4a = 4a^2 + 20a + 16 = (2a + 5)^2 - 9$. It is easy to see that the solution in non-negative integers of the equation $x^2 - 9 = y^2$ is $(3, 0)$. Hence $2a + 5 = \pm 3$. From this we obtain $a = -4, b = -4$ and $a = -1, b = 2$. The latter should be rejected because of the assumption $|a| \geq |b|$.

(2) $x_1 = -2$. Substituting $x_1 = -2$ into $(*)$ we get $b = 4 - 2a$. Hence $b^2 + 4a = 4a^2 - 12a + 16 = (2a - 3)^2 + 7$. It is easy to show that the solution in non-negative integers of the equation $x^2 + 7 = y^2$ is $(3, 4)$. Hence $2a - 3 = \pm 3$. From this we obtain $a = 3, b = -2$.

(3) $x_1 = 1$. Substituting $x_1 = 1$ into $(*)$ we get $b = a + 1$. Hence $b^2 + 4a = a^2 + 6a + 1 = (a + 3)^2 - 8$. It is easy to show that the solution in non-negative integers of the equation $x^2 - 8 = y^2$ is $(3, 1)$. Hence $a + 3 = \pm 3$. From this we obtain $a = -6, b = -5$.

(4) $x_1 = -1$. Substituting $x_1 = -1$ into $(*)$ we get $b = 1 - a$. Then $a^2 + 4b = (a - 2)^2, b^2 + 4a = (a + 1)^2$. Consequently, $a = k, b = 1 - k$ $(k \in Z)$ is a solution.

1 POINT for checking specific values for x_1.

Testing these solutions and by symmetry we obtain the following solutions

$$(-4,-4),(-5,-6),(-6,-5),(0,k^2),(k^2,0),(k,1-k)$$

where k is an arbitrary integer. (Observe that the solution $(3,-2)$ obtained in the second possibility is included in the last solution as a special case.)

1 POINT for writing up the correct answer.

Second Solution and Marking Scheme:

Without loss of generality assume that $|b| \leq |a|$. Then $a^2+4b \leq a^2+4|a| < a^2+4|a|+4 = (|a|+2)^2$. Given that a^2+4b is a perfect square and since a^2+4b y a^2 have same parity then $a^2+4b \neq (|a|+1)^2$, so

$$a^2+4b \leq a^2. \tag{1}$$

2 POINTS for proving (1).

Let us consider three cases.

Case 1. $a^2+4b = a^2$. Then $b=0$ and a must be a perfect square. So $a=k^2$, $b=0$ $(k \in Z)$ is a solution.

Case 2. $a^2+4b = (|a|-2)^2$. Then $b = 1-|a|$, therefore $b^2+4a = a^2-2|a|+4a+1$ must be a perfect square.

If $a>0$ then $b^2+4a = (a+1)^2$ is a perfect square for each $a \in Z$. Consequently $a=k$ and $b=1-k$ $(k \in Z^+)$ is a solution.

If $a=0$ then $b=1$, but from (1) b must be non-positive.

If $a<0$ then $b^2+4a = m^2-6m+1$ must be a perfect square, where $m=-a>0$. For $m \geq 8$

$$(m-3)^2 > m^2-6m+1 > (m-4)^2,$$

therefore $m<8$. If $m=1,2,3,4,5$ then $m^2-6m+1<0$. If $m=6$, $m^2-6m+1=1$ is a perfect square thus $a=-6$ and $b=-5$ is a solution. If $m=7$, $m^2-6m+1=8$ is not a perfect square.

2 POINTS for case 1 and case 2.

Case 3. $a^2+4b \leq (|a|-4)^2$. Since $|b| \leq |a|$ then $b \geq -|a|$, thus $a^2-4|a| \leq a^2+4b \leq (|a|-4)^2$. It follows that $|a| \leq 4$. We have following posibilities:

1 POINT for finding that $|a| \leq 4$ in this case.

(a) $|a|=4$. Then $16+4b=0$ or $b=-4$. Thus $b^2+4a = 16 \pm 16$ must be a perfect square. So $a=-4$ y $b=-4$.

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(b) $|a| = 3$. In this case $a^2 + 4b = 9 + 4b \leq 1$, then $9 + 4b = 0$ or $9 + 4b = 1$. The equation $9 + 4b = 0$ does not have integer solutions. The solution of the second equation is $b = -2$. Then $b^2 + 4a = 4 \pm 12$ must be a perfect square, thus $a = 3$.

(c) $|a| = 2$. $a^2 + 4b = 4 + 4b \leq 4$. Since $4 + 4b$ is even and must be a perfect square then $4 + 4b = 4$ or $4 + 4b = 0$. Therefore $b = 0$ or $b = -1$. If $b = 0$, $b^2 + 4a = \pm 8$ is not a perfect square. If $b = -1$ then $b^2 + 4a = 1 \pm 8$ is a perfect square if $a = 2$. Thus $a = 2$ and $b = -1$ is a solution.

(d) $|a| = 1$. Then $a^2 + 4b = 4b + 1 \leq 9$. Since $4b + 1$ must be an odd perfect square then $4b + 1 = 1$ or $4b + 1 = 9$. So $b = 0$ or $b = 2$. If $b = 0$, $b^2 + 4a = \pm 4$, then $a = 1$. If $b = 2$ then $a = -1$, but this is not possible because $|b| \leq |a|$. Thus $a = 1$ y $b = 0$ is a solution in this case.

(e) $|a| = 0$. Since $|b| \leq |a|$ then $b = 0$.

1 POINT for concluding case 3.

Testing these solutions and by symmetry we obtain the following solutions:

$$(k^2, 0),\ (0, k^2),\ (k, 1-k),\ (-6, -5),\ (-5, -6),\ (-4, -4),$$

where k is an arbitrary integer. Note that if $(k, 1-k)$ is a solution with $k > 0$, then taking $t = 1 - k$, $k = 1 - t$, so $(1 - t, t)$ is solution. Thus by symmetry $(k, 1 - k)$ is a solution for any integer.

1 POINT for writing up the correct answer.

Remark: *1 POINT can be given for checking that $(k, 1 - k)$ is a solution. However NO POINT is given for finding any other particular solution.*

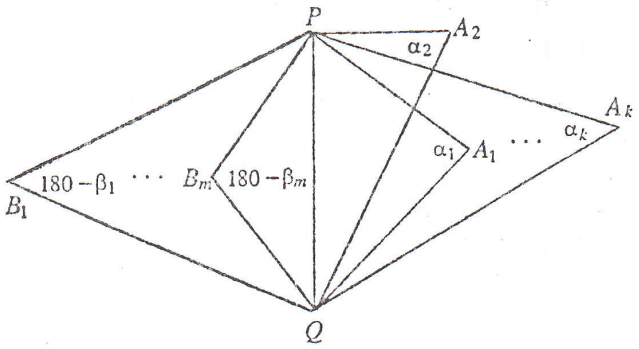
Problem 5. Let S be a set of $2n + 1$ points in the plane such that no three are collinear and no four concyclic. A circle will be called *good* if it has 3 points of S on its circumference, $n - 1$ points in its interior and $n - 1$ in its exterior. Prove that the number of good circles has the same parity as n.

Solution and Marking Scheme:

Lemma 1. Let P and Q be two points of S. The number of good circles that contain P and Q on their circumference is odd.

h

Proof of Lemma 1.



Let N be the number of good circles that pass through P and Q. Number the points on one side of the line PQ by $A_1, A_2, ..., A_k$ and those on the other side by $B_1, B_2, ..., B_m$ in such a way that if $\angle PA_iQ = \alpha_i, \angle PB_jQ = 180 - \beta_j$ then $\alpha_1 > \alpha_2 > ... > \alpha_k$ and $\beta_1 > \beta_2 > ... > \beta_m$.

Note that the angles $\alpha_1, \alpha_2, ..., \alpha_k, \beta_1, \beta_2, ..., \beta_m$ are all distinct since there are no four points in S that are concyclic.

Observe that the circle that passes through P, Q and A_i has A_j in its interior when $\alpha_j > \alpha_i$ that is, when $i > j$; and it contains B_j in its interior when $\alpha_i + 180 - \beta_j > 180$, that is, when $\alpha_i > \beta_j$. Similar conditions apply to the circle that contains P, Q and B_j.

1 POINT for characterizing the points that lie inside a given circle in terms of these angles, or for similar considerations.

Order the angles $\alpha_1, \alpha_2, ..., \alpha_k, \beta_1, \beta_2, ..., \beta_m$ from the greatest to least. Now transform S as follows. Consider a β_j that has an α_i immediately to its left in such an ordering $(... > \alpha_i > \beta_j ...)$. Consider a new set S' that contains the same points as S except for A_i and B_j. These two points will be replaced by A_i' and B_j' that satisfy $\angle PA_i'Q = \beta_j = \alpha_i'$ and $\angle PB_j'Q = 180 - \alpha_i = 180 - \beta_j'$. Thus β_j and α_i have been interchanged and the ordering of the α's and β's has only changed with respect to the relative order of α_i and β_j; we continue to have

$$\alpha_1 > \alpha_2 > ... > \alpha_{i-1} > \alpha_i' > \alpha_{i+1} > ... > \alpha_k$$

and

$$\beta_1 > \beta_2 > ... > \beta_{j-1} > \beta_j' > \beta_{j+1} > ... > \beta_m.$$

1 POINT for this or another useful transformation of the set S.

Analyze the good circles in this new set S'. Clearly, a circle through P, Q, A_r $(r \neq i)$ or through P, Q, B_s $(s \neq j)$ that was good in S will also be good in S', because the order of A_r (or B_s) relative to the rest of the points has not changed, and therefore the number of points in the interior or exterior of this circle has not changed. The only changes that could have taken place are:

a) If the circle P, Q, A_i was good in S, the circle P, Q, A_i' may not be good in S'.

b) If the circle P, Q, B_j was good in S, the circle P, Q, B_j' may not be good in S'.

c) If the circle P, Q, A_i was not good in S, the circle P, Q, A_i' may be good in S'.

d) If the circle P, Q, B_j was not good in S, the circle P, Q, B_j' may be good in S'.

1 POINT for realizing that the transformation can only change the "goodness" of these circles.

But observe that the circle P, Q, A_i contains the points $A_1, A_2, ..., A_{i-1}$, $B_j, B_{j+1}, ..., B_m$ and does not contain the points $A_{i+1}, A_{i+2}, ..., A_k$, $B_1, B_2, ..., B_{j-1}$ in its interior. Then this circle is good if and only if $i + m - j = k - i + j - 1$, which we rewrite as $j - i = \frac{1}{2}(m - k + 1)$. On the other hand, the circle P, Q, B_j contains the points $B_{j+1}, B_{j+2}, ..., B_m, A_1, A_2, ..., A_i$ and does not contain the points $B_1, B_2, ..., B_{j-1}, A_{i+1}, A_{i+2}, ..., A_k$ in its interior. Hence this circle is good if and only if $m - j + i = j - 1 + k - i$, which we rewrite as $j - i = \frac{1}{2}(m - k + 1)$.

Therefore, the circle P, Q, A_i is good if and only if the circle P, Q, B_j is good. Similarly, the circle P, Q, A_i' is good if and only if the circle P, Q, B_j' is good. That is to say, transforming S into S' we lose either 0 or 2 good circles of S and we gain either 0 or 2 good circles in S'.

1 POINT for realizing that the "goodness" of these circles is changed in pairs.

Continuing in this way, we may continue to transform S until we obtain a new set S_0 such that the angles $\alpha_1', \alpha_2', ..., \alpha_k', \beta_1', \beta_2', ..., \beta_m'$ satisfy

$$\beta_1' > \beta_2' > ... > \beta_m' > \alpha_1' > \alpha_2' > ... > \alpha_k'.$$

and such that the number of good circles in S_0 has the same parity as N. We claim that S_0 has exactly one good circle. In this configuration, the circle P, Q, A_i does not contain any B_j and the circle P, Q, B_r does not contain any A_s (for all i, j), because $\alpha_a + (180 - \beta_b) < 180$ for all a, b. Hence, the only possible good circles are P, Q, B_{m-n+1} (which contains the $n - 1$ points $B_{m-n+2}, B_{m-n+3}, ..., B_m$), if $m - n + 1 > 0$, and the circle P, Q, A_n (which contains the $n - 1$ points $A_1, A_2, ..., A_n$), if $n \leq k$. But, since $m + k = 2n - 1$, which we rewrite as $m - n + 1 = n - k$, exactly one of the inequalitites $m - n + 1 > 0$ and $n \leq k$ is satisfied. It follows that one of the points B_{m-n+1} and A_n corresponds to a good circle and the other does not. Hence, S_0 has exactly one good circle, and N is odd. □

1 POINT for showing that this configuration has exactly one good circle.

Now consider the $\binom{2n+1}{2}$ pairs of points in S. Let a_{2k+1} be the number of pairs of points through which exactly $2k + 1$ good circles pass. Then

$$a_1 + a_3 + a_5 + ... = \binom{2n+1}{2}.$$

But then the number of good circles in S is

$$\begin{aligned}\tfrac{1}{3}(a_1+3a_3+5a_5+7a_7+...) &\equiv a_1+3a_3+5a_5+7a_7+...\\ &\equiv a_1+a_3+a_5+a_7+...\\ &\equiv \tbinom{2n+1}{2}\\ &\equiv n(2n+1)\\ &\equiv n \pmod 2.\end{aligned}$$

Here we have taken into account that each good circle is counted 3 times in the expression $a_1+3a_3+5a_5+7a_7+....$ The desired result follows.

2 POINTS for this computation.

Alternative Proof of Lemma 1.

Let $A_1, A_2, \ldots, A_{2n-1}$ be the $2n-1$ given points other than P and Q.

Invert the plane with respect to point P. Let $O, B_1, B_2, \ldots, B_{2n-1}$ be the images of points $Q, A_1, A_2, \ldots, A_{2n-1}$, respectively, under this inversion. Call point B_i "good" if the line OB_i splits the points $B_1, B_2 \ldots, B_{i-1}, B_{i+1}, \ldots, B_{2n-1}$ evenly, leaving $n-1$ of them to each side of it. (Notice that no other B_j can lie on the line OB_i, or else the points P, Q, A_i and A_j would be concyclic.) Then it is clear that the circle through P, Q and A_i is good if and only if point B_i is good. Therefore, it suffices to prove that the number of good points is odd.

1 POINT for inverting and realizing the equivalence between good circles and good points.

Notice that the good points depend only on the relative positions of rays $OB_1, OB_2, \ldots, OB_{2n-1}$, and not on the exact positions of points $B_1, B_2, \ldots, B_{2n-1}$. Therefore we may assume, for simplicity, that $B_1, B_2, \ldots, B_{2n-1}$ lie on the unit circle Γ with center O.

1 POINT for this or a similar simplification.

Let $C_1, C_2, \ldots, C_{2n-1}$ be the points diametrically opposite to $B_1, B_2, \ldots, B_{2n-1}$ in Γ. As remarked earlier, no C_i can coincide with one of the B_j's. We will call the B_i's "white points", and the C_i's "black points". We will refer to these $4n-2$ points as the "colored points".

Now we prove that the number of good points is odd, which will complete the proof of the lemma. We proceed by induction on n. If $n=1$, the result is trivial. Now assume that the result is true for $n=k$, and consider $2k+1$ white points $B_1, B_2, \ldots, B_{2k+1}$ on the circle Γ (no two of which are diametrically opposite), and their diametrically opposite black points $C_1, C_2, \ldots, C_{2k+1}$. Call this configuration of points "configuration 1". It is clear that we must have two consecutive colored points on Γ which have different colors, say B_i and C_j. Now remove points B_i, B_j, C_i and C_j from Γ, to obtain "configuration 2", a configuration with $2k-1$ points of each color.

1 POINT for this or a similar transformation of the set.

It is easy to verify the following two claims:

1. Point B_i is good in configuration 1 if and only if point B_j is good in configuration 1.

1 POINT

2. Let $k \neq i, j$. Then point B_k is good in configuration 1 if and only if it is good in configuration 2.

1 POINT

It follows that, by removing points B_i, B_j, C_i and C_j, the number of good points can either stay the same, or decreases by two. In any case, its parity remains unchanged. Since we know, by the induction hypothesis, that the number of good points in configuration 2 is odd, it follows that the number of good points in configuration 1 is also odd. This completes the proof.

Another Approach to Lemma 1.

One can give another inductive proof of lemma 1, which combines the ideas of the two proofs that we have given. The idea is to start as in the first proof, with the characterization of the points inside a given circle.

1 POINT

Then we transform the set S by removing the points A_i and B_j instead of replacing them by A_i' and B_j'.

1 POINT

It can be shown that every one of the remaining circles going through P and Q, contained exactly one of A_i and B_j. Therefore, the only good circles we could have gained or lost are P, Q, A_i and P, Q, B_j.

2 POINTS

Finally, we show that either both or none of these circles were good, so the parity of the number of good circles isn't changed by this transformation.

1 POINT

Remark: *2 POINTS can be given if the result has been fully proved for a particular case with $n > 1$. (If more than one particular case has been analyzed completly, only 2 POINTS.) These points are awarded only if no progress has been made in the general solution of the problem.*

APMO SOLUTIONS

PROBLEM 1. Let F be the set of all n-tuples $(A_1, A_2, \ldots, A_n)$ where each A_i, $i = 1, 2, \ldots, n$ is a subset of $\{1, 2, \ldots, 1998\}$. Let $|A|$ denote the number of elements of the set A. Find the number

$$\sum_{(A_1, A_2, \ldots, A_n)} |A_1 \cup A_2 \cup \ldots \cup A_n|.$$

MARKING SCHEME:

Let M be a subset of the set $\{1, 2, \ldots, 1998\}$ and let $|M| = k$. Then the set M can be obtained as the union of t sets $A_1, A_2, \ldots, A_t$ in $(2^t - 1)^k$ different ways since each element $x \in M$ can belong to $2^t - 1$ nonempty families of subsets $A_1, A_2, \ldots, A_t$.

3 points for describing the correct counting method

Thus we have

$$\sum_{(A_1, A_2, \ldots, A_t) \in F} |A_1 \cup A_2 \cup \ldots \cup A_t| = \sum_{k=1}^{1998} k \binom{1998}{k} (2^t - 1)^k$$

2 points for setting up the above formula

$$= 1998(2^t - 1) \sum_{k=0}^{1997} \binom{1997}{k} (2^t - 1)^k = 1998(2^t - 1)2^{1997t}.$$

2 points for correctly carrying out the computation

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PROBLEM 2. Show that for any positive integers a and b, $(36a + b)(a + 36b)$ cannot be a power of 2.

MARKING SCHEME:

Suppose that $(36a + b)(a + 36b)$ is a power of 2 for some positive integers a and b. Write $36a + b = 2^m = r$ and $a + 36b = 2^n = s$. Then

$$36r - s = 35 \times 37a \text{ and } 36s - r = 35 \times 37b.$$

Hence

$$1/36 < r/s = 2^{m-n} < 36, \text{ or } -6 < m - n < 6.$$

2 points for the basic setting and observation

Furthermore,

$$4^n(4^{m-n} - 1) = r^2 - s^2 = 35 \times 37(a^2 - b^2).$$

Thus

$$4^{m-n} \equiv 1 (mod 37).$$

2 points for this congruence

Observe that the 9th power of 4 is the smallest power of 4 that is congruent to $1 (mod 37)$. Thus $9|(m - n)$. Also note that $m \neq n$. Hence $|m - n| \geq 9$, which is not possible because $|m - n| < 6$. *3 points for the final conclusion*

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Problem 3. Let a, b, c be positive real numbers. Prove that

$$\left(1+\frac{a}{b}\right)\left(1+\frac{b}{c}\right)\left(1+\frac{c}{a}\right) \geq 2\left(1+\frac{a+b+c}{\sqrt[3]{abc}}\right).$$

MARKING SCHEME:

Let

$$x=\frac{a}{\sqrt[3]{abc}}, \quad y=\frac{b}{\sqrt[3]{abc}}, \quad z=\frac{c}{\sqrt[3]{abc}}.$$

We need to show that

$$\left(1+\frac{x}{y}\right)\left(1+\frac{y}{z}\right)\left(1+\frac{z}{x}\right) \geq 2(1+x+y+z)$$

or, since $xyz=1$,

$$(x+y)(y+z)(z+x) \geq 2+2(x+y+z) \quad (1)$$

2 points for (1) by making change of variables or by assuming $abc=1$ without loss of generality

which is equivalent to

$$(x+y+z)(xy+yz+zx-2)-xyz \geq 2 \quad (2)$$

2 points for reducing to (2)

Since $xyz=1$, the AM-GM inequality implies

$$x+y+z \geq 3 \text{ and } xy+yz+zx \geq 3$$

So, (2) follows. *3 points for successful application of AM-GM inequality*

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ALTERNATE SOLUTION

2 points for (1) by making change of variables or by assuming $abc = 1$ without loss of generality

¿From (1), one can proceed as follows:

$$(x+y)(y+z)(z+x) = 2xyz+x^2(y+z)+y^2(z+x)+z^2(x+y) = 2+x\left(\frac{1}{y}+\frac{1}{z}\right)+y\left(\frac{1}{z}+\frac{1}{x}\right)+z\left(\frac{1}{x}+\frac{1}{y}\right).$$

Thus, (1) is equivalent to

$$x\left(\frac{1}{y}+\frac{1}{z}\right)+y\left(\frac{1}{z}+\frac{1}{x}\right)+z\left(\frac{1}{x}+\frac{1}{y}\right) \geq 2(x+y+z).3$$

2 points for reducing to (3)

Assume that $x \geq y \geq z$. Then

$$\frac{1}{y}+\frac{1}{z} \geq \frac{1}{z}+\frac{1}{x} \geq \frac{1}{x}+\frac{1}{y}$$

hence we can apply Chebyshev's inequality

1 points for checking the hypothesis of Chevyshev's inequality

to get

$$x\left(\frac{1}{y}+\frac{1}{z}\right)+y\left(\frac{1}{z}+\frac{1}{x}\right)+z\left(\frac{1}{x}+\frac{1}{y}\right) \geq \frac{1}{3}(x+y+z)\left(\frac{2}{x}+\frac{2}{y}+\frac{2}{z}\right).$$

¿From the AM-GM inequality and the fact that $xyz = 1$ we obtain

$$\left(\frac{2}{x}+\frac{2}{y}+\frac{2}{z}\right) \geq 6$$

and hence (3) follows. *2 points for successful applications of Chevyshev's and AM-GM inequalities*

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Problem 4. Let ABC be a triangle and D the foot of the altitude from A. Let E and F be on a line passing through D such that AE is perpendicular to BE, AF is perpendicular to CF, and E and F are different from D. Let M and N be the midpoints of the line segments BC and EF, respectively. Prove that AN is perpendicular to NM.

MARKING SCHEME:

Let P be such that $ADMP$ is a rectangle. Choose points Q and R on the line AP such that $QBDA$ and $ADCR$ are rectangles. Points Q, B and D lie on the circle of diameter AB, hence $ADEQ$ is a cyclic quadrilateral. Similarly, R, C and D lie on the circle of diameter AC, hence $ADFR$ is a cyclic quadrilateral.

1 point for proving $ADEQ$ and/or $ADFR$ are cyclic

The two quadrilaterals share a side, and have the same supporting lines for other two sides. Since they are cyclic, the remaining two sides EQ and RF must be parallel. Thus E, Q, R and F are vertices of a trapezoid.

1 point for proving EQ // RF

On the other hand, in rectangle $QBCR$ M is the midpoint of BC, and MP is parallel to QB, so P is the midpoint of QR. Since N is the midpoint of EF, we obtain that, in trapezoid $QEFR$, NP is parallel to QE.

2 points for proving NP //QE

This implies that quadrilateral $ADNP$ is cyclic, having the sides parallel to the sides of $ADFR$. Moreover, A lies on the circle circumscribed to this quadrilateral, because the other three vertices of the rectangle $ADMP$ lie on it. Hence the quadrilateral $ADMN$ is cyclic.

2 points for proving $ADNP$ and $ADMN$ are cyclic

Consequently, $\angle ANM = 180^\circ - \angle ADM = 90^\circ$.

1 point for proving $\angle ANM = 180^\circ - \angle ADM = 90^\circ$

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Problem 5. Determine the largest of all integers n with the property that n is divisible by all positive integers that are less than $\sqrt[3]{n}$.

MARKING SCHEME:

Observation from that $lcm(2,3,4,5,6,7) = 420$ is divisible by every integer less than or equal to $7 = [\sqrt[3]{420}]$ and that $lcm(2,3,4,5,6,7,8) = 840$ is not divisible by $9 = [\sqrt[3]{840}]$, one may guess 420 is the required integer. *2 points for the correct guess*

Let N be the required integer and suppose $N > 420$. Put $t = [\sqrt[3]{N}]$. Then

$$N \leq t(t^3 + 3t + 3) \quad (1)$$

Since $t \geq 7$, $lcm(2,3,4,5,6,7) = 420$ should divide N and hence $N \geq 840$, which implies $t \geq 9$. But then $lcm(2,3,4,5,6,7,8,9) = 2520$ should divide N, which implies $t \geq 13 = [\sqrt[3]{2520}]$.

1 points for $t \geq 13$

Observe that any four consecutive integers are divisible by 8 and that any two out of four consecutive integers have *gcd* either 1, 2, or 3. So, we have $t(t-1)(t-2)(t-3)$ divides $6N$ and in particular.

$$t(t-1)(t-2)(t-3) \leq 6N \quad (2)$$

2 points for $t(t-1)(t-2)(t-3) \leq 6N$

¿From (1) and (2) follows

$$t(t-1)(t-2)(t-3) \leq 6t(t^3 + 3t + 3)\frac{12}{t} + \frac{7}{t^2} + \frac{24}{t^3} \geq 1.$$

Since $t \geq 13$,

$$\frac{12}{t} + \frac{7}{t^2} + \frac{24}{t^3} < 1,$$

which is a contradiction.

2 points for the contradiction

Solutions

Note: The points to be awarded for each part of the solution are indicated on the right side.

Problem 1.

$$1 = \frac{1 \times 2}{2}$$

$$1 + \frac{1}{3} = \frac{2 \times 2}{3}$$

$$1 + \frac{1}{3} + \frac{1}{6} + \ldots + \frac{1}{t_n} = \frac{n \times 2}{n + 1}$$

which is easily shown by induction.

(up to 3 points)

Now S is the sum of the reciprocals of these numbers where the last, $1993006 = \frac{1996 \times 1997}{2} = t_{1996}$. Thus we have

$$S = \frac{1}{2}\left(\frac{2}{1} + \frac{3}{2} + \ldots + \frac{1997}{1996}\right)$$

$$= \frac{1}{2}\left(1996 + \left(1 + \frac{1}{2} + \ldots + \frac{1}{1996}\right)\right)$$

(up to 3 points)

$$> \frac{1}{2}(1996 + 6)$$

(1 point)

$$= 1001$$

Problem 2. Note that $2^n + 2 = 2(2^{n-1} + 1)$ so that n is of the form 2r with r odd. We will consider two cases.

i) $n = 2p$ with p prime. $2p \mid 2^{2p} + 2$, implies that $p \mid 2^{2p-1} + 1$ and hence , hence $p \mid 2^{4p-2} - 1$. On the other hand Fermat's little theorem guarantees that $p \mid 2^{p-1} - 1$. Let d = g.c.d. $(p-1, 4p - 2)$. It follows that $p \mid 2^d - 1$. But $d \mid p - 1$ and $d \mid 4p - 2 = 4(p - 1) + 2$. Hence $d \mid 2$ and since $p - 1$, $4p - 2$ are even $d = 2$. Then $p = 3$ and $n = 6 < 100$.

(up to 2 points)

ii) $n = 2pq$ where p, q are odd primes. $p < q$ and $pq < \frac{1997}{2}$. Now $n \mid 2^n + 2$ implies that $p \mid 2^{n-1} + 1$ and therefore that $p \mid 2^{2n-2} - 1 = 2^{4pq-2} - 1$. Once again by Fermat's theorem we have $p \mid 2^{n-1} - 1$ which implies that $p - 1 \mid 4pq - 2$. The same holds true for q so that

$p \mid 2^{p-1} - 1$

$$q - 1 \mid 4pq - 2 \qquad (1)$$

Both $p - 1$ and $q - 1$ are thus multiples of 2 but not of 4 so that $p \equiv q \equiv 3 \pmod 4$.

(2 points)

Taking $p = 3$, we have $4pq - 2 = 12q - 2$. Now from (1) we have

$$12 = \frac{12q - 12}{q - 1} < \frac{12q - 2}{q - 1} = \frac{12(q - 1) + 10}{q - 1} = 12 + \frac{10}{q - 1} \le 1$$

if $q \ge 11$, and clearly $\frac{12q - 2}{q - 1} = 13$ if $q = 11$. But this gives $n = 2(3)(11) = 66 < 100$. Furthermore $(p, q) = (3, 7)$ does not satisfy (1).

Taking $p = 7$ we observe that $4pq - 2 = 28q - 2$, and from (1) we have

$$28 < \frac{28q - 2}{q - 1} = \frac{28(q - 1) + 26}{q - 1} = 28 + \frac{26}{q - 1} \le 2$$

if $q \ge 27$ and clearly $\frac{28q - 2}{q - 1} = 29$ if $q = 27$. But 27 is not prime and the cases $(p, q) = (7, 11)$, $(7, 19)$ and $(7, 23)$ do not satisfy (1).

Taking $p = 11$, then $4pq - 2 = 44q - 2$, and

$$44 < \frac{44q - 2}{q - 1} \text{ and } \frac{44q - 2}{q - 1} \le 45 \text{ if } q \ge 43.$$

Now clearly $\frac{44q - 2}{q - 1} = 45$ when $q = 43$. In this case we have $n = 2pq = 2(11)(43) = 946$. Furthermore, $\frac{2^{946} + 2}{946}$ is indeed an integer. The cases $(p, q) = (11, 19)$, $(11, 23)$ and $(11, 31)$ do not satisfy (1).

(2 points)

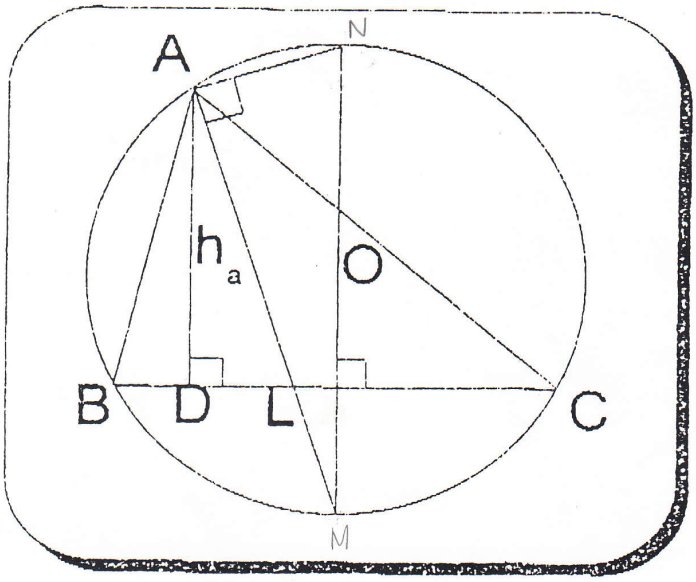
[Additionally for completeness, if $p = 19$ then $4pq - 2 = 76q - 2$ and $76 < \frac{76q - 2}{q - 1} \le 77$ if $q \ge 75$. Now 75 is not prime and for the cases $(p, q) = (19, 23)$, $(19, 31)$, $(19, 43)$ and $(19, 47)$, $q - 1$ is not a divisor of $74 = 2 \times 37$.

Similarly, if p = 23 then $4pq - 2 = 92q - 2$ and $92 < \frac{92q-2}{q-1} \leq 93$ if $q \geq 91$ and $\frac{92q-2}{q-1} = 93$ if $q = 91$. But 91 is not prime and of the cases (p, q) = (23, 31), (23, 43), when q = 31 all of the conditions are satisfied. But, $n = 2pq = 1426$ is not a solution because $\frac{2^{1426}+2}{1426}$ is not an integer.

No other pairs of p, q yield numbers within the required range.]

(1 point)

Problem 3.



$\angle ALD = \frac{1}{2}(\overset{\frown}{MC} + \overset{\frown}{AB})$

$= \frac{1}{2}(\overset{\frown}{BM} + \overset{\frown}{AB})$

$= \angle ANM$

It is known (see Geometry Revisited) or easily derivable that

$$m_a^2 = (AL)^2 = bc\left(1 - \left(\frac{a}{b+c}\right)^2\right).$$

Using Stewart's Theorem:
Let AX be a cevian of length p, dividing BC into segments BX = m and XC = n. Then
$a(p^2 + mn) = b^2m + c^2n.$

(1 point)

Here, we use $m = kc$, $n = kb$, $k = \frac{a}{b+c}$

From Δ ADL ~ Δ MAN we have

$$\frac{AD}{AL} = \frac{AM}{MN} \Rightarrow AD \cdot MN = AL \cdot AM$$

$$h_a \cdot 2R = AL \cdot AM = m_a \cdot M_a$$

$h_a = \frac{2(ABC)}{a}$

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$$\frac{2(ABC)}{a} \cdot 2R = m_a M_a \qquad (ABC) = \frac{abc}{4R}$$

$$\frac{\frac{abc}{4R} \cdot 4R}{a} = m_a M_a$$

$$bc = m_a M_a.$$

So that

$$l_a = \frac{m_a^2}{m_a M_a} = 1 - \left(\frac{a}{b+c}\right)^2$$

with similar expressions for l_b and l_c.

(2 points)

Given that $\sin A = \frac{a}{2R}$, etc. the expression we are working with becomes

$$\frac{l_a}{\sin^2 A} + \frac{l_b}{\sin^2 A} + \frac{l_c}{\sin^2 C} = \frac{4R^2}{a^2}\left(1 - \left(\frac{a}{b+c}\right)^2\right) + \frac{4R^2}{b^2}\left(1 - \left(\frac{b}{a+c}\right)^2\right) + \frac{4R^2}{c^2}\left(1 - \left(\frac{c}{a+b}\right)^2\right)$$

$$= 4R^2\left[\left(\frac{1}{a^2} - \frac{1}{(b+c)^2}\right) + \left(\frac{1}{b^2} - \frac{1}{(a+c)^2}\right) + \left(\frac{1}{c^2} - \frac{1}{(a+b)^2}\right)\right]$$

$$\geq 4R^2\left[\frac{1}{a^2} - \frac{1}{4bc} + \frac{1}{b^2} - \frac{1}{4ac} + \frac{1}{c^2} - \frac{1}{4ab}\right]$$

$$= 2R^2\left[\left(\frac{1}{a^2} + \frac{1}{b^2}\right) + \left(\frac{1}{a^2} + \frac{1}{c^2}\right) + \left(\frac{1}{b^2} + \frac{1}{c^2}\right) - \frac{1}{2ab} - \frac{1}{2ac} - \frac{1}{2bc}\right]$$

$$\geq 2R^2\left[\frac{2}{ab} + \frac{2}{ac} + \frac{2}{bc} - \frac{1}{2ab} - \frac{1}{2ac} - \frac{1}{2bc}\right]$$

$$= 2R^2\left[\frac{3}{2ab} + \frac{3}{2ac} + \frac{3}{2bc}\right]$$

$$= 3R^2\left[\frac{1}{ab} + \frac{1}{ac} + \frac{1}{bc}\right] = 3R^2\left[\frac{a+b+c}{abc}\right]$$

But $abc = 4R(ABC)$ so that this last expression becomes

$$\frac{3R(a+b+c)}{4(ABC)} = \frac{3R \cdot 2s}{4sr} = 3 \cdot \frac{R}{2r} \geq 3$$

(3 points)

since $R \geq 2r$. All of the inequalities are equalities iff $a = b = c$.

(1 point)

Problem 4.

(a) Consider the sequence of triangles on the plane $A_1A_2A_3$, $A_2A_3A_4$, $A_3A_4A_5$, ...
It is easy to see that any pair of them are similar.
Let's prove that triangles $A_1A_2A_3$ and $A_3A_4A_5$ are similar.
Triangles $A_2A_3A_4$ and $A_4A_5A_6$ are similar and their altitudes are A_4A_5 and A_6A_7, then

$$\frac{A_2A_3}{A_4A_5} = \frac{A_4A_5}{A_6A_7}.$$

Triangles $A_3A_4A_5$ and $A_5A_6A_7$ are similar, then

$$\frac{A_4A_5}{A_6A_7} = \frac{A_3A_5}{A_5A_7}$$

Now we can conclude that

$$\frac{A_1A_3}{A_3A_5} = \frac{A_3A_5}{A_4A_5}$$

and triangles $A_1A_2A_3$ and $A_3A_4A_5$ are similar.

Hence, if P is the point where line A_1A_3 meets A_3A_5, $\triangle A_3A_5A_7 = \triangle PA_5A_7 = \triangle A_1A_2A_3$ and $\triangle A_7A_5A_1 = \triangle A_7A_5P = \triangle A_5A_3A_1$, so triangle A_7A_5P has a right angle at P and lines A_1A_5 and A_3A_7 are perpendicular. In the same way lines A_3A_7 and A_5A_9 are perpendicular and lines A_5A_9 and A_7A_{11} are perpendicular, hence A_1, A_5, A_9 are collinear and A_7, A_{11}, A_3 are collinear. It follows that triangle $A_1A_2A_3$ and $A_9A_{10}A_{11}$ are homothetic and the center of homothety is P. Moreover, all triangles from the family $A_1A_2A_3$, $A_9A_{10}A_{11}$, $A_1A_{10}A_{19}$, ...are homothetic. Of course the point P is an interior point to any of these triangles and there is no other point distinct from P that is interior to any of these triangles. So this is the point we are looking for.

(up to 4 points)

(b) Since $\triangle A_1PA_3$ 90^0 then P lies on the circle with diameter A_1A_3. Let $A_1A_3 = 1$, $A_2A_3 = s$, $A_3A_4 = r$, and let $A_1A_2A_3$ be clockwise. Triangles $A_1A_2A_3$ and $A_3A_4A_5$ are similar, thus $A_4A_5 : r = s : 1$, and so $A_4A_5 = rs$. Besides $A_3A_4 = r\sqrt{1+s^2}$ (Pythagoras), and area of triangle $A_1A_3A_5 = \frac{1}{2}r\sqrt{1+s^2} \cdot \sqrt{1+s^2} = \frac{1}{2}s \cdot 1$. Thus $r = \frac{s}{1+s^2}$. By the arithmetic-geometric mean $\frac{s}{1+s^2} \leq \frac{1}{2}$, thus $r \leq \frac{1}{2}$ and the set of all possible values of r consists of two real intervals $\left[-\frac{1}{2}, 0\right)$ and $\left(0, \frac{1}{2}\right]$. $\triangle A_3A_1P$ takes the maximum value when $r = \frac{1}{2}$ thus the locus of P

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consists of two continuous arcs from the circle with diameter A_1A_3 with two extreme positions corresponding to $r = -\frac{1}{2}$ and $r = \frac{1}{2}$.

(up to 3 points)

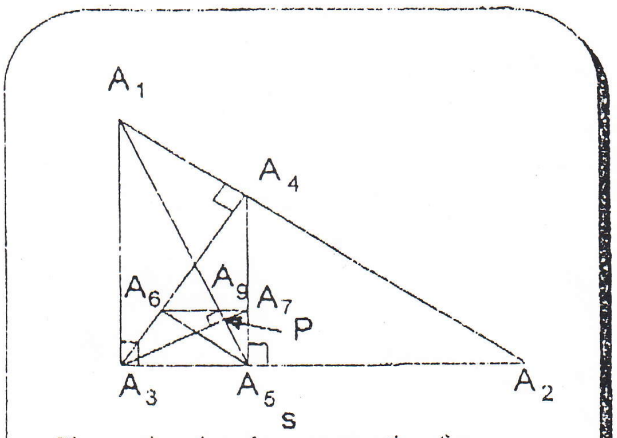

Figure showing the construction for finding point P.

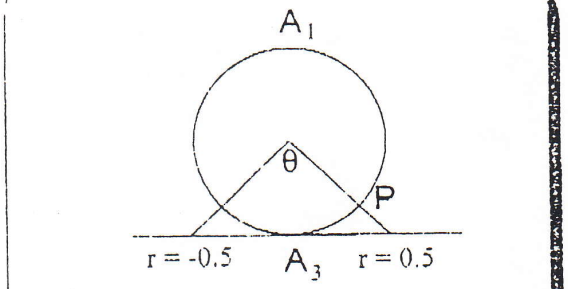

Figure showing the locus of P.

Problem 5.

A redistribution can be written as $(x_1, x_2, \cdots, x_n)$ where x_1 denotes the number of objects transferred from A_i to A_{i+1}. Our objective is to minimize the function

$$F(x_1, x_2, \cdots, x_n) = \sum_{i=1}^{n} |x_1|$$

After redistribution we should have at each A_i, $a_i - x_i + x_{i-1} = N$ for $i \in \{1,2,\ldots,n\}$ where x_0 means x_n. (1 point)

Solving this system of linear equations we obtain:

$x_i = x_1 - \left[(i-1)N - a_2 - a_3 - \ldots - a_i\right]$

for $i \in \{1,2,\ldots,n\}$.

Hence

$$F(x_1, x_2, \ldots, x_n) = |x_1| + |x_1 - (N - a_2)| + |x_1 - 2N - a_2 - a_3| + \ldots + \left|x_1 - \left[(n-1)N - a_2 - a_3 - \ldots - a_n\right]\right|$$

Basically the problem reduces to find the minimum of $F(x) = \sum_{i=1}^{n} |x - \alpha_i|$

where $\alpha_i = (i-1)N - \sum_{j=2}^{i} a_j$. (up to. 3 points)

First rearrange $\alpha_1, \alpha_2, \ldots, \alpha_n$ in non decreasing order. Collecting terms which are equal to one another we write the ordered sequence $\beta_1 < \beta_2 < \cdots < \beta_m$, each β_i occurs k_i times in the family $\{\alpha_1, \alpha_2, \cdots, \alpha_n\}$. Thus $k_1 + k_2 + \cdots + k_m = n$.

Consider the intervals $(-\infty, \beta_1], [\beta_1, \beta_2], \cdots, [\beta_{m-1}, \beta_m], [\beta_m, \infty)$

the graph of $F(x) = \sum_{i=1}^{n} |x - \alpha_i| = \sum_{i=1}^{m} k_i |x - \beta_i|$ is a continuos piece wise linear graph define in the following way:

$$F(x) = \begin{cases} k_1(\beta_1 - x) + k_2(\beta_2 - x) + \cdots + k_m(\beta_m - x) & \text{if } x \in (-\infty, \beta_1] \\ k_1(x - \beta_1) + k_2(\beta_2 - x) + \cdots + k_m(\beta_m - x) & \text{if } x \in [\beta_1, \beta_2] \\ \vdots & \\ k_1(x - \beta_1) + k_2(x - \beta_2) + \cdots + k_m)(x - \beta_m) & \text{if } x \in [\beta_m, \infty) \end{cases}$$

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(up to 4 points)

The slopes of each line segment on each interval are respectively:

$S_0 = -k_1 - k_2 - k_3 - \cdots - k_m$

$S_1 = k_1 - k_2 - k_3 - \cdots - k_m$

$S_2 = k_1 + k_2 - k_3 - \cdots - k_m$

$S_m = k_1 + k_2 + k_3 + \cdots + k_m$

Note that this sequence of increasing numbers goes from a negative to a positive number, hence for some $t \geq 1$ there is an

$$S_t = 0 \text{ or } S_{t-1} < 0 < S_t$$

In the first case the minimum occurs at $x = \beta_t$ or β_{t+1} and in the second case the minimum occurs at $x = \beta_t$

(Up to 7 points)

We can rephrase the computations above in terms of $\alpha_1, \alpha_2, \cdots, \alpha_n$ rather than $\beta_1, \beta_2, \cdots, \beta_m$. After rearranging the $\alpha's$ in non decreasing order, pick $x = \alpha_{\frac{n+1}{2}}$ if n is odd and take $x = \alpha_{\frac{n}{2}}$ or $\alpha_{\frac{n}{2}+1}$ if n is even.

If no justification is given for the choice of x , give up to 4 points.

Solutions

Problem 1. Let M'N' and P'Q' be two segments perpendicular to BD at a distance d. We need to prove that the perimeters AMNCQP and AM'N'CQ'P' are equal. Denote by S' the projection of N into M'N' and by S the projection of P' into PQ. The triangles NS'N' and P'SP are equal. (right triangles such that NS' = P'S and equal angles $\angle P'PS = \angle NN'S'$.) So PP' = NN'. Since it is clear that MM' = NN' , we have MM' = NN' = PP' = QQ'. On the other hand, since SP = S'N', using T' and T the projections of M and Q' into M'N' and PQ, respectively, we have

$$T'M' = S'N' = PS = TQ.$$

So
$$\begin{aligned} P'Q' + M'N' &= ST + M'T' + T'S' + S'N' \\ &= ST + PS + TQ + MN \\ &= PQ + MN. \end{aligned}$$

(Without loss of generality M' lies between M and A)
So the perimeter of AMNCQP is:

$$\begin{gathered} AM + MN + NC + CQ + QP + PA = \\ AM' + M'M + MN + NN' + N'C + CQ + QP + PA = \\ AM' + PP' + MN + QQ' + N'C + CQ + QP + PA = \\ AM' + MN + QP + PP' + PA + QQ' + CQ + N'C = \\ AM' + MN + QT + TS + SP + PP' + PA + QQ' + CQ + N'C = \\ AM' + M'T + T'S' + S'N' + N'C + CQ + QQ' + Q'P' + P'P + PA = \\ AM' + M'N' + N'C + CQ' + Q'P' + P'A, \end{gathered}$$

which is the perimeter of AM'N'CQ'P'.

Points to be given for showing that:

MM' = NN' = PP' = QQ'	1 point
T'M' = S'N' = PS = TQ.	1 point
P'Q' + M'N' = PQ + MN.	2 points
The perimeter of AMNCQP isthe same as the perimeter of AM'N'CQ'P'	3 points

Problem 2. We first prove by induction on n that:

$$\frac{(m+n)!}{(m-n)!} = \prod_{i=1}^{n} (m^2 + m - i^2 + i).$$

1. $\dfrac{(m+1)!}{(m-1)!} = m(m+1) = m^2 + m.$

2. $\dfrac{(m+n+1)!}{(m-n-1)!} = \left(\prod_{i=1}^{n} (m^2 + m - i^2 + i)\right)(m+n+1)(m-n)$ (by induction)

$$= \left(\prod_{i=1}^{n} (m^2 + m - i^2 + i)\right)(m^2 + m - n^2 - n)$$

$$= \prod_{i=1}^{n+1} (m^2 + m - i^2 + i)$$

But $m^2 + m \geq m^2 + m - i^2 + i \geq i^2 + i - i^2 + i = 2i$, for $i \geq m$.
Therefore

$$2^n n! \leq \frac{(m+n)!}{(m-n)!} \leq (m^2 + m)^n.$$

Points to be given for showing that:

the innequlities hold for special values	up to 1 point
$\frac{(m+n)!}{(m-n)!} = \prod_{i=1}^{n} (m^2 + m - i^2 + i)$ for all n	4 points
$m^2 + m \geq m^2 + m - i^2 + i \geq i^2 + i - i^2 + i = 2i$, for $i \geq m$, and therefore the innequalities hold	2 points

Problem 3. Let C be the given circle. Draw four circles C_{12}, C_{23}, C_{34}, C_{41} with centers O_{12}, O_{23}, O_{34}, O_{41}, respectively, on the circle C such that C_{12} passes through P_1 and P_2, C_{23} passes through P_2 and P_3, C_{34} passes through P_3 and P_4, C_{41} passes through P_4 and P_1. Let the other point of intersection of C_{12} and C_{23} be Q_4, the other point of intersection of C_{23} and C_{34} be Q_1, the other point of intersection of C_{34} and C_{41} be Q_2, and the other point of intersection of C_{41} and C_{12} be Q_3.
Then

$$\angle Q_4P_1P_2 = \frac{1}{2}\angle Q_4O_{12}P_2 \text{ and } \angle O_{23}P_1P_2 = \frac{1}{2}\angle P_3O_{12}P_2.$$

Clearly, O_{23}, Q_4 and P_1 are collinear.
It follows that $\angle Q_4Q_3P_2 = \angle Q_4P_1P_2 = \angle O_{23}P_1P_2\ \angle O_{23}O_{41}P_2$. Since also O_{41}, Q_3 and P_2 are collinear, it follows that Q_3Q_4 and $O_{41}O_{23}$ are parallel.
Since O_{ij} bisects The arcs P_iP_j, for (i, j) = (1, 2), (2, 3), (3, 4), (4, 1) we conclude that O_{41} O_{23} and O_{12} O_{34} are perpendicular, and hence Q_3 Q_4 and O_{12} O_{34} are perpendicular.
Since both (O_{12}, Q_3, P_4) and (O_{12}, Q_4, P_3) are collinear triples of points, we have $\angle P_4O_{12}P_3 = \angle Q_3O_{12}Q_4$, and this angle is bisected by O_{12} O_{34}.
Thus Q_3 and Q_4 are reflections through the axis $O_{12}O_{34}$, and so are, by a similar argument Q_1 and Q_2.
We have thus shown: Q_1, Q_2, Q_3, Q_4 form a rectangle. But as Q_4 lies on both the angle bisector $O_{12}P_3$ and the angle bisector $O_{23}P_1$ of the triangle $P_1P_2P_3$, the point Q_4 must coincide with the incenter I_4 of the triangle $P_1P_2P_3$, and by a similar argument, $Q_1 = I_1$, $Q_2 = I_2$ and $Q_3 = I_3$.

Points to be given for showing that

Q_3Q_4 and $O_{41}O_{23}$ are parallel	2 points
Q_3 Q_4 and O_{12} O_{34} are perpendicular	1 points
Q_1, Q_2, Q_3, Q_4 form a rectangle	2 points
$Q_4 = I_4$, $Q_1 = I_1$, $Q_2 = I_2$ and $Q_3 = I_3$	1 point

Problem 4. . We may assume that the n couples will form x male groups and y female groups. Without loss of generality, let $x \geq y$, and

(*) $$x + y = 17.$$

Then, by the pigeonhole theorem, there exists a male group of size $\leq \left[\frac{n}{x}\right]$ and a female group of size $\geq \left[\frac{n}{y}\right]$. By condition (2), we have

(**) $$[\tfrac{n}{y}] - [\tfrac{n}{x}] \leq 1$$

From the conditions $x + y = 17$ and $x \geq y$ follows that $x \geq 9$, and $y \leq 8$, which in turn implies

$$[\tfrac{n}{y}] - [\tfrac{n}{x}] > [\tfrac{n}{8}] - [\tfrac{n}{9}]$$

Therefore, we only need to exclude those n such that $[\tfrac{n}{8}] - [\tfrac{n}{9}] > 1$.Let $n = 9u + s$, $0 \leq s < 9$. Then

$$[\tfrac{n}{8}] - [\tfrac{n}{9}] > 1 \Leftrightarrow [\tfrac{u+s}{8}] > 1.$$

By analyzing this condition it is clear that the only values of n that are allowed are

n = 9, 10, 11, 12, 13, 14, 15, 16, 18, 19, 20, 21, 22, 23, 24, 27, 28, 29, 30, 31, 32, 36, 37, 38, 39, 40, 45, 46, 46, 47, 48, 54, 55, 56, 63, 63, 72.

Conversely, conditions (*) and (**) give rise to a set of discussing groups according to the following description:

Let $[\tfrac{n}{x}] = p$ and $[\tfrac{n}{y}] = q$, then $p \leq q \leq p + 1$

We have

$n = px + \alpha$, $0 \leq \alpha < x$ and $n = qy - \beta$ $0 \leq \beta < y$.

So we can arrange the males in α discussing groups of size $p + 1$ and $x - \alpha$ groups of p elements. The females are distributed in β discussing groups of size $q - 1$, and $(y - \beta)$ discussing groups of size q.

Points to be given for showing that

$x \geq 9$, and $y \leq 8$	1 points
one only needs to exclude those n such that $[\frac{n}{8}] - [\frac{n}{9}] > 1$	2 points
analyzing the condition ian giving the values of n that are allowed	2 points
given p and q as described in the solution one can arrange the males in α discussion groups of size $p + 1$ and $x - \alpha$ groups of p elements and the females in β discussion groups of size $q - 1$, and $(y - \beta)$ groups of size q.	2 points

Problem 5. Without loss of generality we can assume that $a \geq b \geq c$. Note that if $x \geq y > 0$, then $\sqrt{y} \leq 1/2(\sqrt{x} + \sqrt{y})$, i.e.,

$$\frac{1}{2\sqrt{y}}(\sqrt{x} + \sqrt{y}) \geq 1$$

Similarly, if $y \geq x > 0$, then

$$\frac{1}{2\sqrt{y}}(\sqrt{x} + \sqrt{y}) \leq 1$$

Multiplying both inequalities by $\sqrt{x} - \sqrt{y}$ we obtain

$$\sqrt{x} - \sqrt{y} \leq \frac{1}{2\sqrt{y}}(x - y),$$

for every $x, y > 0$. Moreover, it is easily seen that equality occurs if and only if $x = y$. By applying this last inequality we obtain

$$\sqrt{a + b - c} - \sqrt{a} \leq \frac{1}{2\sqrt{a}}(b - c)$$

(**) $$\sqrt{c+a-b} - \sqrt{b} \le \frac{1}{2\sqrt{b}}(c+a-2b)$$
$$\sqrt{b+c-a} - \sqrt{c} \le \frac{1}{2\sqrt{c}}(b-a)$$

and by adding up the left hand and the right hand sides of these inequalities we have

$$\sqrt{a+b-c} + \sqrt{b+c-a} + \sqrt{c+a-b} - \left(\sqrt{a} + \sqrt{b} + \sqrt{c}\right)$$

$$\frac{1}{2}\left(\frac{1}{\sqrt{a}}(b-c) + \frac{1}{\sqrt{b}}(c+a-2b) + \frac{1}{\sqrt{c}}(b-a)\right)$$

$$\frac{1}{2}\left((b-c)\left(\frac{1}{\sqrt{a}} - \frac{1}{\sqrt{b}}\right) + (a-b)\left(\frac{1}{\sqrt{b}} - \frac{1}{\sqrt{c}}\right)\right) \le 0,$$

i.e.,

$$\sqrt{a+b-c} + \sqrt{b+c-a} + \sqrt{c+a-b} \le \sqrt{a} + \sqrt{b} + \sqrt{c},$$

and equality occurs if and only if the three relations in (**) are equalities, i.e., if and only if $a = b = c$.

Points to be given for showing that

$\sqrt{x} - \sqrt{y} \le \frac{1}{2\sqrt{y}}(x-y)$ — 2 points

the relations in (**) hold true — 1 point each

$\sqrt{a+b-c} + \sqrt{b+c-a} + \sqrt{c+a-b} - \left(\sqrt{a} + \sqrt{b} + \sqrt{c}\right) \le$ — 1 point

equality occurs if and only if the three relations in (**) are equalities, i.e., if and only $a = b = c$ — 1 point

SOLUTIONS

Note: On the left side of the page the maximum number of points that may be awarded for every part of the solution is indicated in brackets.

Question 1. Suppose that $(a_1, a_2, ..., a_{1995})$ is a solution of the given system of inequalities. Then

$$\sum_{n=1}^{1995} 2\sqrt{a_n - (n-1)} \geq \sum_{n=1}^{1995} a_n - \sum_{n=1}^{1994}(n-1) + 1 = \sum_{n=1}^{1995} a_n - \sum_{n=1}^{1995}\{(n-1)+1\}$$

i.e.

$$0 \geq \sum_{n=1}^{1995}\{a_n - (n-1) + 1 - 2\sqrt{a_n - (n-1)}\}.$$

[2 points]

Next, note that

$$\left[\sqrt{a_n - (n-1)} - 1\right]^2 = a_n - (n-1) - 2\sqrt{a_n - (n-1)} + 1$$

for n = 1, 2, ..., 1995.

[1 point]

Hence,

$$0 \geq \sum_{n=1}^{1995}\left[a_n - (n-1) - 2\sqrt{a_n - (n-1)} + 1\right] = \sum_{n=1}^{1995}\left[\sqrt{a_n - (n-1)} - 1\right]^2 \geq 0.$$

Therefore, $\sqrt{a_n - (n-1)} = 1$, for n = 1, 2, ..., 1995. It follows that $a_n = n$ for n = 1, 2, ..., 1995.

[2 points]

Conversely, if $\sqrt{a_n - (n-1)} = 1$, for n = 1, 2, ..., 1995, then

$$2\sqrt{n - (n-1)} = 2 = n + 1 - (n-1), \text{ for n = 1, 2, ..., 1994}$$

and

$$2\sqrt{1995 - 1994} = 2 = 1 + 1,$$

which shows that $a_n = n$, for n = 1, 2, ..., 1995, is indeed a solution of the given system of inequalities.

[2 points]

Question 2. The answer is 14.

[1 point]

Denote the required number by M. We observe that the sequence 2·101, 3·97, 5·89, 7·83, 11·79, 13·73, 17·71, 19·67, 23·61 = 1403, 29·59 = 1711, 31·53 = 1643, 37·47 = 1739, 41·43 = 1763 satisfies conditions i) and ii) and contains no prime number. Hence, M > 13.

[3 points]

Now we show that a sequence with 14 elements that satisfies conditions i) and ii) will contain a prime number. We proceed by contradiction. Suppose the elements are $a_1, a_2, ..., a_{14}$. Since none of them is a prime number, each element will contain at least two prime factors. We take any two prime factors from each a_i, and list them in ascending order $p_1 < p_2 < ... < p_{26} < p_{27} < p_{28}$. As the 14th prime is 43, this means $43 \le p_{14}$, $47 \le p_{15}$ and so on. Now 43·47 = 2021 > 1995. This means that p_{14} must pair up with one of the $p_1, p_2 ... p_{13}$ to form a certain a_i. Likewise p_{15} must pair up with one of the $p_1, p_2 ... p_{13}$ to form another a_i, and so on (without repetition). Hence there exist p_i, p_j, $13 < i < j$, that must pair up together to form some a_i. But then $a_i \ge p_i p_j \ge 43 \cdot 47 > 1995$, a contradiction.

[3 points]

Question 3. Let T be the intersection of PQ and RS, T lies outside C, the circle PQRS.

i) Clearly any point on C belongs to the set A.
ii) Let $r = \sqrt{TP \cdot TQ} = \sqrt{TR \cdot TS}$, and consider the circle with center T and radius r. Let V a point on this circle. Since $TV^2 = TP \cdot TQ = TR \cdot TS$, TV is tangent to the circles PQV and RSV. Therefore, PQV is tangent to RSV. That means, V is in the set A.

[4 points]

Conversely, assume V is in A, i.e. PQV is tangent to RSV. If the circles PQV and RSV are the same, then PQV = RSV = PQRS. Otherwise, let the line TV intersect PQV in V_1, and RSV in V_2. Then

$$TP \cdot TQ = TV \cdot TV_1$$
$$TR \cdot TS = TV \cdot TV_2 .$$

Due to the fact that PQR and S are on a circle, we have TP·TQ = TR·TS, thus $TV \cdot TV_1 = TV \cdot TV_2$. Moreover, since T does not lie on C, T ≠ V, which implies $TV_1 = TV_2$, i.e., $V_1 = V_2 = V$.
All this means that TV is tangent to the circles PQV and RSV, therefore V lies on the circle with center T and radius $r = \sqrt{TP \cdot TQ} = \sqrt{TR \cdot TS}$.

[3 points]

Question 4. First, we will show that MS is perpendicular to A'B'. Since SAMB, SBN'A', SA'M'B' and SB'NA are rectangles, it follows that MNM'N' is a rectangle with its sides parallel to AA' and BB'.
Moreover, the perpendicular bisectors of AA' and BB' pass through O, and they coincide with those of MN' and NM'. Therefore, O is the center of the rectangle.
Let I and H be the intersections of MS with AB and A'B'. We then have
∠HSA' = ∠ASI,
∠ASI = ∠SAI,
∠SAI = ∠A'AB = ∠A'B'B.
In the triangle SA'B', ∠A'B'B or ∠A'B'S is the complementary angle of ∠SA'B'.
The angles HSA' and SA'B are complementary angles and the triangle SA'H is a right-angled triangle with right angle at H. Therefore, MS ⊥ A'B'.

[1 point]

Next, we will show that $AB^2 + A'B'^2 = 4R^2$ and that $MN'^2 + N'M'^2$ is constant.

Let D be the second intersection of MN' with the circle, then A'D = AB, since they subtend equal angles. This implies

$$AB^2 + A'B'^2 = A'D^2 + A'B'^2 .$$

But, we know DA' || MH, since ∠BDA' = ∠BAA' = ∠BMH, that means ∠DA'B' = 90^0 and it is inscribed in the circle, therefore D and B' are diametrically opposed, what finally implies

$$AB^2 + A'B'^2 = A'D^2 + A'B'^2 = DB'^2 = (2R)^2 = 4R^2,$$

i.e.

$$AB^2 + A'B'^2 = 4R^2.$$

[2 points]

To see that $MN'^2 + N'M'^2$ is constant consider the following equalities

$$\begin{aligned} MN'^2 = (MB + BN')^2 &= MB^2 + BN'^2 + 2MB \cdot BN' \\ &= SA^2 + SA'^2 + 2SA \cdot SA'^2 \\ M'N'^2 = (N'A' + A'M')^2 &= N'A'^2 + A'M'^2 + 2N'A' \cdot A'M' \\ &= SB^2 + SB'^2 + 2SB \cdot SB'. \end{aligned}$$

By Pythagoras, we have

$$AB^2 + A'B'^2 = (SA^2 + SB^2) + (SA'^2 + SB'^2)$$

This implies,

$$MN'^2 + M'N'^2 = SA^2 + SB^2 + SA'^2 + SB'^2 + 2SA\cdot SA' + 2SBSB'$$
$$= AB^2 + A'B'^2 + 4SA\cdot SA'$$
$$= 8R^2 - 4OS^2.$$

Additionally we know that

$$MN'^2 + M'N'^2 = MM'^2 = 4OM^2.$$

[2 points]

But, $4OM^2 = 8R^2 - 4OS^2$.
Therefore,

$$MN'^2 + M'N'^2 = 4OM^2$$

This last quantity is clearly a constant.

[1 point]

Finally, it is clear that the vertices of the rectangle MNM'N' lie on the circle with center O and radius $OM = \sqrt{2R^2 - OS^2}$. Therefore, the set of points consists of a circle.

[1 point]

Question 5. The minimum value of k is $k^* = 4$.

[1 point]

First, we define a function f from Z to {1, 2, 3, 4} recursively as follows: f(0) = 1. For any positive integer i, f(i) is defined to be the minimum positive integer not in $A_i := \{f(j) : i - j \in \{5, 7, 12\}$ and $-i < j < i\}$, and f(-i) the minimum positive integer not in $B_i := \{f(j) : j + i \in \{5, 7, 12\}$ and $-i < j < i\}$. Note that $|A_i| \le 3$ and $|B_i| \le 3$ for any i. So, f is a function from Z to {1, 2, 3, 4} such that $f(x) \neq f(y)$ whenever $|x - y| \in \{5, 7, 12\}$. This gives that $k^* \le 4$.

[3 points]

Next, we claim that $k^* \ge 4$. Suppose it is not the case. Then there exists a function f from Z to {1, 2, 3} with the property that $f(x) \neq f(y)$ whenever $|x - y| \in \{5, 7, 12\}$. For any integer x, consider the values f(x), f(x - 5), and f(x + 7).These three values are different. Now consider f(x + 2). Since $f(x + 2) \notin \{f(x - 5), f(x + 7)\}$

$$f(x) = f(x + 2) \text{ for any integer } x.$$

Hence,

$$f(x) = f(x + 2) = f(x + 4) = \ldots = f(x + 12),$$

which is impossible. Thus $k^* > 4$.

[3 points]

APMO 1994 – Problems and Solutions

Problem 1
Let $f\colon \mathbb{R} \to \mathbb{R}$ be a function such that

(i) For all $x, y \in \mathbb{R}$,
$$f(x) + f(y) + 1 \geq f(x+y) \geq f(x) + f(y),$$

(ii) For all $x \in [0,1)$, $f(0) \geq f(x)$,

(iii) $-f(-1) = f(1) = 1$.

Find all such functions f.

Answer: $f(x) = \lfloor x \rfloor$, the largest integer that does not exceed x, is the only function.

Solution
Plug $y \to 1$ in (i):

$$f(x) + f(1) + 1 \geq f(x+1) \geq f(x) + f(1) \iff f(x) + 1 \leq f(x+1) \leq f(x) + 2.$$

Now plug $y \to -1$ and $x \to x+1$ in (i):

$$f(x+1) + f(-1) + 1 \geq f(x) \geq f(x+1) + f(-1) \iff f(x) \leq f(x+1) \leq f(x) + 1.$$

Hence $f(x+1) = f(x) + 1$ and we only need to define $f(x)$ on $[0,1)$. Note that $f(1) = f(0) + 1 \implies f(0) = 0$.
Condition (ii) states that $f(x) \leq 0$ in $[0,1)$.
Now plug $y \to 1 - x$ in (i):

$$f(x) + f(1-x) + 1 \leq f(x + (1-x)) \leq f(x) + f(1-x) \implies f(x) + f(1-x) \geq 0.$$

If $x \in (0,1)$ then $1 - x \in (0,1)$ as well, so $f(x) \leq 0$ and $f(1-x) \leq 0$, which implies $f(x) + f(1-x) \leq 0$. Thus, $f(x) = f(1-x) = 0$ for $x \in (0,1)$. This combined with $f(0) = 0$ and $f(x+1) = f(x) + 1$ proves that $f(x) = \lfloor x \rfloor$, which satisfies the problem conditions, as since

$$x + y = \lfloor x \rfloor + \lfloor y \rfloor + \{x\} + \{y\} \text{ and } 0 \leq \{x\} + \{y\} < 2 \implies \lfloor x \rfloor + \lfloor y \rfloor \leq x + y < \lfloor x \rfloor + \lfloor y \rfloor + 2$$

implies

$$\lfloor x \rfloor + \lfloor y \rfloor + 1 \geq \lfloor x + y \rfloor \geq \lfloor x \rfloor + \lfloor y \rfloor.$$

Problem 2
Given a nondegenerate triangle ABC, with circumcentre O, orthocentre H, and circumradius R, prove that $|OH| < 3R$.

Solution 1
Embed ABC in the complex plane, with A, B and C in the circle $|z| = R$, so O is the origin. Represent each point by its lowercase letter. It is well known that $h = a + b + c$, so

$$OH = |a + b + c| \leq |a| + |b| + |c| = 3R.$$

The equality cannot occur because a, b, and c are not collinear, so $OH < 3R$.

Solution 2
Suppose with loss of generality that $\angle A < 90^\circ$. Let BD be an altitude. Then

$$AH = \frac{AD}{\cos(90^\circ - C)} = \frac{AB\cos A}{\sin C} = 2R\cos A.$$

By the triangle inequality,

$$OH < AO + AH = R + 2R\cos A < 3R.$$

Comment: With a bit more work, if a, b, c are the sidelengths of ABC, one can show that

$$OH^2 = 9R^2 - a^2 - b^2 - c^2.$$

In fact, using vectors in a coordinate system with O as origin, by the Euler line

$$\overrightarrow{OH} = 3\overrightarrow{OG} = 3 \cdot \frac{\overrightarrow{OA} + \overrightarrow{OB} + \overrightarrow{OC}}{3} = \overrightarrow{OA} + \overrightarrow{OB} + \overrightarrow{OC}.$$

so

$$OH^2 = \overrightarrow{OH} \cdot \overrightarrow{OH} = (\overrightarrow{OA} + \overrightarrow{OB} + \overrightarrow{OC}) \cdot (\overrightarrow{OA} + \overrightarrow{OB} + \overrightarrow{OC})$$

Expanding and using the fact that $\overrightarrow{OX} \cdot \overrightarrow{OX} = OX^2 = R^2$ for $X \in \{A, B, C\}$, as well as

$$\overrightarrow{OA} \cdot \overrightarrow{OB} = OA \cdot OB \cdot \cos\angle AOB = R^2\cos 2C = R^2(1 - 2\sin^2 C) = R^2\left(1 - 2\left(\frac{c}{2R}\right)^2\right) = R^2 - \frac{c^2}{2},$$

we find that

$$\begin{aligned} OH^2 &= \overrightarrow{OA} \cdot \overrightarrow{OA} + \overrightarrow{OB} \cdot \overrightarrow{OB} + \overrightarrow{OC} \cdot \overrightarrow{OC} + 2\overrightarrow{OA} \cdot \overrightarrow{OB} + 2\overrightarrow{OA} \cdot \overrightarrow{OC} + 2\overrightarrow{OB} \cdot \overrightarrow{OC} \\ &= 3R^2 + (2R^2 - c^2) + (2R^2 - b^2) + (2R^2 - a^2) \\ &= 9R^2 - a^2 - b^2 - c^2, \end{aligned}$$

as required.
This proves that $OH^2 < 9R^2 \implies OH < 3R$, and since a, b, c can be arbitrarily small (fix the circumcircle and choose A, B, C arbitrarily close in this circle), the bound is sharp.

Problem 3
Let n be an integer of the form $a^2 + b^2$, where a and b are relatively prime integers and such that if p is a prime, $p \leq \sqrt{n}$, then p divides ab. Determine all such n.

Answer: $n = 2, 5, 13$.

Solution
A prime p divides ab if and only if divides either a or b. If $n = a^2 + b^2$ is a composite then it has a prime divisor $p \leq \sqrt{n}$, and if p divides a it divides b and vice-versa, which is not possible because a and b are coprime. Therefore n is a prime.
Suppose without loss of generality that $a \geq b$ and consider $a-b$. Note that $a^2+b^2 = (a-b)^2+2ab$.

- If $a = b$ then $a = b = 1$ because a and b are coprime. $n = 2$ is a solution.
- If $a-b = 1$ then a and b are coprime and $a^2+b^2 = (a-b)^2+2ab = 2ab+1 = 2b(b+1)+1 = 2b^2 + 2b + 1$. So any prime factor of any number smaller than $\sqrt{2b^2 + 2b + 1}$ is a divisor of $ab = b(b + 1)$.

  One can check that $b = 1$ and $b = 2$ yields the solutions $n = 1^2 + 2^2 = 5$ (the only prime p is 2) and $n = 2^2 + 3^2 = 13$ (the only primes p are 2 and 3). Suppose that $b > 2$.

  Consider, for instance, the prime factors of $b - 1 \leq \sqrt{2b^2 + 2b + 1}$, which is coprime with b. Any prime must then divide $a = b + 1$. Then it divides $(b + 1) - (b - 1) = 2$, that is, $b - 1$ can only have 2 as a prime factor, that is, $b - 1$ is a power of 2, and since $b - 1 \geq 2$, b is odd.

  Since $2b^2 + 2b + 1 - (b + 2)^2 = b^2 - 2b - 3 = (b - 3)(b + 1) \geq 0$, we can also consider any prime divisor of $b + 2$. Since b is odd, b and $b + 2$ are also coprime, so any prime divisor of $b + 2$ must divide $a = b + 1$. But $b + 1$ and $b + 2$ are also coprime, so there can be no such primes. This is a contradiction, and $b \geq 3$ does not yield any solutions.
- If $a - b > 1$, consider a prime divisor p of $a - b = \sqrt{a^2 - 2ab + b^2} < \sqrt{a^2 + b^2}$. Since p divides one of a and b, p divides both numbers (just add or subtract $a - b$ accordingly.) This is a contradiction.

Hence the only solutions are $n = 2, 5, 13$.

Problem 4

Is there an infinite set of points in the plane such that no three points are collinear, and the distance between any two points is rational?

Answer: Yes.

Solution 1

The answer is *yes* and we present the following construction: the idea is considering points in the unit circle of the form $P_n = (\cos(2n\theta), \sin(2n\theta))$ for an appropriate θ. Then the distance P_mP_n is the length of the chord with central angle $(2m-2n)\theta \bmod \pi$, that is, $2|\sin((m-n)\theta)|$. Our task is then finding θ such that (i) $\sin(k\theta)$ is rational for all $k \in \mathbb{Z}$; (ii) points P_n are all distinct. We claim that $\theta \in (0, \pi/2)$ such that $\cos\theta = \frac{3}{5}$ and therefore $\sin\theta = \frac{4}{5}$ does the job.

Proof of (i): We know that $\sin((n+1)\theta) + \sin((n-1)\theta) = 2\sin(n\theta)\cos\theta$, so if $\sin((n-1)\theta$ and $\sin(n\theta)$ are both rational then $\sin((n+1)\theta)$ also is. Since $\sin(0\theta) = 0$ and $\sin\theta$ are rational, an induction shows that $\sin(n\theta)$ is rational for $n \in \mathbb{Z}_{>0}$; the result is also true if n is negative because sin is an odd function.

Proof of (ii): $P_m = P_n \iff 2n\theta = 2m\theta + 2k\pi$ for some $k \in \mathbb{Z}$, which implies $\sin((n-m)\theta) = \sin(k\pi) = 0$. We show that $\sin(k\theta) \neq 0$ for all $k \neq 0$.

We prove a stronger result: let $\sin(k\theta) = \frac{a_k}{5^k}$. Then

$$\begin{aligned}\sin((k+1)\theta) + \sin((k-1)\theta) = 2\sin(k\theta)\cos\theta &\iff \frac{a_{k+1}}{5^{k+1}} + \frac{a_{k-1}}{5^{k-1}} = 2 \cdot \frac{a_k}{5^k} \cdot \frac{3}{5} \\ &\iff a_{k+1} = 6a_k - 25a_{k-1}.\end{aligned}$$

Since $a_0 = 0$ and $a_1 = 4$, a_k is an integer for $k \geq 0$, and $a_{k+1} \equiv a_k \pmod 5$ for $k \geq 1$ (note that $a_{-1} = -\frac{4}{25}$ is not an integer!). Thus $a_k \equiv 4 \pmod 5$ for all $k \geq 1$, and $\sin(k\theta) = \frac{a_k}{5^k}$ is an irreducible fraction with 5^k as denominator and $a_k \equiv 4 \pmod 5$. This proves (ii) and we are done.

Solution 2

We present a different construction. Consider the (collinear) points

$$P_k = \left(1, \frac{x_k}{y_k}\right),$$

such that the distance OP_k from the origin O,

$$OP_k = \frac{\sqrt{x_k^2 + y_k^2}}{y_k},$$

is rational, and x_k and y_k are integers. Clearly, $P_iP_j = \left|\frac{x_i}{y_i} - \frac{x_j}{y_j}\right|$ is rational.

Perform an inversion with center O and unit radius. It maps the line $x = 1$, which contains all points P_k, to a circle (minus the origin). Let Q_k be the image of P_k under this inversion. Then

$$Q_iQ_j = \frac{1^2 P_iP_j}{OP_i \cdot OP_j}$$

is rational and we are done if we choose x_k and y_k accordingly. But this is not hard, as we can choose the legs of a Pythagorean triple, say

$$x_k = k^2 - 1, \quad y_k = 2k.$$

This implies $OP_k = \frac{k^2+1}{2k}$, and then

$$Q_iQ_j = \frac{\left|\frac{i^2-1}{i} - \frac{j^2-1}{j}\right|}{\frac{i^2+1}{2i} \cdot \frac{j^2+1}{2j}} = \frac{|4(i-j)(ij+1)|}{(i^2+1)(j^2+1)}.$$

Problem 5

You are given three lists A, B, and C. List A contains the numbers of the form 10^k in base 10, with k any integer greater than or equal to 1. Lists B and C contain the same numbers translated into base 2 and 5 respectively:

A	B	C
10	1010	20
100	1100100	400
1000	1111101000	13000
$\vdots$	$\vdots$	$\vdots$

Prove that for every integer $n > 1$, there is exactly one number in exactly one of the lists B or C that has exactly n digits.

Solution

Let b_k and c_k be the number of digits in the kth term in lists B and C, respectively. Then

$$2^{b_k-1} \leq 10^k < 2^{b_k} \iff \log_2 10^k < b_k \leq \log_2 10^k + 1 \iff b_k = \lfloor k \cdot \log_2 10 \rfloor + 1$$

and, similarly

$$c_k = \lfloor k \cdot \log_5 10 \rfloor + 1.$$

Beatty's theorem states that if α and β are irrational positive numbers such that

$$\frac{1}{\alpha} + \frac{1}{\beta} = 1,$$

then the sequences $\lfloor k\alpha \rfloor$ and $\lfloor k\beta \rfloor$, $k = 1, 2, \ldots$, partition the positive integers.
Then, since

$$\frac{1}{\log_2 10} + \frac{1}{\log_5 10} = \log_{10} 2 + \log_{10} 5 = \log_{10}(2 \cdot 5) = 1,$$

the sequences $b_k - 1$ and $c_k - 1$ partition the positive integers, and therefore each integer greater than 1 appears in b_k or c_k exactly once. We are done.

Comment: For the sake of completeness, a proof of Beatty's theorem follows.
Let $x_n = \alpha n$ and $y_n = \beta n$, $n \geq 1$ integer. Note that, since $\alpha m = \beta n$ implies that $\frac{\alpha}{\beta}$ is rational but

$$\frac{\alpha}{\beta} = \alpha \cdot \frac{1}{\beta} = \alpha \left(1 - \frac{1}{\alpha}\right) = \alpha - 1$$

is irrational, the sequences have no common terms, and all terms in both sequences are irrational.
The theorem is equivalent to proving that exactly one term of either x_n of y_n lies in the interval $(N, N+1)$ for each N positive integer. For that purpose we count the number of terms of the union of the two sequences in the interval $(0, N)$: since $n\alpha < N \iff n < \frac{N}{\alpha}$, there are $\left\lfloor \frac{N}{\alpha} \right\rfloor$ terms of x_n in the interval and, similarly, $\left\lfloor \frac{N}{\beta} \right\rfloor$ terms of y_n in the same interval. Since the sequences are disjoint, the total of numbers is

$$T(N) = \left\lfloor \frac{N}{\alpha} \right\rfloor + \left\lfloor \frac{N}{\beta} \right\rfloor.$$

However, $x - 1 < \lfloor x \rfloor < x$ for nonintegers x, so

$$\begin{aligned} \frac{N}{\alpha} - 1 + \frac{N}{\beta} - 1 < T(N) < \frac{N}{\alpha} + \frac{N}{\beta} &\iff N\left(\frac{1}{\alpha} + \frac{1}{\beta}\right) - 2 < T(N) < N\left(\frac{1}{\alpha} + \frac{1}{\beta}\right) \\ &\iff N - 2 < T(N) < N, \end{aligned}$$

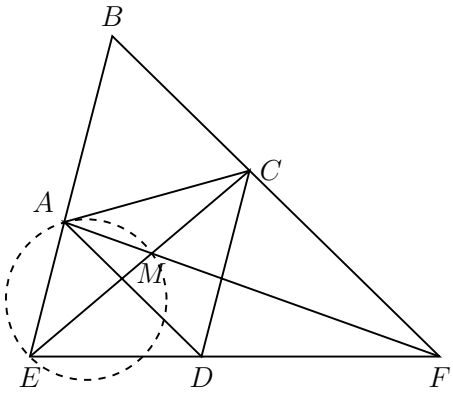
that is, $T(N) = N - 1$.
Therefore the number of terms in $(N, N+1)$ is $T(N+1) - T(N) = N - (N-1) = 1$, and the result follows.

APMO 1993 – Problems and Solutions

Problem 1

Let $ABCD$ be a quadrilateral such that all sides have equal length and angle $\angle ABC$ is 60 degrees. Let ℓ be a line passing through D and not intersecting the quadrilateral (except at D). Let E and F be the points of intersection of ℓ with AB and BC respectively. Let M be the point of intersection of CE and AF.
Prove that $CA^2 = CM \times CE$.

Solution



Triangles AED and CDF are similar, because $AD \parallel CF$ and $AE \parallel CD$. Thus, since ABC and ACD are equilateral triangles,

$$\frac{AE}{CD} = \frac{AD}{CF} \iff \frac{AE}{AC} = \frac{AC}{CF}.$$

The last equality combined with

$$\angle EAC = 180^\circ - \angle BAC = 120^\circ = \angle ACF$$

shows that triangles EAC and ACF are also similar. Therefore $\angle CAM = \angle CAF = \angle AEC$, which implies that line AC is tangent to the circumcircle of AME. By the power of a point, $CA^2 = CM \cdot CE$, and we are done.

Problem 2
Find the total number of different integer values the function

$$f(x) = [x] + [2x] + \left[\frac{5x}{3}\right] + [3x] + [4x]$$

takes for real numbers x with $0 \leq x \leq 100$.
Note: $[t]$ is the largest integer that does not exceed t.

Answer: 734.

Solution
Note that, since $[x + n] = [x] + n$ for any integer n,

$$f(x+3) = [x+3] + [2(x+3)] + \left[\frac{5(x+3)}{3}\right] + [3(x+3)] + [4(x+3)] = f(x) + 35,$$

one only needs to investigate the interval $[0, 3)$.
The numbers in this interval at which at least one of the real numbers $x, 2x, \frac{5x}{3}, 3x, 4x$ is an integer are

- $0, 1, 2$ for x;
- $\frac{n}{2}$, $0 \leq n \leq 5$ for $2x$;
- $\frac{3n}{5}$, $0 \leq n \leq 4$ for $\frac{5x}{3}$;
- $\frac{n}{3}$, $0 \leq n \leq 8$ for $3x$;
- $\frac{n}{4}$, $0 \leq n \leq 11$ for $4x$.

Of these numbers there are

- 3 integers $(0, 1, 2)$;
- 3 irreducible fractions with 2 as denominator (the numerators are $1, 3, 5$);
- 6 irreducible fractions with 3 as denominator (the numerators are $1, 2, 4, 5, 7, 8$);
- 6 irreducible fractions with 4 as denominator (the numerators are $1, 3, 5, 7, 9, 11, 13, 15$);
- 4 irreducible fractions with 5 as denominator (the numerators are $3, 6, 9, 12$).

Therefore $f(x)$ increases 22 times per interval. Since $100 = 33 \cdot 3 + 1$, there are $33 \cdot 22$ changes of value in $[0, 99)$. Finally, there are 8 more changes in $[99, 100]$: 99, 100, $99\frac{1}{2}$, $99\frac{1}{3}$, $99\frac{2}{3}$, $99\frac{1}{4}$, $99\frac{3}{4}$, $99\frac{3}{5}$.
The total is then $33 \cdot 22 + 8 = 734$.

Comment: A more careful inspection shows that the range of f are the numbers congruent modulo 35 to one of

$$0, 1, 2, 4, 5, 6, 7, 11, 12, 13, 14, 16, 17, 18, 19, 23, 24, 25, 26, 28, 29, 30$$

in the interval $[0, f(100)] = [0, 1166]$. Since $1166 \equiv 11 \pmod{35}$, this comprises 33 cycles plus the 8 numbers in the previous list.

Problem 3

Let

$$f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_0 \quad \text{and} \quad g(x) = c_{n+1}x^{n+1} + c_n x^n + \cdots + c_0$$

be non-zero polynomials with real coefficients such that $g(x) = (x+r)f(x)$ for some real number r. If $a = \max(|a_n|, \ldots, |a_0|)$ and $c = \max(|c_{n+1}|, \ldots, |c_0|)$, prove that $\frac{a}{c} \leq n+1$.

Solution

Expanding $(x+r)f(x)$, we find that $c_{n+1} = a_n$, $c_k = a_{k-1} + ra_k$ for $k = 1, 2, \ldots, n$, and $c_0 = ra_0$. Consider three cases:

- $r = 0$. Then $c_0 = 0$ and $c_k = a_{k-1}$ for $k = 1, 2, \ldots, n$, and $a = c \implies \frac{a}{c} = 1 \leq n+1$.
- $|r| \geq 1$. Then

$$|a_0| = \left|\frac{c_0}{r}\right| \leq c,$$
$$|a_1| = \left|\frac{c_1 - a_0}{r}\right| \leq |c_1| + |a_0| \leq 2c,$$

  and inductively if $|a_k| \leq (k+1)c$

$$|a_{k+1}| = \left|\frac{c_{k+1} - a_k}{r}\right| \leq |c_{k+1}| + |a_k| \leq c + (k+1)c = (k+2)c.$$

  Therefore, $|a_k| \leq (k+1)c \leq (n+1)c$ for all k, and $a \leq (n+1)c \iff \frac{a}{c} \leq n+1$.
- $0 < |r| < 1$. Now work *backwards*: $|a_n| = |c_{n+1}| \leq c$,

$$|a_{n-1}| = |c_n - ra_n| \leq |c_n| + |ra_n| < c + c = 2c,$$

  and inductively if $|a_{n-k}| \leq (k+1)c$

$$|a_{n-k-1}| = |c_{n-k} - ra_{n-k}| \leq |c_{n-k}| + |ra_{n-k}| < c + (k+1)c = (k+2)c.$$

  Therefore, $|a_{n-k}| \leq (k+1)c \leq (n+1)c$ for all k, and $a \leq (n+1)c$ again.

Problem 4
Determine all positive integers n for which the equation

$$x^n + (2+x)^n + (2-x)^n = 0$$

has an integer as a solution.

Answer: $n = 1$.

Solution
If n is even, $x^n + (2+x)^n + (2-x)^n > 0$, so n is odd.
For $n = 1$, the equation reduces to $x + (2+x) + (2-x) = 0$, which has the unique solution $x = -4$.
For $n > 1$, notice that x is even, because x, $2-x$, and $2+x$ have all the same parity. Let $x = 2y$, so the equation reduces to

$$y^n + (1+y)^n + (1-y)^n = 0.$$

Looking at this equation modulo 2 yields that $y + (1+y) + (1-y) = y + 2$ is even, so y is even. Using the factorization

$$a^n + b^n = (a+b)(a^{n-1} - a^{n-2}b + \cdots + b^{n-1}) \quad \text{for } n \text{ odd},$$

which has a sum of n terms as the second factor, the equation is now equivalent to

$$y^n + (1+y+1-y)((1+y)^{n-1} - (1+y)^{n-2}(1-y) + \cdots + (1-y)^{n-1}) = 0,$$

or

$$y^n = -2((1+y)^{n-1} - (1+y)^{n-2}(1-y) + \cdots + (1-y)^{n-1}).$$

Each of the n terms in the second factor is odd, and n is odd, so the second factor is odd. Therefore, y^n has only one factor 2, which is a contradiction to the fact that, y being even, y^n has at least $n > 1$ factors 2. Hence there are no solutions if $n > 1$.

Problem 5

Let $P_1, P_2, \ldots, P_{1993} = P_0$ be distinct points in the xy-plane with the following properties:

(i) both coordinates of P_i are integers, for $i = 1, 2, \ldots, 1993$;

(ii) there is no point other than P_i and P_{i+1} on the line segment joining P_i with P_{i+1} whose coordinates are both integers, for $i = 0, 1, \ldots, 1992$.

Prove that for some i, $0 \leq i \leq 1992$, there exists a point Q with coordinates (q_x, q_y) on the line segment joining P_i with P_{i+1} such that both $2q_x$ and $2q_y$ are odd integers.

Solution

Call a point $(x, y) \in \mathbb{Z}^2$ *even* or *odd* according to the parity of $x + y$. Since there are an odd number of points, there are two points $P_i = (a, b)$ and $P_{i+1} = (c, d)$, $0 \leq i \leq 1992$ with the same parity. This implies that $a + b + c + d$ is even. We claim that the midpoint of P_iP_{i+1} is the desired point Q.

In fact, since $a+b+c+d = (a+c)+(b+d)$ is even, a and c have the same parity if and only if b and d also have the same parity. If both happen then the midpoint of P_iP_{i+1}, $Q = \left(\frac{a+c}{2}, \frac{b+d}{2}\right)$, has integer coordinates, which violates condition (ii). Then a and c, as well as b and d, have different parities, and $2q_x = a + c$ and $2q_y = b + d$ are both odd integers.

APMO 1992 – Problems and Solutions

Problem 1

A triangle with sides a, b, and c is given. Denote by s the semiperimeter, that is $s = (a+b+c)/2$. Construct a triangle with sides $s-a$, $s-b$, and $s-c$. This process is repeated until a triangle can no longer be constructed with the sidelengths given.
For which original triangles can this process be repeated indefinitely?

Answer: Only equilateral triangles.

Solution

The perimeter of each new triangle constructed by the process is $(s-a)+(s-b)+(s-c) = 3s-(a+b+c) = 3s-2s = s$, that is, it is halved. Consider a new equivalent process in which a similar triangle with sidelengths $2(s-a), 2(s-b), 2(s-c)$ is constructed, so the perimeter is kept invariant.
Suppose without loss of generality that $a \leq b \leq c$. Then $2(s-c) \leq 2(s-b) \leq 2(s-a)$, and the difference between the largest side and the smallest side changes from $c-a$ to $2(s-a)-2(s-c) = 2(c-a)$, that is, it doubles. Therefore, if $c-a > 0$ then eventually this difference becomes larger than $a+b+c$, and it's immediate that a triangle cannot be constructed with the sidelengths. Hence the only possibility is $c-a = 0 \implies a = b = c$, and it is clear that equilateral triangles can yield an infinite process, because all generated triangles are equilateral.

Problem 2

In a circle C with centre O and radius r, let C_1, C_2 be two circles with centres O_1, O_2 and radii r_1, r_2 respectively, so that each circle C_i is internally tangent to C at A_i and so that C_1, C_2 are externally tangent to each other at A.

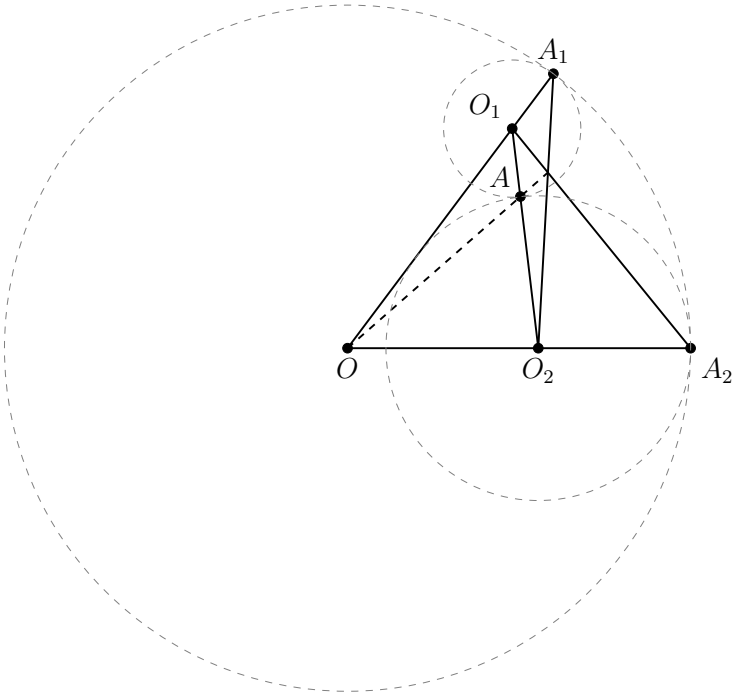
Prove that the three lines OA, O_1A_2, and O_2A_1 are concurrent.

Solution

Because of the tangencies, the following triples of points (two centers and a tangency point) are collinear:

$$O_1; O_2; A, \quad O; O_1; A_1, \quad O; O_2; A_2.$$

Because of that we can ignore the circles and only draw their centers and tangency points.



Now the problem is immediate from Ceva's theorem in triangle OO_1O_2, because

$$\frac{OA_1}{A_1O_1} \cdot \frac{O_1A}{AO_2} \cdot \frac{O_2A_2}{A_2O} = \frac{r}{r_1} \cdot \frac{r_1}{r_2} \cdot \frac{r_2}{r} = 1.$$

Problem 3

Let n be an integer such that $n > 3$. Suppose that we choose three numbers from the set $\{1, 2, \ldots, n\}$. Using each of these three numbers only once and using addition, multiplication, and parenthesis, let us form all possible combinations.

(a) Show that if we choose all three numbers greater than $n/2$, then the values of these combinations are all distinct.

(b) Let p be a prime number such that $p \leq \sqrt{n}$. Show that the number of ways of choosing three numbers so that the smallest one is p and the values of the combinations are not all distinct is precisely the number of positive divisors of $p - 1$.

Solution

In both items, the smallest chosen number is at least 2: in part (a), $n/2 > 1$ and in part (b), p is a prime. So let $1 < x < y < z$ be the chosen numbers. Then all possible combinations are

$$x + y + z, \quad x + yz, \quad xy + z, \quad y + zx, \quad (x + y)z, \quad (z + x)y, \quad (x + y)z, \quad xyz.$$

Since, for $1 < m < n$ and $t > 1$, $(m-1)(n-1) \geq 1 \cdot 2 \implies mn > m + n$, $tn + m - (tm + n) = (t-1)(n-m) > 0 \implies tn + m > tm + n$, and $(t + m)n - (t + n)m = t(n - m) > 0$,

$$x + y + z < z + xy < y + zx < x + yz$$

and

$$(y + z)x < (x + z)y < (x + y)z < xyz.$$

Also, $(y + z)x - (y + zx) = (x - 1)y > 0 \implies (y + z)x > y + zx$ and $(x + z)y - (x + yz) = (y - 1)x > 0 \implies (x + z)y > x + yz$. Therefore the only numbers that can be equal are $x + yz$ and $(y + z)x$. In this case,

$$x + yz = (y + z)x \iff (y - x)(z - x) = x(x - 1).$$

Now we can solve the items.

(a) if $n/2 < x < y < z$ then $z - x < n/2$, and since $y - x < z - x$, $y - x < n/2 - 1$; then

$$(y - x)(z - x) < \frac{n}{2}\left(\frac{n}{2} - 1\right) < x(x - 1),$$

and therefore $x + yz < (y + z)x$.

(b) if $x = p$, then $(y - p)(z - p) = p(p - 1)$. Since $y - p < z - p$, $(y - p)^2 < (y - p)(z - p) = p(p - 1) \implies y - p < p$, that is, p does not divide $y - p$. Then $y - p$ is a divisor d of $p - 1$ and $z - p = \frac{p(p-1)}{d}$. Therefore,

$$x = p, \quad , y = p + d, \quad z = p + \frac{p(p - 1)}{d},$$

which is a solution for every divisor d of $p - 1$ because

$$x = p < y = p + d < 2p \leq p + p \cdot \frac{p - 1}{d} = z.$$

Comment: If $x = 1$ was allowed, then any choice $1, y, z$ would have repeated numbers in the combination, as $1 \cdot y + z = y + 1 \cdot z$.

Problem 4
Determine all pairs (h, s) of positive integers with the following property:
If one draws h horizontal lines and another s lines which satisfy

(i) they are not horizontal,

(ii) no two of them are parallel,

(iii) no three of the $h + s$ lines are concurrent,

then the number of regions formed by these $h + s$ lines is 1992.

Answer: $(995, 1)$, $(176, 10)$, and $(80, 21)$.

Solution
Let $a_{h,s}$ the number of regions formed by h horizontal lines and s another lines as described in the problem. Let $\mathcal{F}_{h,s}$ be the union of the $h + s$ lines and pick any line ℓ. If it intersects the other lines in n (distinct!) points then ℓ is partitioned into $n - 1$ line segments and 2 rays, which delimit regions. Therefore if we remove ℓ the number of regions decreases by exactly $n - 1 + 2 = n + 1$.
Then $a_{0,0} = 1$ (no lines means there is only one region), and since every one of s lines intersects the other $s - 1$ lines, $a_{0,s} = a_{0,s-1} + s$ for $s \geq 0$. Summing yields

$$a_{0,s} = s + (s - 1) + \cdots + 1 + a_{0,0} = \frac{s^2 + s + 2}{2}.$$

Each horizontal line only intersects the s non-horizontal lines, so $a_{h,s} = a_{h-1,s} + s + 1$, which implies

$$a_{h,s} = a_{0,s} + h(s + 1) = \frac{s^2 + s + 2}{2} + h(s + 1).$$

Our final task is solving

$$a_{h,s} = 1992 \iff \frac{s^2 + s + 2}{2} + h(s + 1) = 1992 \iff (s + 1)(s + 2h) = 2 \cdot 1991 = 2 \cdot 11 \cdot 181.$$

The divisors of $2 \cdot 1991$ are $1, 2, 11, 22, 181, 362, 1991, 3982$. Since $s, h > 0$, $2 \leq s + 1 < s + 2h$, so the possibilities for $s + 1$ can only be 2, 11 and 22, yielding the following possibilities for (h, s):

$$(995, 1), \quad (176, 10), \quad \text{and} \quad (80, 21).$$

Problem 5

Find a sequence of maximal length consisting of non-zero integers in which the sum of any seven consecutive terms is positive and that of any eleven consecutive terms is negative.

Answer: The maximum length is 16. There are several possible sequences with this length; one such sequence is $(-7,-7,18,-7,-7,-7,18,-7,-7,18,-7,-7,-7,18,-7,-7)$.

Solution

Suppose it is possible to have more than 16 terms in the sequence. Let $a_1, a_2, \ldots, a_{17}$ be the first 17 terms of the sequence. Consider the following array of terms in the sequence:

$$\begin{array}{ccccccccccc}
a_1 & a_2 & a_3 & a_4 & a_5 & a_6 & a_7 & a_8 & a_9 & a_{10} & a_{11} \\
a_2 & a_3 & a_4 & a_5 & a_6 & a_7 & a_8 & a_9 & a_{10} & a_{11} & a_{12} \\
a_3 & a_4 & a_5 & a_6 & a_7 & a_8 & a_9 & a_{10} & a_{11} & a_{12} & a_{13} \\
a_4 & a_5 & a_6 & a_7 & a_8 & a_9 & a_{10} & a_{11} & a_{12} & a_{13} & a_{14} \\
a_5 & a_6 & a_7 & a_8 & a_9 & a_{10} & a_{11} & a_{12} & a_{13} & a_{14} & a_{15} \\
a_6 & a_7 & a_8 & a_9 & a_{10} & a_{11} & a_{12} & a_{13} & a_{14} & a_{15} & a_{16} \\
a_7 & a_8 & a_9 & a_{10} & a_{11} & a_{12} & a_{13} & a_{14} & a_{15} & a_{16} & a_{17}
\end{array}$$

Let S the sum of the numbers in the array. If we sum by rows we obtain negative sums in each row, so $S < 0$; however, it we sum by columns we obtain positive sums in each column, so $S > 0$, a contradiction. This implies that the sequence cannot have more than 16 terms.
One idea to find a suitable sequence with 16 terms is considering cycles of 7 numbers. For instance, one can try

$$-a,-a,b,-a,-a,-a,b,-a,-a,b,-a,-a,-a,b,-a,-a.$$

The sum of every seven consecutive numbers is $-5a+2b$ and the sum of every eleven consecutive numbers is $-8a+3b$, so $-5a+2b>0$ and $-8a+3b<0$, that is,

$$\frac{5a}{2} < b < \frac{8a}{3} \iff 15a < 6b < 16a.$$

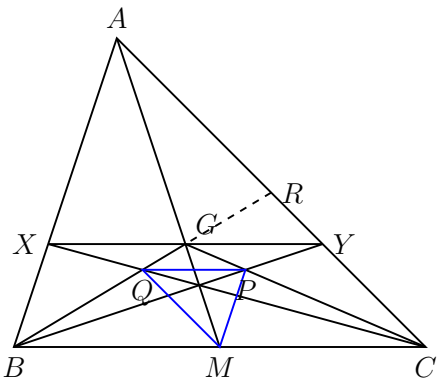
Then we can choose, say, $a=7$ and $105<6b<112 \iff b=18$. A valid sequence is then

$$-7,-7,18,-7,-7,-7,18,-7,-7,18,-7,-7,-7,18,-7,-7.$$

APMO 1991 – Problems and Solutions

Problem 1
Let G be the centroid of triangle ABC and M be the midpoint of BC. Let X be on AB and Y on AC such that the points X, Y, and G are collinear and XY and BC are parallel. Suppose that XC and GB intersect at Q and YB and GC intersect at P. Show that triangle MPQ is similar to triangle ABC.

Solution 1
Let R be the midpoint of AC; so BR is a median and contains the centroid G.

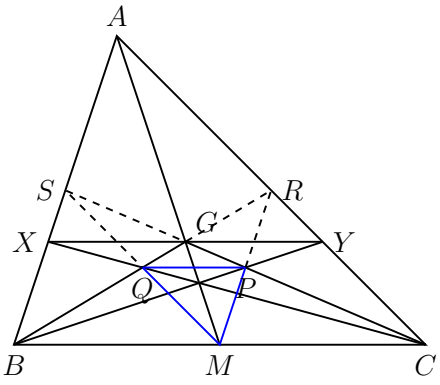


It is well known that $\frac{AG}{AM} = \frac{2}{3}$; thus the ratio of the similarity between AXY and ABC is $\frac{2}{3}$. Hence $GX = \frac{1}{2}XY = \frac{1}{3}BC$.
Now look at the similarity between triangles QBC and QGX:

$$\frac{QG}{QB} = \frac{GX}{BC} = \frac{1}{3} \implies QB = 3QG \implies QB = \frac{3}{4}BG = \frac{3}{4}\cdot\frac{2}{3}BR = \frac{1}{2}BR.$$

Finally, since $\frac{BM}{BC} = \frac{BQ}{BR}$, MQ is a midline in BCR. Therefore $MQ = \frac{1}{2}CR = \frac{1}{4}AC$ and $MQ \parallel AC$. Similarly, $MP = \frac{1}{4}AB$ and $MP \parallel AB$. This is sufficient to establish that MPQ and ABC are similar (with similarity ratio $\frac{1}{4}$).

Solution 2
Let S and R be the midpoints of AB and AC, respectively. Since G is the centroid, it lies in the medians BR and CS.



Due to the similarity between triangles QBC and QGX (which is true because $GX \parallel BC$), there is an inverse homothety with center Q and ratio $-\frac{XG}{BC} = \frac{XY}{2BC}$ that takes B to G and C to X. This homothety takes the midpoint M of BC to the midpoint K of GX.

Now consider the homothety that takes B to X and C to G. This new homothety, with ratio $\frac{XY}{2BC}$, also takes M to K. Hence lines BX (which contains side AB), CG (which contains the median CS), and MK have a common point, which is S. Thus Q lies on midline MS.
The same reasoning proves that P lies on midline MR. Since all homothety ratios are the same, $\frac{MQ}{MS} = \frac{MP}{MR}$, which shows that MPQ is similar to MRS, which in turn is similar to ABC, and we are done.

Problem 2

Suppose there are 997 points given in a plane. If every two points are joined by a line segment with its midpoint coloured in red, show that there are at least 1991 red points in the plane. Can you find a special case with exactly 1991 red points?

Solution

Embed the points in the cartesian plane such that no two points have the same y-coordinate. Let $P_1, P_2, \ldots, P_{997}$ be the points and $y_1 < y_2 < \ldots < y_{997}$ be their respective y-coordinates. Then the y-coordinate of the midpoint of P_iP_{i+1}, $i = 1, 2, \ldots, 996$ is $\frac{y_i+y_{i+1}}{2}$ and the y-coordinate of the midpoint of P_iP_{i+2}, $i = 1, 2, \ldots, 995$ is $\frac{y_i+y_{i+2}}{2}$. Since

$$\frac{y_1+y_2}{2} < \frac{y_1+y_3}{2} < \frac{y_2+y_3}{2} < \frac{y_2+y_4}{2} < \cdots < \frac{y_{995}+y_{997}}{2} < \frac{y_{996}+y_{997}}{2},$$

there are at least $996 + 995 = 1991$ distinct midpoints, and therefore at least 1991 red points. The equality case happens if we take $P_i = (0, 2i)$, $i = 1, 2, \ldots, 997$. The midpoints are $(0, i+j)$, $1 \le i < j \le 997$, which are the points $(0, k)$ with $1 + 2 = 3 \le k \le 996 + 997 = 1993$, a total of $1993 - 3 + 1 = 1991$ red points.

Problem 3

Let $a_1, a_2, \ldots, a_n$, $b_1, b_2, \ldots, b_n$ be positive real numbers such that $a_1 + a_2 + \cdots + a_n = b_1 + b_2 + \cdots + b_n$. Show that

$$\frac{a_1^2}{a_1 + b_1} + \frac{a_2^2}{a_2 + b_2} + \cdots + \frac{a_n^2}{a_n + b_n} \geq \frac{a_1 + a_2 + \cdots + a_n}{2}.$$

Solution

By the Cauchy-Schwartz inequality,

$$\left(\frac{a_1^2}{a_1 + b_1} + \frac{a_2^2}{a_2 + b_2} + \cdots + \frac{a_n^2}{a_n + b_n}\right)((a_1+b_1)+(a_2+b_2)+\cdots+(a_n+b_n)) \geq (a_1+a_2+\cdots+a_n)^2.$$

Since $((a_1 + b_1) + (a_2 + b_2) + \cdots + (a_n + b_n)) = 2(a_1 + a_2 + \cdots + a_n)$,

$$\frac{a_1^2}{a_1 + b_1} + \frac{a_2^2}{a_2 + b_2} + \cdots + \frac{a_n^2}{a_n + b_n} \geq \frac{(a_1 + a_2 + \cdots + a_n)^2}{2(a_1 + a_2 + \cdots + a_n)} = \frac{a_1 + a_2 + \cdots + a_n}{2}.$$

Problem 4

During a break, n children at school sit in a circle around their teacher to play a game. The teacher walks clockwise close to the children and hands out candies to some of them according to the following rule. He selects one child and gives him a candy, then he skips the next child and gives a candy to the next one, then he skips 2 and gives a candy to the next one, then he skips 3, and soon. Determine the values of n for which eventually, perhaps after many rounds, all children will have at least one candy each.

Answer: All powers of 2.

Solution 1

Number the children from 0 to $n-1$. Then the teacher hands candy to children in positions $f(x) = 1 + 2 + \cdots + x \bmod n = \frac{x(x+1)}{2} \bmod n$. Our task is to find the range of $f\colon \mathbb{Z}_n \to \mathbb{Z}_n$, and to verify whether the range is $\mathbb{Z}_n$, that is, whether f is a bijection.

If $n = 2^a m$, $m > 1$ odd, look at $f(x)$ modulo m. Since m is odd, $m \mid f(x) \iff m \mid x(x+1)$. Then, for instance, $f(x) \equiv 0 \pmod m$ for $x = 0$ and $x = m-1$. This means that $f(x)$ is not a bijection modulo m, and there exists t such that $f(x) \not\equiv t \pmod m$ for all x. By the Chinese Remainder Theorem,

$$f(x) \equiv t \pmod n \iff \begin{cases} f(x) \equiv t \pmod{2^a} \\ f(x) \equiv t \pmod m \end{cases}$$

Therefore, f is not a bijection modulo n.

If $n = 2^a$, then

$$f(x) - f(y) = \frac{1}{2}(x(x+1) - y(y+1)) = \frac{1}{2}(x^2 - y^2 + x - y) = \frac{(x-y)(x+y+1)}{2}.$$

and

$$f(x) \equiv f(y) \pmod{2^a} \iff (x-y)(x+y+1) \equiv 0 \pmod{2^{a+1}}. \qquad (*)$$

If x and y have the same parity, $x+y+1$ is odd and $(*)$ is equivalent to $x \equiv y \pmod{2^{a+1}}$. If x and y have different parity,

$$(*) \iff x + y + 1 \equiv 0 \pmod{2^{a+1}}.$$

However, $1 \le x + y + 1 \le 2(2^a - 1) + 1 = 2^{a+1} - 1$, so $x + y + 1$ is not a multiple of 2^{a+1}. Therefore f is a bijection if n is a power of 2.

Solution 2

We give a full description of a_n, the size of the range of f.

Since congruences modulo n are defined, via Chinese Remainder Theorem, by congruences modulo p^α for all prime divisors p of n and α being the number of factors p in the factorization of n, $a_n = \prod_{p^\alpha \| n} a_{p^\alpha}$.

Refer to the first solution to check the case $p = 2$: $a_{2^\alpha} = 2^\alpha$.

For an odd prime p,

$$f(x) = \frac{x(x+1)}{2} = \frac{(2x+1)^2 - 1}{8},$$

and since p is odd, there is a bijection between the range of f and the quadratic residues modulo p^α, namely $t \mapsto 8t + 1$. So a_{p^α} is the number of quadratic residues modulo p^α.

Let g be a primitive root of p^α. Then there are $\frac{1}{2}\phi(p^\alpha) = \frac{p-1}{2} \cdot p^{\alpha-1}$ quadratic residues that are coprime with p: $1, g^2, g^4, \ldots, g^{\phi(p^n)-2}$. If p divides a quadratic residue kp, that is, $x^2 \equiv kp \pmod{p^\alpha}$, $\alpha \ge 2$, then p divides x and, therefore, also k. Hence p^2 divides this quadratic residue, and these quadratic residues are p^2 times each quadratic residue of $p^{\alpha-2}$. Thus

$$a_{p^\alpha} = \frac{p-1}{2} \cdot p^{\alpha-1} + a_{p^{\alpha-2}}.$$

Since $a_p = \frac{p-1}{2} + 1$ and $a_{p^2} = \frac{p-1}{2} \cdot p + 1$, telescoping yields

$$a_{p^{2t}} = \frac{p-1}{2}(p^{2t-1} + p^{2t-3} + \cdots + p) + 1 = \frac{p(p^{2t}-1)}{2(p+1)} + 1$$

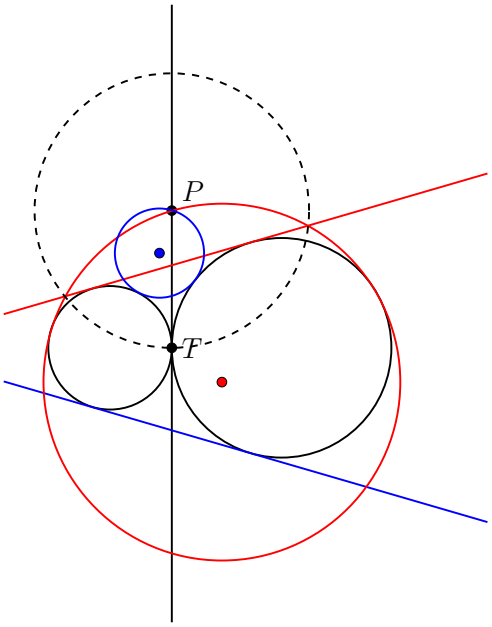
and

$$a_{p^{2t-1}} = \frac{p-1}{2}(p^{2t-2} + p^{2t-4} + \cdots + 1) + 1 = \frac{p^{2t}-1}{2(p+1)} + 1$$

Now the problem is immediate: if n is divisible by an odd prime p, $a_{p^\alpha} < p^\alpha$ for all α, and since $a_t \leq t$ for all t, $a_n < n$.

Problem 5
Given are two tangent circles and a point P on their common tangent perpendicular to the lines joining their centres. Construct with ruler and compass all the circles that are tangent to these two circles and pass through the point P.

Solution
Throughout this problem, we will assume that the given circles are *externally* tangent, since the problem does not have a solution otherwise.
Let Γ_1 and Γ_2 be the given circles and T be their tangency point. Suppose ω is a circle that is tangent to Γ_1 and Γ_2 and passes through P.
Now invert about point P, with radius PT. Let any line through P that cuts Γ_1 do so at points X and Y. The power of P with respect to Γ_1 is $PT^2 = PX \cdot PY$, so X and Y are swapped by this inversion. Therefore Γ_1 is mapped to itself in this inversion. The same applies to Γ_2. Since circle ω passes through P, it is mapped to a line tangent to the images of Γ_1 (itself) and Γ_2 (also itself), that is, a common tangent line. This common tangent cannot be PT, as PT is also mapped to itself. Since Γ_1 and Γ_2 have exactly other two common tangent lines, there are two solutions: the inverses of the tangent lines.



We proceed with the construction with the aid of some macro constructions that will be detailed later.

Step 1. Draw the common tangents to Γ_1 and Γ_2.

Step 2. For each common tangent t, draw the projection P_t of P onto t.

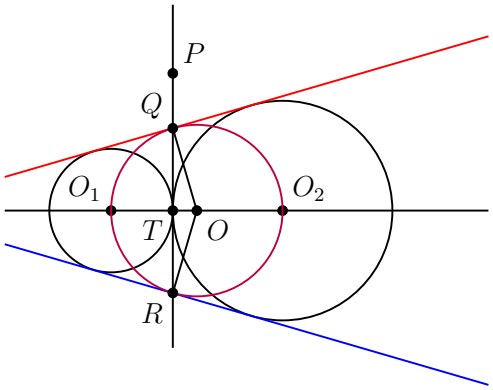
Step 3. Find the inverse P_1 of P_t with respect to the circle with center P and radius PT.

Step 4. ω_t is the circle with diameter PP_1.

Let's work out the details for steps 1 and 3. Steps 2 and 4 are immediate.
Step 1. In this particular case in which Γ_1 and Γ_2 are externally tangent, there is a small shortcut:

- Draw the circle with diameter on the two centers O_1 of Γ_1 and O_2 of Γ_2, and find its center O.

- Let this circle meet common tangent line OP at points Q, R. The required lines are the perpendicular to OQ at Q and the perpendicular to OR at R.



Let's show why this construction works. Let R_i be the radius of circle Γ_i and suppose without loss of generality that $R_1 \leq R_2$. Note that $OQ = \frac{1}{2}O_1O_2 = \frac{R_1+R_2}{2}$, $OT = OO_1 - R_1 = \frac{R_2-R_1}{2}$, so

$$\sin \angle TQO = \frac{OT}{OQ} = \frac{R_2 - R_1}{R_1 + R_2},$$

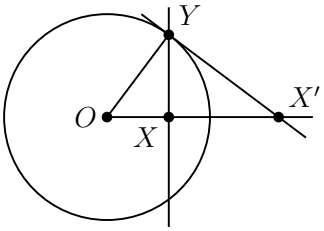
which is also the sine of the angle between O_1O_2 and the common tangent lines.
Let t be the perpendicular to OQ through Q. Then $\angle(t, O_1O_2) = \angle(OQ, QT) = \angle TQO$, and t is parallel to a common tangent line. Since

$$d(O,t) = OQ = \frac{R_1 + R_2}{2} = \frac{d(O_1,t) + d(O_2,t)}{2},$$

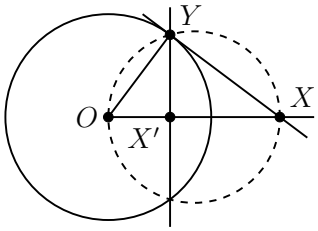
and O is the midpoint of O_1O_2, O is also at the same distance from t and the common tangent line, so these two lines coincide.

Step 3. Finding the inverse of a point X given the inversion circle Ω with center O is a well known procedure, but we describe it here for the sake of completeness.

- If X lies in Ω, then its inverse is X.
- If X lies in the interior of Ω, draw ray OX, then the perpendicular line ℓ to OX at X. Let ℓ meet Ω at a point Y. The inverse of X is the intersection X' of OX and the line perpendicular to OY at Y. This is because OYX' is a right triangle with altitude YX, and therefore $OX \cdot OX' = OY^2$.
- If X is in the exterior of Ω, draw ray OX and one of the tangent lines ℓ from X to Ω (just connect X to one of the intersections of Ω and the circle with diameter OX). Let ℓ touch Ω at a point Y. The inverse of X is the projection X' of Y onto OX. This is because OYX' is a right triangle with altitude YX', and therefore $OX \cdot OX' = OY^2$.



X is inside Ω



X is outside Ω

SOLUTIONS TO THE PROBLEMS OF THE 1990 ASIAN PACIFIC MATHEMATICS OLYMPIAD

Question 1

In $\triangle ABC$, let D, E, F be the midpoints of BC, AC, AB respectively and let G be the centroid of the triangle.

For each value of $\angle BAC$, how many non-similar triangles are there in which $AEGF$ is a cyclic quadrilateral?

FIRST SOLUTION

Let I be the intersection of AG and EF.

Let $\delta = AI.IG - FI.IE$. Then

$$AI = AD/2, \quad IG = AD/6, \quad FI = BC/4 = IE. \tag{1}$$

Further, applying the cosine rule to triangles ABD, ACD we get

$$\begin{aligned} AB^2 &= BC^2/4 + AD^2 - AD.BC.\cos\angle BDA, \\ AC^2 &= BC^2/4 + AD^2 + AD.BC.\cos\angle BDA, \\ \text{so } AD^2 &= \left(AB^2 + AC^2 - BC^2/2\right)/2. \end{aligned}$$

Hence

$$\begin{aligned} \delta &= \left(AB^2 + AC^2 - 2BC^2\right)/24 \\ &= \left(4AB.AC.\cos\angle BAC - AB^2 - AC^2\right). \end{aligned}$$

Now $AEFG$ is a cyclic quadrilateral if and only if $\delta = 0$, i.e. if and only if

$$\begin{aligned} \cos\angle BAC &= \left(AB^2 + AB^2\right)/\left(4.AB.AC\right) \\ &= \left(AB/AC + AC/AB\right)/4. \end{aligned}$$

Now $AB/AC + AC/AB \geq 2$. Hence $\cos\angle BAC \geq 1/2$ and so $\angle BAC \leq 60°$.

For $\angle BAC > 60°$ there is no triangle with the required property.

For $\angle BAC = 60°$ there exists, within similarity, precisely one triangle (which is equilateral) having the required property.

For $\angle BAC < 60°$ there exists, within similarity, again precisely one triangle having the required property (even though for fixed median AE there are two, but one arises from the other by interchanging point B with point C, thus proving them to be similar).

SECOND SOLUTION (Mr Marcus Brazil, La Trobe University, Bundoora, Melbourne, Australia):

We require, as above,

$$AI.IG = EI.IF,$$

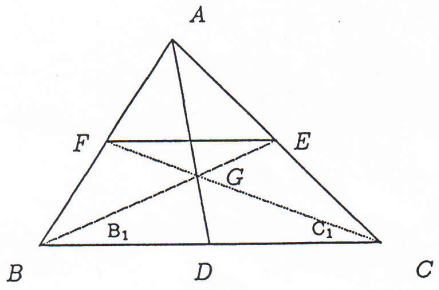
(which by (1) is equivalent to $AD^2/3 = CD^2$, i.e. $AD = CD.\sqrt{3}$).

Let, without loss of generality, $CD = 1$. Then A lies on the circle of radius $\sqrt{3}$ with centre D. If CD and DA are perpendicular, the angle BAC is 60°, otherwise it must be less.

In this case, for each angle BAC there are two solutions, which are congruent.

THIRD SOLUTION (provided by the Canadian Problems Committee)

In the figure as shown below, we first show that it is necessary that $\angle A$ is less than 90° if the quadrilateral $AEGF$ is cyclic.



Now, since $EF \parallel BC$, we get

$$\begin{aligned} \angle EGF &= 180° - (B_1 + C_1) \\ &\geq 180° - (B + C) \\ &= A\,. \end{aligned}$$

1 Thus, if $AEGF$ is cyclic, we would have $\angle EGF + \angle A = 180°$. Therefore it is necessary that $0 < \angle A \leq 90°$.

Continuation "A"

Let O be the circumcentre of ΔAFE. Without loss of generality, let the radius of this circle be 1.

We then let $A = 1, F = z = e^{i\theta}$ and $E = ze^{2i\alpha} = e^{i(\theta+2\alpha)}$.

Then $\angle A = \alpha$, $0 < \alpha \leq 90°$, and $0 < \theta < 360° - 2\alpha$.

Thus,

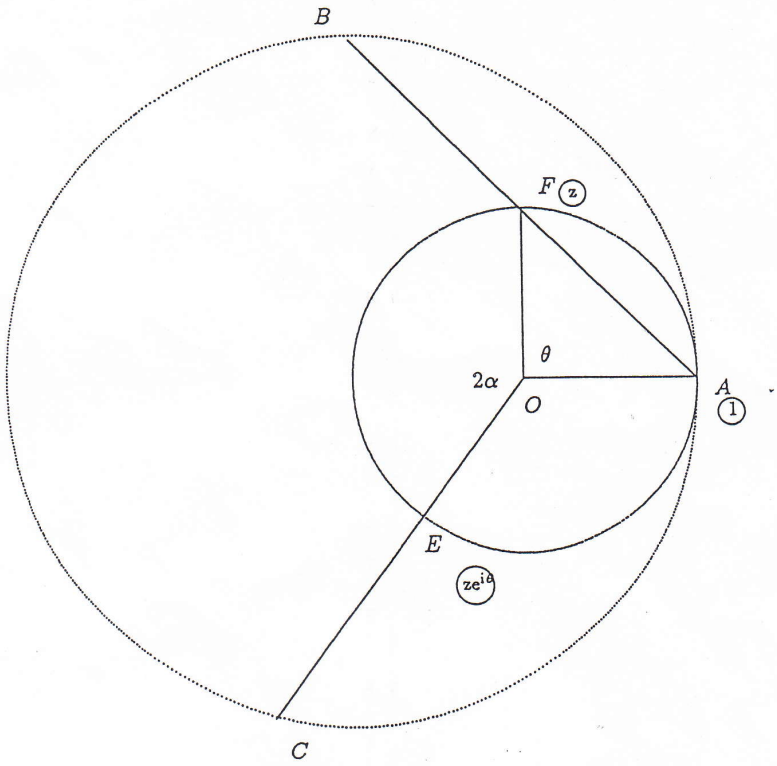
$$B = 2z - 1$$

and

$$\begin{aligned} G &= \frac{1}{3}(2z-1) + \frac{2}{3}\left(z\,e^{2i\alpha}\right) \\ &= \frac{1}{3}\left(2e^{i\theta} + 2e^{i(\theta+2\alpha)} - 1\right). \end{aligned}$$

For quadrilateral $AFGE$ to be cyclic, it is now necessary that

3 $$|G| = 1.$$



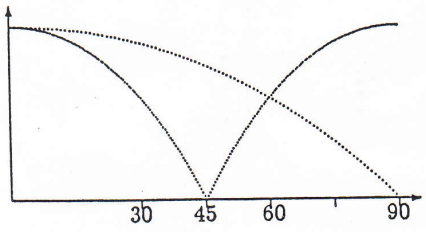
For $|G| = 1$, we must have

$$\begin{aligned} 9 &= (2\cos(\theta) + 2\cos(\theta + 2\alpha) - 1)^2 + (2\sin(\theta) + 2\sin(\theta + 2\alpha))^2 \\ &= 4\left(\cos^2(\theta) + \sin^2(\theta)\right) + 4\left(\cos^2(\theta + 2\alpha) + \sin^2(\theta + 2\alpha)\right) + 1 \\ &\quad +8\left(\cos(\theta)\cos(\theta + 2\alpha) + \sin(\theta)\sin(\theta + 2\alpha)\right) - 4\cos(\theta) - 4\cos(\theta + 2\alpha) \\ &= 9 + 8\cos(2\alpha) - 8\cos(\alpha)\cos(\theta + \alpha) \end{aligned}$$

so that

$$\cos(\theta + \alpha) = \frac{\cos(2\alpha)}{\cos(\alpha)}.$$

5 Now, $\left|\frac{\cos(2\alpha)}{\cos(\alpha)}\right| \leq 1$ if and only if $\alpha \in (0, 60°]$ in the range of α under consideration, that is $\alpha \in (0, 90°]$. There is equality if and only if $\alpha = 60°$.



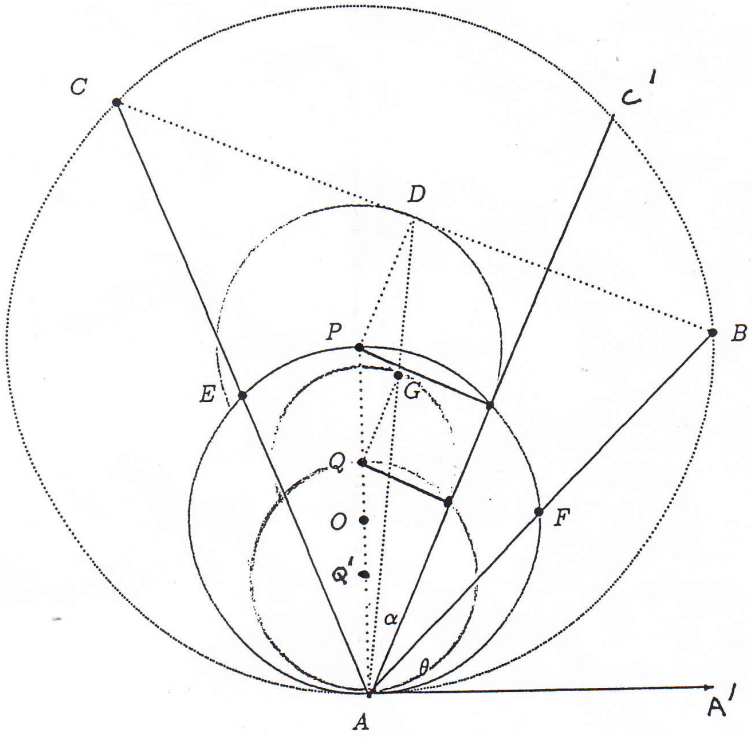
7 Note there is only one solution. The apparent other solution is the mirror image of the first. We are solving for $\alpha + \theta$. The other solution is $360° - \alpha - \theta$.

Continuation "B"

Let O be the circumcentre of triangle AEF. Let AP be a diameter of this circle. Construct the circle with centre P and radius AP. Then B and C lie on this circle.

It is clear that the problem is solved if we allow the angle $\angle BAC = \alpha$ to vary and restrict B and C to the constructed circle.

Let θ be the angle from the drawn axis. Then θ lies in the range $(0, 180° - \alpha)$. We must not forget the necessary restriction of α, that is $\alpha \in (0, 90°]$.



$A'\hat{A}C' = \alpha$
$= C\hat{A}B$

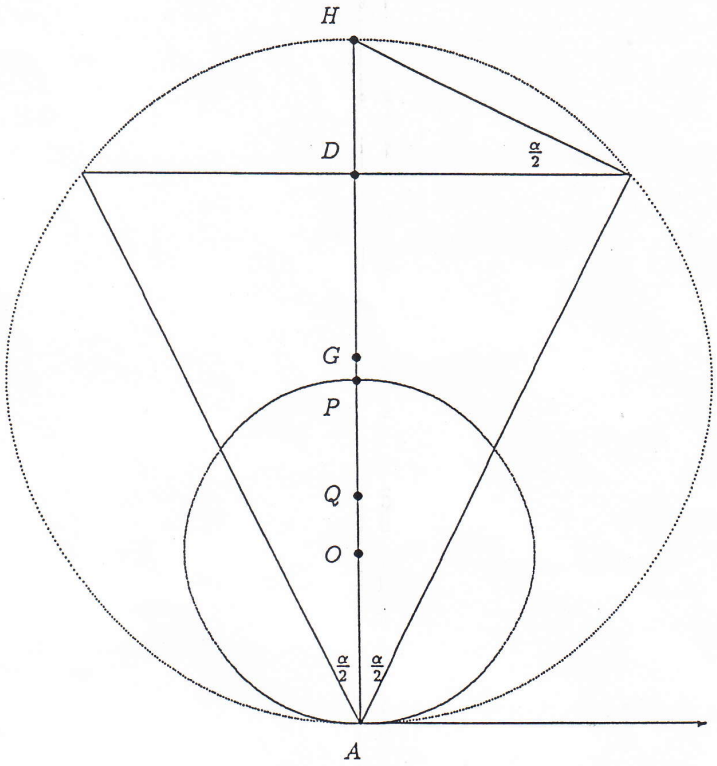
Now, D lies on an arc of a circle, centre P, radius PD exterior to the circle, centre O, radius AO.

By similarity, G, lies on an arc of a circle, centre Q, radius QG where $AQ = \frac{2}{3}AP$ and $QG = \frac{2}{3}PD$.

3 For the quadrilateral $AFGE$ to be cyclic, we must have that the radius QG is greater than or equal to QP.

The easiest way to calulate these radii is to consider the case in which the diameter AP bisects the angle $\angle BAC$.

Thus we re-draw the diagram as below. Let AH be a diameter of the larger circle.



Thus we have $AH = 4$ and by similar triangles,

$$\frac{AD}{AB} = \frac{AB}{AH} = \cos\left(\frac{\alpha}{2}\right),$$

so that

$$\begin{aligned} AD &= 4\cos^2\left(\frac{\alpha}{2}\right) \\ &= 2 + 2\cos(\alpha). \end{aligned}$$

Thus $PD = 2\cos(\alpha)$ and $QG = \frac{2}{3}\,2\cos(\alpha) = \frac{4}{3}\cos(\alpha)$.

The necessary condition for a cyclic quadrilateral is then

$$\frac{4}{3}(1 + \cos(\alpha)) \geq 2,$$

5 which reduces to

$$\cos(\alpha) \geq \frac{1}{2}.$$

7 Thus it is clear that there is precisely one (up to similarity) solution for $0 < \alpha \leq 60°$ and no solutions otherwise.

5

The 1990 Asian Pacific Mathematical Olympiad

Question 2

Let $a_1, a_2, \ldots, a_n$ be positive real numbers, and let S_k be the sum of products of $a_1, a_2, \ldots, a_n$ taken k at a time.
Show that

$$S_k S_{n-k} \geq \binom{n}{k}^2 a_1 a_2 \ldots a_n, \quad \text{for} \quad k = 1, 2, \ldots, n-1$$

.

FIRST SOLUTION

$$\binom{n}{k} a_1 a_2 \ldots a_n$$

2 $$= \sum_{1 \leq i_1 < i_2 < \ldots < i_k \leq n} a_{i_1} a_{i_2} \ldots a_{i_k} . a_1 a_2 \ldots a_n / a_{i_1} a_{i_2} \ldots a_{i_k}$$

(and using the Cauchy-Schwarz inequality)

4 $$\leq \left(\sum_{1 \leq i_1 < i_2 < \ldots < i_k \leq n} a_{i_1} a_{i_2} \ldots a_{i_k} \right)^{\frac{1}{2}} \cdot \left(\sum_{1 \leq i_1 < i_2 < \ldots < i_k \leq n} a_1 a_2 \ldots a_n / a_{i_1} a_{i_2} \ldots a_{i_k} \right)^{\frac{1}{2}}$$

5 $$= S_k^{\frac{1}{2}} . S_{n-k}^{\frac{1}{2}}.$$

Therefore

7 $$\binom{n}{k}^2 a_1 a_2 \ldots a_n \leq S_k S_{n-k},$$

q.e.d.

SECOND SOLUTION (provided by the Canadian Problems Committee).

Write S_k as $\sum_{i=1}^{\binom{n}{k}} t_i$. Then

2 $$S_{n-k} = \left(\prod_{m=1}^{n} a_m \right) \left(\sum_{i=1}^{\binom{n}{k}} \frac{1}{t_i} \right),$$

$$\text{so that} \quad S_k S_{n-k} = \left(\prod_{m=1}^{n} a_m \right) \left(\sum_{i=1}^{\binom{n}{k}} t_i \right) \left(\sum_{j=1}^{\binom{n}{k}} \frac{1}{t_j} \right)$$

$$= \left(\prod_{m=1}^{n} a_m \right) \left[\sum_{i=1}^{\binom{n}{k}} 1 + \sum_{i=1}^{\binom{n}{k}} \sum_{j=1 i \neq j}^{\binom{n}{k}} \frac{t_i}{t_j} \right]$$

4 $$= \left(\prod_{m=1}^{n} a_m \right) \left[\binom{n}{k} + \sum_{i,j} \left(\frac{t_i}{t_j} + \frac{t_j}{t_i} \right) \right].$$

As there are

$$\frac{\binom{n}{k}^2 - \binom{n}{k}}{2}$$

terms in the sum

$$S_k S_{n-k} \geq \left(\prod_{m=1}^{n} a_m\right)\left[\binom{n}{k} + 2 \cdot \frac{\binom{n}{k}^2 - \binom{n}{k}}{2}\right]$$

$$= \binom{n}{k}^2 \left(\prod_{m=1}^{n} a_m\right)$$

since $\frac{t_i}{t_j} + \frac{t_j}{t_i} \geq 2$ for $t_i, t_j > 0$.

Question 3

Consider all the triangles ABC which have a fixed base AB and whose altitude from C is a constant h. For which of these triangles is the product of its altitudes a maximum?

SOLUTION:

Let h_a and h_b be the altitudes from A and B, respectively. Then

$$\begin{aligned} AB.h.AC.h_b.BC.h_a &= 8.(\text{area of } \triangle ABC)^3 \\ &= (AB.h)^3, \end{aligned}$$

which is a constant. So the product $h.h_a.h_b$ attains its maximum when the product $AC.BC$ attains its minimum.

Since

$$\begin{aligned} (\sin C).AC.BC &= BC.h_a \\ &= 2.\text{area of } \triangle ABC, \end{aligned}$$

which is a constant, $AC.BC$ attains its minimum when $\sin C$ reaches its maximum.

There are two cases:

(a) $h \leq AB/2$. Then there exists a triangle ABC which has a right angle at C, and for precisely such a triangle $\sin C$ attains its maximum, namely 1.

(b) $h > AB/2$. In this case the angle at C is acute and assumes its maximum when the triangle is isosceles.

Note that a solution using calculus obviously exists.

Question 4

A set of 1990 persons is divided into non-intersecting subsets in such a way that

(a) no one in a subset knows all the others in the subset;

(b) among any three persons in a subset, there are always at least two who do not know each other; and

(c) for any two persons in a subset who do not know each other, there is exactly one person in the same subset knowing both of them.

(i) Prove that within each subset, every person has the same number of acquaintances.

(ii) Determine the maximum possible number of subsets.

Note: it is understood that if a person A knows person B, then person B will know person A; an acquaintance is someone who is known. Every person is assumed to know one's self.

SOLUTION:

(i) Let S be a subset of persons satisfying conditions (a), (b) and (c). Let $x \in S$ be one who knows the maximum number of persons in S.

Assume that x knows $x_1, x_2, \ldots, x_n$. By (b), x_i and x_j are strangers if $i \neq j$. For each x_i, let N_i be the set of persons in S who know x_i but not x. Note that, for $i \neq j$, N_i has no person in common with N_j, otherwise there would be more than one person knowing x_i and x_j, contradicting (c).

By (a) we may assume that N_1 is not empty. Let $y_1 \in N_1$. By (c), for each $k > 1$, there is exactly one person y_k in N_k who knows y_1. This means that y_1 knows n persons, namely $x_1, y_2, \ldots, y_n$.

Because n is the maximal number of persons in S a person in S can know, y_1 knows exactly n persons in S. By precisely the same reasoning we find that each person in N_i, $i = 1, 2, \ldots, n$, knows exactly n persons in S.

Letting y_1 take the role of x in our argument, we see that also each x_i knows exactly n persons. Note that, by (c), every person in S other than $x, x_1, \ldots, x_n$, must be in some N_j. Therefore every person in S knows exactly n persons in S and thus has the same number of acquaintances in S.

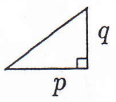
(ii) To maximize the number of subsets, we have to minimize the size of each group. The smallest possible subset is one in which every person knows exactly two persons, and hence there must be exactly five persons in the subset, forming a cycle where two persons stand side by side only if they know each other. Therefore the maximum possible number of subsets is $1990/5 = 398$.

Question 5

Show that for every integer $n \geq 6$, there exists a convex hexagon which can be dissected into exactly n congruent triangles.

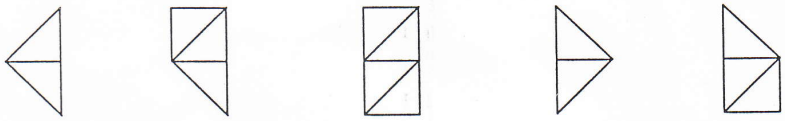
FIRST SOLUTION (provided by the Canadian Problems Committee).

The basic building blocks will be right angled triangles with sides p, q (which are positive integers) adjacent to the right angle.



In the first instance, we take $p = q = 1$ and construct five basic building blocks: L_1, L_2, M, R_1 and R_2.

[3]

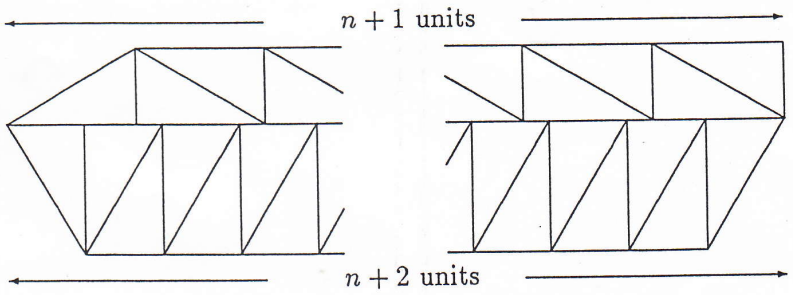


L_1 L_2 M R_1 R_2

We shall now build convex hexagons by taking, on the left, one of the blocks L_i, attaching n copies of the block M, and finally attaching one of the blocks R_j. We must therefore exclude the case when $(i,j) = (2,1)$ for this does not generate a hexagon. Further, for $(i,j) = (1,1)$ or $(i,j) = (1,2)$, we require that $n \geq 1$, whereas for $(i,j) = (2,2)$, we only need require that $n \geq 0$.

Thus, with the obvious interpretation:
$L_1 + nM + R_1$ gives a convex hexagon containing $2+4n+2 = 4n+4$ $(n \geq 1)$ congruent triangles;
$L_1 + nM + R_2$ gives a convex hexagon containing $2+4n+3 = 4n+5$ $(n \geq 1)$ congruent triangles; and
$L_2 + nM + R_2$ gives a convex hexagon containing $3+4n+3 = 4n+6$ $(n \geq 0)$ congruent triangles, or $4n+2$ $(n \geq 1)$ congruent triangles.

We shall now modify the lengths of the sides of the right triangle to obtain the case of $4n+3$ $(n \geq 1)$ congruent triangles.



So we have $2n+1$ triangles in the top part and $2n+2$ triangles in the bottom part. In order to match, we need

$$(n+1)p = (n+2)q,$$

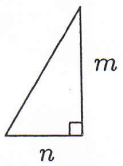
so we take

$$q = n+1 \qquad \text{and} \qquad p = n+2.$$

This completes the solution.

SECOND SOLUTION (provided by the Canadian Problems Committee):

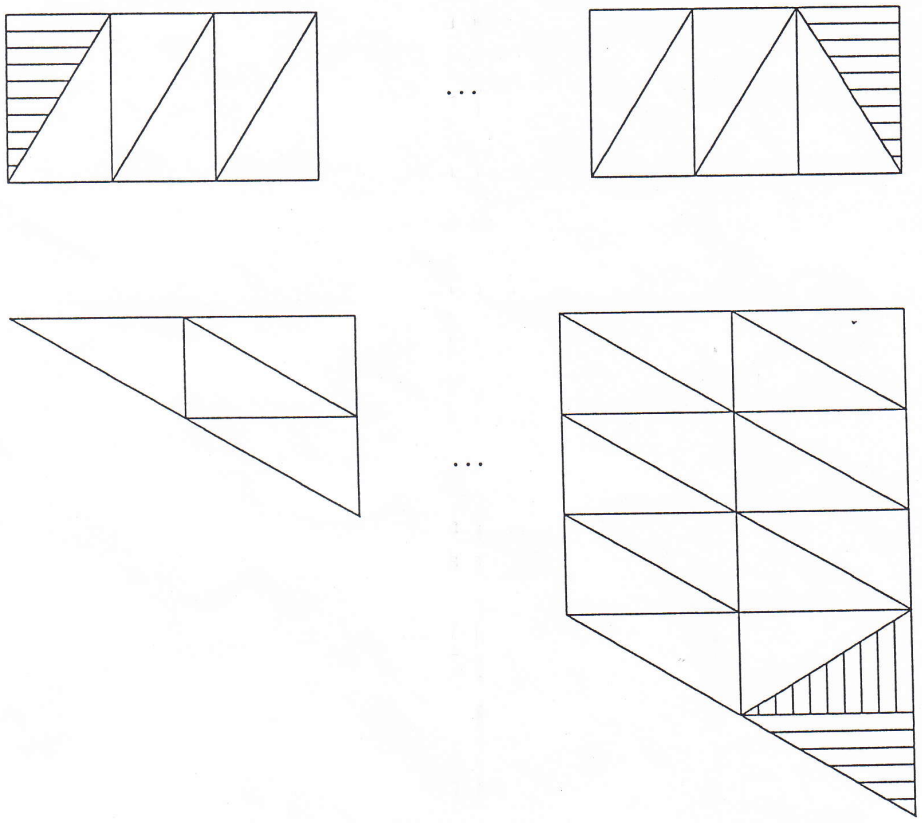
The basic building blocks will be right angled triangles with sides m, n (which are positive integers) adjacent to the right angle.



We construct an "UPPER CONFIGURATION", being a rectangle consisting of m building block units of pairs of triangles with the side of length n as base. This gives a base length of nm across the configuration.

We further construct a "LOWER CONFIGURATION", being a triangle with base up, consisting along the base of n building block units. Again, we have a base length of mn across the configuration.

Two triangles in the upper configuration are shaded horizontally. One triangle in the lower configuration is also shaded horizontally. Another triangle in the lower configuration is shaded vertically.



3

Now consider the figure obtained by joining the two configurations along the base line of common length nm. To create the classes of hexagons defined below, it is necessary that both $n \geq 3$ and $m \geq 3$.

We create a class of convex hexagons (class 1) by omitting the three triangles that are shaded horizontally. The other class of convex hexagons (class 2) is obtained by omitting all shaded triangles.

Now count the total number of triangles in the full configuration.

The upper configuration gives $2m$ triangles. The lower configuration gives

$$\sum_{k=1}^{n}(2k-1) \;=\; n^2 \quad \text{triangles.}$$

Thus the total number of triangles in a hexagon in class 1 is

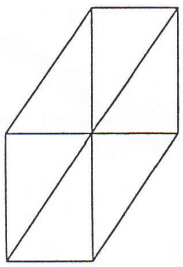
$$2m-2+n^2-1\,,$$

and the total number of triangle in a hexagon in class 2 is

$$2m-2+n^2-2\,.$$

These, together with the restrictions on n and m, generate all positive integers greater than or equal to 11.

For the integers $6, 7, 8, 9$ and 10, we give specific examples:



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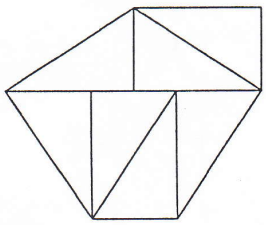

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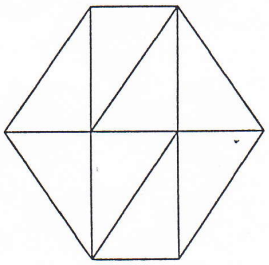

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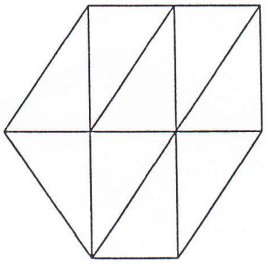

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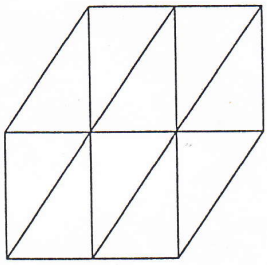

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This completes the solution.

ANSWER KEY

There are $\binom{n}{k}$ products of the a_i taken k at a time. Amongst these products any given a_i will appear $\binom{n-1}{k-1}$ times, since $\binom{n-1}{k-1}$ is the number of ways of choosing the other factors of the product. So the AM/GM inequality gives

$$\frac{S_k}{\binom{n}{k}} \geq \left[\prod_{i=1}^{n} a_i^{\binom{n-1}{k-1}}\right]^{\frac{1}{\binom{n}{k}}}.$$

But $\binom{n}{k} = \frac{n}{k}\binom{n-1}{k-1}$, leading to

$$S_k \geq \binom{n}{k}\left(\prod_{i=1}^{n} a_i\right)^{\frac{k}{n}}.$$

Hence

$$S_k S_{n-k} \geq \binom{n}{k}\left(\prod_{i=1}^{n} a_i\right)^{\frac{k}{n}}\binom{n}{n-k}\left(\prod_{i=1}^{n} a_i\right)^{\frac{n-k}{n}} = \binom{n}{k}^2\left(\prod_{1}^{n} a_i\right).$$

APMO 1989 – Problems and Solutions

Problem 1
Let $x_1, x_2, \ldots, x_n$ be positive real numbers, and let

$$S = x_1 + x_2 + \cdots + x_n.$$

Prove that

$$(1+x_1)(1+x_2)\cdots(1+x_n) \le 1 + S + \frac{S^2}{2!} + \frac{S^3}{3!} + \cdots + \frac{S^n}{n!}.$$

Solution 1
Let σ_k be the *kth symmetric polynomial*, namely

$$\sigma_k = \sum_{\substack{|S|=k \\ S\subseteq\{1,2,\ldots,n\}}} \prod_{i\in S} x_i,$$

and more explicitly

$$\sigma_1 = S, \quad \sigma_2 = x_1x_2 + x_1x_3 + \cdots + x_{n-1}x_n, \quad \text{and so on.}$$

Then

$$(1+x_1)(1+x_2)\cdots(1+x_n) = 1 + \sigma_1 + \sigma_2 + \cdots + \sigma_n.$$

The expansion of

$$S^k = (x_1+x_2+\cdots+x_n)^k = \underbrace{(x_1+x_2+\cdots+x_n)(x_1+x_2+\cdots+x_n)\cdots(x_1+x_2+\cdots+x_n)}_{k \text{ times}}$$

has at least $k!$ occurrences of $\prod_{i\in S} x_i$ for each subset S with k indices from $\{1,2,\ldots,n\}$. In fact, if π is a permutation of S, we can choose each $x_{\pi(i)}$ from the ith factor of $(x_1+x_2+\cdots+x_n)^k$. Then each term appears at least $k!$ times, and

$$S^k \ge k!\sigma_k \iff \sigma_k \le \frac{S^k}{k!}.$$

Summing the obtained inequalities for $k = 1, 2, \ldots, n$ yields the result.

Solution 2
By AM-GM,

$$(1+x_1)(1+x_2)\cdots(1+x_n) \le \left(\frac{(1+x_1)+(1+x_2)+\cdots+(1+x_n)}{n}\right)^n = \left(1+\frac{S}{n}\right)^n.$$

By the binomial theorem,

$$\left(1+\frac{S}{n}\right)^n = \sum_{k=0}^{n} \binom{n}{k}\left(\frac{S}{n}\right)^k = \sum_{k=0}^{n} \frac{1}{k!}\frac{n(n-1)\ldots(n-k+1)}{n^k} S^k \le \sum_{k=0}^{n} \frac{S^k}{k!},$$

and the result follows.

Comment: Maclaurin's inequality states that

$$\frac{\sigma_1}{n} \ge \sqrt{\frac{\sigma_2}{\binom{n}{2}}} \ge \cdots \ge \sqrt[k]{\frac{\sigma_k}{\binom{n}{k}}} \ge \cdots \ge \sqrt[n]{\frac{\sigma_n}{\binom{n}{n}}}.$$

Then $\sigma_k \le \binom{n}{k}\frac{S^k}{n^k} = \frac{1}{k!}\frac{n(n-1)\ldots(n-k+1)}{n^k} S^k \le \frac{S^k}{k!}$.

Problem 2
Prove that the equation

$$6(6a^2 + 3b^2 + c^2) = 5n^2$$

has no solutions in integers except $a = b = c = n = 0$.

Solution
We can suppose without loss of generality that $a, b, c, n \geq 0$. Let (a, b, c, n) be a solution with minimum sum $a + b + c + n$. Suppose, for the sake of contradiction, that $a + b + c + n > 0$. Since 6 divides $5n^2$, n is a multiple of 6. Let $n = 6n_0$. Then the equation reduces to

$$6a^2 + 3b^2 + c^2 = 30n_0^2.$$

The number c is a multiple of 3, so let $c = 3c_0$. The equation now reduces to

$$2a^2 + b^2 + 3c_0^2 = 10n_0^2.$$

Now look at the equation modulo 8:

$$b^2 + 3c_0^2 \equiv 2(n_0^2 - a^2) \pmod 8.$$

Integers b and c_0 have the same parity. Either way, since x^2 is congruent to 0 or 1 modulo 4, $b^2 + 3c_0^2$ is a multiple of 4, so $n_0^2 - a^2 = (n_0 - a)(n_0 + a)$ is even, and therefore also a multiple of 4, since $n_0 - a$ and $n_0 + a$ have the same parity. Hence $2(n_0^2 - a^2)$ is a multiple of 8, and

$$b^2 + 3c_0^2 \equiv 0 \pmod 8.$$

If b and c_0 are both odd, $b^2 + 3c_0^2 \equiv 4 \pmod 8$, which is impossible. Then b and c_0 are both even. Let $b = 2b_0$ and $c_0 = 2c_1$, and we find

$$a^2 + 2b_0^2 + 6c_1^2 = 5n_0^2.$$

Look at the last equation modulo 8:

$$a^2 + 3n_0^2 \equiv 2(c_1^2 - b_0^2) \pmod 8.$$

A similar argument shows that a and n_0 are both even.
We have proven that a, b, c, n are all even. Then, dividing the original equation by 4 we find

$$6(6(a/2)^2 + 3(b/2)^2 + (c/2)^2) = 5(n/2)^2,$$

and we find that $(a/2, b/2, c/2, n/2)$ is a new solution with smaller sum. This is a contradiction, and the only solution is $(a, b, c, n) = (0, 0, 0, 0)$.

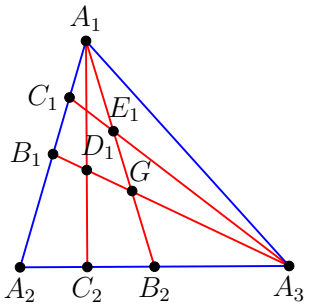
Problem 3

Let A_1, A_2, A_3 be three points in the plane, and for convenience,let $A_4 = A_1$, $A_5 = A_2$. For $n = 1, 2$, and 3, suppose that B_n is the midpoint of A_nA_{n+1}, and suppose that C_n is the midpoint of A_nB_n. Suppose that A_nC_{n+1} and B_nA_{n+2} meet at D_n, and that A_nB_{n+1} and C_nA_{n+2} meet at E_n. Calculate the ratio of the area of triangle $D_1D_2D_3$ to the area of triangle $E_1E_2E_3$.

Answer: $\dfrac{25}{49}$.

Solution

Let G be the centroid of triangle ABC, and also the intersection point of A_1B_2, A_2B_3, and A_3B_1.



By Menelao's theorem on triangle $B_1A_2A_3$ and line $A_1D_1C_2$,

$$\frac{A_1B_1}{A_1A_2}\cdot\frac{D_1A_3}{D_1B_1}\cdot\frac{C_2A_2}{C_2A_3}=1 \iff \frac{D_1A_3}{D_1B_1}=2\cdot 3=6 \iff \frac{D_1B_1}{A_3B_1}=\frac{1}{7}.$$

Since $A_3G=\frac{2}{3}A_3B_1$, if $A_3B_1=21t$ then $GA_3=14t$, $D_1B_1=\frac{21t}{7}=3t$, $A_3D_1=18t$, and $GD_1=A_3D_1-A_3G=18t-14t=4t$, and

$$\frac{GD_1}{GA_3}=\frac{4}{14}=\frac{2}{7}.$$

Similar results hold for the other medians, therefore $D_1D_2D_3$ and $A_1A_2A_3$ are homothetic with center G and ratio $-\frac{2}{7}$.
By Menelao's theorem on triangle $A_1A_2B_2$ and line $C_1E_1A_3$,

$$\frac{C_1A_1}{C_1A_2}\cdot\frac{E_1B_2}{E_1A_1}\cdot\frac{A_3A_2}{A_3B_2}=1 \iff \frac{E_1B_2}{E_1A_1}=3\cdot\frac{1}{2}=\frac{3}{2} \iff \frac{A_1E_1}{A_1B_2}=\frac{2}{5}.$$

If $A_1B_2=15u$, then $A_1G=\frac{2}{3}\cdot 15u=10u$ and $GE_1=A_1G-A_1E_1=10u-\frac{2}{5}\cdot 15u=4u$, and

$$\frac{GE_1}{GA_1}=\frac{4}{10}=\frac{2}{5}.$$

Similar results hold for the other medians, therefore $E_1E_2E_3$ and $A_1A_2A_3$ are homothetic with center G and ratio $\frac{2}{5}$.
Then $D_1D_2D_3$ and $E_1E_2E_3$ are homothetic with center G and ratio $-\frac{2}{7}:\frac{2}{5}=-\frac{5}{7}$, and the ratio of their area is $\left(\frac{5}{7}\right)^2=\frac{25}{49}$.

Problem 4

Let S be a set consisting of m pairs (a, b) of positive integers with the property that $1 \leq a < b \leq n$. Show that there are at least

$$4m\frac{(m - \frac{n^2}{4})}{3n}$$

triples (a, b, c) such that (a, b), (a, c), and (b, c) belong to S.

Solution

Call a triple (a, b, c) *good* if and only if (a, b), (a, c), and (b, c) all belong to S. For i in $\{1, 2, \ldots, n\}$, let d_i be the number of pairs in S that contain i, and let D_i be the set of numbers paired with i in S (so $|D_i| = d_i$). Consider a pair $(i, j) \in S$. Our goal is to estimate the number of integers k such that any permutation of $\{i, j, k\}$ is good, that is, $|D_i \cap D_j|$. Note that $i \notin D_i$ and $j \notin D_j$, so $i, j \notin D_i \cap D_j$; thus any $k \in D_i \cap D_j$ is different from both i and j, and $\{i, j, k\}$ has three elements as required. Now, since $D_i \cup D_j \subseteq \{1, 2, \ldots, n\}$,

$$|D_i \cap D_j| = |D_i| + |D_j| - |D_i \cup D_j| \leq d_i + d_j - n.$$

Summing all the results, and having in mind that each good triple is counted three times (one for each two of the three numbers), the number of good triples T is at least

$$T \geq \frac{1}{3} \sum_{(i,j)\in S} (d_i + d_j - n).$$

Each term d_i appears each time i is in a pair from S, that is, d_i times; there are m pairs in S, so n is subtracted m times. By the Cauchy-Schwartz inequality

$$T \geq \frac{1}{3}\left(\sum_{i=1}^{n} d_i^2 - mn\right) \geq \frac{1}{3}\left(\frac{\left(\sum_{i=1}^{n} d_i\right)^2}{n} - mn\right).$$

Finally, the sum $\sum_{i=1}^{n} d_i$ is $2m$, since d_i counts the number of pairs containing i, and each pair (i, j) is counted twice: once in d_i and once in d_j. Therefore

$$T \geq \frac{1}{3}\left(\frac{(2m)^2}{n} - mn\right) = 4m\frac{(m - \frac{n^2}{4})}{3n}.$$

Comment: This is a celebrated graph theory fact named *Goodman's bound*, after A. M. Goodman's method published in 1959. The generalized version of the problem is still studied to this day.

Problem 5
Determine all functions f from the reals to the reals for which

(1) $f(x)$ is strictly increasing,

(2) $f(x) + g(x) = 2x$ for all real x, where $g(x)$ is the composition inverse function to $f(x)$.

(*Note*: f and g are said to be composition inverses if $f(g(x)) = x$ and $g(f(x)) = x$ for all real x.)

Answer: $f(x) = x + c$, $c \in \mathbb{R}$ constant.

Solution
Denote by f_n the nth iterate of f, that is, $f_n(x) = \underbrace{f(f(\ldots f}_{n \text{ times}}(x)))$.
Plug $x \to f_{n+1}(x)$ in (2): since $g(f_{n+1}(x)) = g(f(f_n(x))) = f_n(x)$,

$$f_{n+2}(x) + f_n(x) = 2f_{n+1}(x),$$

that is,

$$f_{n+2}(x) - f_{n+1}(x) = f_{n+1}(x) - f_n(x).$$

Therefore $f_n(x) - f_{n-1}(x)$ does not depend on n, and is equal to $f(x) - x$. Summing the corresponding results for smaller values of n we find

$$f_n(x) - x = n(f(x) - x).$$

Since g has the same properties as f,

$$g_n(x) - x = n(g(x) - x) = -n(f(x) - x).$$

Finally, g is also increasing, because since f is increasing $g(x) > g(y) \implies f(g(x)) > f(g(y)) \implies x > y$. An induction proves that f_n and g_n are also increasing functions.
Let $x > y$ be real numbers. Since f_n and g_n are increasing,

$$x + n(f(x) - x) > y + n(f(y) - y) \iff n[(f(x) - x) - (f(y) - y)] > y - x$$

and

$$x - n(f(x) - x) > y - n(f(y) - y) \iff n[(f(x) - x) - (f(y) - y)] < x - y.$$

Summing it up,

$$|n[(f(x) - x) - (f(y) - y)]| < x - y \quad \text{for all } n \in \mathbb{Z}_{>0}.$$

Suppose that $a = f(x) - x$ and $b = f(y) - y$ are distinct. Then, for all positive integers n,

$$|n(a - b)| < x - y,$$

which is false for a sufficiently large n. Hence $a = b$, and $f(x) - x$ is a constant c for all $x \in \mathbb{R}$, that is, $f(x) = x + c$.
It is immediate that $f(x) = x + c$ satisfies the problem, as $g(x) = x - c$.

DANO

ВЫСШАЯ ШКОЛА ЭКОНОМИКИ

Т БАНК

Вариант 1

1. По выборке из 100 значений посчитали выборочную дисперсию. После этого все значения увеличили в 2 раза. Как изменится выборочная дисперсия, если вычислить ее по новым значениям выборки?

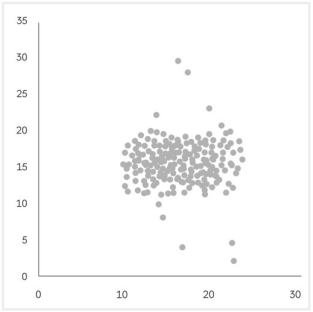
   A. Не изменится

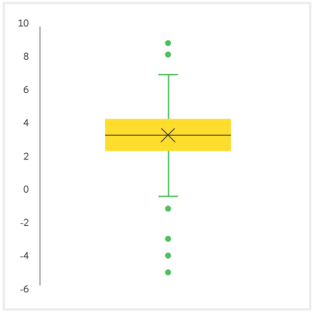
   B. Увеличится в 2 раза

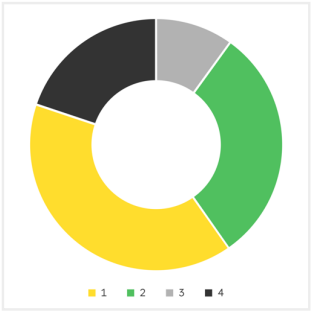
   C. Увеличится в 4 раза

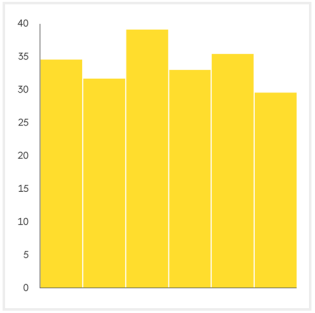
   D. Уменьшится в 2 раза

2. Сопоставьте названия типов визуализации с графиками:

1. 

2. 

3. 

4. 

1. Диаграмма рассеяния
2. Диаграмма размаха
3. Секторная диаграмма
4. Гистограмма

3. Исследователи проверили гипотезу о среднем значении с использованием T-теста. Уровень значимости установлен равным 0,05. Как интерпретировать такое значение уровня значимости?

   A. Вероятность того, что альтернативная гипотеза верна, равна 5%

   B. Допустимая вероятность отвержения нулевой гипотезы, если она на самом деле верна, равна 5%

   C. Если нулевая гипотеза верна, в 5% случаев мы допускаем, что можем ошибочно отвергнуть ее при многократном повторении теста на семплированных данных из одной генеральной совокупности

   D. Вероятность ошибки второго рода равна 5%

4. Сотрудники отдела прогнозирования и анализа данных изучили ежемесячные данные по ценам на гибискусы и гортензии за 2024 год. Средние цены оказались одинаковыми. Значения дисперсий: 144 ₽^{2} для гибискуса, и 225 ₽^{2} для гортензии. Какое из утверждений верное?

   A. Цена гибискуса в 2024 году была более изменчива, чем цена гортензии

   B. Цена гортензии в 2024 году была более изменчива, чем цена гибискуса

   C. Ежемесячные цены на гортензию и гибискус всегда одинаковы

   D. Стандартные отклонения цен на гибискус и гортензию одинаковы

5. Студент построил 95%-й доверительный интервал для средней цены на паровой утюг и получил оценку от 1 500 до 7 500 ₽. Какая гипотеза не будет отвергнута на уровне значимости 5% при использовании двусторонней альтернативной гипотезы?

   A. Дисперсия цены на паровой утюг равна 3 000 ₽

   B. Средняя цена парового утюга равна 2 700 ₽

   C. Средняя цена парового утюга равна 1 300 ₽

   D. Стандартное отклонение цены на паровой утюг равно 3 000 ₽

6. 15 учеников прошли тест. Среднее значение набранных баллов у 10 учеников получилось 65. Оказалось, что 5 работ преподаватель забыл внести в ведомость. После их внесения среднее по всем 15 работам стало равно 90 баллам. Максимум за тест можно набрать 100 баллов. Какие выводы можно сделать о добавленных пяти работах?

   A. Преподаватель забыл добавить работы отличников, набравших максимум

   B. Преподаватель неправильно рассчитал среднее

   C. Среднее число баллов в добавленных пяти работах равно 90

   D. Стандартное отклонение набранных баллов одним учеником в среднем равно 25 баллам

7. Сопоставьте следующие данные о студентах, собранные в рамках опроса, и типы признаков, которыми могут быть описаны эти данные.

   Пол (мужской/женский) — категориальный

   Средний балл (от 0 до 10) — непрерывный

   Количество пропущенных занятий (целое число) — дискретный

   Уровень удовлетворенности (цифра от 1 до 5) — порядковый

8. Оля из города N опросила своих друзей в социальных сетях, готовы ли они купить лабубу. У Оли 700 друзей в соцсетях, из них 643 проживают с ней в одном городе. Результаты опроса такие: 441 человек готовы купить лабубу, 59 человек — нет, 100 затруднились ответить. Какие выводы можно сделать по результатам опроса?

   A. В среднем 73,5% населения города N готовы купить лабубу

   B. В среднем 63% населения города N готовы купить лабубу

   C. Полученной информации недостаточно для открытия магазина на всероссийском маркетплейсе

   D. В городе N примерно 68% готовы купить лабубу

9. Ваш друг рассматривает инвестиционные проекты для вложения сбережений в размере 1 000 коинов. Он рассматривает банковский депозит на 6 месяцев под годовую ставку доходности 12% и частный проект с ежемесячной доходностью в ближайшие 6 месяцев +2%, +3%, -5%, +1%, +0,5%, +5%. Куда вы посоветуете другу вложить свои сбережения? Предполагается, что процентный доход ежемесячно выплачивается сразу после начисления (капитализация процентного дохода отсутствует).

   A. Вложиться в частный проект: доход в нем за полгода будет больше, чем на депозитном вкладе

   B. Открыть депозит: он принесет больший доход, чем частный проект

   C. Оба варианта равнозначны

   D. Недостаточно информации для принятия решения, так как нужно учитывать капитализацию

10. Заводы автомобильной краски сговорились и последние 10 лет производили преимущественно черную краску. Доля объема поставленной автоконцернам черной краски составила 80%. На остальные 20 цветов пришлись оставшиеся 20%. Конверсия в покупку автомобиля каждого цвета на рынке одинаковая. Маркетологи опубликовали исследование, в котором привели набор тезисов.

    Выберите один или несколько верных тезисов маркетологов.

    A. Потребители предпочитают черный цвет автомобиля остальной палитре

    B. Доля красных машин составляет 1% или более

    C. Потребителям не нравятся яркие цвета автомобилей

D. Большинство продающихся автомобилей черного цвета

E. Нет верных ответов среди вышеперечисленных вариантов

11. В колл-центре желтого банка работают только два сотрудника: Маша и Степа. Маша тратит на одного клиента 1 минуту, Степа — 3 минуты. Время рабочей смены — 3 часа. Сколько в среднем длится время обслуживания одного клиента в колл-центре?

A. 2 минуты

B. 45 секунд

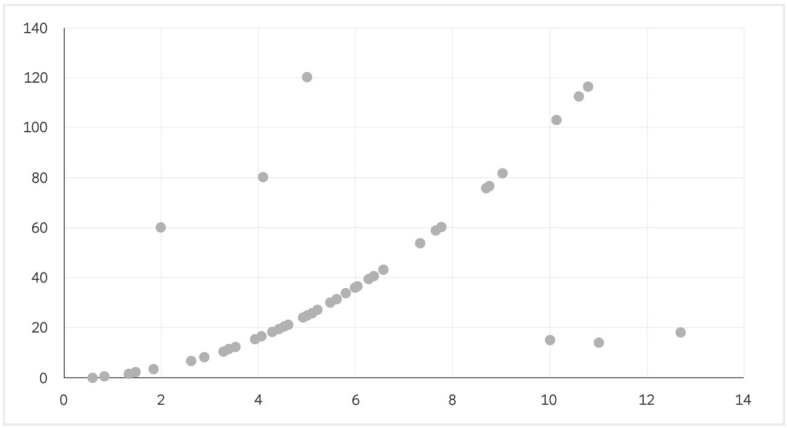
C. 1 минуту

D. 1 минуту 30 секунд

12. В лавке у разбойника лежит 15 острых ножей, расположенных так, что каждый следующий нож на 1 см длиннее предыдущего. Разбойник рассказал, что среднее арифметическое длины первых 11 ножей равно 12, а среднее арифметическое последних 11 ножей — 16. Какова длина наиболее длинного ножа у разбойника?

Ответ: 21

13. Директор цеха составил график зависимости оплаты труда сотрудников всех профессий в цеху (в тысячах рублей по вертикальной оси) от длины рабочей смены (в часах по горизонтальной оси). Выберите верные утверждения, которые соответствуют построенному графику.



A. Между длиной рабочей смены и оплатой существует линейная положительная связь

B. Коэффициент корреляции Спирмена между длиной рабочей смены и оплатой отрицательный

C. В цеху трое перерабатывают и им недоплачивают, а трое недорабатывают и им переплачивают

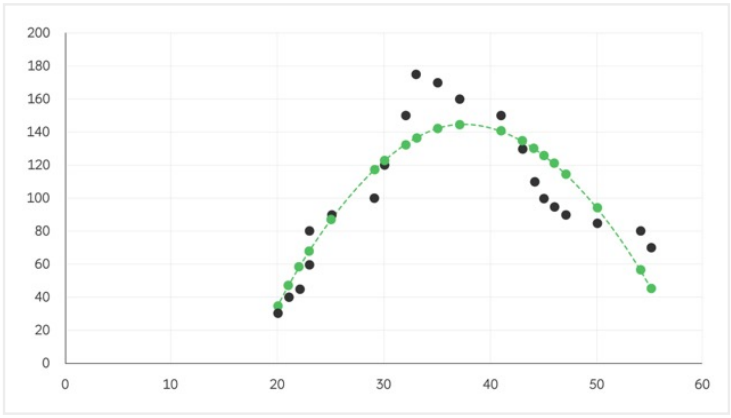
D. Зависимость между оплатой труда и длительностью рабочей смены можно описать нелинейной функцией, исключив выбросы

14. Аналитики компании решили оценить зависимость заработной платы сотрудников от их возраста. Для этого они оценили регрессионную модель:

    Зарплата = –344.78 + 25.76 * возраст – 0.34 * возраст^2

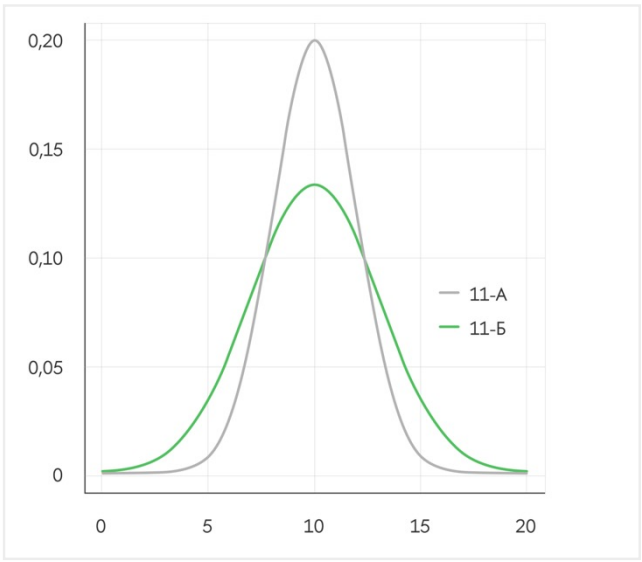
    Коэффициент детерминации R^2 модели оказался равен 0,79.

    Результаты аналитики представили на графике: черным цветом — фактические данные, зеленым цветом — оцененную модель.



Ответ: 38

15. Саша построил распределения количества игр в библиотеке магазина «Стем» для учащихся 11 «А» и 11 «Б». В результате он получил такой график распределений.



Какие выводы может сделать Саша из этого графика?

A. В среднем в обоих классах по 10 игр у каждого учащегося

B. В 11 «Б» разброс числа игр у учащихся больше, чем в 11 «А»

C. Медиана в 11 «А» не равна моде в 11 «Б»

D. Больше половины учащихся 11 «А» имеют в своей библиотеке 7 и более игр

16. Фанатам творчества Михаила Афанасьевича Булгакова задали вопрос: сколько литров подсолнечного масла нужно разлить Аннушке в зависимости от роста повстречавшегося ей незнакомца? Результаты опроса представили в таблице с усредненными значениями.

Рост	**Средний объем масла (по данным опроса)**
Низкий	1,5 л
Средний	2 л
Высокий	1 л
Очень высокий	3 л

При помощи каких статистических инструментов можно подтвердить или опровергнуть наличие зависимости между объемом пролитого масла и ростом незнакомца?

A. Коэффициент корреляции Спирмена

B. F-тест на равенство дисперсий

C. Коэффициент корреляции R^2

D. T-тест

17. Аналитик получил датасет со следующими переменными: пол, возраст, количество братьев и (или) сестер, стиль вождения. Какие из этих переменных имеет смысл использовать в качестве зависимых, а какие — в качестве регрессоров?

A. Пол — зависимая переменная, а остальные в качестве регрессоров

B. Возраст — зависимая переменная, а остальные в качестве регрессоров

C. Количество братьев и (или) сестер — зависимая переменная, а остальные в качестве регрессоров

D. Стиль вождения в качестве зависимой переменной, а остальные в качестве регрессоров

18. Вам нужно визуализировать распределение брендов смартфонов у целевой аудитории вашего блога. Какой (-ие) тип (-ы) визуального представления вы выберете?

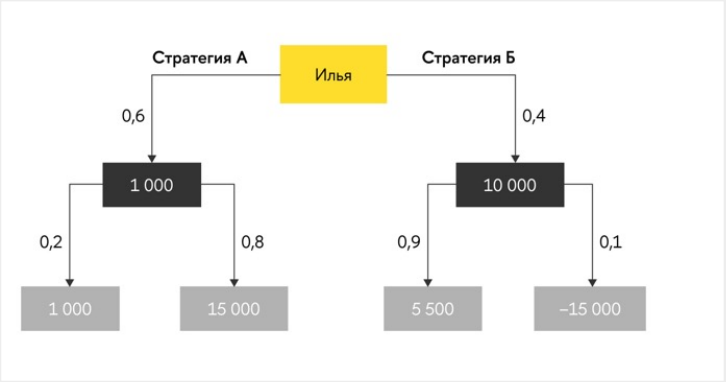
A. Диаграмма рассеяния

B. Столбчатая диаграмма

C. Гистограмма

D. Круговая диаграмма

19. Илья руководит стартапом по анализу данных. Ему нужно выбрать одну из дальнейших стратегий развития стартапа: А или Б. Обычно руководители стартапов с вероятностью 0,6 выбирают стратегию А и с вероятностью 0,4 — стратегию Б.



Каждая стратегия состоит из двух этапов. Стратегия А на первом этапе принесет 1 000 бонусов, а на втором — с вероятностью 0,2 еще 1 000 бонусов или с вероятностью 0,8 еще 15 000 бонусов. Стратегия Б на первом этапе принесет 10 000 бонусов, а на втором — с вероятностью 0,9 еще 5 500 бонусов или с вероятностью 0,1 Илья потеряет 15 000 бонусов.

Какую стратегию стоит выбрать Илье?

A. А

B. Стратегии равноценны

C. Б

D. Недостаточно информации для принятия решения

20. Зарплата аналитика за последние 4 года менялась так: 150 000, 200 000, 230 000, 300 000 ₽. Чему равен средний темп прироста зарплаты аналитика в год в процентах? Укажите ответ одним числом без дополнительных знаков и символов, округлив до целого значения.

Ответ: 26

21. Перед вами таблица сопряженности двух признаков: пол и сфера работы.

	Образование	**Медицина**	**Финансы**
Муж.	350	215	460
Жен.	500	210	370

Какие выводы об имеющейся выборке можно сделать на основе этой таблицы?

A. Вероятность, что случайно выбранный человек из выборки работает в сфере финансов, выше, чем вероятность, что он работает в образовании

B. Мужчин меньше в выборке, чем женщин

C. Сфера образования и финансы одинаково представлены в выборке

D. Больше трети мужчин выборки работают в сфере образования

E. Вероятность работать в медицине выше среди мужчин, чем среди женщин

22. Ангелина работает врачом и сталкивается с пациентами, от которых можно заразиться тяжелыми заболеваниями. Вероятность остаться здоровой при контакте с зараженным равна 0,95. Раз в 6 дней в больницу, где работает Ангелина, попадает пациент с такой болезнью, и Ангелина с ним контактирует. Определите, через сколько дней вероятность, что Ангелина заразится болезнью, станет выше, чем вероятность не заразиться.

    Укажите ответ одним числом без дополнительных знаков и символов, округлив в случае необходимости до целого значения.

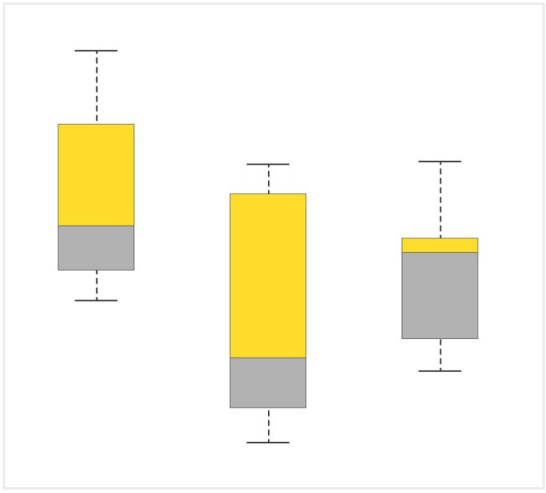
    Ответ: 84

23. В соревнованиях по программированию участвовали 120 человек. Медиана набранных баллов среди участников — 50 баллов, а среднее значение баллов — 45. Организаторы решили добавить 20 дополнительных баллов самым лучшим участникам, чтобы поощрить их.

    Какое минимальное число участников нужно наградить дополнительными баллами, чтобы средний балл стал равен медиане (50 баллов)?

    Ответ: 30

24. Перед вами графики Boxplot для трех выборок с параметром длины усов = 1,5 × IQR.



    Выберите верные утверждения, опираясь на эти графики:

    — Дисперсия третьей выборки меньше, чем первых двух

    — Медианное значение во второй выборке наименьшее из представленных

    — В выборках нет выбросов, превышающих 1,5 × IQR

    — Максимальное из всех значений по трем выборкам находится в первой из них

25. В школьном кружке по анализу данных провели опрос среди 100 учащихся. Каждый ученик должен был ответить на два вопроса:

    1) Любите ли вы математику?

    2) Любите ли вы информатику?

    Анкеты обработали и получили следующие результаты:

    — 60 человек ответили, что любят математику;

    — 55 человек ответили, что любят информатику;

    — 20 человек ответили, что не любят ни математику, ни информатику.

    Сколько учеников любят и математику, и информатику?

    Ответ: 35

26. Компания занимается онлайн-продажами. Каждый клиент проходит несколько этапов:

    1. Посещение сайта.
    2. Добавление товара в корзину.
    3. Оформление заказа.
    4. Оплата заказа.

    За один день аналитики собрали информацию:

    — На сайт зашло 10 000 пользователей.

    — 30% из всех, кто заходил на сайт, оформили заказ.

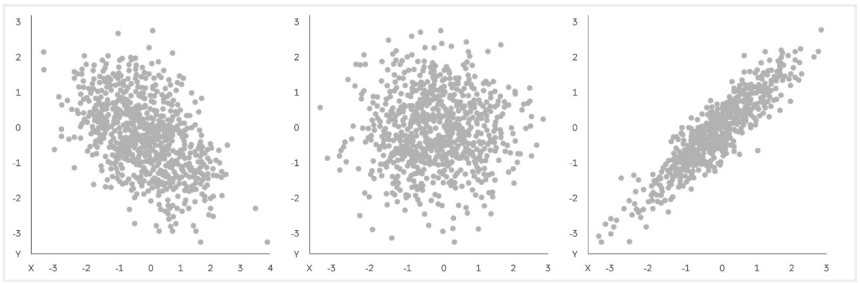
    — 2 500 клиентов оплатили заказ.

    — Только 50% тех, кто добавил товар в корзину, оплатили его.

    Рассчитайте долю оформивших заказ клиентов среди пользователей, добавивших товар в корзину. Ответ округлите до второго знака после запятой.

    Ответ: 0,6 или 60

27. Перед вами три диаграммы рассеяния. Сопоставьте их с коэффициентами корреляции, которые получены на этих выборках. В ответ запишите буквы, в порядке расположения картинок.



А. 0,5

Б. 0

В. 0,8

Г. −0,5

Ответ: 1Г, 2Б, 3В, 4А

28. Перед вами иллюстрация половозрастной пирамиды России с указанием работающего (занятого) населения. Выберите верные утверждения для этой картинки.

— Доля занятых женщин в возрасте 30—34 года выше, чем у мужчин в том же возрасте

— Среди мужчин трудовых мигрантов больше, чем среди женщин

— Максимальная доля работающего населения среди мужчин приходится на возрастную группу от 25 до 29 лет

— В возрасте 75—79 лет количество женщин примерно вдвое больше, чем мужчин

— После 75 лет население состоит из женщин менее чем на 50%

29. Выберите из списка все корректные методы борьбы с выбросами в данных

A. Бутстрап

B. Удаление 1% значений

C. Удаление значений, которые отклоняются сильнее чем 3 × IQR

D. Стандартизация исходной метрики

DANO

ВЫСШАЯ ШКОЛА ЭКОНОМИКИ

Т БАНК

Вариант 2

1. По выборке из 144 значений посчитали выборочную дисперсию. После этого все значения увеличили в 10 раз. Как изменится выборочная дисперсия, если вычислить ее по новым значениям выборки?

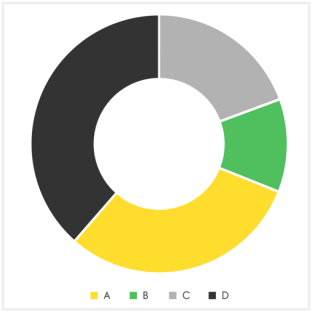
   A. Не изменится

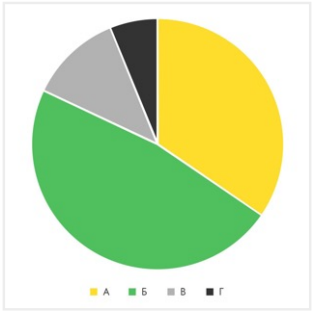
   B. Увеличится в 10 раз

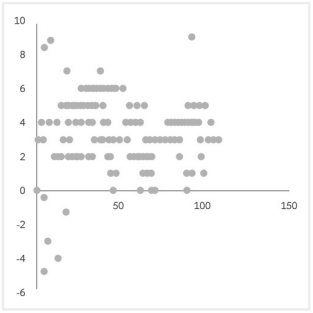
   C. Увеличится в 100 раз

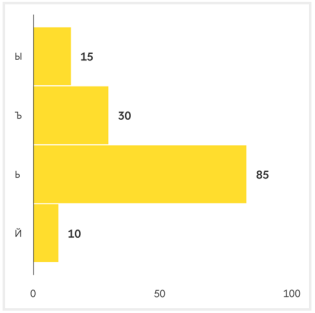
   D. Изменится в 0,1 раза

2. Сопоставьте названия типов визуализации с графиками:

1. 

2. 

3. 

4. 

1. Кольцевая диаграмма
2. Секторная диаграмма
3. Диаграмма рассеяния
4. Гистограмма

3. Исследователи проверили гипотезу о среднем значении с использованием Z-теста. Уровень значимости установлен равным 0,01. Как интерпретировать такое значение уровня значимости?

   A. Вероятность того, что альтернативная гипотеза верна, равна 1%

   B. Допустимая вероятность отвержения альтернативной гипотезы, если она на самом деле верна, равна 1%

   C. Если нулевая гипотеза не верна, в 1% случаев мы допускаем, что можем ошибочно принять ее при многократном повторении теста на семплированных данных из одной генеральной совокупности

   D. Вероятность ошибки первого рода равна 1%

   E. Если тест показал p-value = 0,1, вероятность того, что нулевая гипотеза верна, равна 1%

4. Сотрудники отдела прогнозирования и анализа данных изучили ежемесячные данные по ценам на рис и гречку за 2021 год. Средние цены оказались одинаковыми. Значения дисперсий: 64 ₽^{2} для риса и 82 ₽^{2} для гречки. Какое из утверждений верное?

   A. Цена риса в 2021 году была более изменчива, чем цена гречки

   B. Цена гречки в 2021 году была более изменчива, чем цена риса

   C. Ежемесячные цены на рис и гречку всегда одинаковы

   D. Стандартные отклонения цен на рис и гречку одинаковы

5. Влад построил 80%-й доверительный интервал для средней цены пары кроссовок «Майк» и получил оценку от 7 000 до 15 000 ₽. Какая гипотеза не отвергнется на 20%-м уровне значимости при использовании двусторонней альтернативной гипотезы?

   A. Дисперсия цены пары кроссовок «Майк» равна 4 000 ₽

   B. Средняя цена пары кроссовок «Майк» равна 2 700 ₽

   C. Средняя цена пары кроссовок «Майк» равна 9 535 ₽

   D. Стандартное отклонение цены пары кроссовок «Майк» равно 4 000 ₽

6. В автошколе после тестирования инструктор оценил среднее значение баллов за тест, набранных 10 сдавшими, и получил 85 баллов. Оказалось, что 5 работ инструктор забыл внести в ведомость. После их внесения среднее по всем 15 работам оказалось равно 90 баллам. Максимум за тест можно было набрать 100 баллов. Что можно сказать о добавленных пяти работах?

   A. Инструктор забыл добавить работы учеников, набравших максимум баллов

   B. Инструктор неправильно рассчитал среднее

   C. Среднее число баллов в добавленных пяти работах равно 99

   D. Стандартное отклонение набранных баллов одним учеником в среднем равно 10 баллам

7. Сопоставьте следующие данные о сотрудниках, собранные в рамках опроса, и типы признаков, которыми могут быть описаны эти данные.

   Должность (менеджер/разработчик/аналитик) — категориальный

   Процент надбавки от фиксированной заработной платы — непрерывный

   Количество рабочих дней в командировках — дискретный

   Уровень стресса (целое число от 1 до 10) — порядковый

8. Алексей из Москвы провел опрос среди своих подписчиков в соцсети, чтобы узнать, будут ли они покупать новый кофейный напиток. Всего у Алексея 1 000 подписчиков, 850 из которых живут в Москве. Результаты опроса такие:

   — 620 человек ответили «Да, куплю»

   — 150 человек сказали «Нет»

   — 230 человек затруднились с ответом

   Какие выводы можно сделать по результатам этого опроса?

   A. В среднем в России 62% потребителей готовы купить новый кофейный напиток

   B. В среднем в России 15% потребителей готовы купить новый кофейный напиток

   C. В Москве примерно 73% потребителей готовы купить новый кофейный напиток

   D. Полученных данных недостаточно для запуска продаж по всей стране

9. У Кати есть 10 000 ₽, и она думает об открытии депозита на 6 месяцев под годовую ставку 20%. Сотрудники банка предложили ей инновационный продукт, ежемесячная доходность которого будет такая: +10%, +1%, –3%, +12%, +1%, –2%. Куда стоит вложить средства Кате? Предполагается, что процентный доход ежемесячно выплачивается после начисления (капитализация процентного дохода отсутствует).

   A. Стоит выбрать инновационный продукт: так Катя сможет заработать через полгода больше, чем на вкладе в депозит

   B. Стоит открыть депозит: он принесет больший доход, чем инновационный продукт

   C. Все равно, куда вкладываться

   D. Недостаточно информации для принятия решения, так как нужно учитывать капитализацию

10. Производители чехлов для смартфонов последние 8 лет по негласному соглашению выпускали преимущественно черные чехлы. В результате доля черных чехлов в поставках ретейлерам составила 85%, а на все остальные цвета (белый, синий, красный и др.) пришлось только 15%. Конверсия в покупку чехлов каждого цвета одинаковая. Аналитики рынка опубликовали исследование с несколькими утверждениями. Какие из этих утверждений можно считать достоверными на основе приведенных данных?

    A. Покупатели предпочитают черные чехлы для смартфонов всем другим цветам

    B. Доля синих чехлов составляет 2% или более

C. Потребителям не нравятся яркие чехлы для телефонов

D. Большинство проданных чехлов черного цвета

E. Нет верных ответов среди вышеперечисленных вариантов

11. На космической станции связи работают два робота-оператора: R2 и D2. R2 тратит 2 минуты, чтобы обработать один запрос. D2 тратит на один запрос 6 минут. Их рабочая смена длится 4 часа. Сколько в среднем длится обработка запроса на космической станции?

A. 3 минуты

B. 3 минуты 30 секунд

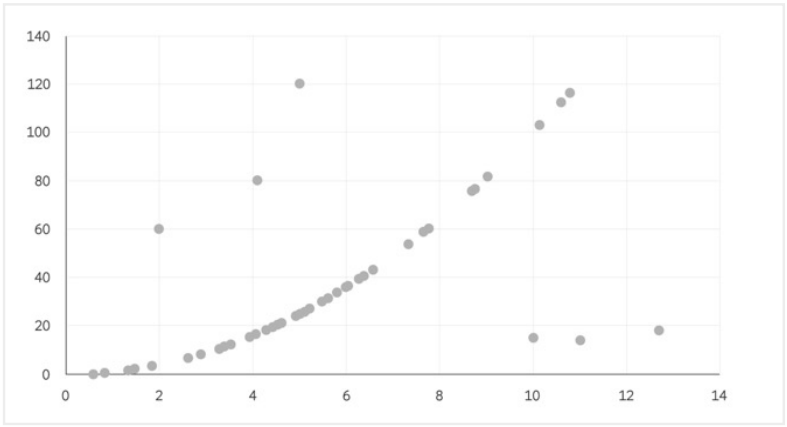
C. 5 минут

D. 1 минуту 30 секунд

12. В сказочной стране Конфетляндии у волшебника есть 15 волшебных конфет, выстроенных в ряд так, что каждая следующая конфета весит на 1 г больше предыдущей. Известно, что средний вес первых 11 конфет равен 10 г, а средний вес последних 11 конфет — 14 г. Найдите вес самой тяжелой конфеты в этом ряду.

Ответ: 19

13. Директор цеха по производству шпулек составил график зависимости оплаты труда сотрудников всех профессий в цеху (в тысячах рублей по вертикальной оси) от длины рабочей смены (в часах по горизонтальной оси). Выберите верные утверждения, которые соответствуют построенному графику.



A. Между длиной рабочей смены и оплатой есть линейная отрицательная связь

B. Коэффициент корреляции Спирмена между длиной рабочей смены и оплатой нулевой

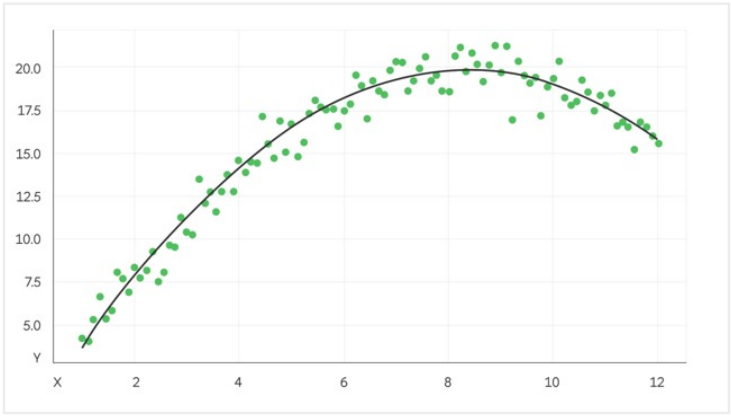
C. В цеху трое перерабатывают и им недоплачивают

D. Зависимость между оплатой труда и длительностью рабочей смены можно описать нелинейной функцией, исключив выбросы

14. Аграрная компания провела исследование, чтобы определить, как количество внесенных удобрений (килограмм на гектар) влияет на урожайность винограда (килограмм с гектара). Для анализа данных оценили квадратичную регрессионную модель:

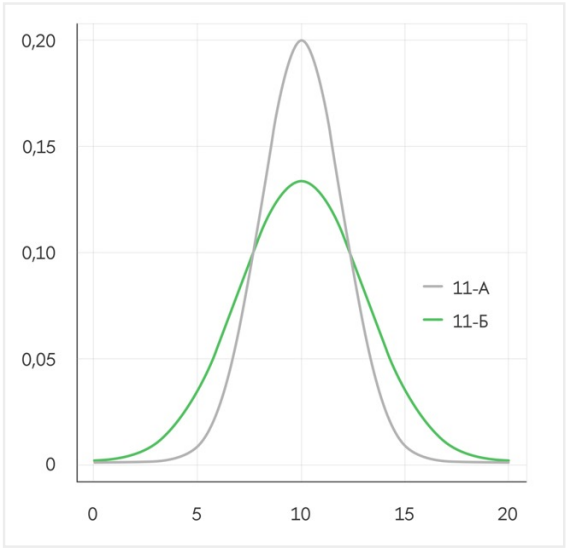
    Урожайность = $-0{,}3 \times \text{Удобрения}^2 + 5 \times \text{Удобрения} - 1$

    Коэффициент детерминации $R^2 = 0{,}82$. Определите оптимальное количество удобрений, при котором урожайность достигает максимума, согласно оцененной модели. На графике зеленым цветом показаны фактические данные, черным — оцененная зависимость.



Ответ: 8

15. Дима построил распределения количества игр в библиотеке магазина «Стем» для учащихся 11 «А» и 11 «Б». Получился такой график распределений.



A. В обоих классах среднее число игр у одного учащегося равно 10

B. В 11 «Б» разброс числа игр у учащихся меньше, чем в 11 «А»

C. Медиана в 11 «А» равна моде в 11 «Б»

D. Больше половины учащихся 11 «А» обладают 8 и более играми

16. Любителей творчества Чарльза Диккенса спросили: сколько чашек чая выпивает мисс Хэвишем в зависимости от количества гостей на ее знаменитых чаепитиях? В результате получилась следующая таблица с усредненными значениями

Количество гостей	**Среднее количество выпитых чашек чая (оцененное по данным опроса)**
Мало	5 чашек
Средне	4 чашки
Много	3 чашки
Очень много	2 чашки

При помощи каких статистических инструментов можно подтвердить или опровергнуть наличие зависимости между количеством выпитых чашек чая и количеством гостей на чаепитии у мисс Хэвишем?

A. Коэффициент корреляции Спирмена

B. F-тест на равенство дисперсий

C. Z-тест

D. U-тест Манна — Уитни

17. Исследователь получил набор данных о вкусовых предпочтениях мороженого анкетируемых со следующими переменными: цвет упаковки, возраст потребителя, уровень дохода и предпочитаемый вкус мороженого. Какие из этих переменных имеет смысл использовать в качестве зависимых, а какие в качестве независимых?

A. Цвет упаковки как независимая переменная, а остальные как зависимые

B. Возраст потребителя как зависимая переменная, а остальные как независимые

C. Уровень дохода как независимая переменная, а остальные как зависимые

D. Предпочитаемый вкус мороженого как зависимая переменная, а остальные как независимые

18. Специалист по флоре тропического леса хочет визуализировать распределение различных видов растений в исследуемой области. Какой тип визуализации лучше всего подходит для этой задачи?

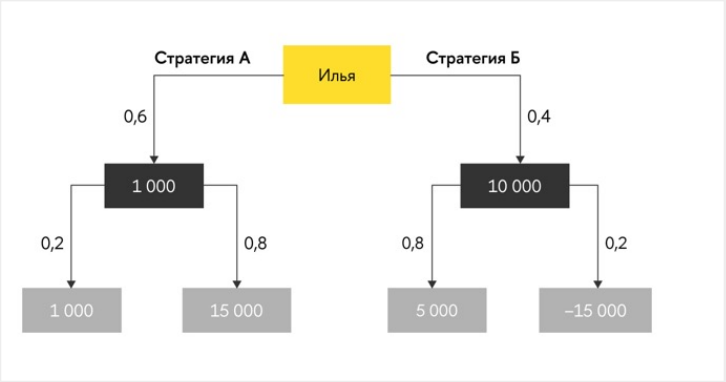
A. Тепловая карта

B. Линейный график

C. Диаграмма Венна

D. Столбчатая диаграмма

19. Илья руководит стартапом по анализу данных. Ему нужно выбрать одну из дальнейших стратегий развития стартапа: А или Б. Обычно руководители стартапов с вероятностью 0,6 выбирают стратегию А и с вероятностью 0,4 — стратегию Б.



Каждая стратегия состоит из двух этапов. Стратегия А на первом этапе принесет 1 000 бонусов, а на втором — с вероятностью 0,2 еще 1 000 бонусов или с вероятностью 0,8 еще 15 000 бонусов. Стратегия Б на первом этапе принесет 10 000 бонусов, а на втором — с вероятностью 0,8 еще 5 000 бонусов или с вероятностью 0,2 Илья потеряет 15 000 бонусов.

Какую стратегию стоит выбрать Илье?

A. А

B. Стратегии равноценны

C. Б

D. Недостаточно информации для принятия решения

20. Владелец небольшого кафе фиксировал количество посетителей каждый год в течение последних 4 лет и получил такие данные по годам: 1 200 посетителей, 1 500, 1 800, 2 250. Чему равен средний темп прироста числа посетителей кафе за год в процентах? Укажите ответ одним числом без дополнительных знаков и символов, округлите ответ до целого.

Ответ: 23

21. Аналитики исследовали, каких домашних животных предпочитают жители города. Получилась такая таблица сопряженности, отражающая связь между типом жилья и предпочтением вида питомца.

Тип жилья	**Кошки**	**Собаки**	**Рыбки**
Квартира	250	180	120
Частный дом	150	320	80

Известно также, что все жители города держат какого-нибудь питомца!

A. Вероятность, что случайно выбранный житель держит собаку, выше, чем вероятность, что он держит кошку

B. Среди жителей частных домов больше тех, кто держит рыбок, чем среди жителей квартир

C. Кошки и рыбки одинаково популярны среди всех жителей

D. Больше половины жителей квартир предпочитают держать кошек

E. Вероятность держать собаку выше среди жителей частных домов, чем среди жителей квартир

22. Леша — профессиональный киберспортсмен и ежедневно участвует в онлайн-турнирах. Вероятность того, что его аккаунт не будет взломан при попытке взлома, составляет 0,98. Раз в 4 дня фиксируется попытка взлома аккаунтов всех участников турнира. Через сколько дней вероятность того, что аккаунт Алексея будет взломан, станет выше, чем вероятность, что этого не произойдет?

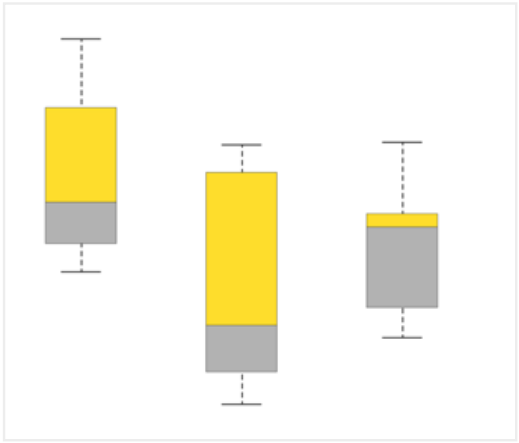
Укажите ответ одним числом без дополнительных знаков и символов, округлив в случае необходимости до целого значения.

Ответ: 140

23. В выборке из 100 значений медиана равна 50, а среднее — 45. Вася может увеличить X чисел в выборке на 50. Какого минимального X достаточно, чтобы среднее стало равно медиане, равной 50?

Ответ: 10

24. Перед вами графики Boxplot для трех выборок с параметром длины усов = 1,5 × IQR. Выберите верные утверждения.



— Дисперсия второй выборки больше, чем двух других

— Медианное значение в третьей выборке наименьшее из представленных графиков

— В выборках нет выбросов, превышающих 1,5 × IQR

— Максимальное значение из всех значений по трем выборкам находится в первой выборке

25. 100 ученикам школы задали два вопроса:

    1) Занимаетесь ли вы спортом?

    2) Играете ли вы на музыкальных инструментах?

    По результатам опроса 50 учеников занимаются спортом; 45 учеников играют на музыкальных инструментах; 25 учеников не занимаются ни спортом, ни музыкой.

    Сколько учеников и занимаются спортом, и играют на музыкальных инструментах?

    Ответ: 20

26. Компания проводит онлайн-продажи, и каждый клиент проходит несколько этапов:

    1. Посещение сайта.
    2. Добавление товара в корзину.
    3. Оформление заказа.
    4. Оплата заказа.

    Аналитики собрали следующую информацию за один день:

    — На сайт зашло 5 000 пользователей.

    — Из всех, кто заходил на сайт, 20% пользователей оформили заказ.

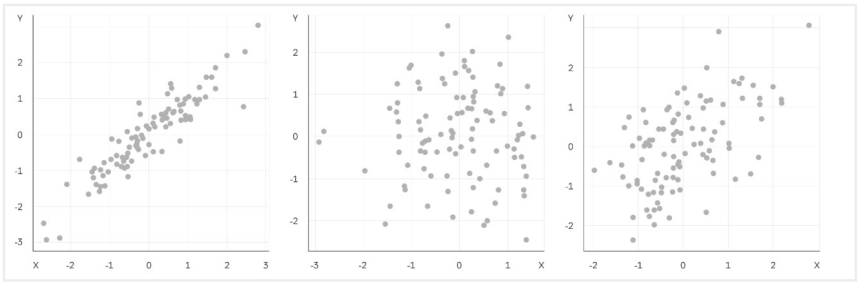
    — 500 клиентов оплатили заказ.

    — Из всех, кто добавил товар в корзину, только 50% его оплатили.

    Рассчитайте долю оформивших заказ клиентов среди пользователей, добавивших товар в корзину. Ответ округлите до второго знака после запятой.

    Ответ: 1 или 100

27. Даны три диаграммы рассеяния. Сопоставьте их с коэффициентами корреляции, которые получены на этих выборках. В ответ запишите буквы в порядке расположения картинок.



    А. –0,4

    Б. 0

    В. 0,9

    Г. 0,4

Ответ: 1В, 2Б, 3Г, 4А

28. Перед вами иллюстрация половозрастной пирамиды России с указанием работающего (занятого) населения. Выберите верные утверждения для этой картинки.

— Доля занятых женщин в возрасте 30—34 года ниже, чем у мужчин в том же возрасте

— Среди мужчин трудовых мигрантов меньше, чем среди женщин

— Максимальная доля работающего населения среди мужчин приходится на возрастную группу от 30 до 34 лет

— В возрасте 75—79 лет количество женщин примерно вдвое больше, чем мужчин

— После 75 лет население состоит из женщин более чем на 50%

29. Выберите все варианты ответов, в которых перечислены корректные методы борьбы с выбросами в данных.

A. Бутстрап

B. Использование T-тестов вместо Z-тестов

C. Удаление значений, которые отклоняются сильнее чем 1,5 × IQR

D. Нормализация исходных данных

DANO

ВЫСШАЯ ШКОЛА ЭКОНОМИКИ

Т БАНК

Вариант 3

1. По выборке из 225 значений посчитали выборочную дисперсию. После этого все значения увеличили в 3 раза. Как изменится выборочная дисперсия, если вычислить ее по новым значениям выборки?

   A. Не изменится

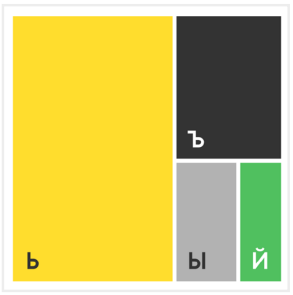
   B. Увеличится в 3 раза

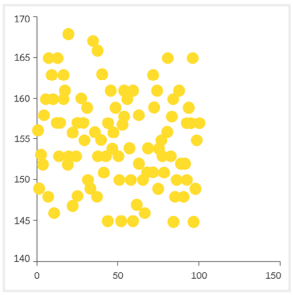
   C. Увеличится в 6 раз

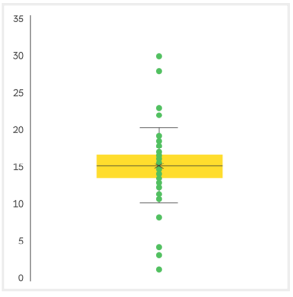
   D. Увеличится в 9 раз

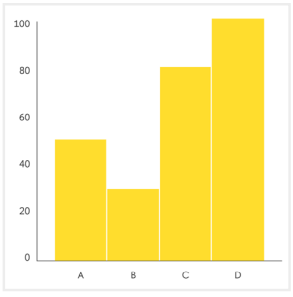
   E. Изменится примерно в 1,732 раза

2. Сопоставьте названия типов визуализации с графиками:

1. 

2. 

3. 

4. 

1 Древовидная диаграмма

2. Диаграмма рассеяния

3. Ящик с усами

4. Гистограмма

3. Исследователи проверили гипотезу о среднем значении с использованием Т-теста. Уровень статистической значимости установлен равным 0,05. Выберите верные варианты ответов.

   A. Вероятность того, что альтернативная гипотеза верна, равна 5%

   B. Допустимая вероятность принятия нулевой гипотезы, если она на самом деле не верна, равна 95%

   C. Если нулевая гипотеза верна, в 5% случаев мы допускаем, что можем ошибочно отвергнуть ее при многократном повторении теста на семплированных данных из одной генеральной совокупности

   D. Вероятность ошибки первого рода равна 5%

   E. Если тест показал p-value = 0,054, вероятность того, что альтернативная гипотеза верна, равна 5%

4. Сотрудники отдела прогнозирования и анализа данных изучили ежемесячные данные по ценам на пшеницу и ячмень за 2022 год. Средние цены оказались одинаковыми. Значения дисперсий: 100 ₽^{2} для пшеницы и 121 ₽^{2} для ячменя. Какое утверждение верное?

   A. Цена пшеницы в 2022 году была более изменчива, чем цена ячменя

   B. Цена ячменя в 2022 году была более изменчива, чем цена пшеницы

   C. Ежемесячные цены на пшеницу и ячмень всегда одинаковы

   D. Стандартные отклонения цен на пшеницу и ячмень одинаковы

5. Маша построила 99%-й доверительный интервал для средней цены на электрический чайник и получила оценку от 1 000 до 5 000 ₽. Какая гипотеза не будет отвергнута на уровне значимости 1% при использовании двусторонней альтернативной гипотезы?

   A. Средняя цена электрического чайника равна 4 500 ₽

   B. Средняя цена электрического чайника равна 9 000 ₽

   C. Стандартное отклонение цены на электрический чайник равно 2 000 ₽

   D. Нет верного варианта ответа

6. В автошколе после тестирования инструктор оценил среднее значение баллов за тест, набранных 10 сдавшими, и получил 80 баллов. Оказалось, что 5 работ инструктор забыл внести в ведомость. После их внесения среднее по всем 15 работам оказалось равно 90 баллам. Максимум за тест можно было набрать 100 баллов. Что можно сказать о добавленных пяти работах?

   A. Инструктор забыл добавить работы учеников, набравших максимум баллов

   B. Инструктор неправильно рассчитал среднее

   C. Среднее число баллов в добавленных пяти работах равно 99

   D. Стандартное отклонение набранных баллов одним учеником в среднем равно 10 баллам

7. Сопоставьте следующие данные о квартирах, собранные в рамках опроса собственников, с типами признаков, которыми они могут быть описаны.

   Количество балконов — дискретный

   Тип дома (кирпичный или не кирпичный) — категориальный

   Степень удаленности от метро (по шкале от 1 до 3) — порядковый

   Высота потолков (в см) — непрерывный

8. Машенька решила узнать у своих друзей в социальной сети, хотят ли они купить эликсир невидимости. Всего у Машеньки 256 онлайн-друзей, из которых 225 живут с ней в одном городе. По результатам опроса 113 друзей готовы купить эликсир, 19 друзей отказались, а 10 затруднились ответить. Какие выводы можно сделать из этого опроса?

   A. В среднем в России 76,3% людей готовы купить эликсир невидимости

   B. В среднем в России 44% людей готовы купить эликсир невидимости

   C. В Машином городе примерно 50% людей готовы купить эликсир невидимости

   D. Нет верного ответа

9. Пирожок-Мастер рассматривает инвестиционные проекты для вложения своих сбережений в размере 5 000 золотых монет. Ему предложили два варианта: банковский депозит на 6 месяцев под годовую ставку доходности 10% и проект по выращиванию волшебных бобов, ежемесячная доходность которого в ближайшие 6 месяцев оценивается как +1%, +2%, –3%, +0,5%, +0,2%, +4%. Куда стоит вложить свои сбережения Пирожку-Мастеру? Предполагается, что процентный доход ежемесячно выплачивается сразу Пирожку-Мастеру и не вкладывается им в проект (капитализация процентного дохода отсутствует).

   A. Стоит поддержать проект по выращиванию волшебных бобов: там можно больше заработать через полгода, чем на депозите

   B. Стоит открыть депозит: он принесет больше, чем проект по выращиванию волшебных бобов

   C. Все равно, куда вкладываться: доходность вариантов одинаковая

   D. Недостаточно информации для принятия решения, так как нужно учитывать капитализацию

10. В сказочном королевстве заговор: последние 10 лет все гномы-мастера выпускают в основном только зонтики с узором в горошек. В результате доля зонтиков в горошек на полках в магазинах составила 90%, а на все остальные узоры пришлось только 10%. Конверсия в покупку зонтиков с разными расцветками одинаковая. Королевские аналитики опубликовали исследование, в котором привели набор тезисов. Выберите один или несколько верных тезисов аналитиков.

    A. Жители королевства предпочитают зонтики в горошек всем остальным узорам

    B. Доля зонтиков в полоску составляет не менее 1%

    C. Жителям королевства не нравятся яркие узоры на зонтиках

D. Большинство продающихся зонтиков в королевстве имеют узор в горошек

E. Нет верных ответов среди вышеперечисленных вариантов

11. В колл-центре работают два оператора: Алина и Борис. Алина тратит на одного клиента 10 минут, а Борис, который любит подробно отвечать на каждый вопрос, — 40 минут. Их рабочая смена длится 4 часа. Сколько в среднем длится обслуживание одного клиента в колл-центре?

A. 20 минут

B. 16 минут

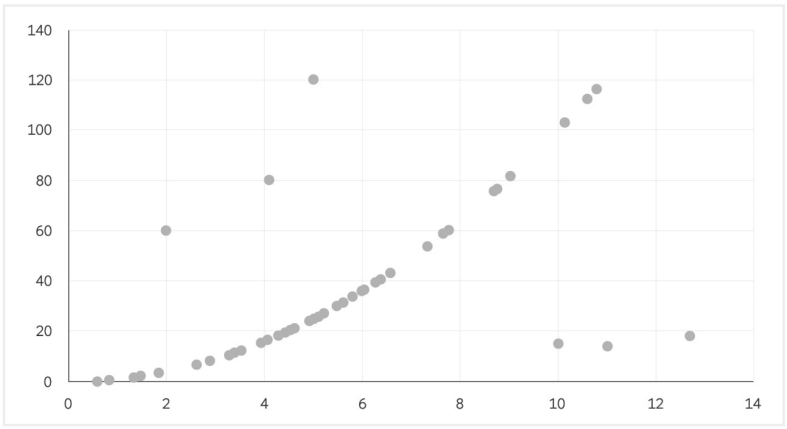
C. 8 минут

D. 30 минут

12. В личной библиотеке Даши есть 17 книг, выстроенных на полке так, что каждая следующая книга на 1 страницу толще предыдущей. Даша рассказала, что среднее арифметическое числа страниц первых 7 книг из них равно 47, а среднее арифметическое следующих 7 — 54. Сколько страниц в самой толстой книге в библиотеке Даши?

Ответ: 60

13. Директор цеха составил график зависимости оплаты труда сотрудников всех профессий в цеху (в тысячах рублей по вертикальной оси) от длины рабочей смены (в часах по горизонтальной оси). Выберите верные утверждения, которые соответствуют построенному графику.



A. Между длиной рабочей смены и оплатой есть нелинейная положительная связь

B. Коэффициент корреляции Спирмена между длиной рабочей смены и оплатой равен нулю

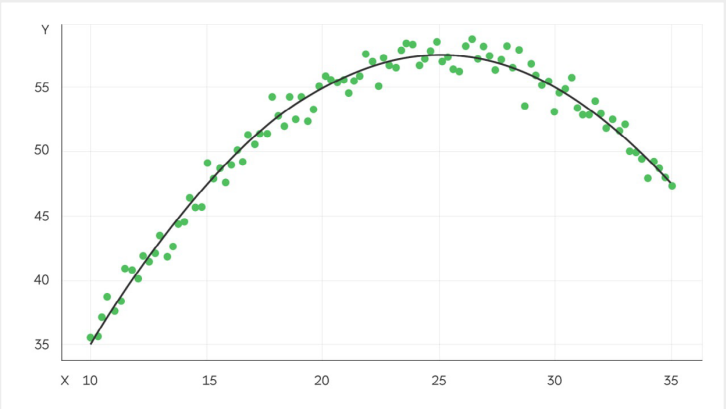
C. В цеху трое недорабатывают и им переплачивают

D. Зависимость между оплатой труда и длительностью рабочей смены нельзя описать нелинейной функцией, исключив выбросы

14. Спортивные аналитики изучили, как температура воздуха во время забега влияет на среднюю скорость марафонцев (в километрах в час). Для этого они оценили квадратичную модель:

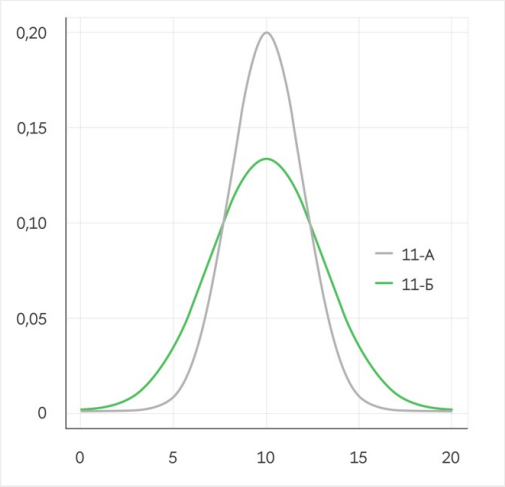
Средняя скорость = $-0{,}1 \times \text{Температура}^2 + 5 \times \text{Температура} - 5$

Коэффициент детерминации $R^2 = 0{,}76$. Определите температуру, при которой средняя скорость марафонцев достигает максимума. На графике: зеленым — наблюдаемые значения, черным — оцененная модель.



Ответ: 25

15. Денис построил распределения количества игр в библиотеке магазина «Стем» для учащихся 11 «А» и 11 «Б». В результате он получил такой график распределений.



A. Среднее число игр у одного учащегося равно 5 в обоих классах

B. В 11 «Б» разброс числа игр у учащихся больше, чем в 11 «А»

C. Мода в 11 «А» равна медиане в 11 «Б»

D. Больше половины учащихся 11 «А» обладают 15 и более играми

16. В сказочном лесу провели исследование и обнаружили: число ягод, собранных эльфами, зависит от времени суток. Данные были собраны с помощью волшебных часов, которые автоматически записывают количество ягод, собранных в разное время суток. В результате получили такое распределение средних значений.

Время суток	**Среднее количество собранных ягод**
Утро	15
День	12
Вечер	10
Ночь	8

При помощи каких статистических инструментов можно подтвердить или опровергнуть наличие зависимости между количеством собранных ягод и временем суток?

A. Коэффициент корреляции Спирмена

B. Тест Стьюдента

C. Тест Манна-Уитни

D. Логистическая регрессия

17. Ученый получил набор данных со следующими переменными: тип тренировки, возраст спортсмена, спортивные достижения и количество тренировок в неделю. Какие из этих переменных имеет смысл использовать в качестве зависимых, а какие — в качестве независимых?

A. Тип тренировки как независимая переменная, а остальные как зависимые

B. Возраст спортсмена как зависимая переменная, а остальные как независимые

C. Спортивные достижения как зависимая переменная, а остальные как независимые

D. Количество тренировок в неделю как независимая переменная, а остальные как зависимые

18. Ученый-эколог изучает зависимость уровня загрязнения воздуха от количества автомобилей на дорогах города. Он собрал данные о концентрации вредных веществ в воздухе и количестве автомобилей в разные дни недели. Какой тип визуализации лучше остальных подходит для анализа этой зависимости?

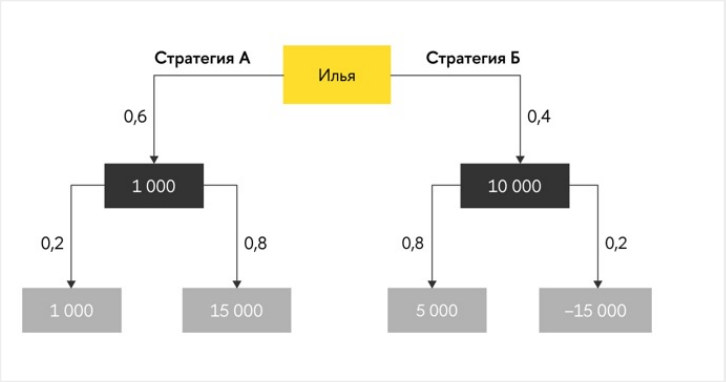
A. Диаграмма рассеяния

B. Столбчатая диаграмма

C. Гистограмма

D. Круговая диаграмма

19. Илья руководит стартапом по анализу данных. Ему нужно выбрать одну из дальнейших стратегий развития стартапа: А или Б. Обычно руководители стартапов с вероятностью 0,6 выбирают стратегию А и с вероятностью 0,4 — стратегию Б.



Каждая стратегия состоит из двух этапов. Стратегия А на первом этапе принесет 1 000 бонусов, а на втором — с вероятностью 0,2 еще 1 000 бонусов или с вероятностью 0,8 еще 15 000 бонусов. Стратегия Б на первом этапе принесет 10 000 бонусов, а на втором — с вероятностью 0,8 еще 5 500 бонусов или с вероятностью 0,2 Илья потеряет 15 000 бонусов.

Какую стратегию стоит выбрать Илье?

A. А

B. Стратегии равноценны

C. Б

D. Недостаточно информации для принятия решения

20. В Зеленограде последние 4 года фиксировали количество посаженных деревьев в рамках городской программы озеленения:

2021 год — 500 деревьев;

2022 год — 650 деревьев;

2023 год — 780 деревьев;

2024 год — 975 деревьев.

Чему равен средний темп прироста количества посаженных деревьев за год в процентах? Укажите ответ одним числом без дополнительных знаков и символов, округлите его до целого.

Ответ: 25

21. Аналитики изучали предпочтения транспорта среди жителей города. По результатам они составили таблицу сопряженности, которая отражает связь между возрастной группой и предпочитаемым видом транспорта.

Возрастная группа	Метро	Наземный транспорт	Личный автомобиль
Молодежь (18—25)	300	150	100
Взрослые (26—50)	250	200	300
Пожилые (51+)	100	250	50

Какие выводы можно сделать на основе данной таблицы?

A. Вероятность, что случайно выбранный житель предпочитает наземный транспорт, выше, чем вероятность, что он предпочитает метро

B. Среди молодежи меньше тех, кто пользуется личным автомобилем, чем среди взрослых

C. Метро и личный автомобиль одинаково популярны среди всех жителей

D. Больше половины пожилых людей предпочитают наземный транспорт

E. Вероятность пользоваться личным автомобилем выше среди взрослых, чем среди молодежи

22. Иван — фермер, который выращивает редкий сорт яблок. Иногда в сад прилетают вредители и пытаются уничтожить урожай. Вероятность того, что попытка уничтожения урожая окажется неуспешной, составляет 0,92. Раз в 5 дней в его сад прилетают вредители, способные нанести ущерб. Через сколько дней вероятность того, что урожай Ивана будет уничтожен, превысит вероятность, что он останется невредимым?

    Укажите ответ одним числом без дополнительных знаков и символов, округлив в случае необходимости до целого значения.

    Ответ: 45

23. В компании работает 60 сотрудников. Медиана заработных плат составляет 60 тысяч рублей, а средняя зарплата — 58 тысяч рублей. Руководство решило повысить зарплату X сотрудникам с самой низкой зарплатой на 10 тысяч рублей, чтобы улучшить средние показатели по компании.

    Какому минимальному количеству сотрудников (X) нужно повысить зарплату, чтобы средняя стала равна медиане (60 тысяч рублей)?

    Ответ: 12

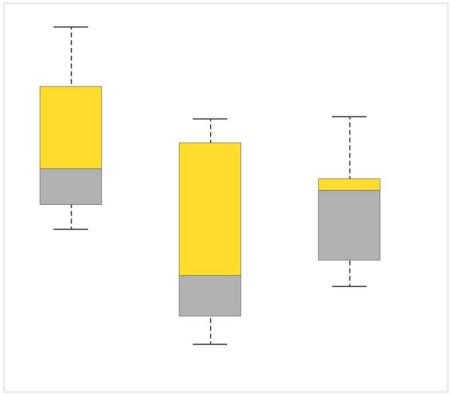
24. Перед вами графики Boxplot для трех выборок с параметром длины усов = 1,5 × IQR. Выберите верные утверждения.

    — Среднее значение сильнее всего отклоняется от медианы в третьей выборке

    — Медианное значение в третьей выборке наибольшее из представленных графиков

    — В выборках есть выбросы, превышающие 1,5 × IQR

    — Минимальное значение из всех значений по трем выборкам находится во второй выборке



25. В библиотеке провели опрос среди 120 посетителей. Им задали два вопроса:

    1) Любите ли вы читать книги?

    2) Увлекаетесь ли вы рисованием?

    По результатам опроса 95 человек любят читать книги; 40 человек увлекаются рисованием; 15 человек не любят ни читать, ни рисовать.

    Сколько посетителей любят и читать, и рисовать?

    Ответ: 30

26. Компания проводит онлайн-продажи, и каждый клиент проходит несколько этапов:

    1. Посещение сайта.
    2. Добавление товара в корзину.
    3. Оформление заказа.
    4. Оплата заказа.

    Аналитики собрали следующую информацию за один день:

    — На сайт зашло 10 000 пользователей.

    — Из всех, кто заходил на сайт, 80% пользователей оформили заказ.

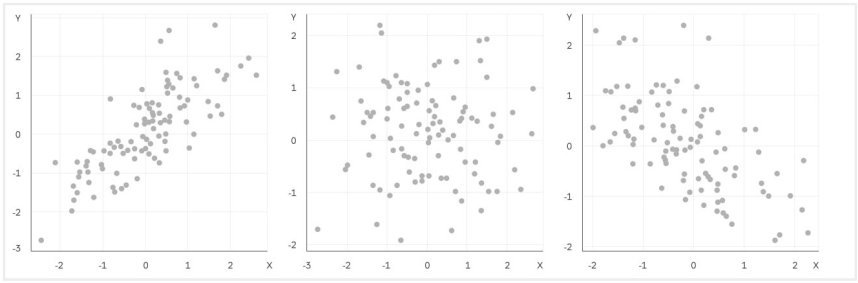
    — 4 500 клиентов оплатили заказ.

    — Из всех, кто добавил товар в корзину, только 50% его оплатили.

    Рассчитайте долю оформивших заказ клиентов среди пользователей, добавивших товар в корзину. Ответ округлите до второго знака после запятой.

    Ответ: 0,89 или 88,88 или 88,89

27. Перед вами три диаграммы рассеяния. Сопоставьте их с коэффициентами корреляции, которые получены на этих выборках. В ответ запишите буквы в порядке расположения картинок.



А. 0,26

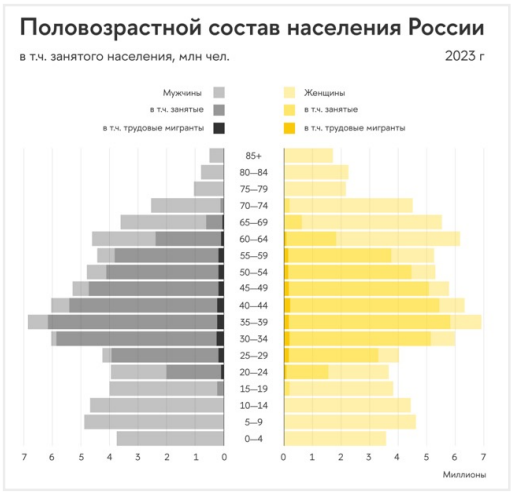
Б. 0

В. 0,75

Г. –0,56

Ответ: 1В, 2Б, 3Г, 4А

28. Перед вами иллюстрация половозрастной пирамиды России с указанием работающего (занятого) населения. Выберите верные утверждения для этой картинки.



— Доля занятых женщин в возрасте 55—59 лет ниже, чем у мужчин в том же возрасте

— Среди мужчин трудовых мигрантов больше, чем среди женщин

— Максимальная доля работающего населения среди мужчин приходится на возрастную группу от 25 до 29 лет

— В возрасте 60—64 лет количество женщин примерно вдвое больше, чем мужчин

— После 75 лет население состоит из женщин менее чем на 50%

29. Выберите все варианты ответов, в которых перечислены корректные методы борьбы с выбросами в данных.

A. Бутстрап

B. Стандартизация данных

C. Удаление значений, которые отклоняются сильнее чем 1,5 × IQR

D. Дисперсионный анализ (ANOVA)

DANO

ВЫСШАЯ ШКОЛА ЭКОНОМИКИ

Т БАНК

Вариант 4

1. По выборке из 256 значений посчитали выборочную дисперсию. После этого все значения увеличили в 9 раз. Как изменится выборочная дисперсия, если вычислить ее по новым значениям выборки?

   A. Не изменится

   B. Увеличится в 81 раз

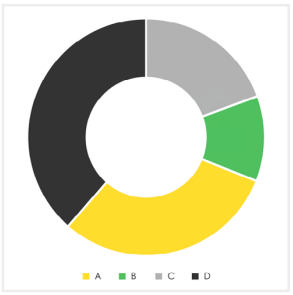
   C. Увеличится в 9 раз

   D. Уменьшится в 18 раз

   E. Нет верного варианта ответа среди вышеперечисленных

2. Сопоставьте названия типов визуализации с графиками:

1.



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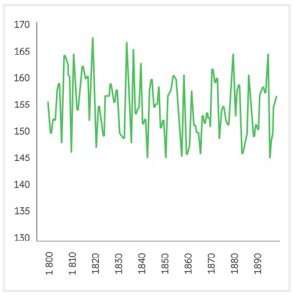

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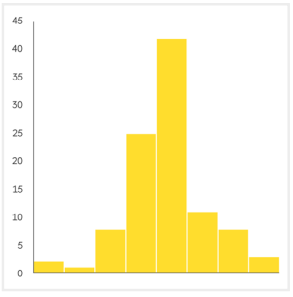


4.

1. Кольцевая диаграмма
2. График временного ряда
3. Гистограмма
4. Древовидная диаграмма

3. Исследователи проверили гипотезу о среднем значении с использованием Z-теста. Уровень значимости установлен равным 0,10. Как интерпретировать такое значение уровня значимости?

   A. Вероятность того, что альтернативная гипотеза верна, равна 10%

   B. Допустимая вероятность отвержения нулевой гипотезы, если она на самом деле верна, равна 10%

   C. Если нулевая гипотеза верна, в 10% случаев мы можем ошибочно отвергнуть ее при многократном повторении теста на семплированных данных из одной генеральной совокупности

   D. Вероятность ошибки второго рода равна 10%

   E. Если тест показал p-value = 0,97, вероятность того, что нулевая гипотеза верна, равна 10%

4. Сотрудники отдела прогнозирования и анализа данных изучили ежемесячные данные по ценам на рис и гречку за 2023 год. Средние цены оказались одинаковыми. Значения дисперсий: 49 ₽^{2} для сахара и 36 ₽^{2} для соли. Какое утверждение верное?

   A. Цена сахара в 2023 году была более изменчива, чем цена соли

   B. Цена соли в 2023 году была более изменчива, чем цена сахара

   C. Ежемесячные цены на сахар и соль всегда одинаковы

   D. Стандартные отклонения цен на сахар и соль одинаковы

5. Коля построил 97%-й доверительный интервал для средней цены на микроволновую печь и получил оценку от 3 000 до 9 000 ₽. Какая гипотеза не отвергается на 3%-м уровне значимости при использовании двусторонней альтернативной гипотезы?

   A. Дисперсия цены на микроволновую печь равна 4 000 ₽

   B. Средняя цена микроволновой печи равна 5 500 ₽

   C. Средняя цена микроволновой печи равна 2 500 ₽

   D. Нет верного варианта ответа

6. В автошколе после тестирования инструктор оценил среднее значение баллов за тест, набранных 10 сдавшими, и получил 85,5 балла. Оказалось, что 5 работ инструктор забыл внести в ведомость. После их внесения среднее по всем 15 работам оказалось равно 90 баллам. Максимум за тест можно было набрать 100 баллов. Что можно сказать о добавленных пяти работах?

   A. Инструктор забыл добавить работы учеников, набравших максимум

   B. Инструктор неправильно рассчитал среднее

   C. Среднее число баллов в добавленных пяти работах равно 99

   D. Стандартное отклонение набранных баллов одним учеником в среднем равно 10 баллам

7. Сопоставьте данные о питомцах, собранные в рамках опроса в ветклинике, с типами признаков, которыми они могут быть описаны.

   Пол питомца (самец/самка) — бинарный

   Вид животного (кошка/собака/...) — категориальный

   Размер (для определения переноски: крупные/средние/мелкие — порядковый

   Вес питомца — непрерывный

8. Петя из Солнечногорска провел опрос среди своих друзей в социальных сетях, чтобы узнать, хотят ли они купить новый VR-гаджет. У Пети 515 онлайн-друзей, 127 из которых живут с ним в одном городе. По результатам опроса 44 друга готовы купить гаджет, 59 друзей ответили «нет», а 24 затруднились ответить. Какие выводы можно сделать из этого опроса?

   A. В среднем в России 8,5% людей готовы купить новый VR-гаджет

   B. В среднем в России 34% людей готовы купить новый VR-гаджет

   C. В Солнечногорске примерно 34% людей готовы купить новый VR-гаджет

   D. Петя может выдвинуть гипотезу, что в среднем большинство людей не готовы покупать новый VR-гаджет

9. Волшебник Мерлин задумался, куда бы ему спрятать свои 5 000 золотых драконьих монет, чтобы они приумножились. У него есть два варианта: волшебный сундук, который за 6 месяцев увеличивает содержимое на 8% годовых, и проект по выращиванию говорящих тыкв с ежемесячной доходностью +1%, +2%, –3%, +0,5%, +0,2%, +4% за каждый из следующих шести месяцев. Куда стоит вложить свои монеты Мерлину, если доход он будет получать ежемесячно и сразу тратить на волшебные зелья (капитализация процентного дохода отсутствует)?

   A. Стоит поддержать проект по выращиванию говорящих тыкв: на нем можно заработать больше, чем с волшебным сундуком

   B. Стоит выбрать волшебный сундук: он принесет больше монет, чем проект с тыквами

   C. Все равно, куда вкладывать монеты: доходность вариантов одинаковая

   D. Недостаточно информации для принятия решения, так как нужно учитывать капитализацию

10. В Грибном царстве последние 8 лет все эльфы-производители шапок выпускали преимущественно красные шапки. В результате доля красных шапок в поставках магазинам составила 83%, а на все остальные цвета (зеленый, синий, желтый и др.) пришлось только 17%. Конверсия в покупку шапок каждого цвета одинаковая. Лесные мудрецы опубликовали исследование с несколькими утверждениями. Какие из этих утверждений можно считать достоверными на основе приведенных данных?

    A. Жители царства предпочитают красные шапки всем другим цветам

    B. Доля зеленых шапок составляет 2% или более

    C. Большинство шапок в царстве красные

D. Большинство продающихся шапок красного цвета

E. Нет верных ответов среди вышеперечисленных вариантов

11. В подводном колл-центре работают два сотрудника: осьминог Олли и рыба-клоун Немо. Олли благодаря своим восьми рукам справляется с одним звонком за 3 минуты. Немо, который любит рассказывать анекдоты, тратит на один звонок 9 минут. Их рабочая смена длится 6 часов. Сколько в среднем длится обслуживание одного клиента в колл-центре?

A. 2 минуты

B. 3 минуты

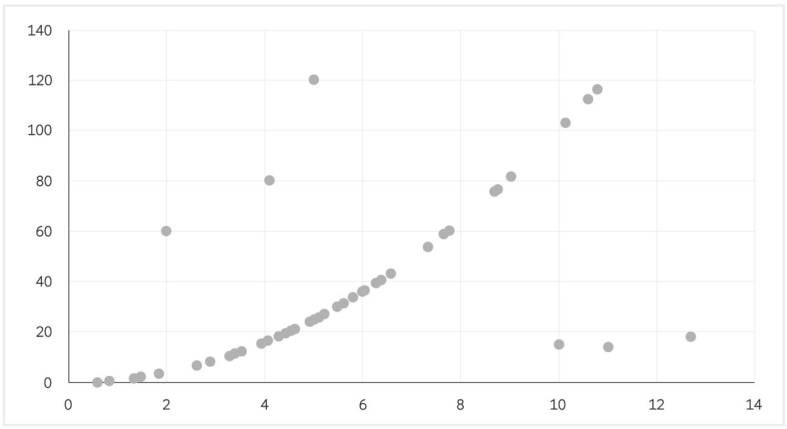
C. 4 минуты 30 секунд

D. 6 минут

12. В тайном хранилище у археолога найдено 17 загадочных артефактов, выстроенных в ряд так, что возраст каждого следующего артефакта на 1 год больше предыдущего. Известно, что средний возраст первых 11 артефактов равен 149 годам, а средний возраст последних 11 артефактов равен 155 годам. Найдите возраст самого старого артефакта в этом ряду.

Ответ: 160

13. Директор цеха составил график зависимости оплаты труда сотрудников (в тысячах рублей по вертикальной оси) от длины рабочей смены (в часах по горизонтальной оси). Выберите верные утверждения, которые соответствуют построенному графику.



A. Между длиной рабочей смены и оплатой есть линейная отрицательная связь

B. Коэффициент корреляции Спирмена между длиной рабочей смены и оплатой отрицательный

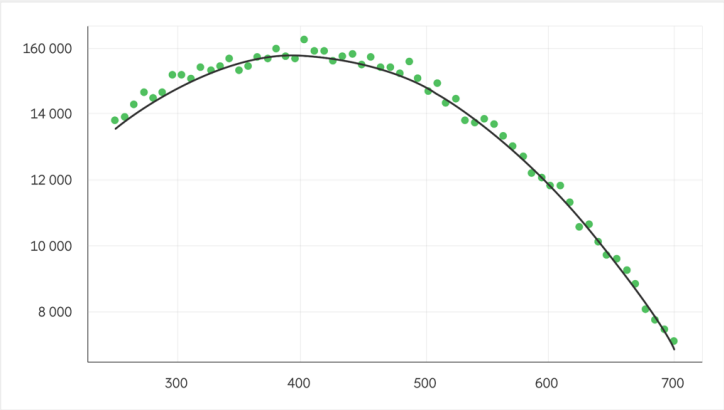
C. Меньше 25% сотрудников получают больше 100 тысяч рублей

D. Зависимость между оплатой труда и длительностью рабочей смены можно описать нелинейной функцией, исключив выбросы

14. Сеть кофеен решила оценить, как цена чашки капучино (в рублях) влияет на ежемесячную прибыль одной точки (в тысячах рублей). Для этого была построена квадратичная регрессионная модель:

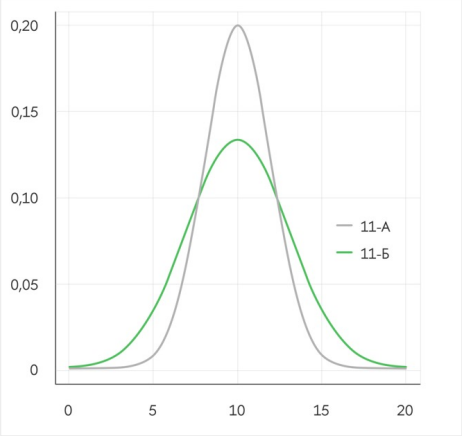
Прибыль = –0,1 × Стоимость чашки^2 + 80 × Стоимость чашки – 50

Коэффициент детерминации $R^2 = 0{,}81$. Определите цену на кофе, при которой прибыль будет максимальной, согласно оцененной модели. На графике зеленым цветом представлены фактические данные, черным — результат модели. Ответ округлите до целого.



Ответ: 400

15. Слава построил распределения количества игр в библиотеке магазина «Стем» для учащихся 11 «А» и 11 «Б». В результате он получил такой график распределений.



Что из этого графика может заключить Слава об учащихся?

A. В обоих классах среднее число игр у одного учащегося равно 5

B. В 11 «Б» разброс числа игр у учащихся меньше, чем в 11 «А»

C. Мода в 11 «А» равна медиане в 11 «Б»

D. Меньше половины учащихся 11 «А» обладают 11 и менее играми

16. В ходе исследования в волшебной лаборатории обнаружили: количество искр, создаваемых волшебной палочкой, зависит от типа используемого камня. Данные были собраны с помощью магических датчиков, которые автоматически фиксируют количество искр для каждого типа камня. В результате получили такое распределение средних значений.

Тип камня	**Среднее количество искр**
Простой булыжник	50
Полудрагоценный камень	60
Драгоценный камень	70

При помощи каких статистических инструментов можно подтвердить или опровергнуть наличие зависимости между числом искр и типом драгоценного камня?

A. Коэффициент корреляции Спирмена

B. Линейная регрессия

C. Логистическая регрессия

D. Т-тест

17. Аналитик получил набор данных со следующими переменными: пол слушателя, возраст слушателя, предпочтение музыкального стиля и уровень образования. Какие из этих переменных имеет смысл использовать в качестве зависимых, а какие — в качестве независимых?

A. Пол слушателя как независимая переменная, а остальные как зависимые

B. Возраст слушателя как зависимая переменная, а остальные как независимые

C. Предпочтение музыкального стиля как зависимая переменная, а остальные как независимые

D. Уровень образования как независимая переменная, а остальные как зависимые

18. Директор школы хочет наглядно представить распределение бюджета между различными направлениями: закупкой учебников, ремонтом классов, зарплатой учителей и внеклассными мероприятиями. Какой тип визуализации лучше всего подойдет для демонстрации доли каждого направления в общем бюджете?

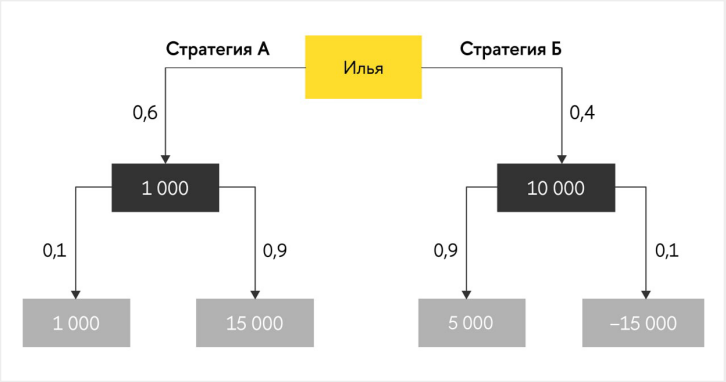
A. Диаграмма рассеяния

B. Линейный график

C. Пузырьковая диаграмма

D. Круговая диаграмма

19. Илья руководит стартапом по анализу данных. Ему нужно выбрать одну из дальнейших стратегий развития стартапа: А или Б. Обычно руководители стартапов с вероятностью 0,6 выбирают стратегию А и с вероятностью 0,4 — стратегию Б.



Каждая стратегия состоит из двух этапов. Стратегия А на первом этапе принесет 1 000 бонусов, а на втором — с вероятностью 0,1 еще 1 000 бонусов или с вероятностью 0,9 еще 15 000 бонусов. Стратегия Б на первом этапе принесет 10 000 бонусов, а на втором — с вероятностью 0,9 еще 5 000 бонусов или с вероятностью 0,1 Илья потеряет 15 000 бонусов.

Какую стратегию стоит выбрать Илье?

A. А

B. Стратегии равноценны

C. Б

D. Недостаточно информации для принятия решения

20. В ходе тренировок спортсмен показывал следующие результаты в метании копья за последние 4 сезона:

в первом сезоне — 41 м;

во втором сезоне — 45 м;

в третьем сезоне — 51 м;

в четвертом сезоне — 52 м.

Чему равен средний темп прироста дальности полета копья в процентах? Укажите ответ одним числом без дополнительных знаков и символов, округлите до целого.

Ответ: 8

21. В туристическом агентстве провели анализ предпочтений клиентов по видам отдыха и сезонам, когда они предпочитают отдыхать. Составили такую таблицу сопряженности.

Сезон	Пляжный	Горнолыжный	Экскурсионный
Лето	450	20	130
Зима	50	350	100
Межсезонье	100	50	250

Какие выводы можно сделать на основе данной таблицы?

A. Вероятность, что случайно выбранный клиент предпочитает пляжный отдых, выше, чем вероятность, что он выберет горнолыжный отдых

B. В межсезонье меньше клиентов, чем зимой

C. Горнолыжный и экскурсионный отдых одинаково популярны среди всех клиентов

D. Больше половины клиентов, выбирающих отдых летом, предпочитают пляжный отдых

E. Вероятность выбрать горнолыжный отдых выше зимой, чем летом

22. Инженер Наталья работает на орбитальной станции, где существует риск поломки оборудования из-за космического излучения. Вероятность того, что оборудование не выйдет из строя во время периода критической нагрузки, составляет 0,97. Критическая нагрузка на системы станции фиксируется раз в 7 дней. Через сколько дней вероятность того, что оборудование выйдет из строя, станет выше, чем вероятность, что оно останется работоспособным?

Укажите ответ одним числом без дополнительных знаков и символов, округлив в случае необходимости до целого значения.

Ответ: 161

23. На производственной линии за день упаковывается 100 коробок с продукцией. Медиана веса коробок составляет 500 г, а средний вес — 497 г. Инженер по качеству может добавить по 10 г продукции в X коробок, чтобы привести средний вес к медиане.

Какое минимальное количество коробок (X) нужно дополнить, чтобы средний вес стал равен медиане (500 г)?

Ответ: 30

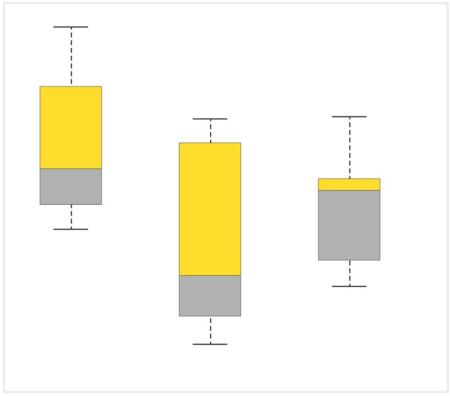
24. Перед вами графики Boxplot для трех выборок с параметром длины усов = 1,5 × IQR. Выберите верные утверждения.

— Среднее значение сильнее всего отклоняется от медианы во второй выборке

— Медианное значение в первой выборке наибольшее из представленных графиков

— В выборках есть выбросы, превышающие 1,5 × IQR

— Минимальное значение из всех значений по трем выборкам находится в третьей выборке



25. В туристическом клубе провели опрос среди 100 участников. Им задали два вопроса:

    1) Любите ли вы путешествовать?

    2) Увлекаетесь ли вы фотографией?

    По результатам опроса 70 человек любят путешествовать; 60 человек увлекаются фотографией; 10 человек не любят ни путешествовать, ни фотографировать.

    Сколько участников любят и путешествовать, и фотографировать?

    Ответ: 40

26. Компания проводит онлайн-продажи, и каждый клиент проходит несколько этапов:

    1. Посещение сайта.
    2. Добавление товара в корзину.
    3. Оформление заказа.
    4. Оплата заказа.

    Аналитики собрали следующую информацию за один день:

    — На сайт зашло 7 000 пользователей.

    — Из всех, кто заходил на сайт, 12% пользователей оформили заказ.

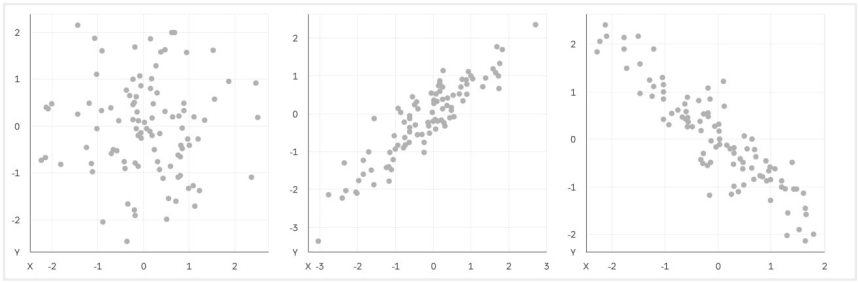
    — 351 клиент оплатил заказ.

    — Из всех, кто добавил товар в корзину, только 30% его оплатили.

    Рассчитайте долю оформивших заказ клиентов среди пользователей, добавивших товар в корзину. Ответ округлите до второго знака после запятой.

    Ответ: 0,72 или 71,79

27. Даны три диаграммы рассеяния. Сопоставьте их с коэффициентами корреляции, которые получены на этих выборках. В ответ запишите буквы в порядке расположения картинок.



А. 0

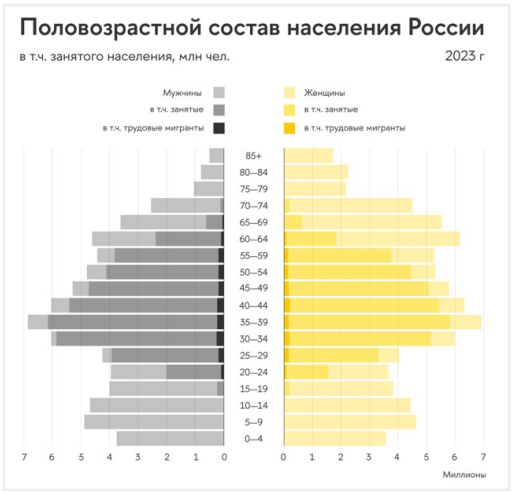
Б. 0,5

В. 0,9

Г. –0,9

Ответ: 1А, 2В, 3Г, 4Б

28. Перед вами иллюстрация половозрастной пирамиды России с указанием работающего (занятого) населения. Выберите верные утверждения для этой картинки.



— Доля занятых женщин в возрасте 55—59 лет ниже, чем у мужчин в том же возрасте

— Среди мужчин трудовых мигрантов больше, чем среди женщин

— Максимальная доля работающего населения среди мужчин приходится на возрастную группу от 25 до 29 лет

— В возрасте 60—64 лет количество женщин примерно вдвое больше, чем мужчин

— После 75 лет население состоит из женщин менее чем на 50%

29. Выберите все варианты ответов, в которых перечислены корректные методы борьбы с выбросами в данных.

A. Удаление 20% значений

B. Стандартизация данных

C. Удаление значений, которые отклоняются сильнее чем 1,5 × IQR

D. Использование T-тестов вместо Z-тестов